Welcome!

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WordPub

BECE

Mathematics

Past Questions & Answers

1990 - 2012

Junior High School
Years 1, 2 & 3

Compiled by:

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PUBLISHERS

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- Junior High School **students** – you are the ultimate reason for this work.

DEDICATION

To the Lord **Jesus Christ**, our Saviour and soon-coming King

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April 2012

MATHEMATICS 1

Objective Test
1 hour

1. If P = {2, 3, 5, 7}and Q = {2, 4, 6, 8}, find P∩Q

A. {2}
B. {3}
C. {4}
D. {5}

2. Which of the following numbers is an integer?

A. $-\frac{5}{4}$
B. $-\frac{2}{3}$
C. 0.5
D. 1

3. Find the Lowest Common Multiple (LCM) of $2^2 \times 3 \times 5^2$ and $2^3 \times 3^2 \times 5$

A. $2^2 \times 3 \times 5$
B. $2^2 \times 3^3 \times 5^2$
C. $2^3 \times 3 \times 5$
D. $2^3 \times 3^2 \times 5^2$

4. How many diagonals are in a rectangle?

A. 1
B. 2
C. 3
D. 4

5. Simplify $-4(3-5)+10-3(7+4)+30$

A. - 1
B. 15
C. 56
D. 65

6. An iron rod 15 m long is divided into 12 equal parts. How long is each part?

A. 0.80 m
B. 1.25 m
C. 1.50 m
D. 3.00 m

7. Convert 42 to a base two numeral.

A. 1001010_{two}
B. 1010010_{two}
C. 1010100_{two}
D. 101010_{two}

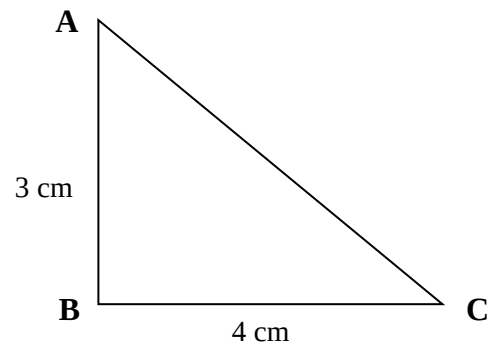
8. Simplify $\frac{5^7 \times 5^4}{5^2}$

A. 5^7
B. 5^8
C. 5^9
D. 5^{13}

9. A tank contains 400 litres of water. If 100 litres is used, what percentage is left?

A. 25%
B. 30%
C. 40%
D. 75%

10.



NOT DRAWN TO SCALE

Triangle ABC is a right-angled triangle. Find the length of AC.

A. 1 cm
B. 5 cm
C. 7 cm
D. 12 cm

11. Arrange the following fractions in descending order of magnitude: $\frac{2}{3}, \frac{5}{7}, \frac{2}{5}, \frac{1}{2}$

A. $\frac{5}{7}, \frac{2}{5}, \frac{2}{3}, \frac{1}{2}$
B. $\frac{5}{7}, \frac{2}{3}, \frac{1}{2}, \frac{2}{5}$
C. $\frac{1}{2}, \frac{2}{5}, \frac{5}{7}, \frac{2}{3}$
D. $\frac{1}{2}, \frac{5}{7}, \frac{2}{3}, \frac{2}{5}$

12. Find the image of 3 under the mapping, $x \rightarrow 10 - 2x$

A. 4
B. 5
C. 8

D. 16

13. Simplify $\frac{1}{3} + \frac{1}{9} + \frac{1}{27}$

A. $\frac{5}{27}$

B. $\frac{7}{27}$

C. $\frac{11}{27}$

D. $\frac{13}{27}$

14. If $2x = 5(x - 2) + 7$, find the value of x

A. $-5\frac{2}{3}$

B. –1

C. 1

D. $5\frac{2}{3}$

The table below shows the day and night temperatures of a town during a week.
Use it to answer Questions **15** *and* **16**

Week day	Temperatures (°C)	
	Day	Night
Monday	33	24
Tuesday	29	25
Wednesday	32	23
Thursday	34	26
Friday	32	24
Saturday	30	24
Sunday	30	25

15. Find, correct to **one** decimal place, the average day temperature for the week

A. 24.4 °C

B. 30.2 °C

C. 31.4 °C

D. 32.2 °C

16. On which day was the change in temperature the **least**?

A. Monday

B. Saturday

C. Sunday

D. Tuesday

17. A box contains 30 identical balls of which 16 are white and the rest yellow. If a girl picks a ball at random from the box, what is the probability that it is a yellow ball?

A. $\frac{1}{16}$

B. $\frac{7}{15}$

C. $\frac{8}{15}$

D. $\frac{7}{8}$

18. Find the truth set of $\frac{1}{4}(x + 3) \leq 2x - 1$

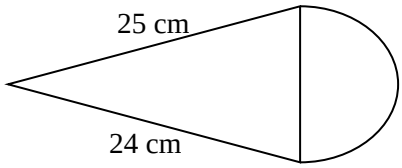
A. $\{x: x \leq -3\}$

B. $\{x: x \leq -1\}$

C. $\{x: x \geq 1\}$

D. $\{x: x \geq 3\}$

19. The perimeter of the figure below is 71 cm. Find the diameter of the semi-circular portion. [Take = π]



NOT DRAWN TO SCALE

A. 1.0 cm

B. 3.5 cm

C. 7.0 cm

D. 14.0 cm

20. Simplify $\frac{3x}{4} - \frac{x-y}{3}$

A. $\frac{5x-4y}{12}$

B. $\frac{13x-4y}{12}$

C. $\frac{5x+4y}{12}$

D. $\frac{13x+4y}{12}$

21. Kojo is 20% heavier than Afua. If Kojo weighs 6 kg, what is Afua's weight?

A. 4.8 kg

B. 5.0 kg

C. 6.0 kg

D. 7.2 kg

22. Find the volume of a cylinder of height 3 cm and radius 2 cm.

A. 6π cm^3

B. 12π cm^3

C. 18π cm^3

D. 24π cm^3

23. Given the points S(5, -2) and T(3, 2), calculate the gradient of the line ST.

A. - 2

B. $-\frac{3}{5}$

C. $\frac{1}{2}$

D. 2

24. Kofi invested GH¢ 150,000 at 2.5% per annum simple interest. How long will it take this amount to yield an interest of GH¢11,250.00?

A. 2 years

B. 3 years

C. 4 years

D. 5 years

25. Express 3.75 as a mixed fraction.

A. $3\frac{1}{5}$

B. $3\frac{1}{4}$

C. $3\frac{1}{3}$

D. $3\frac{3}{4}$

26. A map is drawn to the scale 1:100,000. What distance in kilometres is represented by 5 cm on the map?

A. 0.5 km

B. 5.0 km

C. 50.0 km

D. 500.0 km

27. Given that **r** = $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$ and **s** = $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$, find **r** – 2**s**

A. $\begin{pmatrix} -5 \\ 1 \end{pmatrix}$

B. $\begin{pmatrix} -5 \\ 10 \end{pmatrix}$

C. $\begin{pmatrix} -2 \\ 10 \end{pmatrix}$

D. $\begin{pmatrix} -1 \\ 10 \end{pmatrix}$

28. Esi went to the market and bought 500 g of meat, 850 g of fish and 900 g of eggs. What is the total weight of the items she bought in kilograms?

A. 2.20 kg

B. 2.25 kg

C. 2.35 kg

D. 22.50 kg

29. A watch gains $1\frac{1}{2}$ minutes per hour. What is the total time gained from 12 noon to 12 midnight in a day?

A. 9 minutes

B. 15 minutes

C. 18 minutes
D. 36 minutes

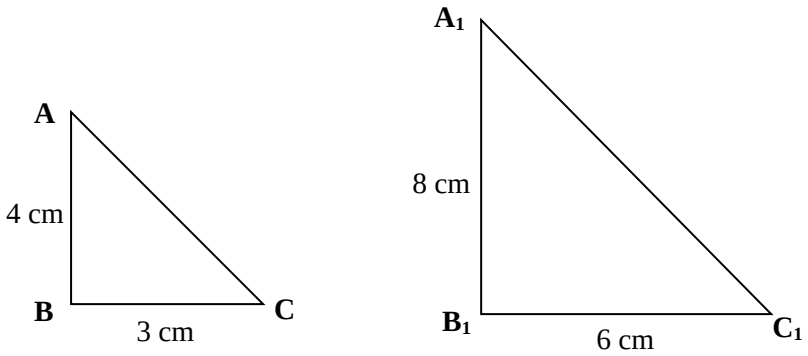
30. A printing machine prints 600 books in 3 hours. How many books will the machine print in 5 hours?
A. 360 books
B. 1000 books
C. 1800 books
D. 3000 books

31. The bearing of Atoru from Busase is 275°. What is the bearing of Busase from Atoru?
A. 180°
B. 175°
C. 095°
D. 075°

32. In a class of 24 pupils, 10 study French only and 8 study English only. If each pupil studies at least one of the two subjects, how many study English?
A. 12
B. 14
C. 16
D. 18

33. Convert 84 to a base five numeral.
A. 4130_{five}
B. 3014_{five}
C. 314_{five}
D. 114_{five}

34. In the diagrams below, triangle $A_1B_1C_1$ is an enlargement of triangle ABC. Determine the scale factor.



NOT DRAWN TO SCALE

A. 0.50
B. 0.75
C. 2.00
D. 4.00

35. Find the least number that must be added to 308 to make it divisible by 19.
A. 4
B. 7

C. 15
D. 18

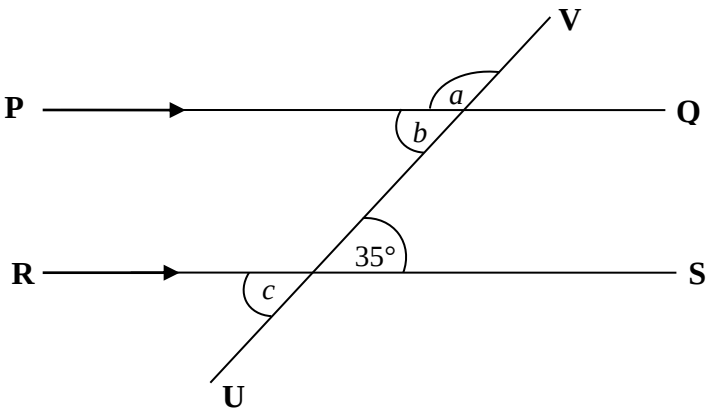
36. In a school of 940 pupils, the number of girls exceeds the number of boys by 150. How many girls are there in the school?

A. 620
B. 545
C. 470
D. 395

37. Which of the following fractions is equivalent to $\frac{3}{5}$?

A. $\frac{21}{30}$
B. $\frac{12}{20}$
C. $\frac{15}{45}$
D. $\frac{6}{15}$

In the diagram below, line PQ is parallel to RS and UV is a line drawn through PQ and RS.
Use the diagram to answer Questions **38** *and* **39**.



NOT DRAWN TO SCALE

38. Find angle *a*.

A. 35°
B. 55°
C. 135°
D. 145°

39. Angle ***b*** and angle ***c*** are

A. alternate angles
B. vertically opposite angles
C. corresponding angles
D. interior opposite angles

40. Expand $-x(3-2x)$

A. $-2x^2 - 3x$

B. $2x^2 - 3x$
C. $-2x^2 + 3x$
D. $2x^2 + 3x$

April 2012

MATHEMATICS 1

OBJECTIVE TEST

SOLUTIONS

1. A. {2}

2. D. 1

3. D. $2^3 \times 3^2 \times 5^2$

4. B. 2

5. B. 15

6. B. 1.25 m

7. D. 101010_{two}

8. C. 5^9

9. D. 75%

10. B. 5 cm

11. B. $\frac{5}{7}, \frac{2}{3}, \frac{1}{2}, \frac{2}{5}$

12. A. 4

13. D. $\frac{13}{27}$

14. C. 1

15. C. 31.4 °C

16. D. Tuesday

17. B. $\frac{7}{15}$

18. C. $\{x: x \geq 1\}$

19. D. 14.0 cm

20. C. $\frac{5x+4y}{12}$

21. B. 5.0 kg

22. B. 12π cm^3

23. A. - 2

24. B. 3 years

25. D. $3\frac{3}{4}$

26. B. 5.0 km

27. B. $\begin{pmatrix}-5\\10\end{pmatrix}$

28. B. 2.25 kg

29. C. 18 minutes

30. B. 1000 books

31. C. 095°

32. B. 14

33. C. 314_{five}

34. C. 2.00

35. C. 15

36. B. 545

37. B. $\frac{12}{20}$

38. D. 145°

39. C. corresponding angles

40. B. $2x^2 - 3x$

April 2012

MATHEMATICS 2

Essay
1 hour

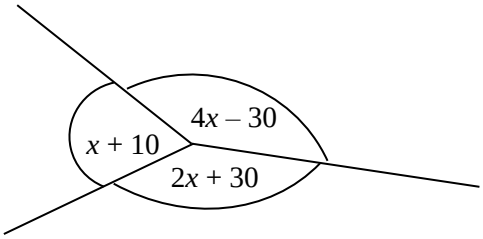
1. **(a)** Evaluate $\frac{0.035 \times 1.02}{0.00015}$, leaving the answer in standard form.

(b) An amount of GH¢4,200.00 was shared between Aba and Kwame. If Aba had $\frac{5}{7}$ of the amount,

(i) how much did Kwame receive?

(ii) what percentage of Aba's share did Kwame receive?

(c) Find the value of x in the diagram below.



2. **(a)** A car consumes a gallon of petrol for every 30 km drive. The driver of the car set out on a journey of 420 km with 10 gallons of petrol in the fuel tank.

(i) How many more gallons of petrol will be needed to complete the journey?

(ii) Find the cost of the petrol used for the journey of 420 km if a gallon of petrol costs GH¢5.50

(b) The average number of spectators at a football competition for the first five days was 3,144. The attendance on the sixth day was 3,990. Find the

(i) The total attendance on the first five days

(i) The average attendance for the 6 days

(c) The area enclosed by a square garden is 121 m^2. What is the distance around the garden?

3. **(a)** The table below shows the number of students who scored more than 80% in the listed subjects

Subject	Number of students
Biology	26
Physics	30
Chemistry	32
French	38
Geography	24
History	30

(i) Draw a pie chart for the distribution

(ii) What is the probability that a student chosen at random from the distribution offers Chemistry?

(b) A woman bought 210 oranges for GH¢7.50. She sold all of them at 3 for 15 Gp. Find the

(i) total selling price of the oranges

(ii) percentage profit

4. The marks scored by some students in a Mathematics test are as follows:

3	3	5	6	3	4	7	8	3	4
5	4	7	4	3	7	4	6	4	8
4	5	6	3	8	4	5	6	4	5

(a) Construct a frequency distribution table for the scores.

(b) Using the table, find for the distribution, the

(i) mode;

(ii) mean, correct to one decimal place

(iii) median

5. **(a)** **(i)** Find the least Common Multiple (L.C.M.) of 9, 18 and 16.

(ii) Arrange $\frac{8}{9}, \frac{7}{18}$ *and* $\frac{10}{16}$ in **ascending** order of magnitude

(b) Using a ruler and a pair of compasses only,

(i) construct a triangle *PQR* with length *PQ* = 10 cm, angles *QPR* = 45° and *PQR* = 60°.

(ii) Construct the perpendicular bisectors of *PR* and *RQ* to meet at *T*.

(iii) Measure the length of *TP*.

6. **(a)** **(i)** Using a scale of 2 cm to 1 unit on both axes, draw two perpendicular axes Ox and Oy on a graph sheet.

(ii) Mark on the same graph sheet, the x-axis from -5 to 5 and y-axis from -6 to 6.

(iii) Plot the points $P(4, 2)$, $Q(2, 5)$ and $R(2, 2)$. Join the points P, Q, R to form a triangle PQR

(iv) Using the x-axis as a mirror line, draw the image $P_1Q_1R_1$ of the triangle PQR such that $P \rightarrow P_1$, $Q \rightarrow Q_1$, $R \rightarrow R_1$

(v) Write down the coordinates of P_1, Q_1 and R_1.

(vi) Translate triangle PQR by the vector $\begin{pmatrix} -1 \\ -1 \end{pmatrix}$ such that $P \rightarrow P_2$, $Q \rightarrow Q_2$, $R \rightarrow R_2$

(vii) Label the vertices of triangle $P_2Q_2R_2$

April 2012

MATHEMATICS 2

OBJECTIVE TEST

SOLUTIONS

1. (a) $\dfrac{0.035 \times 1.02}{0.00015}$

Method 1 (Converting to common fractions)

$$\frac{0.035 \times 1.02}{0.00015}$$

$$= (0.035 \times 1.02) \div 0.00015$$

$$= \left(\frac{35}{1000} \times \frac{102}{100}\right) \div \frac{15}{100000}$$

$$= \left(\frac{35}{1000} \times \frac{102}{100}\right) \times \left(\frac{100000}{15}\right)$$

$$= \frac{35 \times 102}{15}$$

$$= \frac{7 \times 34}{1}$$

$$= 238 = \underline{2.38 \times 10^2}$$

NB:
$\frac{X}{d} = X \div d$

Eg,
$\frac{12}{3} = 12 \div 3$

1. (a) $\dfrac{0.035 \times 1.02}{0.00015}$

Method 2 (Converting to whole numbers)

Multiplying both numerator and denominator by 100000
(to convert to whole numbers)

$$\frac{0.035 \times 1.02}{0.00015} \times \frac{100000}{100000}$$

NB
Multiplying by 100,000 shifts the decimal point 5 places to the right (in both numerator and denominator)

$$= \frac{35 \times 102}{15}$$

$$= \frac{7 \times 34}{1}$$

$$= 238$$

$$= \underline{2.38 \times 10^2}$$

1. (b) (i) **Method 1**

Aba's share = $\frac{5}{7}$ of the amount,

$$= \frac{5}{7} \times GHc\,4{,}200$$

= 5 × GHc 600

= GHc 3,000

NB:
7 divides ('goes into') 4,200 to give 600

Therefore **Kwame's** share = GHc 4,200 – GHc 3,000

= **GHc 1,200**

(i) **Method 2**

Kwame's fraction = $\frac{7}{7} - \frac{5}{7} = \frac{2}{7}$

Therefore **Kwame's** share = $\frac{2}{7} \times GHc\,4{,}200$

= 2 × GHc 600

= **GHc 1,200**

(ii) Percentage of Aba's share that Kwame received

$$= \frac{\text{Kwame's share}}{\text{Aba's share}} \times 100\%$$

$$= \frac{1200}{3000} \times 100\%$$

= **40%**

1.(c) $(4x - 30°) + (2x + 30°) + (x + 10°) = 360°$

$4x + 2x + x - 30° + 30° + 10° = 360°$

$7x + 10° = 360°$

$7x = 360° - 10°$

$7x = 350°$

$\frac{7x}{7} = \frac{350^o}{7}$

x = 50°

NB:
One complete revolution (circle) = 360°

2. (a) (i) Let g = no. of gallons needed for the entire journey

Approach 1 (Equivalent Fractions)

	Distance (km)	Petrol (Gallon)
	30	1
The entire journey	420	g

From the table above, $\frac{30}{420} = \frac{1}{g}$

Cross multiplying, we have $30 \times g = 1 \times 420$

$\Rightarrow \quad \frac{30g}{30} = \frac{1 \times 420}{30}$

$\Rightarrow \quad g = 14$

Amount needed for the entire journey = 14 gallons

He already has 10 gallons; therefore he needs **4 gallons more**

2. (a) (i)

Approach 2 (The rule: If more, less divides …)

If 30 km → 1

Then 420 km → ? (more)

If more, less (i.e., 30) divides; ∴ we have

$\frac{420}{30} \times 1 = 14$

Amount needed for the entire journey = 14 gallons

He already has 10 gallons, therefore he needs **4 gallons more**

2. (a) (i)

Approach 3 (Equivalent ratios)

30 : 1 = 420 : g

$\Rightarrow \quad \frac{30}{1} = \frac{420}{g}$

$\Rightarrow \quad 30 \text{ x } g = 420 \text{ x } 1$

$$\Rightarrow \quad \frac{30 \times g}{30} = \frac{420 \times 1}{30}$$

$$\Rightarrow \quad g = 14$$

Amount needed for the entire journey = 14 gallons
He already has 10 gallons, therefore he needs **4 gallons more**

2. (a) (ii) **Let THE COST OF PETROL USED FOR THE JOURNEY = c**

Approach 1 (Common Logic)

If 1 gallon → GHc 5.50

Then 14 gallons → 14 × GHc 5.50 = **GHc 77.00**

Approach 2 (Equivalent Fractions)

	Petrol (gallons)	Cost (GHc)
	1	5.50
The entire journey	14	c

From the table above, $\frac{1}{14} = \frac{5.50}{c}$

Cross multiplying, we have $1 \times c = 5.50 \times 14$

$$\Rightarrow \quad c = 77$$

Cost of petrol needed for the journey = **GHc 77.00**

2. (b) (i) **Total attendance on first five days** = 5 × average attendance

= 5 × 3144

= **15720 spectators**

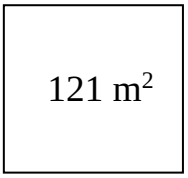
(ii) Average attendance for the 6 days $= \frac{\text{Total attendance for the 6 days}}{\text{No. of days}}$

$$= \frac{15720 + 3990}{6}$$

$$= \frac{19710}{6}$$

= **3285 spectators**

2. (c)



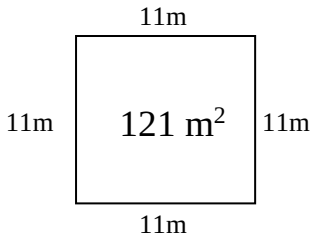
Area of the square = $121m^2$

$L \times L = 121m^2$

$L^2 = 121m^2$

$L = \sqrt{121m^2}$

$L = 11$ m



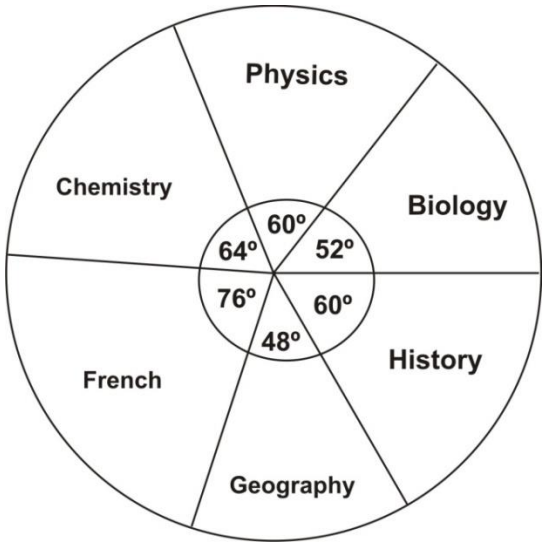
The distance around the garden (perimeter) = 4×11 m

= **44 m**

3. (a) (i)

Subject	**No. of students**	**Angle of sector**
Biology	26	$\frac{26}{180} \times 360^o = 52^o$
Physics	30	$\frac{30}{180} \times 360^o = 60^o$
Chemistry	32	$\frac{32}{180} \times 360^o = 64^o$
French	38	$\frac{38}{180} \times 360^o = 76^o$
Geography	24	$\frac{24}{180} \times 360^o = 48^o$
History	30	$\frac{30}{180} \times 360^o = 48^o$
TOTAL	**180**	**360°**

Pie chart showing the number of students who scored more than 80% in the listed subjects



3 (a) (ii) **Probability** = $\dfrac{\text{No. of students who offer Chemistry}}{\text{Total no.}}$

= $\dfrac{32}{180}$

= $\dfrac{8}{45}$

Approach 2 (using angle of sector)

Probability = $\dfrac{\text{Angle for Chemistry}}{\text{Total Angle.}}$

= $\dfrac{64^o}{360^o}$

= $\dfrac{8}{45}$

3 (b) (i)

Approach 1 (Unitary Approach – the value of one item)

If 3 oranges sell for 15 Gp,

then 1 orange sells for $\dfrac{15}{3}$ = 5 *Gp*

hence, 210 oranges would sell for 210 × 5 Gp

= 1050 Gp = GHc 10.50

Approach 2 (Equivalent Fractions)

No. of oranges	SP (in Gp)
3	15
210	y

From the table above, $\frac{3}{210} = \frac{15}{y}$

Cross multiplying, we have $3 \times y = 15 \times 210$

$\Rightarrow \quad \frac{3y}{3} = \frac{15 \times 210}{3}$

$\Rightarrow \quad y = 5 \times 210$

$\Rightarrow \quad y = 1050 \text{ Gp}$

$= \underline{\text{GHc } 10.50}$

Approach 3 (The rule: If more, less divides …)

If 3 → 15 Gp

Then 210 → ? (more)

If more, less (i.e., 4) divides; ∴ we have

$\frac{210}{3} \times 15\text{Gp} \quad \Rightarrow \quad 70 \times 15\text{Gp}$

$= \underline{1050 \text{ Gp}} \quad = \underline{\text{GHc } 10.50}$

Approach 4 (Equivalent ratios)

$3 : 15 = 210 : y$

$\Rightarrow \quad \frac{3y}{3} = \frac{15 \times 210}{3}$

$= \underline{1050 \text{ Gp}} \quad = \underline{\text{GHc } 10.50}$

3. (b) (ii) Profit = Total SP – Total CP

= GHc 10.50 – GHc 7.50

= GHc 3.00

Percentage Profit = $\frac{\text{Profit}}{CP} \times 100\%$

$$= \frac{3}{7.50} \times 100\%$$

$$= \frac{300}{7.50}\ \% \quad = \frac{3000}{75}\ \%$$

$$= \underline{\mathbf{40\%}}$$

4. (a) Frequency distribution table

Mark (x)	Tally	Frequency (f)
3	𝍸 /	6
4	𝍸 ////	9
5	𝍸	5
6	////	4
7	///	3
8	///	3

4. (b) (i) Mode = The mark with the highest frequency = 4

(ii) Mean

Mark (x)	Tally	Frequency (f)	fx
3	𝍸 /	6	18
4	𝍸 ////	9	36
5	𝍸	5	25
6	////	4	24
7	///	3	21
8	///	3	24
		$\Sigma f = 30$	$\Sigma fx = 148$

The mean mark $= \frac{\Sigma f x}{\Sigma f} = \frac{148}{30}$

Mean $= 4.9$ (correct to 1 d.p.)

4.(b) (iii) Median

Approach1

Listing all the given marks in ascending order, we have
3,3,3,3,3,3, 4,4,4,4,4,4,4,4, **4, 5**, 5,5,55, 6,6,6,6, 7,7,7, 8,8,8

The middle marks are 4 and 5

$\therefore$ The median $= \frac{4 + 5}{2} = \frac{9}{2} =$ 4.5 or 4

Approach 2

The median

= the mark at the ½(Σf+1)th position

$\Rightarrow$ the ½ (30+1)th position

= ½ (31) = 15½th position

The mark at the 15½th position

= the average of the marks at the 15th and 16th positions

$= \frac{4 + 5}{2} = \frac{9}{2} =$ 4.5 or 4

5. (a) (i) LCM of 9, 18 and 16

Approach 1 (Successive division by prime factors)

2	**9**	**18**	**16**
2	9	9	8
2	9	9	4
2	9	9	2
3	9	9	1
3	3	3	1
	1	1	1

NB
2 cannot divide ('go into') 9 without a remainder, therefore, 9 is repeated.
2 'into' 18 = 9,
2 'into' 16 = 8

The process is repeated till we obtain 1 for all.

The LCM = $2 \times 2 \times 2 \times 2 \times 3 \times 3$

$= 16 \times 9 = \underline{\underline{144}}$

(a) (i) **Approach 2 (Listing multiples)**

9 = {9, 18, 27, 36, 45, 54, 63, 72, 81, 90, 99, 108, 117, 126, 135, **144**, …}

18 = {18, 36, 54, 72, 90, 108, 126, **144**, …}

16 = {16, 32, 48, 64, 80, 96, 112, 128, **144**, …}

Hence the Least Common Multiple **(LCM)** = **144**

5. (a) (ii) $\frac{8}{9}, \frac{7}{18}$ *and* $\frac{10}{16}$ in ascending order of magnitude

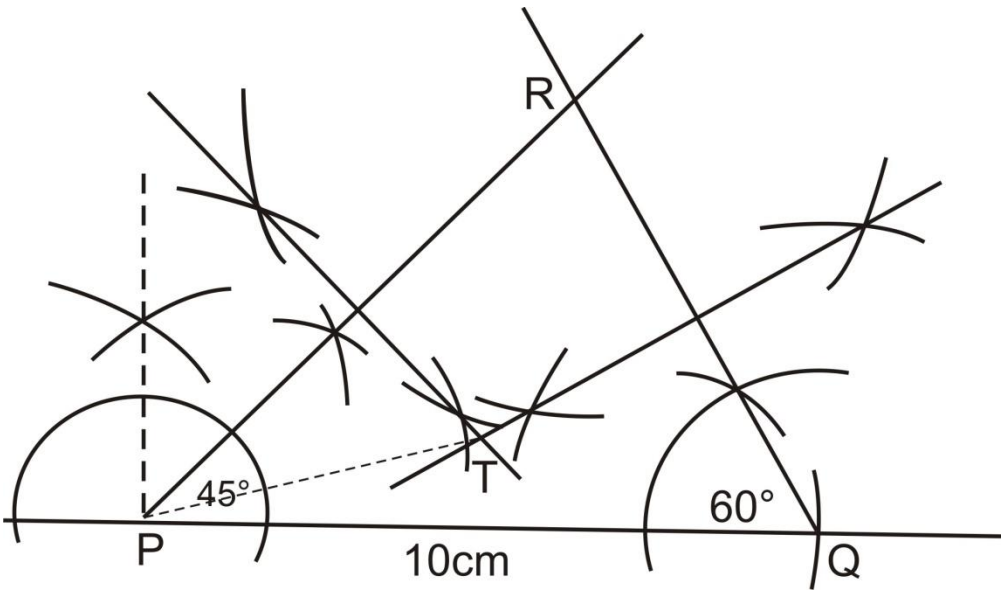
$= \frac{8}{9}, \frac{7}{18}, \frac{10}{16}$

$= \frac{16(8),\ 8(7),\ 9(10)}{144}$

$= \frac{128,\ 56,\ 90}{144}$

Therefore in ascending order, we have $\underline{\underline{\frac{7}{18}, \frac{10}{16}, \frac{8}{9}}}$

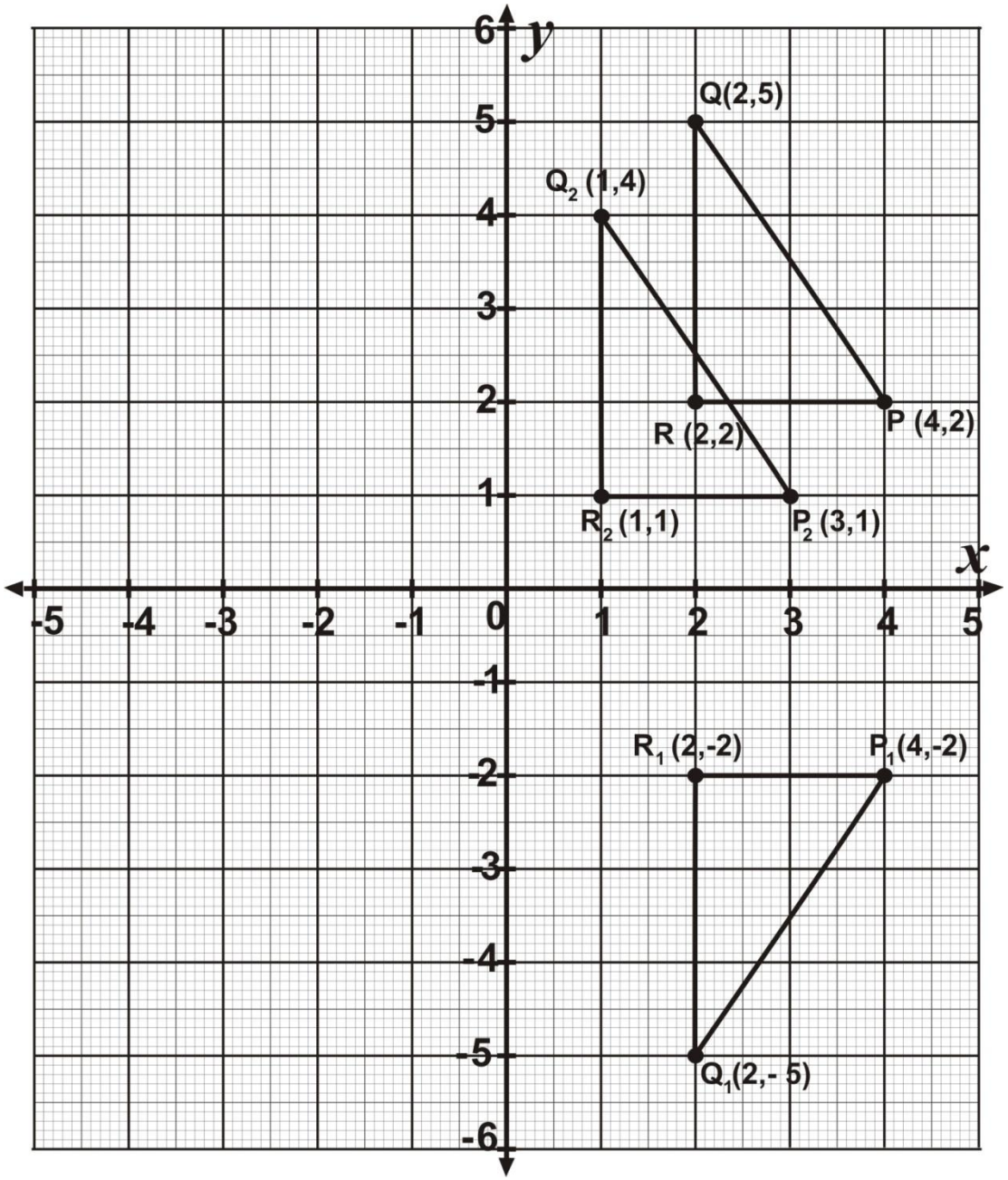
5. (b) (i), (ii)



The length of TP = 5.2 cm (approx.)

6. (a)

Approach 1 (By Inspection / Construction)



Approach 2 (The rule / formula)

(a) (iv) Reflecting (x, y) in the x-axis

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} x \\ -y \end{pmatrix}$$

$$\Rightarrow \quad OP\begin{pmatrix} 4 \\ 2 \end{pmatrix} \rightarrow OP_1\begin{pmatrix} 4 \\ -2 \end{pmatrix}, \quad \therefore P_1(4,-2)$$

$\Rightarrow \quad OQ\begin{pmatrix}2\\5\end{pmatrix} \rightarrow OQ_1\begin{pmatrix}2\\-5\end{pmatrix}, \quad \therefore Q_1(2,-5)$

$\Rightarrow \quad OR\begin{pmatrix}2\\2\end{pmatrix} \rightarrow OC_1\begin{pmatrix}2\\-2\end{pmatrix}, \quad \therefore C_1(2,-2)$

$\therefore$ Plot and join $P_1(4, -2)$, $Q_1(2, -5)$ and $C_1(2, -2)$ as the image of triangle PQR under a reflection in the x- axis

(*as shown in the diagram above*)

(a) (vi) Translating PQR by vector $\begin{pmatrix}-1\\-1\end{pmatrix}$

Translating (x, y) by vector $\begin{pmatrix}a\\b\end{pmatrix}$

$$\begin{pmatrix}x\\y\end{pmatrix} \rightarrow \begin{pmatrix}x+a\\y+b\end{pmatrix}$$

$$OP\begin{pmatrix}4\\2\end{pmatrix} \rightarrow OP_2\begin{pmatrix}4+-1\\2+-1\end{pmatrix} = \begin{pmatrix}3\\1\end{pmatrix}, \quad \therefore P_2(3, 1)$$

$$OQ\begin{pmatrix}2\\5\end{pmatrix} \rightarrow OQ_2\begin{pmatrix}2+-1\\5+-1\end{pmatrix} = \begin{pmatrix}1\\4\end{pmatrix}, \quad \therefore Q_2(1, 4)$$

$$OR\begin{pmatrix}2\\2\end{pmatrix} \rightarrow OR_2\begin{pmatrix}2+-1\\2+-1\end{pmatrix} = \begin{pmatrix}1\\1\end{pmatrix}, \quad \therefore R_2(1, 1)$$

$\therefore$ Plot and join $P_2(3, 1)$, $Q_2(1, 4)$ and $R_2(1, 1)$ as the image of triangle PQR under a translation by vector $\begin{pmatrix}-1\\-1\end{pmatrix}$

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MATHEMATICS 1

OBJECTIVE TEST

1 hour

1. Which of the following is the set of prime factors of 12?
A. {2, 3}
B. {1, 2, 3}
C. {1, 2, 4, 6}
D. {2, 3, 4, 6}

2. Expand 3a (a – 4b)
A. 3a – 12ab
B. $3a^2$ – 12ab
C. $3a^2$ – 12b
D. $3a^2$ – 12a

3. Express 5 as a percentage of 4
A. 125%
B. 120%
C. 25%
D. 20%

4. Express 2700 as a product of prime numbers.
A. $2^2 \times 3^2 \times 5^2$
B. $2 \times 3^3 \times 5^2$
C. $2^2 \times 3^3 \times 5^2$
D. $2 \times 3^2 \times 5^3$

5. The ratio of mangoes to oranges in a basket is 3:2. If there are 36 mangoes, how many oranges are in the basket?
A. 90
B. 60
C. 24
D. 12

6. Express 0.125 as a fraction in its lowest form.
A. $\frac{1}{8}$
B. $\frac{1}{9}$
C. $\frac{1}{12}$
D. $\frac{1}{16}$

7. Convert 222_{five} to a number in base ten.

A. 30
B. 52
C. 60
D. 62

8. If A = {18, 19, 20} and B = {15, 16, 17}, find A∩B

A. {15, 16, 17, 18, 19, 20}
B. {15, 16, 18, 19}
C. {18, 19}
D. {}

9. Simplify $3^9 \div 3^3$

A. 3^{27}
B. 3^{12}
C. 3^6
D. 3^3

10. An article which costs GH¢ 25.00 was sold for GH¢ 35.00. Find the percentage profit made.

A. 10%
B. 28%
C. 40%
D. 70%

11. Factorize completely $b^2 + fb - mb - fm$

A. $(b - f)(b - m)$
B. $(b + f)(b - m)$
C. $(b + f)(m - b)$
D. $(b + f)(m + b)$

12. Simplify: -13 – (-3) + (-10)

A. -26
B. -20
C. -10
D. - 6

13. Find the HCF of $3^3 \times 5^2$ and $3^2 \times 5^4$

A. $3^2 \times 5^2$
B. $3^3 \times 5^2$
C. $3^2 \times 5^4$
D. $3^5 \times 5^6$

14. State the rule for the mapping

x	1	2	3	4
↓	↓	↓	↓	↓
y	15	30	45	60

A. $x \rightarrow 15x$

B. $x \rightarrow 15 + x$

C. $x \rightarrow \frac{15}{x}$

D. $x \rightarrow 10 + 5x$

15. Solve the inequality $x - \frac{1}{3} \geq \frac{2}{3} - x$

A. $x \leq \frac{1}{2}$

B. $x \leq \frac{2}{3}$

C. $x \geq \frac{1}{2}$

D. $x \geq \frac{2}{3}$

16. Find the area of a square, if its perimeter is 28 cm.

A. 784 cm^2

B. 196 cm^2

C. 49 cm^2

D. 14 cm^2

17. Simplify: $\frac{1}{3}\left(\frac{1}{2} - \frac{1}{3}\right) - \frac{1}{3}\left(\frac{1}{3} - \frac{1}{2}\right)$

A. $-\frac{1}{9}$

B. $-\frac{1}{18}$

C. $\frac{1}{18}$

D. $\frac{1}{9}$

18. Make n the subject of the relation $\theta = 180 - \frac{360}{n}$

A. $\frac{\theta+180}{2}$

B. $\frac{\theta-180}{2}$

C. $\frac{360}{180-\theta}$

D. $\frac{360}{180+\theta}$

19. If $R = \frac{h}{2} + \frac{d^2}{8h}$, find R when d = 8 and h = 6.

A. $3\frac{1}{6}$

B. $4\frac{1}{3}$

C. $4\frac{3}{4}$

D. $4\frac{9}{16}$

20. Eight copies of a book cost GH¢ 16.00. Find the cost of 5 copies.

A. GH¢ 2.00
B. GH¢ 3.20
C. GH¢ 5.00
D. GH¢ 10.00

21. Solve the equation $\frac{1}{5}(2+y) = \frac{1}{2}(y-1)$

A. -3
B. $\frac{3}{4}$
C. $\frac{5}{3}$
D. 3

22. The gradient of the straight line that passes through points A(3,2) and B(4,8) is

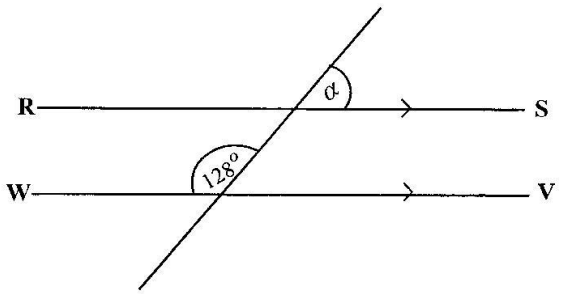
A. $-\frac{1}{6}$
B. $-\frac{1}{2}$
C. 2
D. 6

23. A car is travelling at 60 km per hour. How far does it travel in 2 ½ hours?

A. 30 km
B. 60 km
C. 120 km
D. 150 km

24. In the diagram below RS and WV are parallel lines. The value of the angle marked α is

A. 38°
B. 52°
C. 58°
D. 64°



NOT DRAWN TO SCALE

25. Given that $\boldsymbol{a} = \begin{pmatrix} 5 \\ 2n \end{pmatrix}$ and $\boldsymbol{b} = \begin{pmatrix} 2n-1 \\ 6 \end{pmatrix}$. If $\boldsymbol{a} = \boldsymbol{b}$, find the values of *n*.

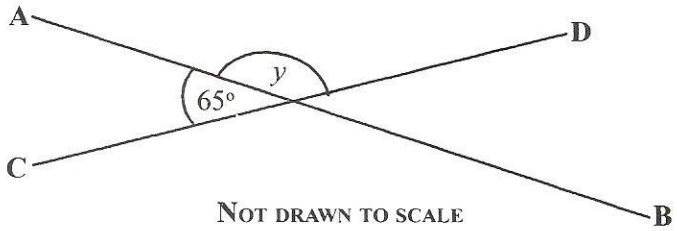
A. 6
B. 3
C. 2
D. 1

26. Find the volume of a cube of side 5 cm.

A. 10 cm^3
B. 15 cm^3
C. 25 cm^3
D. 125 cm^3

27. In the diagram below, **AB** and **CD** are two intersecting straight lines. Find the value of the angle marked y.

A. 130°
B. 115°
C. 65°
D. 60°

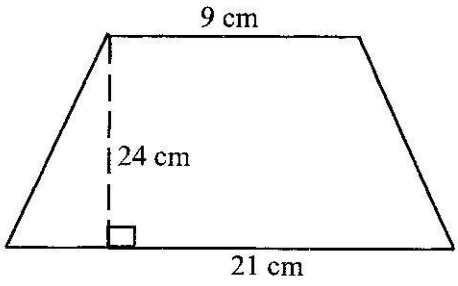


NOT DRAWN TO SCALE

28. Kwame and Ama shared an amount of money in the ratio 5:4 respectively. If Kwame had GH¢ 9.00, how much did they share?

A. GH¢ 16.20
B. GH¢ 36.00
C. GH¢ 45.00
D. GH¢ 81.00

29.



NOT DRAWN TO SCALE

The area of the trapezium above is

A. 120 cm^2
B. 180 cm^2
C. 256 cm^2
D. 360 cm^2

30. If $r = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$ and $s = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$, calculate 2**r** – 3**s**

A. $\begin{pmatrix} -10 \\ -25 \end{pmatrix}$

B. $\begin{pmatrix} -2 \\ -25 \end{pmatrix}$

C. $\begin{pmatrix} 10 \\ -25 \end{pmatrix}$

D. $\begin{pmatrix} 10 \\ 25 \end{pmatrix}$

31. There are 10 red and 15 green balls in a bag. Find the probability of selecting at random a red ball from the bag.

A. $\frac{3}{5}$

B. $\frac{2}{5}$

C. $\frac{1}{10}$

D. $\frac{1}{25}$

The table below gives the distribution of ages of students in a class. Use it to answer Questions **32** – **34**

Ages (years)	13	14	15	16	17
Number of students	3	10	6	7	4

32. How many students are in the class?
A. 20
B. 30
C. 45
D. 75

33. What is the modal age?
A. 14
B. 15
C. 16
D. 17

34. If a student is chosen at random from the class, what is the probability that the student is 15 years old?
A. $\frac{1}{5}$

B. $\frac{1}{3}$

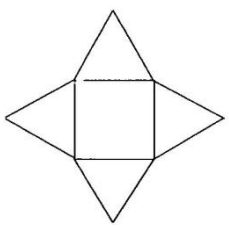
C. $\frac{1}{2}$

D. $\frac{2}{3}$

35. A length of a ribbon is 16.8 m long. How many ribbons 0.36 m long can be cut from it?
A. 0.46
B. 4.60
C. 46
D. 460

36. A refrigerator was sold for GH¢ 200.00 at a loss of 10%. Find the cost price.
A. GH¢ 180.00
B. GH¢ 190.00
C. GH¢ 220.00
D. GH¢ 222.22

37. The diagram below is the net of a



A. cone
B. cuboid
C. rectangular prism
D. pyramid

38. What is the value of 7 in the number 832713?

A. Seven thousand
B. Seven hundred
C. Seventy
D. Seven

39. Write 3560 in standard form.

A. 3.56×10^{-4}
B. 3.56×10^{-3}
C. 3.56×10^{3}
D. 3.56×10^{4}

40. Correct 0.02751 to three decimal places

A. 0.027
B. 0.028
C. 0.03
D. 0.28

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MATHEMATICS 1

OBJECTIVE TEST

SOLUTIONS

1. A. {2, 3}

2. B. $3a^2 - 12ab$

3. A. 125%

4. C. $2^2 \times 3^3 \times 5^2$

5. C. 24

6. A. $\frac{1}{8}$

7. D. 62

8. D. {}

9. C. 3^6

10. C. 40%

11. B. $(b + f)(b - m)$

12. B. -20

13. A. $3^2 \times 5^2$

14. A. $x \rightarrow 15x$

15. C. $x \geq \frac{1}{2}$

16. C. 49 cm^2

17. D. $\frac{1}{9}$

18. C. $\frac{360}{180 - \theta}$

19. B. $4\frac{1}{3}$

20. D. GH¢ 10.00

21. D. 3

22. D. 6

23. D. 150 km

24. B. 52°

25. B. 3

26. D. 125 cm^3

27. B. 115°

28. A. GH¢ 16.20

29. D. 360 cm^2

30. C. $\begin{pmatrix} 10 \\ -25 \end{pmatrix}$

31. B. $\frac{2}{5}$

32. B. 30

33. A. 14

34. A. $\frac{1}{5}$

35. C. 46

36. D. GH¢ 222.22

37. D. pyramid

38. B. Seven hundred

39. C. 3.56×10^3

40. B. 0.028

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MATHEMATICS 2

ESSAY
1 hour

[60 marks]

Answer **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is **not** *allowed*

Marks will **not** *be awarded for correct answers without corresponding working.*

All questions carry equal marks

1. (a) In a school of 255 students, 80 of them study Arabic only and 125 study French only. Each student studies at least one of the two subjects

(i) Draw a Venn diagram to represent the information

(ii) How many students study

(α) both subjects?

(β) French?

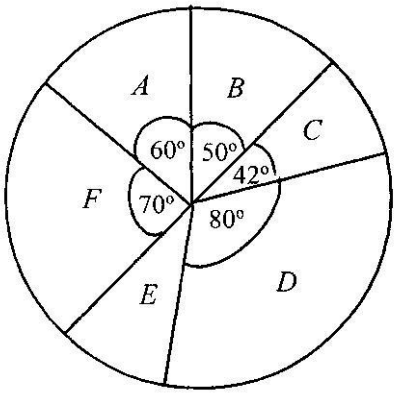
(b) Make h the subject of $v = \frac{1}{3}\pi r^2 h$

(c) A bookseller bought 80 copies of books at GH¢ 3.50 per copy. He sold each of them at GH¢ 4.20. Find

(i) the total cost price

(ii) his percentage profit

2. (a) The pie chart below shows the distribution of exercise books to six schools ABCDE and F in a town. School D was given 8000 exercise books.



NOT DRAWN TO SCALE

(i) How many exercise books were given to each of the rest of the schools?

(ii) What is the average number of exercise books given to the schools?

(iii) How many schools had less than the average number of exercise books?

(b) Solve the inequality below and illustrate the answer on the number line

$$\frac{1}{3}x+1 \geq \frac{1}{2}x+\frac{1}{4}(2-x)$$

3. (a) Using a ruler and a pair of compasses only, construct

(i) triangle ABC such that $|AB| = 8$cm, angle CBA = 45° and CAB = 60°.

(ii) the bisector of angle ACB to meet AB at T

(b) Measure

(i) $|CT|$;

(ii) angle CTB.

(c) A boy spent $\frac{3}{8}$ of his money and had GH¢ 15.00 left. How much did he have?

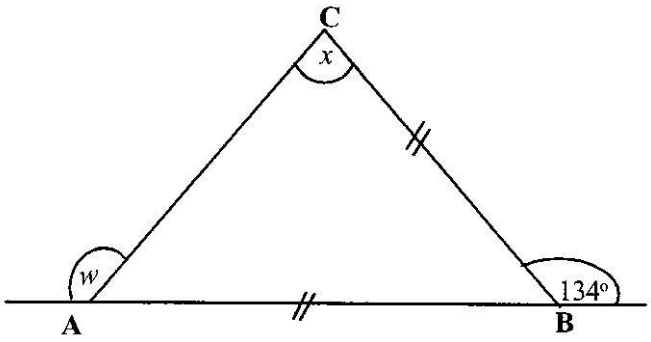
4. (a) The perimeter of a rectangular plot of land whose length is (2x+5) m and width (x – 10) is 80 m. Find the

(i) value of x;

(ii) area of the plot;

(iii) cost of weeding the plot at GH¢ 0.24 per m^2.

(b) Find the value of x and w in the diagram below if $|AB| = |BC|$



NOT DRAWN TO SCALE

5. (a) Given that $\mathbf{a} = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ -6 \end{pmatrix}$, calculate

(i) $\mathbf{a} + 2\mathbf{b}$;

(ii) $\frac{1}{2}(2\mathbf{a} - \mathbf{b})$

(b) The number of pupils in a primary school is given in the table below:

Class	One	Two	Three	Four	Five	Six
Number of pupils	24	35	35	20	21	45

(i) Find the number of pupils in the school

(ii) What is the mean number of pupils in a class?

(iii) What percentage of pupils is in class six?

(c) Convert 312_{five} to a base ten numeral

6. (a) Copy and complete the table for the relation $y = \frac{x}{20}$, where y is the cost (in Ghana cedis) and x is the weight (in grammes) of rice sold in a market.

x (weight in grammes)	50	100	150	200	250	300
y (cost in GH¢)		5.00			12.50	

(b) (i) On a graph sheet, draw two perpendicular axes OX and OY.

(ii) Using a scale of 2 cm to 50 grammes on the x-axis and 2 cm to GH¢ 2.00 on the y-axis draw the graph of the relation $y = \frac{x}{20}$.

(c) Using the graph, find

(i) the cost of 175 grammes of rice

(ii) the weight of rice that can be bought with GH¢14.00

(d) Factorize: $3a^2 - 8bc - 12ac + 2ba$

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MATHEMATICS 2

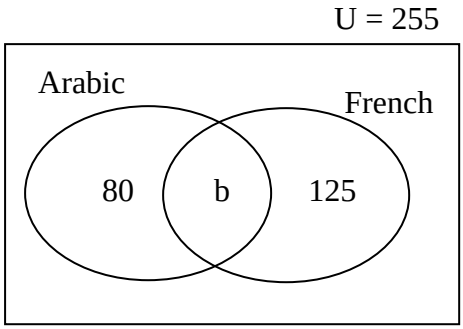
ESSAY

SOLUTIONS

1. **(a)** **(i)** Let U = Number of students in the school

b = Number of students who study both Arabic and French

The Venn diagram is shown below



(ii) (α) From the Venn diagram above,

$80 + b + 125 = 255$

$\Rightarrow \quad b + 205 = 255$

$\Rightarrow \quad b = 255 - 205$ [Subtracting 205 from both sides (carrying + 205 to the other side)]

$\Rightarrow \quad b = 50$

$\Rightarrow$ <u>50 students study both Arabic and French</u>.

(β) Number of French students $= b + 125$

$= 50 + 125$

$= \underline{175}$

(b) $v = \frac{1}{3}\pi r^2 h$

$$3 \times v = 3 \times \frac{1}{3}\pi r^2 h$$ [Multiplying both sides by 3 (to clear fraction)]

$$\Rightarrow \quad 3v = \pi r^2 h$$ [Simplifying]

$$\Rightarrow \quad \frac{3v}{\pi r^2} = \frac{\pi r^2 h}{\pi r^2}$$ [Dividing both sides by πr^2]

$$\Rightarrow \quad \frac{3v}{\pi r^2} = h$$

$$\Rightarrow \quad h = \frac{3v}{\pi r^2}$$

(c) (i) Total cost price = 80 × GHc 3.50

= <u>GHc 280.00</u>

(ii) Profit on each book = GHc 4.20 – GHc 3.50

= GHc 0.70 (or 70 Gp)

Cost price of each book = GHc 3.50 (or 350 Gp)

Percentage profit = $\frac{Profit}{Original\ Cost} \times 100\%$ [NB: 100%, **not** 100]

= $\frac{70}{350} \times 100\%$ [Substituting values (of profit and original cost)]

= $\frac{1}{5} \times 100\%$ [Simplifying]

= <u>20%</u>

2. (a) (i) Since School D's share = 80° ≡ 8000 books

By inspection of the above correspondence, it is obvious that

there is a factor of 100 (ie, 80 × **100** = 8000). Hence,

School A's share = 60° ≡ 6000 books

School B's share = 50° ≡ 5000 books

School C's share = 42° ≡ 4200 books

School F's share = 70° ≡ 7000 books

Now, School E's angle = 360° - (80°+42+50°+60°+70°)

= 360° - 302°

= 58°

Hence, School E's share = 5800 books

(ii) Average number of books given

$$= \frac{Total\ number\ of\ books\ given}{Number\ of\ schools}$$

$$= \frac{8000 + 6000 + 5000 + 4200 + 7000 + 5800}{6} \text{ books}$$

$$= \frac{36000}{6} \text{ books}$$

= 6000 books

(iii) Schools B, C and E had less than 6000 books

Therefore 3 schools had less than the average number of books.

(b) $\frac{1}{3}x + 1 \geq \frac{1}{2}x + \frac{1}{4}(2-x)$

$\Rightarrow 12\left(\frac{1}{3}x\right) + 12(1) \geq 12\left(\frac{1}{2}x\right) + 12\left[\frac{1}{4}(2-x)\right]$ Multiplying through by 12 (the LCM of the denominators)

$\Rightarrow 4x + 12 \geq 6x + 3(2-x)$ Simplifying

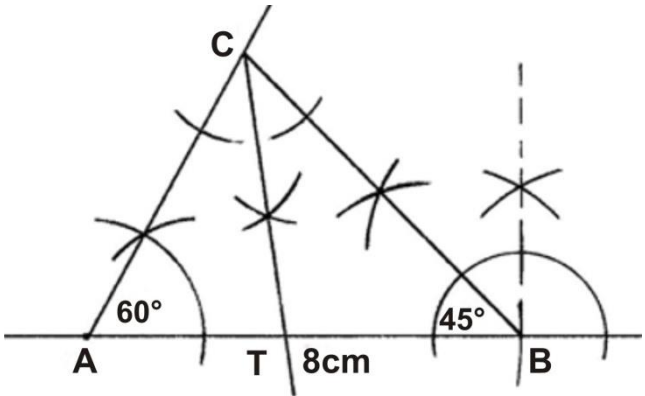
$\Rightarrow 4x + 12 \geq 6x + 6 - 3x$

$\Rightarrow 4x + 12 \geq 3x + 6$

$\Rightarrow 4x - 3x \geq 6 - 12$ Grouping like terms on one side

$\Rightarrow x \geq -6$

3. (a)



(b) (i) $|CT| \approx 5.1$ cm

(ii) angle CTB $\approx$ 97° (or 98°)

(c)

		Fraction	Amount (GH¢)
	Spent	$\frac{3}{8}$	**?**
✓	***Left***	$\frac{5}{8}$	**15**
✓	***Total*** (M. salary)	$\frac{8}{8}$	**m**

NB: Fraction left
= 'Total fraction' – Fraction spent
$= 1 - \frac{3}{8}$
$= \frac{8}{8} - \frac{3}{8} = \frac{5}{8}$

From the above table, we have $\frac{\frac{5}{8}}{\frac{8}{8}} = \frac{15}{m}$ [From the '*Left*' and '*Total*' rows]

$\Rightarrow \frac{5}{8} \times m = 15 \times \frac{8}{8}$ [Cross-multiplying]

$\Rightarrow \frac{5}{8} \times m = 15$

$\Rightarrow \frac{8}{5} \times \frac{5}{8} \times m = 15 \times \frac{8}{5}$ [Multiplying both sides by $\frac{8}{5}$ (to remove $\frac{5}{8}$)]

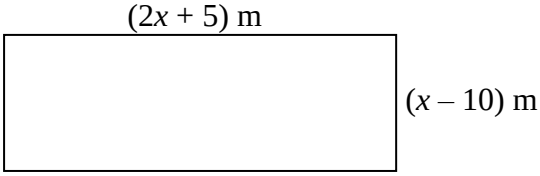
$\Rightarrow m = 15 \times \frac{8}{5}$

$\Rightarrow m = 3 \times 8$ [Simplifying]

$\Rightarrow \underline{\underline{m = 24}}$

Therefore he had GHc 24.00

4. (a)



(i) Since Perimeter = 80 m

⇒ 2 (Length) + 2 (width) = 80 m

⇒ $2(2x + 5) + 2(x - 10) = 80$ Substituting the values of length and width

⇒ $4x + 10 + 2x - 20 = 80$ Expanding

⇒ $4x + 2x + 10 - 20 = 80$ Grouping like terms

⇒ $6x - 10 = 80$ Simplifying

⇒ $6x = 80 + 10$ Adding 10 to both sides (carrying -10 to the other side)

⇒ $6x = 90$

⇒ $\frac{6x}{6} = \frac{90}{6}$ Dividing both sides by 6 (the co-efficient of x)

⇒ $x = 90 \div 6$

⇒ $\underline{\underline{x = 15}}$

(ii) Area of the plot = Length × Width

= $[2x + 5] \times [x - 10]$ Substituting the expressions of length and width

= $[2(15) + 5] \times [15 - 10]$ Substituting the value of x (15)

= $[30 + 5] \times [15 - 10]$ Simplifying

= $35m^2 \times 5m^2$

= $\underline{\underline{175\ m^2}}$

(iii) If $1m^2$ costs GHc 0.24

Then, $175\ m^2$ costs $175 \times$ GHc 0.24

⇒ $175\ m^2$ costs $\underline{\underline{\text{GHc } 42.00}}$

⇒ $\underline{\underline{\text{The cost of weeding the plot is GHc } 42.00}}$

(b) $\angle ABC = 180° - 134°$ (Sum of angles at a point on a straight line = 180°)

$= 46°$

Since $|AB| = |BC|$

⇒ $\angle BCA = x = \angle BAC$ (Base angles of isosceles triangle equal)

Now, $\angle BCA + \angle BAC + 46° = 180°$ (Sum of angles in a triangle = 180°)

$x + x + 46° = 180°$ (Substituting: $\angle BCA = x$, $\angle BAC = x$)

$2x + 46 = 180°$ [Simplifying]

$2x = 180° - 46°$ [Grouping like terms]

$2x = 134°$

$\frac{2x}{2} = \frac{134°}{2}$ [Dividing both sides by 2 (the co-efficient of x)]

$x = 134° \div 2$

$\underline{x = 67°}$

Therefore $\angle BAC = 67°$

$\Rightarrow \angle BAC + w = 180°$ (Sum of angles at a point on a straight line)

$\Rightarrow 67° + w = 180°$

$\Rightarrow w = 180° - 67°$

$\Rightarrow \underline{w = 113°}$

5. (a) (i) $\mathbf{a} + 2\mathbf{b}$

$= \begin{pmatrix}-3\\3\end{pmatrix} + 2\begin{pmatrix}4\\-6\end{pmatrix}$ Substituting

$= \begin{pmatrix}-3\\3\end{pmatrix} + \begin{pmatrix}8\\-12\end{pmatrix}$ Simplifying

$= \begin{pmatrix}-3+8\\3+(-12)\end{pmatrix}$

$= \begin{pmatrix}5\\-9\end{pmatrix}$

(ii) $\frac{1}{2}(2\boldsymbol{a} - \boldsymbol{b})$

$= \frac{1}{2}\left[2\begin{pmatrix}-3\\3\end{pmatrix} - \begin{pmatrix}4\\-6\end{pmatrix}\right]$

$= \frac{1}{2}\left[\begin{pmatrix}-6\\6\end{pmatrix} - \begin{pmatrix}4\\-6\end{pmatrix}\right]$

$= \frac{1}{2}\left[\begin{pmatrix}-6-4\\6-(-6)\end{pmatrix}\right]$

$= \frac{1}{2}\begin{pmatrix}-10\\12\end{pmatrix}$

$= \begin{pmatrix}\frac{1}{2}\times -10\\\frac{1}{2}\times 12\end{pmatrix}$

= $\begin{pmatrix} -5 \\ 6 \end{pmatrix}$

(b) (i) The number of pupils in the school

= 24 + 35 + 35 + 20 + 21 + 45

= 180

There are 180 pupils in the school

(ii) The mean number of pupils in a class

= $\frac{Total\ number\ of\ pupils\ in\ the\ school}{Number\ of\ classes}$

= $\frac{24 + 35 + 35 + 20 + 21 + 45}{6}$

= $\frac{180}{6}$

= 30 pupils

(iii) The percentage of pupils in class six

= $\frac{No.\ of\ pupils\ in\ class\ six}{Total\ no.\ of\ pupils} \times 100\%$

= $\frac{45}{180} \times 100\%$

= $\frac{1}{4} \times 100\%$

= 25%

(c) 312_{five} to a base ten numeral

3	**1**	**2**
5^2	5^1	5^0

$\Rightarrow \quad (3 \times 5^2) + (1 \times 5^1) + (2 \times 5^0)$

$= \quad (3 \times 25) + (1 \times 5) + (2 \times 1)$

$= \quad 75 + 5 + 2$

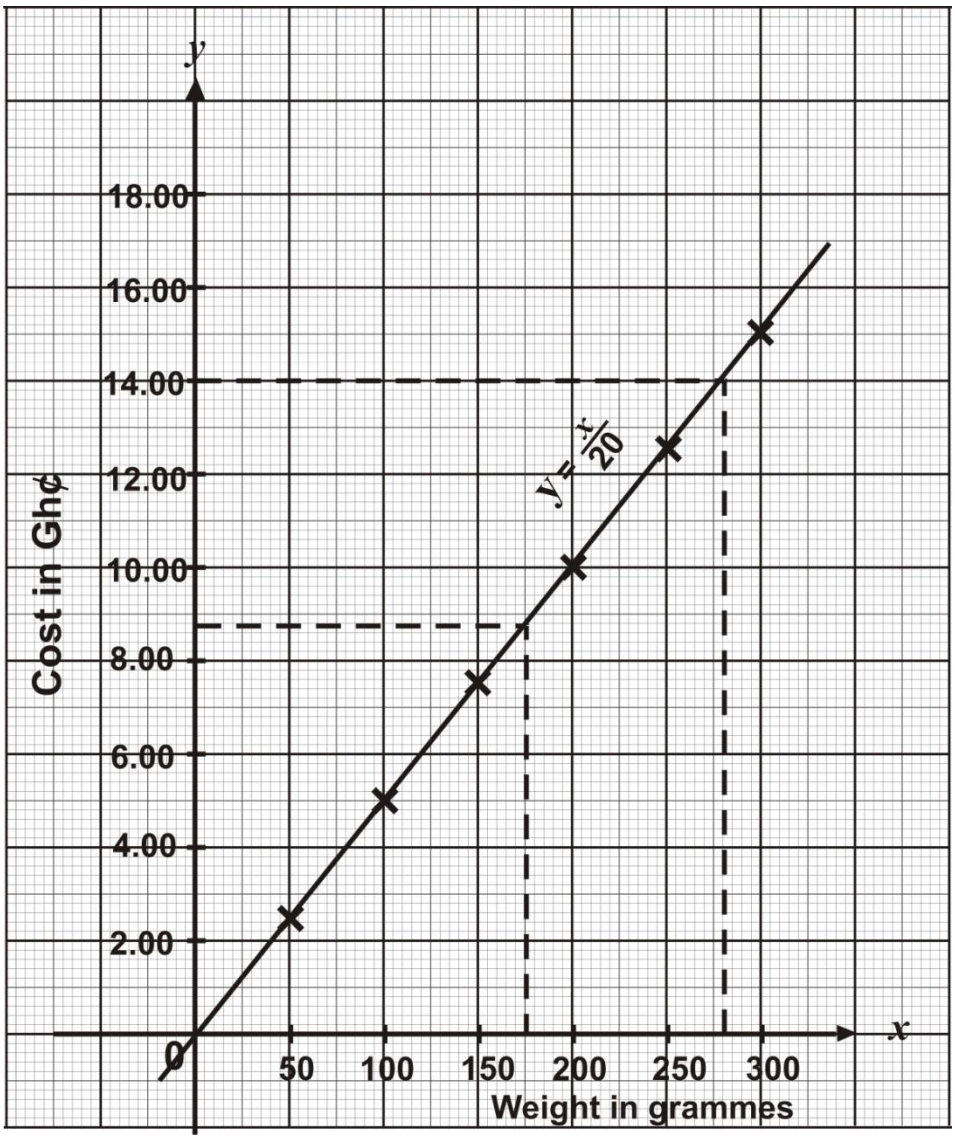
$= \quad \underline{\underline{82}} \quad (\text{or } \underline{\underline{82}}_{\text{ten}})$

6. **(a)** $y = \frac{x}{20}$

$\frac{50}{20} = 2.50, \quad \frac{150}{20} = 7.50, \quad \frac{200}{20} = 10.00, \quad \frac{300}{20} = 15.00$

x (weight in grammes)	50	100	150	200	250	300
y (cost in GHc)	**2.50**	5.00	**7.50**	**10.00**	12.50	**15.00**

(b) (i)



(c) **(i)** The cost of 175 grammes of rice = GHc 8.75

(ii) The weight or rice that can be bought with GHc 14.00 = 280 grammes

(d) $3a^2 - 8bc - 12ac + 2ba$

= $3a^2 - 12ac + 2ba - 8bc$ [Re-arranging terms]

= $3a(a - 4c) + 2b(a - 4c)$ [Factorizing]

= $\underline{(a - 4c)\,(3a + 2b)}$

April 2010

MATHEMATICS 1

OBJECTIVE TEST
1 hour

1. Which of the following sets is well defined?

A. {Man, Kofi, Red, 14}

B. {Ink, Mango, Green, Nail}

C. {Car, Road, Glass, Book}

D. {Seth, Mary, Jacob, Evelyn}

2. If set B is a subset of set A, then

A) sets A and B have the same number of elements

B) some members of set B can be found in set A

C) no member of set B is in set A

D) all the members of set B are in set A

3. The least common multiple (LCM) of 16, 30 and 36 is

A) 3 B) 6 C) 240 D) 720

4. The sum of 5 and x divided by 4 is equal to 3.25. Find the value of x.

A) 8 B) 7 C) 2¼ D) $-3\frac{4}{13}$

5. The numbers 32, 33, 34, …, …, 42 form a sequence in base 5.

Find the missing numbers

A) 35, 36 B) 30, 31 C) 40, 41 D) 31, 41

6. Write down all the integers in the set A = {-10, -4, 0, ¼, 2½, 45, 100}

A. {-10, -4, 0, 45, 100}

B. {-10, - 4}

C. {0, 45, 100}

D. {¼, 2½}

7. Find the total cost of 25 pens and 75 books if each pen costs GH¢ 0.20 and each book costs GH¢ 0.30.

A) GH¢22.50 B) GH¢23.50 C) GH¢27.50 D) GH¢50.00

8. Simplify - 27 + 18 – (10 – 14) – (-2)

A) – 3 B) – 7 C) – 11 D) – 35

9. Arrange the following numbers from the lowest to the highest: 0.5, 3, -5, 0.

A) 0, 0.5, -5, 3 B) 0, -5, 0.5, 3 C) -5, 0, 0.5, 3 D) -5, 0.5, 0, 3

10. Find how many pieces of cloth 5½ m long that can be cut from a roll of cloth 121 m long.

A) 665½ B) 115½ C) 66 D) 22

11. Find the value of 124.3 + 0.275 + 74.06, correcting your answer to one decimal place.

A) 198.6 B) 198.7 C) 892.0 D) 892.4

12. Esi and Kwasi are 12 and 8 years old respectively. They share 60 mangoes in the ratio of their ages. How many mangoes does Esi get?

A) 42 B) 40 C) 36 D) 18

13. It takes 6 students 1 hour to sweep their school compound. How long will it take 15 students to sweep the same compound?

A) 24 minutes B) 12 minutes C) 3 hours D) 2 hours

14. A housing agent makes a commission of GH¢ 103,500 when he sells a house for GH¢ 690,000. Calculate the percentage of his commission.

A) 15.0% B) 10.0% C) 7.5% D) 5.0%

15. A simple interest of GH¢ 37,500.00 is earned on an amount of GH¢ 500,000.00 for 3 years. Find the rate of interest per annum.

A) 20.0% B) 10.0% C) 5.0% D) 2.5%

16. Simplify: $(8x^2y^3)(\frac{3}{8}xy^4)$

A) $3x^3y^7$ B) $3x^2y^7$ C) $3x^3y^4$ D) $3xy$

17. The scores of 10 students in an examination are given as follows:
45, 12, 75, 81, 54, 51, 24, 67, 19 and 39.

What is the median of the scores?

A) 39 B) 48 C) 51 D) 54

18. A pie chart is to be drawn from the data in the following table:

Cassava	20%
Yam	17%
Plantain	28%
Maize	35%

What will be the value of the angle of the sector for maize?

A) 126.0° B) 100.8° C) 72.0° D) 61.2°

19. Eighteen cards are numbered from 11 to 29. If one card is chosen at random, what is the probability that it contains the digit 2?

A) $\frac{3}{9}$ B) $\frac{7}{18}$ C) $\frac{5}{9}$ D) $\frac{11}{18}$

20. Find the value of *x*, if $\frac{x}{4} + 1 = 5$.

A) 24 B) 20 C) 19 D) 16

21. Factorize: $xy + 5x + 2y + 10$

A. $(x + 5)(2y + 10)$

B. $(x + 2)(y + 10)$

C. $(x + 5)(y + 2)$

D. $(x + 2)(y + 5)$

22. If $x \in \{2, 3, 4, 5\}$, find the truth set of $2x + 1 < 8$

A. {2,3,4} B. {2,3} C. {3,4} D. {4,5}

23. Solve the inequality: $7x - (10x + 3) \geq -9$

A) $x \geq 2$ B) $x \leq 4$ C) $x \geq 4$ D) $x \leq 2$

24. Find the rule of the mapping:

1	2	3	4	5	...	x
↓	↓	↓	↓	↓		↓
7	11	15	19	23	...	y

A) $x \rightarrow 4x-3$ B) $x \rightarrow 3-4x$ C) $x \rightarrow 4x+3$ D) $x \rightarrow 4x+5$

25. Find the circumference of a circle whose area is equal to $64\ \pi\ cm^2$.

A) $32\ \pi\ cm^2$ B) $16\ \pi\ cm^2$ C) $8\ \pi\ cm^2$ D) $4\ \pi\ cm^2$

26. Which of the following geometric figures is the plane shape of a cube?

A) Circle
B) Rectangle
C) Square
D) Triangle

27. How many lines of symmetry has a rectangle?

A) 4 B) 3 C) 2 D) 1

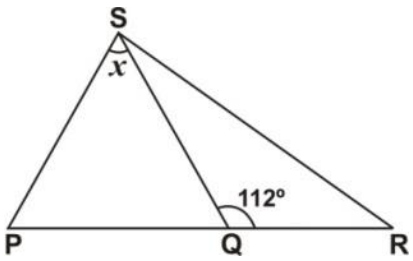
28. A rectangular box has length 20 cm, width 6 cm and height 4 cm. Find how many cubes of side 2 cm that will fit into the box.

A) 120 B) 60 C) 30 D) 15

29. The interior angle of a regular polygon is 120°. How many sides has this polygon?

A) 3 B) 4 C) 5 D) 6

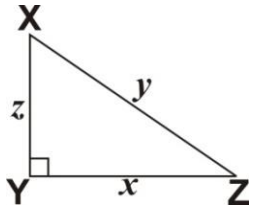
30.



In the diagram above, length of PS = length of SQ and angle SQR = 112°. Find the value of x.

A) 68° B) 56° C) 46° D) 44°

31. **XYZ** is a right-angled triangle with length of sides as shown.



Which of the following equations gives the value of z^2?

A) $z^2 = (x^2 + y^2)$

B) $z^2 = (x - y)$

C) $z^2 = (y^2 - x^2)$

D) $z^2 = (x^2 - y^2)$

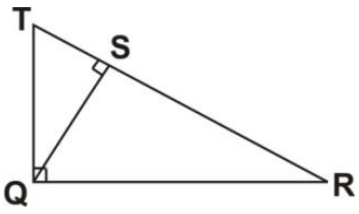
32. Express 7 min. 30 sec. as a percentage of 1 hour.

A) 2.5% B) 7.5% C) 11.7% D) 12.5%

33. The point (4,5) is translated to the point (3,1). What is the translation vector?

A. $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$ B. $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ C. $\begin{pmatrix} 1 \\ -4 \end{pmatrix}$ D. $\begin{pmatrix} -1 \\ -4 \end{pmatrix}$

34. In the diagram below, triangle QRT is the enlargement of QST.



Which side of triangle QRT corresponds to side QT of triangle QST?

A) TS B) TR C) QR D) SR

35.

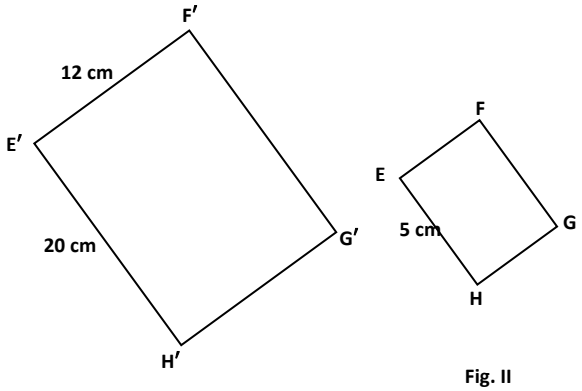


Fig. I

Fig. II

Not drawn to scale

In the diagrams above Fig. I is an enlargement of Fig. II.

Find the side EF of Fig. II

A) 20 cm B) 5 cm C) 4 cm D) 3 cm

36. Express 4037 in standard form

A) 4.037×10^{-4}

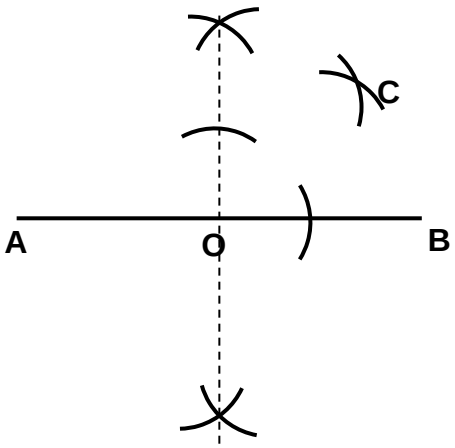
B) 4.037×10^{-3}

C) 4.037×10^{3}

D) 4.037×10^{4}

37. Which of the following angles can be constructed by using the arcs at point C in the diagram below?

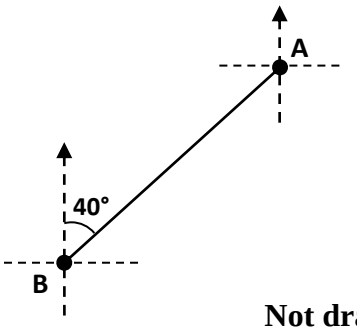
A) 30° B) 45° C) 60° D) 75°



38. Given that vector **a** = $\begin{pmatrix} -5 \\ 12 \end{pmatrix}$ and **b** = $\begin{pmatrix} 10x \\ 12 \end{pmatrix}$ find the value of x if **a** = **b**.

A) -2 B) - ½ C) ½ D) 2

39.



Not drawn to scale

In the diagram above, the bearing of point **B** from **A** is

A) 340° B) 220° C) 140° D) 50°

40. Ama is 9 years older than Kwame. If Kwame is 18 years old, find the ratio of the age of Kwame to that of Ama.

A) 3 : 2 B) 1 : 3 C) 2 : 3 D) 2 : 1

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MATHEMATICS 1

OBJECTIVE TEST

SOLUTIONS

1. D.{Seth, Mary, Jacob, Evelyn}

2. D) all the members of set B are in set A

3. D) 720

4. A) 8

5. C) 40, 41

6. A. {-10, -4, 0, 45, 100}

7. C) GH¢27.50

8. A) – 3

9. C) -5, 0, 0.5,

10. D) 22

11. A) 198.6

12. C) 36

13. A) 24 minutes

14. A) 15.0%

15. D) 2.5%

16. A) $3x^3y^7$

17. B) 48

18. A) 126.0°

19. $^{11}/_{19}$ (NB: Not included in given options)

20. D) 16

21. D. $(x + 2)(y + 5)$

22. B. {2,3}

23. D) $x \leq 2$

24. C) $x \rightarrow 4x+3$

25. B) $16\ \pi\ \text{cm}^2$

26. C) Square

27. C) 2

28. B) 60

29. D) 6

30. D) 44°

31. C) $z^2 = (y^2 - x^2)$

32. D) 12.5%

33. D. $\begin{pmatrix} -1 \\ -4 \end{pmatrix}$

34. B) TR

35. D) 3 cm

36. C) 4.037×10^3

37. B) 45°

38. B) - ½

39. B) 220°

40. C) 2: 3

April 2010

MATHEMATICS 2

ESSAY
1 hour

[60 marks]

Answer **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is **not** *allowed*

Marks will **not** *be awarded for correct answers without corresponding working.*

All questions carry equal marks

1. (a) Factorize: $(m + n)(2x – y) – x(m + n)$

(b) A and B are subsets of a universal set

U = {1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18}

Such that A = {even numbers} and B = {multiples of 3}

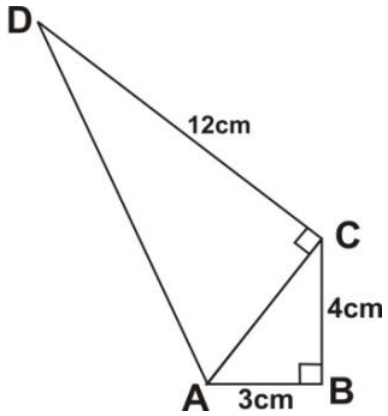
(i) List the elements of the sets A, B, (A∩B), (AUB) and (AUB)′

(ii) Illustrate the information in (i) on a Venn diagram

(c) Find the values of x and y in the vector equation

$$\begin{pmatrix}5\\3\end{pmatrix} + 2\begin{pmatrix}x\\y\end{pmatrix} - \begin{pmatrix}1\\-7\end{pmatrix} = 0$$

2. (a) Find the sum of 2,483.65, 701.532 and 102.7, giving your answer to one decimal place.



(b) In the quadrilateral ABCD above, |AB| = 3 cm, |BC| = 4 cm, |CD| = 12 cm and angle ABC = 90°. Calculate:

(i) the perimeter of ABCD

(ii) the area of ABCD

3. (a) Evaluate: $\frac{2^7 \times 3^4 \times 5^3}{2^3 \times 3^2 \times 5^2}$, leaving your answer in standard form.

(b) Kwame rode a bicycle for a distance of *x* km and walked for another ½ hour at a rate of 6 km/hour. If Kwame covered a total distance of 10 km, find the distance *x* he covered by bicycle.

(c) A rectangular tank of length 22 cm, width 9 cm and height 16 cm is filled with water. The water is poured into a cylindrical container of radius 6 cm.
Calculate the :

(i) volume of the rectangular tank

(ii) depth of water in the cylindrical container.

[Take $\pi = {}^{22}/_{7}$]

4. (a) Simplify: $7\frac{2}{3} - 4\frac{5}{6} + 2\frac{3}{8}$

(b) The area of a trapezium is 31.5 cm^2. If the parallel sides are of lengths 7.3 cm and 5.3 cm, calculate the perpendicular distance between them.

(c) The marks scored by four students in a Mathematics test are as follows:

Esi - 92

Seth - 85

Mary - 65

Efe - x

(i) Write down an expression for the mean (average) of the marks.

(ii) If the mean is less than 80, write a linear inequality for the information

(iii) Find the possible marks Efe scored in the test. Represent your answer on the number line.

5. (a) Solve: $\frac{4x-3}{2} = \frac{8x-10}{8} + 2\frac{3}{4}$

(b) Using a scale of 2 cm to 1 unit on both axes, draw two perpendicular lines OX and OY on a graph sheet for the x – axis from -5 to 5 and the y – axis from -6 to 6.

(i) Plot the points A(2, 3) and B(-3, 4) and join them with a long straight line

(ii) Plot on the same graph sheet, the points C(4, 2) and D(-2, -3) and join them with a long straight line to meet the line through AB

(iii) Measure the angle between the lines through AB and CD.

(iv) Find the coordinates of the point at which the lines through AB and CD meet.

6. The table below shows the frequency distribution of the number of letters in the surnames of some students in a school.

No. of letters	4	5	6	7	8	9	10
No. of students	7	3	2	8	5	3	1

(a) From the distribution, determine

(i) the mode

(ii) the mean

(b) If a student is selected at random, find the probability that his/ her name will contain more than 7 letters.

(c) Draw a bar chart for the distribution

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MATHEMATICS 2

ESSAY

SOLUTIONS

1. **(a)** $(m + n)(2x – y) – x(m + n)$

Method 1

$= (m + n)[2x – y – x]$ — Factorizing (m + n) out

$= (m + n)(2x – x – y)$

$= \underline{(m + n)(x – y)}$ — **NB: $(m + n)(x – y) = (x – y)(m + n)$**

Method 2

$(m + n)(2x – y) – x(m + n)$ — Expanding

$= 2mx – my + 2nx – ny – mx – nx$ — Grouping like terms & simplifying

$= 2mx – mx + 2nx – nx – my – ny$ — Factorizing

$= mx + nx – my – ny$ — Factorizing (m+n) out

$= x(m + n) – y(m + n)$

$= \underline{(m + n)(x – y)}$

1 (b) (i)

$A = \{2, 4, 6, 8, 10, 12, 14, 16, 18\}$

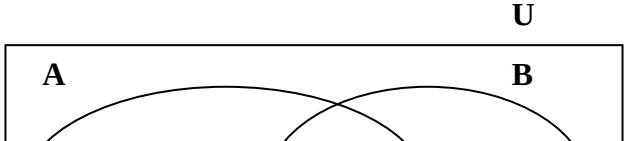
$B = \{3, 6, 9, 12, 15, 18\}$

$A \cap B = \{6, 12, 18\}$

$A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18\}$

$(A \cup B)' = \{1, 5, 7, 11, 13, 17\}$

1(b) (ii)



1 (c) $\begin{pmatrix}5\\3\end{pmatrix} + 2\begin{pmatrix}x\\y\end{pmatrix} - \begin{pmatrix}1\\-7\end{pmatrix} = 0$

Using the horizontal (x) component, we have

$5 + 2x - 1 = 0$

$\Rightarrow \quad 2x = 1 - 5$

$\Rightarrow \quad \frac{2x}{2} = \frac{-4}{2}$

$\Rightarrow \quad \underline{\underline{x = -2}}$

Solving for x

Using the vertical (y) component, we have

$3 + 2y - (-7) = 0$

$\Rightarrow \quad 3 + 2y + 7 = 0$

$\Rightarrow \quad 2y + 10 = 0$

$\Rightarrow \quad 2y = -10$

$\Rightarrow \quad y = \frac{-10}{2}$

$\Rightarrow \quad \underline{\underline{y = -5}}$

Solving for y

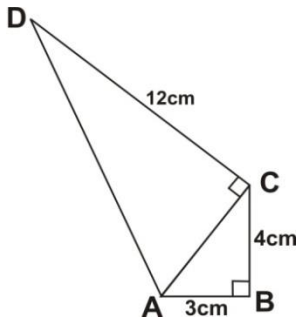
2. **(a)** 2,483.65 + 701.532 + 102.7

$$\begin{array}{r} 2483.65 \\ 701.532 \\ +\ 102.7 \\ \hline 3287.882 \end{array}$$

Ensure that the places of the addends are in line

$\approx$ $\underline{3287.9}$ (one decimal place)

2 (b)



(i) Side **AC** is the hypotenuse of triangle ABC.

From the Pythagorean theorem,

$$|AC|^2 = |AB|^2 + |BC|^2$$

$$\Rightarrow \quad |AC|^2 = (3cm)^2 + (4cm)^2$$

$$\Rightarrow \quad |AC|^2 = 9cm^2 + 16cm^2$$

$$\Rightarrow \quad |AC|^2 = 25cm^2$$

$$\Rightarrow \quad |AC| = \sqrt{25cm^2}$$

$$\Rightarrow \quad |AC| = 5\,cm$$

Now, side **AD** is the hypotenuse of triangle ACD

From the Pythagorean theorem,

$$|AD|^2 = |AC|^2 + |CD|^2$$

$$\Rightarrow \quad |AD| = \sqrt{(5cm)^2 + (12cm)^2}$$

$$\Rightarrow \quad |AD| = \sqrt{169cm^2}$$

$$\Rightarrow \quad |AD| = 13cm$$

Hence the perimeter of ABCD

$$= |AB| + |BC| + |CD| + |DA|$$

$$= 3\,cm + 4cm + 12cm + 13cm$$

$$= \underline{\underline{32\,\text{cm}}}$$

The perimeter of ABCD is 32 cm

2 (b) (ii) Area of (ABCD = ΔABC + ΔACD)

$$= \tfrac{1}{2}(b_1h_1) + \tfrac{1}{2}(b_2h_2)$$

$$= \tfrac{1}{2}(3cm)(4cm) + \tfrac{1}{2}(5cm)(12cm)$$

$= \quad 6cm^2 + 30cm^2$

$= \quad \underline{\underline{36cm^2}}$

The area of ABCD is 36 cm^2

3. **(a)** $\dfrac{2^7 \times 3^4 \times 5^3}{2^3 \times 3^2 \times 5^2}$

$= \quad \dfrac{2^7}{2^3} \times \dfrac{3^4}{3^2} \times \dfrac{5^3}{5^2}$

$= \quad 2^4 \times 3^2 \times 5$

$= \quad 8 \times 9 \times 5$

$= \quad 360$

$= \quad \underline{\underline{3.6 \times 10^2}}$

3 (b) Distance ridden = x km

Distance walked = $\frac{1}{2}$ h × 6km/h = 3km

Total distance = 10 km

Dist. ridden + dist. walked = total dist.

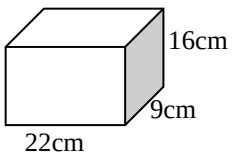
⇒ x km + 3 km = 10 km

⇒ x km = 10 km – 3km

⇒ x km = 7 km

The distance Kwame covered by bicycle is 7 km

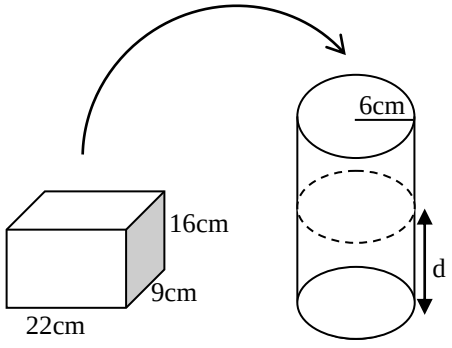
3 (c) (i)



Volume = length × width × height

= $22cm \times 9cm \times 16cm$

= $\underline{3168cm^3}$



Not drawn to scale

Let d = the depth of water in the cylinder

3 (c) (ii) **Method 1** (Using calculated volume of rectangular tank)

Vol. of water in rectangular tank = Vol. of water in cylinder

$\Rightarrow \quad 3168\text{cm}^3 = \pi r^2 \times d$

$\Rightarrow \quad 3168\text{cm}^3 = {}^{22}/_{7} \times (6\text{cm})^2 \times \text{d}$ Substituting

$\Rightarrow \quad 3168\text{cm}^3 = {}^{22}/_{7} \times 36\text{cm}^2 \times \text{d}$

$\Rightarrow \quad \frac{3168\ cm^3}{\frac{22}{7} \times 36\text{cm}^2} = \text{d}$ Dividing both sides by ${}^{22}/_{7} \times 36\text{cm}^2$

$\Rightarrow \quad \frac{3168}{{}^{792}/_{7}}\ cm = \text{d}$ Simplifying

$\Rightarrow \quad 3168 \div \frac{792}{7}\text{cm} = \text{d}$

$\Rightarrow \quad 3168 \times \frac{7}{792}\text{cm} = \text{d}$ You may avoid the tedious simplification here by using Method 2 below

$\Rightarrow \quad 28\text{ cm} = \text{d}$

Hence the depth of water in the cylindrical container = 28cm

3(c)(ii) **Method 2** (Using the given dimensions of rectangular tank)

Vol. of water in cuboid = Vol. of water in cylinder

$\Rightarrow \quad l \times w \times h_{\text{ cuboid}} = \pi r^2 \times d_{\text{ cylinder}}$

$[22\text{cm} \times 9\text{cm}] \times 16\text{cm} = [{}^{22}/_{7} \times (6\text{cm})^2] \times \text{d}$ *Substituting and solving for d*

$\Rightarrow \quad [22\text{cm} \times 9\text{cm}] \times 16\text{cm} = (\frac{22}{7} \times 6\text{cm} \times 6\text{cm}) \times \text{d}$

$$\Rightarrow \quad \frac{22\text{cm}\times 9\text{cm}\times 16\text{cm}}{\frac{22}{7}\times 6\text{cm}\times 6\text{cm}} = d \qquad \text{Simplifying}$$

$$\Rightarrow \quad \frac{22\text{cm}\times 9\text{cm}\times 16\text{cm}\times 7}{22\times 6\text{cm}\times 6\text{cm}} = d \qquad \text{Simplifying (by 'cancellation')}$$

$$\Rightarrow \quad \underline{\underline{28\text{cm}}} = d$$

$\therefore$ The depth (d) of water in the cylindrical container = $\underline{\underline{28\text{cm.}}}$

4. (a)

Method 1 (Evaluating whole number and fractions separately)

$$7\tfrac{2}{3} - 4\tfrac{5}{6} + 2\tfrac{3}{8}$$

$$\Rightarrow \quad 7-4+2 + \tfrac{2}{3}-\tfrac{5}{6}+\tfrac{3}{8}$$

$$\Rightarrow \quad 5 + \frac{16 - 20 + 9}{24}$$

$$\Rightarrow \quad 5 + \tfrac{5}{24}$$

$$= \quad \underline{\underline{5\tfrac{5}{24}}}$$

4 (a) **Method 2** (Changing mixed fractions to improper fractions)

$$7\tfrac{2}{3} - 4\tfrac{5}{6} + 2\tfrac{3}{8}$$

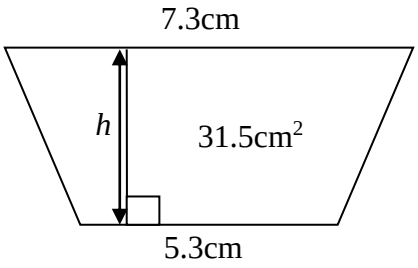
$$\Rightarrow \quad \tfrac{23}{3} - \tfrac{29}{6} + \tfrac{19}{8}$$

$$\Rightarrow \quad \frac{184 - 116 + 57}{24}$$

$$\Rightarrow \quad \frac{125}{24}$$

$$= \quad \underline{\underline{5\tfrac{5}{24}}}$$

4 (b)



Let h = perpendicular dist. between parallel sides

Method 1 (Substituting first)

Area = ½ (sum of parallel sides) × h

$\Rightarrow \quad 31.5\text{cm}^2 = \frac{1}{2}(7.3\text{cm} + 5.3\text{cm}) \times h$

Mulitiplying both sides by 2 to remove fraction

$\Rightarrow \quad 2 \times 31.5\ \text{cm}^2 = 2 \times \frac{1}{2}(7.3\text{cm} + 5.3\text{cm}) \times h$

$\Rightarrow \quad 63\text{cm}^2 = 12.6\text{cm} \times h$

$\Rightarrow \quad \frac{63\text{cm}^2}{12.6\text{cm}} = \frac{12.6\text{cm} \times h}{12.6\text{cm}}$

Dividing both sides by 12.6 cm to find 'h'

$\frac{630\text{cm}^2}{126\text{cm}} = h$

Multiplying both numerator and denominator by 10 and breaking down by cancellation

$\Rightarrow \quad \underline{\underline{5\text{ cm} = h}}$

4 (b) **Method 2** (Making h the subject first)

Let A = Area of trapezium

$c + d$ = sum of parallel sides

h = perpendicular dist. between parallel sides

$\Rightarrow \quad A = \frac{1}{2}(c + d)\ h$

$\Rightarrow \quad 2 \times A = 2 \times \frac{1}{2}(c + d)\ h$

Mulitiplying both sides by 2 to remove fraction

$\Rightarrow \quad 2\ A = (c + d)\ h$

$$\Rightarrow \quad \frac{2\,A}{c\,+\,d} = h$$

Dividing both sides by 'c+d' to make 'h' the subject

$$\Rightarrow \quad h = \frac{2\,A}{c\,+\,d}$$

Switching positions

$$\Rightarrow \quad h = \frac{2 \times 31.5\,cm^2}{7.3\,cm\,+\,5.3\,cm}$$

Substituting values to find 'h'

$$h = \frac{63\,cm^2}{12.6\,cm} = \underline{\underline{5\,cm}}$$

The perpendicular distance between the parallel sides is 5 cm.

4 (c)

(i) The mean mark $= \underline{\underline{\dfrac{92+85+65+x}{4}}}$

(ii) If the mean is less than 80 then $\underline{\underline{\dfrac{92+85+65+x}{4} < 80}}$, $\{x: x \geq 0\}$

(iii) $$\frac{92+85+65+x}{4} < 80$$

$$\Rightarrow \quad \frac{242+x}{4} < 80$$

$$\Rightarrow \quad 4 \times \left(\frac{242+x}{4}\right) < 4 \times 80$$

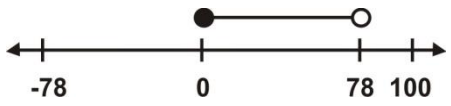
$$\Rightarrow \quad 242 + x < 320$$

$$\Rightarrow \quad x < 320 - 242$$

$$\Rightarrow \quad x < 78 \qquad \{x: x < 78,\ x \geq 0\}$$

The possible marks (x) that Efe scored in the test

$= \quad \mathbf{0 \leq x < 78}$



5. (a)

Method 1 (Clearing fractions first)

$$\frac{4x-3}{2} = \frac{8x-10}{8} + 2\tfrac{3}{4}$$

$$\Rightarrow \quad \frac{4x-3}{2} = \frac{8x-10}{8} + \frac{11}{4}$$

Changing mixed fraction $2\frac{3}{4}$ to improper fraction

$$8\times\left(\frac{4x-3}{2}\right) = 8\times\left(\frac{8x-10}{8}\right) + 8\times\frac{11}{4}$$

Multiplying through by 8 (to clear fractions):

$$\Rightarrow \quad 4(4x-3) = 8x-10 + 2(11)$$

Expanding and Simplifying

$$\Rightarrow \quad 16x-12 = 8x-10 + 22$$

$$\Rightarrow \quad 16x-8x = 12-10 + 22$$

Grouping like terms on one side

$$\Rightarrow \quad 8x = 24$$

Dividing both sides by 8 to find x

$$\Rightarrow \quad x = \frac{24}{8} = \underline{3}$$

5 (a)

Method 2 (Grouping and simplifying terms containing the variable first)

$$\frac{4x-3}{2} = \frac{8x-10}{8} + 2\tfrac{3}{4}$$

$$\Rightarrow \quad \frac{4x-3}{2} - \frac{8x-10}{8} = 2\tfrac{3}{4}$$

Grouping the terms containing the variable

$$\Rightarrow \quad \frac{4(4x-3)-(8x-10)}{8} = \frac{11}{4}$$

Simplifying

$$\Rightarrow \quad \frac{16x-12-8x+10}{8} = \frac{11}{4}$$

$$\Rightarrow \quad \frac{16x-8x-12+10}{8} = \frac{11}{4}$$

$$\Rightarrow \quad \frac{8x-2}{8} = \frac{11}{4}$$

$$\Rightarrow \quad 4(8x-2) = 8(11)$$

Cross multiplying

$$\Rightarrow \quad 32x-8 = 88$$

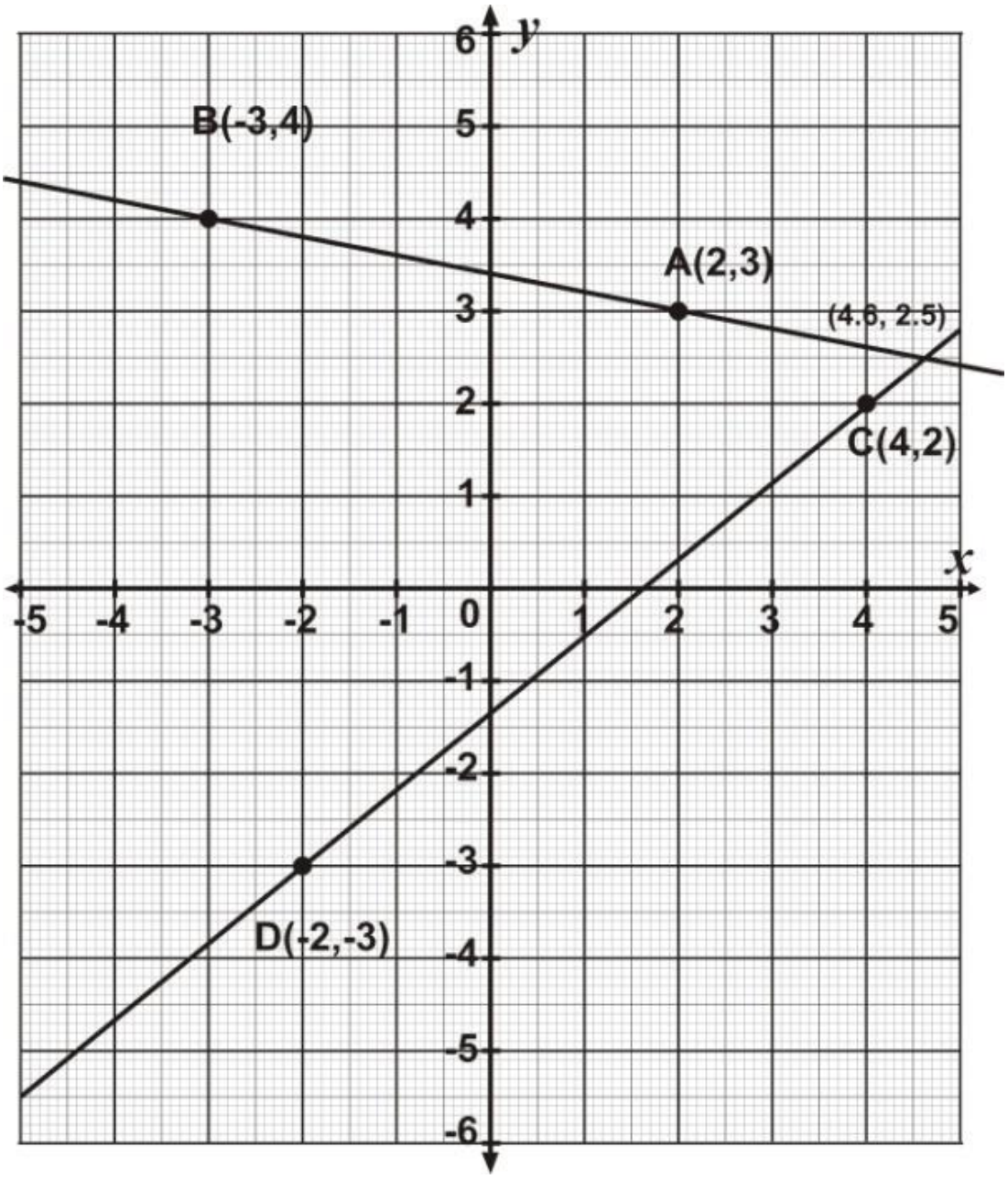
$$\Rightarrow \quad 32x = 88 + 8$$

Simplifying and finding x

$$\Rightarrow \quad \frac{32x}{32} = \frac{88+8}{32} = \frac{96}{32}$$

$$\Rightarrow \quad \underline{\underline{x = 3}}$$

5 (b)



5 (b) (iii) The acute angle between lines AB and CD $\approx$ <u>50°</u>

5(b) (iv) The coordinates of the point at which the lines through AB and CD meet

$\approx$ <u>(4.6, 2.5)</u>

6.

(a) (i) The Mode = The most-occurring number of letters

(the number of letters with the highest frequency)

⇒ The mode = 7 letters

(ii)

No. of letters ***(x)***	**No. of students (*f*)**	***f x***
4	7	28
5	3	15
6	2	12
7	8	56
8	5	40
9	3	27
10	1	10
	$\Sigma f = 29$	$\Sigma fx = 188$

The mean $= \frac{\Sigma f x}{\Sigma f} = \frac{188}{29}$

$= 6\frac{14}{29} \approx 6.483$

6 (b) P(surname has more than 7 letters)

$= \frac{\text{Number of students with more than 7 letters in surname}}{\text{Total number of students}}$

$= \frac{5+3+1}{7+3+2+8+5+3+1}$

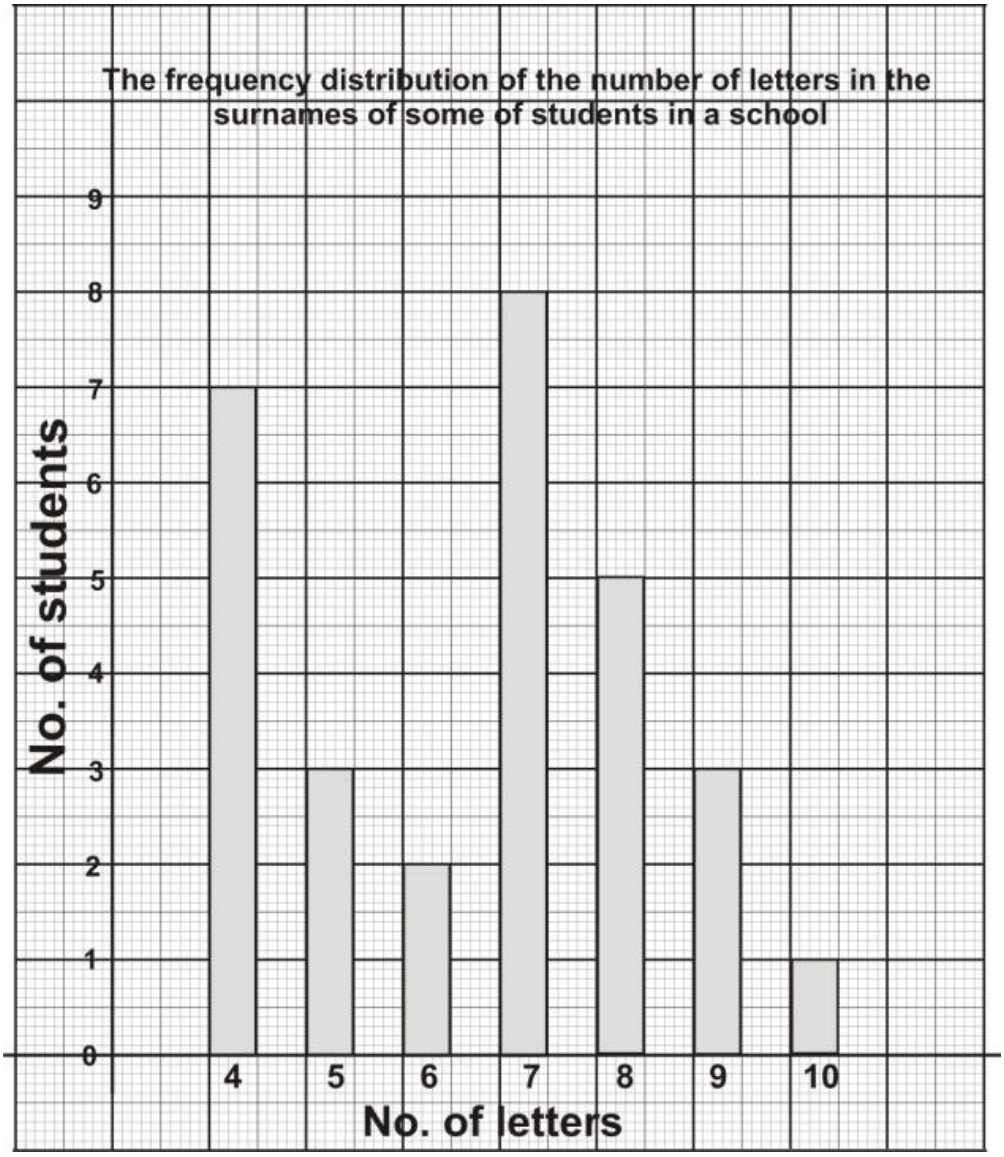
$= \frac{9}{29}$

The probability that a student selected at random has a surname with more than 7 letters

$= 9/29$

6 (c)

Vertical AxisScale: 2cm to 1 student


The frequency distribution of the number of letters in the surnames of some of students in a school
No. of students
0
1
2
3
4
5
6
7
8
9
4
5
6
7
8
9
10
No. of letters

April 2009

MATHEMATICS 1

OBJECTIVE TEST
1 hour

1. Given that A = {1,2,3,4,5,6} and B = {3,6,9,12}, find A∪B

A. {3,6} B. {6,12} C. {2,3,4,5,9,12} D.{1,2,3,4,5,6,9,12}

2. Write 2340000 in standard form.

A) 2.34×10^{-6}

B) 2.34×10^{-5}

C) 2.34×10^{6}

D) 2.34×10^{7}

3. How many faces has a cuboid?

A) 12 B) 8 C) 6 D) 4

4. Subtract 125.47 from 203.90

A) 78.57 B) 78.43 C) -121.57 D) -122.38

5. Simplify (5m+3n) – (2m-n)

A) 5m-4n B)3m+4n C)3m-2n D)7m+4n

6. Which of the following sets is equal to {1,2,3,4} ?

A. {2,4,1,5} B. {2,1,4,3,} C. {1,2,3,4,…} D. {2,3,4,5,…}

7. In a town of 42,800 inhabitants, 48% are male. The rest are female. How many more females are there than males?

A) 22,256 B) 20,544 C)1,712 D)1,612

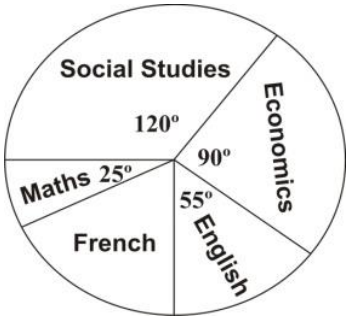
8. What set does the following graph represent?



A. {x: x<2} B. {x: x≤2} C. {x: x>2} D. {x: x≥2}

The pie chart below shows the performance of Kate in her final examination.

Use the diagram to answer Questions **9** *and* **10**



9. What is the angle for French?

A) 120° B) 100° C) 70° D) 35°

10. If Kate scored 60% in Social Studies, what was her score in Economics?

A) 45% B) 54% C) 72% D) 90%

11. A teacher is to be chosen at random from a group of 6 male and 2 female teachers for a patron of a Mathematics Club. What is the probability that a female teacher is chosen?

A) ¾ B) ⅔ C) ⅓ D) ¼

12. Solve the equation $2x - 3(x-1) = 6$

A) 3 B) -3 C) -7 D) -9

13. If m = 3 and n = -3, evaluate ½(3m-n)

A) 6 B) 3 C) 0 D) -6

14. Which of the following sets of angles form the interior angles of a right angled triangle?

A. {20°, 50°, 90°}

B. {80°, 60°, 90°}

C. {45°, 45°, 90°}

D. {65°, 90°, 35°}

15. State the rule for the mapping

x	0	1	2	3	4
y	0	¼	½	¾	1

A) $x \rightarrow \frac{x}{4}$ B) $x \rightarrow \frac{x}{2}$ C) $x \rightarrow \frac{3x}{4}$ D) $x \rightarrow \frac{1}{2x}$

16. A rectangle has an area of 36 cm^2 and a width of 3 cm. Find its perimeter.

A) 12 cm B) 18 cm C) 24 cm D) 30 cm

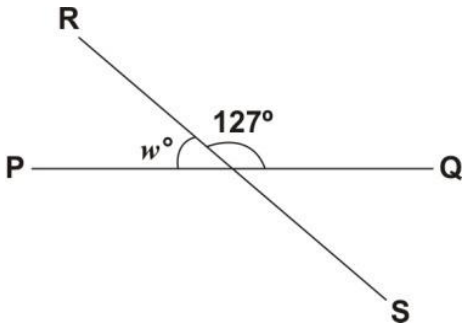
17. Find the value of $4 + x^o$

A) 3 B) 4 C) 5 D) 6

18. A van travels 154km in 1¾hours. Find its speed in km/h.

A) 77km/h B) 88km/h C) 100km/h D)269.5km

19. In the figure below, PQ and RS are straight lines. Find the value of the angle marked w^o.



A) 27° B) 53° C) 63° D) 127°

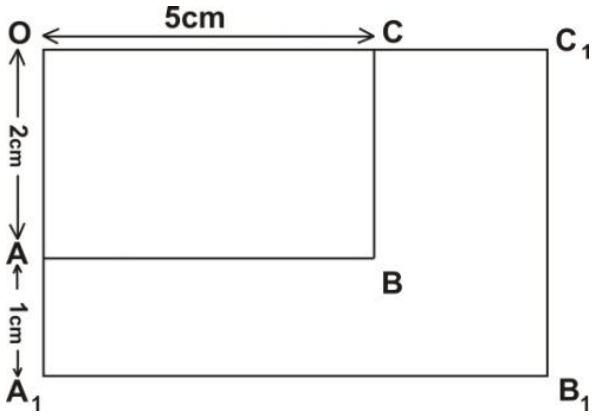
20. If $x + 5 = -7$, find the value of $\frac{x}{6}$

A) -5 B) -2 C)-⅓ D) 2

In the diagram below, rectangle OABC is enlarged into rectangle $OA_1B_1C_1$ from centre O.

|OC|=5cm, |OA|=2cm and $|AA_1| = 1$cm

Use the diagram to answer Questions **21** *and* **22**



21. Find the scale factor of the enlargement.

A) 1 B) 1.5 C) 2 D) 2.5

22. Calculate OC_1

A) 7.5cm B) 8cm C) 9cm D) 12cm

23. The height of a flag pole in a scale drawing is 5cm. If the scale is 1cm to 3m, what is the actual height of the pole?

A) 10 m B) 15 m C) 8 m D) 5 m

24. Factorize: $ax + 3x + a + 3$

A. x(a+x) B. (x-a)(x+3) C. x(a-x) D. (a+3)(x+1)

25. If $y = \dfrac{x+8}{x-4}$, find the value of y when $x = 2$

A) 5 B) -5 C) 4 D) -4

26. Find the greatest common factor (GCF) of 90, 126 and 72.

A) 6 B) 9 C) 18 D) 24

27. Given that $1 : 3 = x : 21$, find the value of x.

A) 4 B) 5 C) 7 D) 63

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28. Find the difference between 432_{five} and 143_{five}

A)234_{five} B) 334_{five} C) 1130_{five} D) 1310_{five}

29. Simplify: $14\frac{1}{3} - 2\frac{3}{8} + 5\frac{7}{12}$

A) $6\frac{3}{8}$ B) $7\frac{5}{12}$ C) $17\frac{5}{12}$ D) $17\frac{13}{24}$

30. Nine bottles of equal capacity hold 4½ litres of water. How much do *x* bottles hold?

A) ½ *x* litres

B) 2*x* litres

C) 20½ *x* litres

D) 40½ *x* litres

31. A sales girl receives a 5% commission on all she sells. Find how much she has to sell to receive GH¢15.00

A) GH¢750.00 B) GH¢300.00 C) GH¢75.00 D)GH¢30.00

32. Arrange the following fractions in ascending order: $\frac{7}{12}$, $\frac{3}{5}$, $\frac{7}{15}$, $\frac{3}{4}$

A) $\frac{3}{5}, \frac{7}{15}, \frac{7}{12}, \frac{3}{4}$

B) $\frac{7}{12}, \frac{7}{15}, \frac{3}{5}, \frac{3}{4}$

C) $\frac{7}{15}, \frac{7}{12}, \frac{3}{5}, \frac{3}{4}$

D) $\frac{3}{5}, \frac{3}{4}, \frac{7}{12}, \frac{7}{15}$

33. Add 2.5 to the product of 4.2 and 0.2

A) 13.4 B)10.9 C)3.34 D)1.34

34. What is the value of four in the number 7073.48?

A) four tenth B) four C) forty D) four hundred

The table below shows the ages of children at a birthday party.

Ages (yrs)	1	2	3	4	5	6	7	8	9	10
No. of children	3	4	2	5	4	4	6	4	2	1

Use this table to answer Questions **35**, **36** *and* **37**

35. What is the modal age?

A) 10 B) 7 C) 5 D) 4

36. A child is selected at random. What is the probability that he or she is 4 years old?

A) $^4/_{35}$ B) $^1/_7$ C) $^2/_5$ D) $^4/_5$

37. How many children are 7 or more years old?

A) 28 B) 22 C) 13 D) 6

38. Write the vector $\begin{pmatrix} 6cm\ W \\ 4cm\ S \end{pmatrix}$ using Cartesian components.

A. $\begin{pmatrix} -6 \\ -4 \end{pmatrix}$ B. $\begin{pmatrix} -4 \\ -6 \end{pmatrix}$ C. $\begin{pmatrix} -6 \\ 4 \end{pmatrix}$ D. $\begin{pmatrix} 6 \\ -4 \end{pmatrix}$

39. Simplify: $5 - 7 + 2(3 - 8)$

A) -12 B) -8 C) -5 D) -4

40. If $\mathbf{r} = \begin{pmatrix} -10 \\ -3 \end{pmatrix}$ and $\mathbf{n} = \begin{pmatrix} 8 \\ -6 \end{pmatrix}$, find $\mathbf{n}+\mathbf{r}$

A. $\begin{pmatrix} -2 \\ -9 \end{pmatrix}$ B. $\begin{pmatrix} 2 \\ -9 \end{pmatrix}$ C. $\begin{pmatrix} 18 \\ 9 \end{pmatrix}$ D. $\begin{pmatrix} -2 \\ -3 \end{pmatrix}$

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MATHEMATICS 1

OBJECTIVE TEST

SOLUTIONS

1. D.{1,2,3,4,5,6,9,12}

2. C) 2.34×10^6

3. C) 6

4. B) 78.43

5. B) 3m+4n

6. B. {2,1,4,3,}

7. C)1,712

8. B. $\{x: x\leq 2\}$

9. C) 70°

10. A) 45%

11. D) ¼

12. B) -3

13. A) 6

14. C. {45°, 45°, 90°}

15. A) $x\rightarrow \frac{x}{4}$

16. D) 30 cm

17. C) 5

18. B) 88km/h

19. B) 53°

20. B) -2

21. B) 1.5

22. A) 7.5cm

23. B) 15 m

24. D. $(a+3)(x+1)$

25. B) -5

26. C) 18

27. C) 7

28. A) 234_{five}

29. D) $17\ {}^{13}/_{24}$

30. A) ½ x litres

31. B) GH¢300.00

32. C) ${}^{7}/_{15}$, ${}^{7}/_{12}$, ${}^{3}/_{5}$, ¾

33. C)3.34

34. A) four tenth

35. B) 7

36. B) ${}^{1}/_{7}$

37. C) 13

38. A. $\begin{pmatrix} -6 \\ -4 \end{pmatrix}$

39. A) -12

40. A. $\begin{pmatrix} -2 \\ -9 \end{pmatrix}$

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April 2009

MATHEMATICS 2

ESSAY
1 hour

[60 marks]

Answer **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is **not** *allowed*

Marks will **not** *be awarded for correct answers without corresponding working.*

All questions carry equal marks

1. (a) Simplify: $\frac{1200\times1260}{800}$ and write your answer in standard form

(b) A plot of land measures 25m by 12m. A portion of this plot measuring 8m by 8m is used for the cultivation of vegetables. Find the area of the plot **not** cultivated

(c) The table below shows the performance of Aisha in her final examination.

Subject	**Score**
English Language	54%
Mathematics	36%
Ga	68%
Science	50%
Social Studies	32%

Draw a pie chart to represent this information

2. The table below shows the scores of some students in an examination.

Scores	0	1	2	3	4	5	6	7	8	9	10

Frequency	3	5	3	2	7	6	5	4	2	2	1

From the table find

(a) how many students wrote the examination

(b) the modal score

(c) the number of students that scored 7 or more

(d) the mean score correct to one decimal place

3. (a) (i) Using a scale of 2cm to 1 unit on both axes, draw two perpendicular axes OX and OY on a graph sheet.

(ii) Mark on the same graph sheet the x-axis from -5 to 5 and y-axis from -6 to 6

(iii) Plot the points A(2,5), B(2,2) and C(4,2). Join the points A, B and C to form a triangle ABC

(iv) Using the y-axis as mirror line, draw the image triangle $A_1B_1C_1$ of the triangle ABC such that $A \rightarrow A_1$, $B \rightarrow B_1$ and $C \rightarrow C_1$. Write down the coordinates of A_1, B_1 and C_1

(v) Draw the image triangle $A_2B_2C_2$ of triangle ABC under anticlockwise rotation of 180° about the origin where $A \rightarrow A_2$, $B \rightarrow B_2$ and $C \rightarrow C_2$. Write down the coordinates of A_2, B_2 and C_2

(b) Given that $a = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$, $b = \begin{pmatrix} -5 \\ -7 \end{pmatrix}$ *and* $c = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$

Evaluate $2a - 3c + b$

4. (a) The ratio of sheep to goats on a farm is 4:7. If there are 1,428 sheep, find how many goats are on the farm.

(b) Using a ruler and a pair of compasses only, construct a triangle ABC with |AB| = 6cm, |AC| = 8cm and angle BAC = 30°.

Construct the bisector of angle ACB to meet line AB at D.

(i) Measure |AD| and |BD|

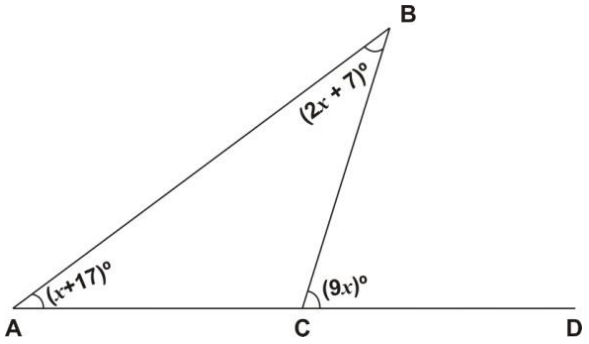
(ii) Write down the ratio |AD|**:**|BD|.

5. (a) The diagram is a triangle ABC with the side AC produced to D.

Find:

(i) The value of x

(ii) angle ACB



(b) The simple interest formula $I = \frac{PTR}{100}$ gives the interest, I on a principal, P invested at a rate, R per annum for time, T years.

(i) Find the simple interest on GH¢3,600.00 at 15% per annum for 2 years.

(ii) Make R the subject of the simple interest formula.

(iii) At what rate per annum will GH¢6,000.00 earn GH¢2,400.00 simple interest in 2 years?

6. (a) Given that ¢10,000.00 = GH¢1.00.

Complete the following table relating cedis (x) to Ghana cedis (y)

¢ (x)	10,000	50,000	150,000	250,000	350,000	450,000
GH¢(y)	1					45

(b) (i) On a graph sheet draw two perpendicular axes OX and OY.

(ii) Using a scale of 2cm to ¢50,000.00 on the Ox axis and 2cm to GH¢5.00 on the Oyo axis. Mark the Ox axis from 0 to ¢450,000 and OY axis from 0 to GH¢50.00

(c) Plot the points and join them with a straight line

(d) From your graph find the value of

(i) GH¢8.00 in cedis (¢)

(ii) GH¢35.00 in cedis (¢)

(iii) ¢260,000.00 in Ghana Cedis (GH¢)

April 2009

MATHEMATICS 2

ESSAY

SOLUTIONS

1 (a)

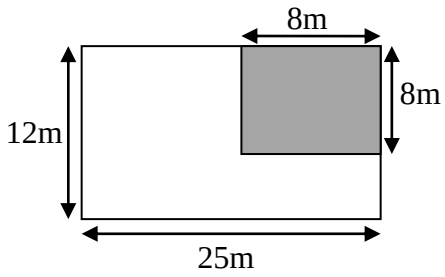
$$\frac{1200\times1260}{800} = \frac{12\times1260}{8}$$

$$= 1890$$

$$= \underline{1.89 \times 10^3}$$

Simplifying by 'cancellation'

1 (b)



Total area of land = 25 m × 12 m = 300 m^2

Area cultivated = 8 m × 8 m = 64 m^2

Area not cultivated = Total area – Area cultivated

= 300 m^2 – 64 m^2

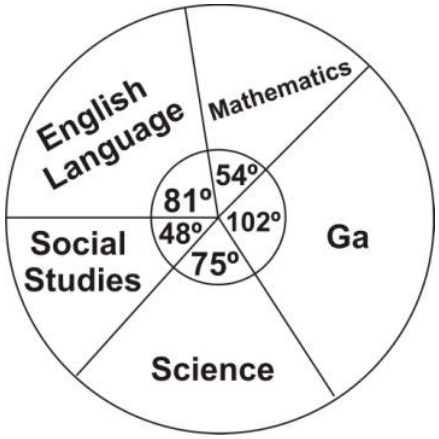
= $\underline{236\ m^2}$

1 (c)

Subject	Score	Angle of sector
English Language	54%	$\frac{54}{240}\times 360^o = 81^o$
Mathematics	36%	$\frac{36}{240}\times 360^o = 54^o$
Ga	68%	$\frac{68}{240}\times 360^o = 102^o$

Science	50%	$\frac{50}{240} \times 360^{o} = 75^{o}$
Social Studies	32%	$\frac{32}{240} \times 360^{o} = 48^{o}$
TOTAL	**240**	**360°**

Pie chart showing the performance of Aisha in her final examination



2 (a) 3 + 5 + 3 + 2 + 7 + 6 + 5 + 4 + 2 + 2 + 1 = 40

⇒ 40 students wrote the examination

2(b) The modal score is the score that occurs most

(i.e., has the highest frequency)

⇒ Modal score = 4

2(c) 4 + 2 + 2 + 1 = 9

⇒ 9 students scored 7 or more.

2(d) **Method 1**

$$\text{The Mean} = \frac{\text{The sum of scores}}{\text{The total number of students}}$$

$$= \frac{(0\times3)+(1\times5)+(2\times3)+(3\times2)+(4\times7)+(5\times6)+(6\times5)+(7\times4)+(8\times2)+(9\times2)+(10\times1)}{3+5+3+2+7+6+5+4+2+2+1}$$

$$= \frac{0+5+6+6+28+30+30+28+16+18+10}{40}$$

$$= \frac{177}{40} = \underline{\underline{4\frac{17}{40}}} \quad \text{or} \quad \underline{\underline{4.425}}$$

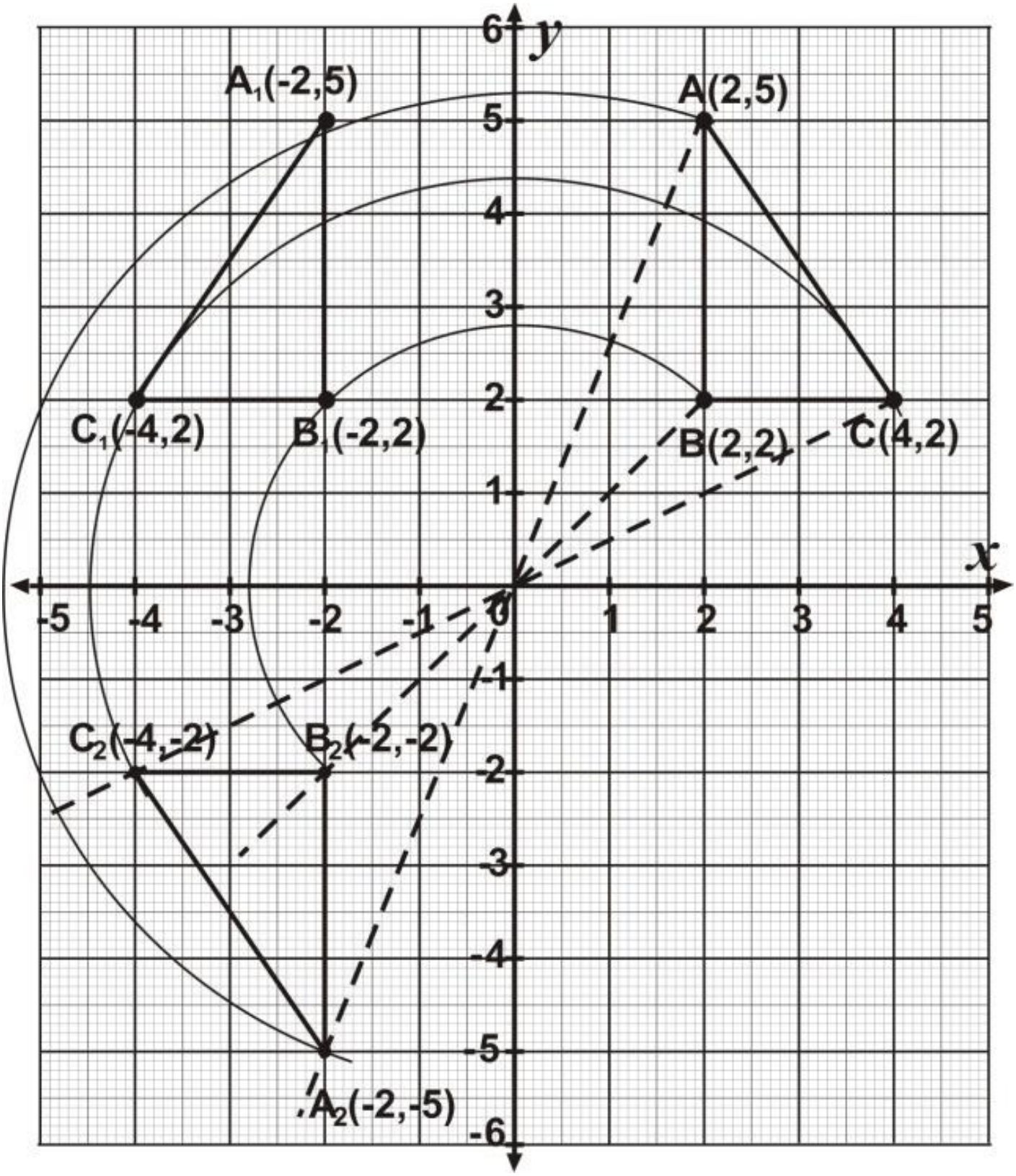
2(d) **Method 2**

Scores (x)	**Frequency (f)**	**f x**
0	3	0
1	5	5
2	3	6
3	2	6
4	7	28
5	6	30
6	5	30
7	4	28
8	2	16
9	2	18
10	1	10
	$\Sigma f=40$	$\Sigma fx=177$

The mean $= \frac{\Sigma f\ x}{\Sigma f} = \frac{177}{40} = \underline{\underline{4\frac{17}{40}}}$ or $\underline{\underline{4.425}}$

3(a)

Approach 1 (By Inspection / Construction)



3(a)

Approach 2 (The rule / formula)

3 (a) (iv) Reflecting (x, y) in the y-axis

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} -x \\ y \end{pmatrix}$$

$$\Rightarrow \quad OA\begin{pmatrix} 2 \\ 5 \end{pmatrix} \rightarrow OA_1\begin{pmatrix} -2 \\ 5 \end{pmatrix}, \quad \therefore\ A_1(-2,5)$$

$$\Rightarrow \quad OB\begin{pmatrix} 2 \\ 2 \end{pmatrix} \rightarrow OB_1\begin{pmatrix} -2 \\ 2 \end{pmatrix}, \quad \therefore\ B_1(-2,2)$$

$$\Rightarrow \quad OC\begin{pmatrix} 4 \\ 2 \end{pmatrix} \rightarrow OC_1\begin{pmatrix} -4 \\ 2 \end{pmatrix}, \quad \therefore\ C_1(-4,2)$$

∴ Plot and join A_1(-2,5), B_1(-2,2) and C_1(-4,2) as the image of triangle ABC under a reflection in the y axis as shown above.

3 (a) (v) Rotating (x, y) through 180° about the origin

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} -x \\ -y \end{pmatrix}$$

$$OA\begin{pmatrix} 2 \\ 5 \end{pmatrix} \rightarrow OA_2\begin{pmatrix} -2 \\ -5 \end{pmatrix}, \quad \therefore\ A_2(-2,-5)$$

$$OB\begin{pmatrix} 2 \\ 2 \end{pmatrix} \rightarrow OB_2\begin{pmatrix} -2 \\ -2 \end{pmatrix}, \quad \therefore B_2(-2,-2)$$

$$OC\begin{pmatrix} 4 \\ 2 \end{pmatrix} \rightarrow OC_2\begin{pmatrix} -4 \\ -2 \end{pmatrix}, \quad \therefore\ C_2(-4,-2)$$

∴ Plot and join A_2(-2,-5), B_2(-2,-2) and C_2(-4,-2) as the image of triangle ABC under a rotation through 180° about the origin as shown above.

3 (b)

$$2\boldsymbol{a} - 3\boldsymbol{c} + \boldsymbol{b}$$

$$\Rightarrow \quad 2\begin{pmatrix} -3 \\ 2 \end{pmatrix} - 3\begin{pmatrix} -4 \\ -1 \end{pmatrix} + \begin{pmatrix} -5 \\ -7 \end{pmatrix}$$

$$\Rightarrow \quad \begin{pmatrix} 2\times(-3) \\ 2\times 2 \end{pmatrix} - \begin{pmatrix} 3\times(-4) \\ 3\times(-1) \end{pmatrix} + \begin{pmatrix} -5 \\ -7 \end{pmatrix}$$

$$\Rightarrow \quad \begin{pmatrix} -6 \\ 4 \end{pmatrix} - \begin{pmatrix} -12 \\ -3 \end{pmatrix} + \begin{pmatrix} -5 \\ -7 \end{pmatrix}$$

$$\Rightarrow \quad \begin{pmatrix} -6-(-12)+(-5) \\ 4-(-3)+(-7) \end{pmatrix}$$

$$\Rightarrow \quad \begin{pmatrix} -6+12-5 \\ 4+3-7 \end{pmatrix}$$

$$\Rightarrow \quad \underline{\underline{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}}$$

4 (a) **Method 1** (Unitary Approach – the value of one item)

If the sheep ratio, 4, corresponds to 1428,

then ratio, 1, corresponds to $\frac{1428}{4} = 357$

hence, goat ratio, 7, corresponds to $7 \times 357 = \underline{2499 \text{ goats}}$

4 (a) **Method 2** (Equivalent Fractions)

	Ratio	**Actual no.**
Sheep	4	1428
Goats	7	g

From the table above, $\frac{4}{7} = \frac{1428}{g}$

Cross-multiplying

$\Rightarrow \quad 4 \times g = 1428 \times 7$

Dividing both side by 4 to find g

$\Rightarrow \quad \frac{4g}{4} = \frac{1428 \times 7}{4}$

$\Rightarrow \quad g = 357 \times 7$

$\Rightarrow \quad = \underline{2499 \text{ goats}}$

4 (a) **Method 3** (The rule: If more, less divides …)

If 4 → 1428

Then 7 → ? (more)

If more, less (i.e., 4) divides; ∴ we have

$\frac{7}{4} \times 1428 \quad \Rightarrow 7 \times 357 = \underline{2499 \text{ goats}}$

4 (a) **Method 4** (Equivalent ratios)

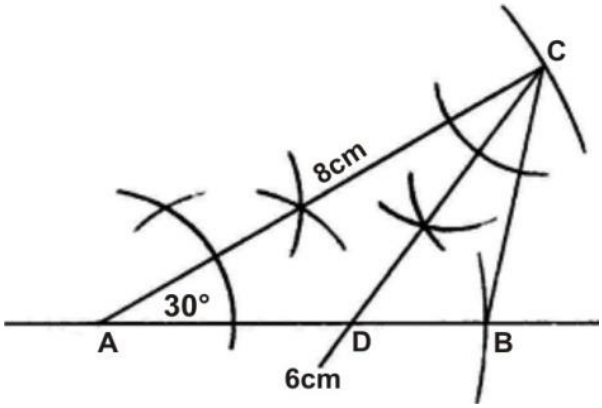
$4 : 7 = 1428 : g$

$\Rightarrow \quad \frac{4g}{4} = \frac{7 \times 1428}{4}$

$\Rightarrow \quad g = 2499$

$\Rightarrow \quad$ 2499 goats

4(b)



(i) $|AD| \approx 4$ cm, $|BD| \approx 2$ cm

(ii) $|AD| : IBD| = 4\text{cm} : 2\text{cm}$

$= 2 : 1$

5 (a)(i) **Method 1** **(Exterior angle = two opposite interior angles)**

From the diagram,

$\Rightarrow \quad (x+17)^o + (2x+7)^o = (9x)^o$

$\Rightarrow \quad x^o + 17^o + (2x)^o + 7^o = (9x)^o$

$\Rightarrow \quad 17^o + 7^o = (9x)^o - x^o - (2x)^o$

$\Rightarrow \quad 24^o = (6x)^o$

$\Rightarrow \quad \frac{24}{6} = \frac{6x}{6}$

Exterior angle (9x)° = two opposite interior angles (x+17)° and (2x+7)°

Solving for x

$\Rightarrow \quad 4 = x$

$\Rightarrow \quad x = 4$

5 (a)(i) Method 2 (Sum of interior angles of triangle = 180°)

From the diagram, $\angle ACB + (9x)° = 180°$

Sum of angles at a point on a straight line = 180°

$\angle ACB = 180° - (9x)°$

$\Rightarrow 180° - (9x)° + (x+17)° + (2x+7)° = 180°$

Sum of interior angles of triangle = 180°

$\Rightarrow 180° - 180° + (x+17)° + (2x+7)° = (9x)°$

$\Rightarrow \quad x^o + 17^o + (2x)^o + 7^o = (9x)^o$

$\Rightarrow \quad 17^o + 7^o = (9x)^o - x^o - (2x)^o$

$\Rightarrow \quad 24^o = (6x)^o$

$\Rightarrow \quad \frac{24}{6} = \frac{6x}{6}$

$\Rightarrow \quad 4 = x$

$\Rightarrow \quad x = 4$

5 (a)(ii)

Sum of angles at a point on a straight line = 180°

$\angle ACB + (9x)° = 180°$

$\angle ACB + (9\times 4)° = 180°$

$\angle ACB + 36° = 180°$

$\angle ACB = 180° - 36°$

Angle ACB = 144°

5 (b) (i) $I = \frac{PTR}{100} = \frac{3{,}600 \times 2 \times 15}{100}$

$= 36 \times 2 \times 15 = 1080$

Simple interest = GH¢1,080.00

5 (b) (ii) Method 1

$$\mathrm{I} = \frac{PTR}{100}$$

$$100 \times I = \frac{PTR}{100} \times 100$$

Multiplying both sides by 100 to clear fraction

$$\Rightarrow \quad 100 \times I = PTR$$

Dividing both sides by PT to make R the subject

$$\frac{100 \times I}{PT} = \frac{PTR}{PT}$$

$$\Rightarrow \quad \frac{100 \times I}{PT} = R$$

Switching positions

$$\Rightarrow \quad R = \frac{100 \times I}{PT}$$

$$\Rightarrow \quad \underline{\underline{R = \frac{100\ I}{PT}}}$$

5 (b) (ii) **Method 2**

$$\mathrm{I} = \frac{PTR}{100}$$

$$100 \times \mathrm{I} = PTR$$

Cross-multiplying

$$\Rightarrow \quad \frac{100 \times \mathrm{I}}{\mathrm{PT}} = \frac{PTR}{PT}$$

Dividing both sides by PT

$$\Rightarrow \quad \frac{100 \times I}{PT} = R$$

Switching positions

$$\Rightarrow \quad R = \frac{100 \times I}{PT}$$

$$\Rightarrow \quad \underline{\underline{R = \frac{100\ I}{PT}}}$$

5 (b) (ii) **Method 3**

$$\mathrm{I} = \frac{PTR}{100}$$

Multiplying both sides by $\frac{100}{PT}$ to remove $\frac{PT}{100}$

$$\frac{100}{PT} \times I = \frac{100}{PT} \times \frac{PTR}{100}$$

$$\Rightarrow \quad \frac{100 \times I}{PT} = R$$

Switching positions

$$\Rightarrow \quad R = \frac{100 \times I}{PT}$$

$$\Rightarrow \qquad R = \frac{100\ I}{PT}$$

5 (b) (iii) **Approach 1** **(Substitute values in the expression for R)**

Using the result obtained in (ii) above

$$R = \frac{100\ I}{PT}$$

Substituting and simplifying to find R

$$\Rightarrow \quad R = \frac{100 \times 2{,}400}{6{,}000 \times 2}$$

$$\Rightarrow \quad R = \frac{100 \times 2{,}400}{6{,}000 \times 2} = 20$$

$$\Rightarrow \text{The rate} = \underline{20\%}$$

Approach 2 **(Substitute values in the given expression for Simple Interest)**

Substituting in $I = \frac{PTR}{100}$, we have

Substituting and solving for R

$$2{,}400 = \frac{6{,}000 \times 2 \times R}{100}$$

Simplifying to find R

$$\Rightarrow \quad \frac{2{,}400 \times 100}{6{,}000 \times 2} = R$$

$$\Rightarrow \quad R = 20$$

$$\Rightarrow \quad \text{The rate} = \underline{20\%}$$

NB:

In actual fact, the simple interest, I, on a principal, P, at a rate, R, over a time, T, is given by I = PTR, (not $\frac{PTR}{100}$)

[the $\frac{}{100}$ is on account of the rate being usually expressed as a percentage]

Hence, if for instance, P = GH¢6000, T = 3 years and R = 15%

Then Simple interest $= P \times T \times R$

$$= 6000 \times 3 \times 15\%$$

$$= 6000 \times 3 \times \frac{15}{100}$$

$$= \underline{\text{GH¢ 2700}}$$

Not $\frac{P \times T \times R}{100}$,

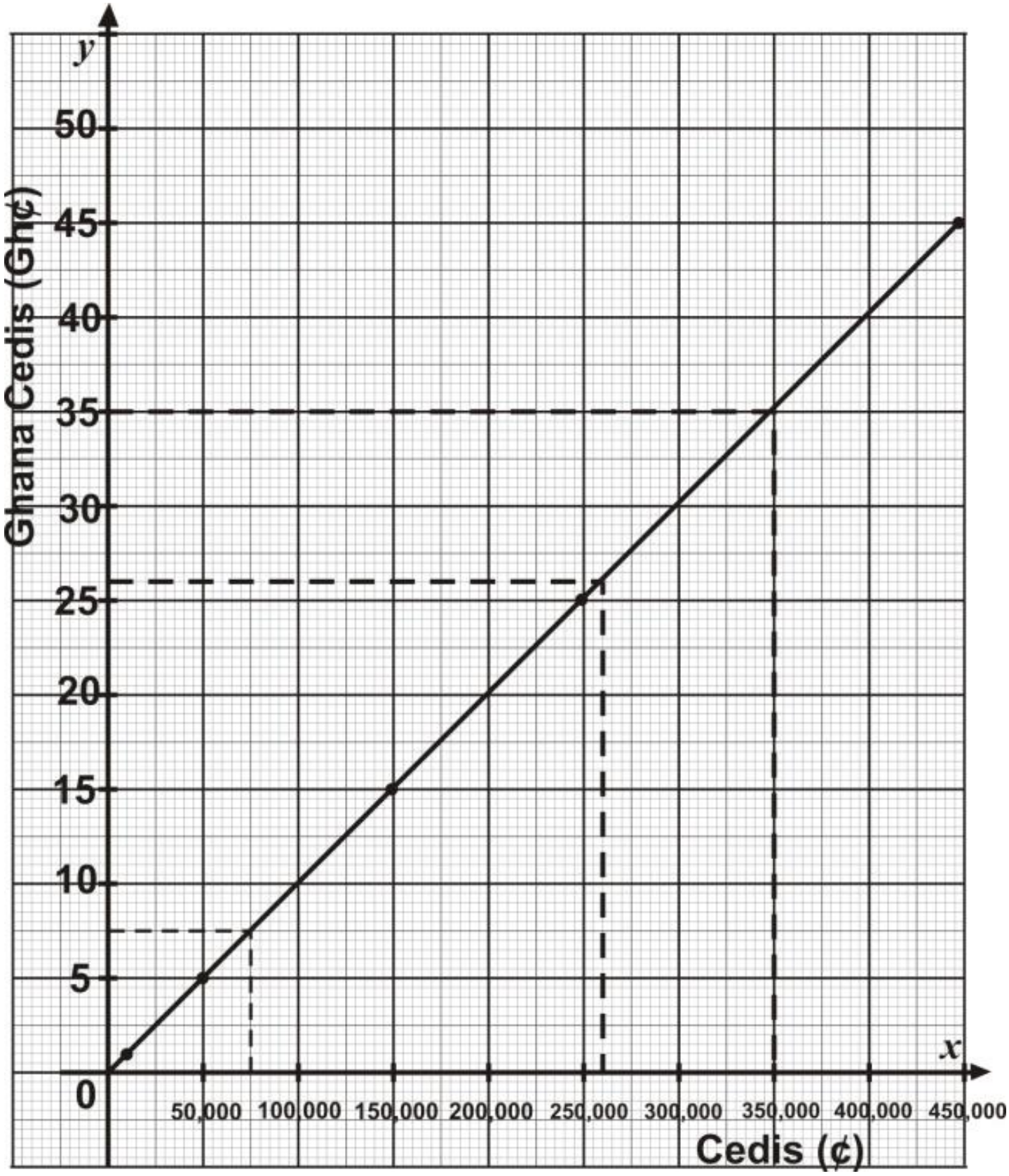
$$\Rightarrow \frac{6000 \times 3 \times 15\%}{100}$$

6(a) Using the idea of simple proportion, the table is completed as shown below:

¢ (x)	10,000	50,000	150,000	250,000	350,000	450,000
GH¢(y)	1	5	15	25	35	45

6 **(b), (c)** The straight line graph is shown below

0201054428.



6 (d) From the graph, the value of

(i) GH¢8.00 = ¢ 80,000.00

(ii) GH¢35.00 = ¢350,000.00

(iii) ¢260,000.00 = GH¢26,000.00

April 2008

MATHEMATICS 1

OBJECTIVE TEST
1 hour

1. M = {1,2,3,4,5, …, 20}, Q = {3,4,5,6,7,8} and R = {2,3,5,7}

If Q and R are subsets of M, find Q∩R.

A. {3,5} B.{5,7} C.{3,5,7} D.{,2,3,5,7}

2. List the members of the set $\{2 \leq x \leq 5\}$

A.{2,5} B.{2,3,4} C.{2,3,5} D.{2,3,4,5}

3. Round 8921465 to the nearest hundred

A) 8921000 B) 8921400 C) 8921460 D) 8921500

4. Write 98 as a product of its prime factors

A) 2×7 B) $2^2\times7$ C) 2×7^2 D)$2^2\times7^2$

5. Evaluate 4(8-2) + 5(3-8)

A) -31 B) -1 C) 37 D) 49

6. Arrange the following in descending order of magnitude: 0.32, $^2/_5$, 27%, $^1/_3$

A) 0.32, $^2/_5$, 27%, $^1/_3$

B) 0.32, $^1/_3$, $^2/_5$,27%

C) 27%, 0.32, $^1/_3$, $^2/_5$

D) $^2/_5$, $^1/_3$, 0.32,27%

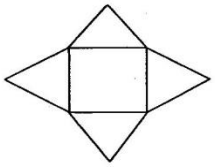
7. The ratio 8:12 is equivalent to *y*:9. What is the value of *y*?

A) 4 B) 5 C) 6 D) 7

8. Write 0.55 as a fraction in its lowest term

A) $^{11}/_{200}$ B) $^{11}/_{20}$ C) $^{11}/_2$ D) $^{11}/_5$

9. Which solid figure can be made from the net below?



A) Triangular prism
B) Square pyramid
C) Triangular pyramid
D) Cuboid

10. Simplify $5w + 7p^2 - 4w + 3p^2$

A) $9w+10p^2$ B) $w+10p^2$ C) $w+4p^2$ D) $9w+4p^2$

11. Osei bought a hat for GH¢5.00. He sold it to Yaovi at a profit of 20%. How much did Yaovi pay for the hat?

A) GH¢4.80 B) GH¢5.50 C) GH¢6.00 D)GH¢7.00

12. Simplify $2^9 \div 2^3$

A) 2^3 B) 2^6 C) 2^{12} D) 2^{27}

13. Find the image of -3 under the mapping $x \rightarrow 2(x+3)$

A) 0 B) 2 C) 6 D) 12

14. A football field is 120 m long and 75 m wide. What is the perimeter of the field?

A) 195 m B) 390 m C) 780 m D) 900 m

15. A tank holds 240 litres of water. How much water is in the tank when it is $^4/_5$ full?

A) 60 litres B) 132 litres C) 192 litres D)240 litres

16. A man has 6x sheep and 5y goats. He sells 3x sheep and 2y goats. How many animals are left after the sales?

A) 3x-3y B) 3x+3y C) 9x-5y D) 9x+5y

The table below shows the average rainfall in a town from March 2003 to August 2003.

Month	Mar	Apr	May	Jun	Jul	Aug
Rainfall (mm)	96	147	281	452	265	139

17. What was the total amount of rainfall in May, June and July?

A) 696mm B) 930mm C) 998mm D) 1020mm

18. What was the mean rainfall in the town over the six months?

A) 230mm B) 281mm C) 366mm D) 452mm

19. A trader buys a dozen pens at GH¢4.80 and sells them at 48Gp each. Find her percentage profit.

A) 5% B) 10% C) 15% D) 20%

20. Find the angle through which the minute hand of a clock moves from 5:15pm to 5:25pm

A) 30° B) 45° C) 60° D) 120°

21. If E = {prime numbers between 10 and 20} and F = {odd numbers between 0 and 16}, find E∩F.

A. {11} B. {11,13} C. {3,11,13} D.{3,11,13,15}

22. Write 17_{ten} as a base two numeral.

A) 1001 B) 10001 C) 11001 D) 11011

23. What fraction of a revolution is 72°

A) $^1/_6$ B) $^1/_5$ C) $^2/_5$ D) $^5/_8$

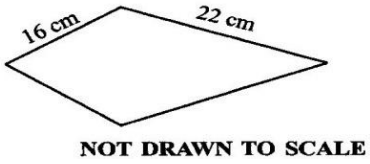
24. Express 6 days is to 3 weeks as a ratio in its simplest form

A) 1:2 B) 2:1 C) 2:7 D) 7:2

25. The letters in the word HIPPOPOTAMUS are placed in a box. What is the probability of taking out a letter that is a vowel?

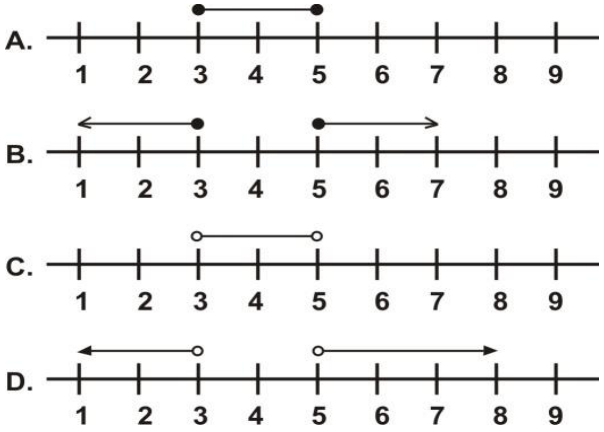
A) $^{1}/_{12}$ B) $^{3}/_{12}$ C) $^{5}/_{12}$ D) $^{7}/_{12}$

26. Find the perimeter of the kite below.



A) 38cm B) 64cm C) 76cm D) 88cm

27. Illustrate $3 < x < 5$ on the number line, where $x \in$ {rational numbers}



28. How many lines of symmetry has an equilateral triangle?

A) 1 B) 2 C) 3 D) 4

29. If $\mathbf{p} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -2 \\ 7 \end{pmatrix}$, find $4\mathbf{p} - 2\mathbf{q}$.

A. $\begin{pmatrix} 12 \\ -2 \end{pmatrix}$ B. $\begin{pmatrix} 12 \\ 2 \end{pmatrix}$ C. $\begin{pmatrix} 20 \\ -2 \end{pmatrix}$ D. $\begin{pmatrix} 20 \\ 2 \end{pmatrix}$

30. Which of the following is not a quadrilateral?

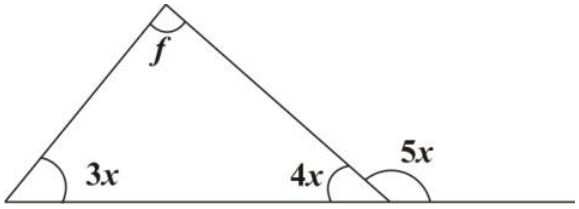
A) Hexagon

B) kite

C) Rectangle

D) Trapezium

Use the figure below to answer Questions **31** and **32**



31. Find the value of x

A) 20° B) 30° C) 40° D) 50°

32. What is the value of f?

A) 10° B) 20° C) 40° D) 80°

33. Solve $\frac{4k}{9} = 12$

A) 23 B) 25 C) 27 D) 29

34. Expand (a + 4) (a + 6)

A) 2a+24 B) a^2+6a+10 C) a^2+10a+10 D)a^2+10a+24

35. Find 12½ % of GH¢80.00

A) GH¢8.00 B) GH¢10.00 C) GH¢12.00 D)GH¢12.50

36. Write the number 34.1 in standard form

A) 3.41×10^{-2}

B) 3.41×10^{-1}

C) 3.41×10^{0}

D) 3.41×10

37. Simplify $\frac{30}{5(-2)}$

A) -10 B) -6 C) -3 D) 3

38. Find the highest common factor of 15 and 21.

A) 1 B) 3 C) 5 D) 7

39. The difference between two numbers is 168. If the smaller number is 113, find the other number

A) 223 B) 271 C) 281 D) 291

40. If $\mathbf{r} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ and $\mathbf{t} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$, evaluate $\mathbf{r} + \mathbf{t}$.

A. $\begin{pmatrix} 0 \\ -2 \end{pmatrix}$ B. $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ C. $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ D. $\begin{pmatrix} 4 \\ 8 \end{pmatrix}$

April 2008

MATHEMATICS 1

OBJECTIVE TEST

SOLUTIONS

1. C. {3,5,7}

2. D. {2,3,4,5}

3. D) 8921500

4. C) 2×7^2

5. B) -1

6. D) $^2/_5$, $^1/_3$, 0.32, 27%

7. C) 6

8. B) $^{11}/_{20}$

9. B) Square pyramid

10. B) $w+10p^2$

11. C) GH¢6.00

12. B) 2^6

13. A) 0

14. B) 390 m

15. C) 192 litres

16. B) $3x+3y$

17. C) 998mm

18. A) 230mm

19. D) 20%

20. C) 60°

21. B. {11,13}

22. B) 10001

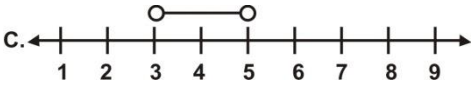
23. B) 1/5

24. C) 2:7

25. C) $^{5}/_{12}$

26. C) 76 cm

27. C.



28. C) 3

29. C. $\begin{pmatrix} 20 \\ -2 \end{pmatrix}$

30. A) Hexagon

31. A) 20°

32. C) 40°

33. C) 27

34. D) $a^2+10a+24$

35. B) GH¢10.00

36. D) 3.41×10

37. C) -3

38. B) 3

39. C) 281

B. $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$

40.

April 2008

MATHEMATICS 2

ESSAY
1 hour

[60 marks]

Answer **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is **not** *allowed*

Marks will **not** *be awarded for correct answers without corresponding working.*

All questions carry equal marks

1. (a) E and F are subsets of the universal set U such that

U = {natural numbers less than 15},

E = {even numbers between 1 and 15} and

F = {multiples of 4 between 9 and 15}.

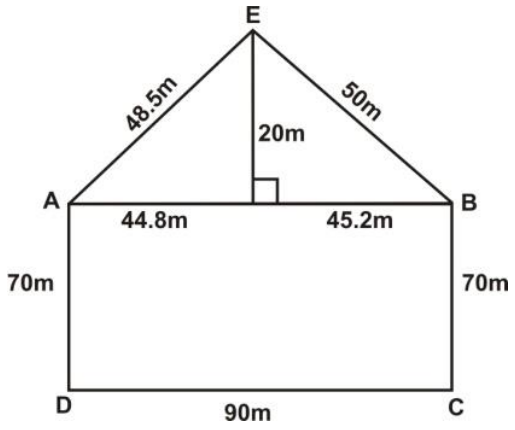
(i) List the elements of U, E and F

(ii) Draw a Venn diagram to show the sets U, E and F

(b) In a school, $^{7}/_{10}$ of the pupils like Mathematics. Half of those pupils who like Mathematics are girls. If there are 240 pupils altogether in the school, how many girls like Mathematics?

(c) A typist charges 28Gp for the first five sheets and 8Gp for each additional sheet she types. How much will she earn, if she types 36 sheets?

2.



The diagram AEBCD shows the shape of Mr. Awuah's garden, which is made up of a rectangular portion ABCD and a triangular portion AEB.

|AB| = |DC| = 90m, |AD|=|BC|=70m, |AE|=48.5m and |EB|=50m. The height of the triangle is 20m.

(a) Find the

(i) Area of ABCD

(ii) Area of AEB

(iii) Total area of the garden

(iv) Perimeter of the garden.

(b) Find the value of x if $\frac{3x-2}{5}$ is greater than $\frac{1-4x}{10}$ by 5.

3. (a) A traffic survey gave the results shown in the table below.

Vehicle	Car	Lorry	Bus	Bicycle
Frequency	15	12	8	25

(i) Represent the information on a pie chart.

(ii) What percentage of the vehicles were lorries?

(b) Akosua was granted a loan of GH¢96.00. The interest rate was 24% per annum. Calculate the

(i) Interest at the end of the year

(ii) Total amount she had to pay at the end of the year.

(iii) Amount she still owes, if Akosua was able to pay only GH¢60.00 at the end of the year

4. (a) Copy and complete the table of values for the relations

0201054428

$y_1 = 2x+5$ and $y_2 = 3-2x$ for x from -4 to3

x	-4	-3	-2	-1	0	1	2	3
$y_1=2x+5$	-3			3		7		11
$y_2=3-2x$	11	9		5				-3

(b) (i) Using a scale of 2cm to 1 unit on the x - axis and 2cm to 2 units on the y-axis, draw two perpendicular axes OX and OY on a graph sheet.

(ii) On the same graph sheet draw the graphs of the relations $y_1 = 2x+5$ and $y_2 = 3-2x$

(c) Find the coordinates of the point where y_1 and y_2 meet.

(d) The vectors $p = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, $q = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ and $r = ½ (q + p)$. Find the vector r

5. (a) Using a ruler and a pair of compasses only,

(i) construct triangle ABC with sides |AB|=7cm, |BC|=8cm and |AC|=9cm;

(ii) draw the perpendicular bisectors of the three sides;

(iii) locate the point of the intersection, O, of the perpendicular bisectors;

(iv) With centre O and radius OA, draw a circle to pass through the vertices of the triangle.

(b) Measure and write down the radius of the circle you have drawn in a) (iv)

(c) Find the product of $(2x – 3)$ and $(x – 1)$

6. (a) The marks obtained by 20 pupils in a test were as follows:

4	8	7	6
2	1	7	4
3	7	6	4
7	5	2	7
5	4	8	3

(i) Construct a frequency distribution table for this data

(ii) What is the mode of the distribution?

(iii) Calculate the mean mark

(iv) What percentage of the pupils passed, if the pass mark is 6?

(v) What is the probability that a pupil selected at random scored **not** more than 5 marks?

(b) Simplify $7\frac{2}{3} - 4\frac{5}{6} + 2\frac{3}{8}$

April 2008

MATHEMATICS 2

ESSAY

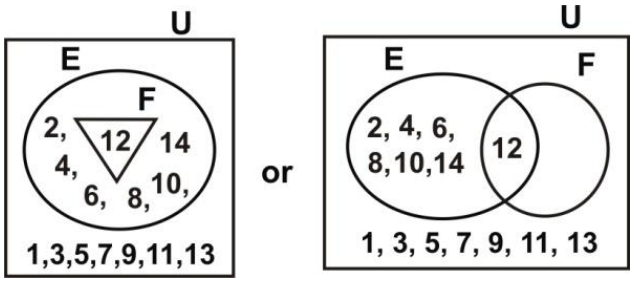
SOLUTIONS

1 (a) (i) U = {1, 2, 3, 4, …, 14}

E = {2, 4, 6, 8, 10, 12, 14}

F = {12}

(ii)



1 (b) **Method 1**

$^{7}/_{10}$ of 240 pupils like Mathematics

⇒ $^{7}/_{10}$ × 240 pupils like Mathematics

= 7 × 24 = 168

⇒ 168 pupils like Mathematics

But half of 168 pupils who like Mathematics are girls

⇒ ½ of 168 pupils are girls who like Mathematics

⇒ ½ × 168 pupils

⇒ 84 girls like Mathematics

1 (b) **Method 2**

Half of pupils who like Mathematics are girls

⇒ ½ of $^{7}/_{10}$ of 240 pupils

⇒ ½ × $^{7}/_{10}$ × 240

⇒ $^{7}/_{20}$ × 240

⇒ 84

⇒ 84 girls like Mathematics

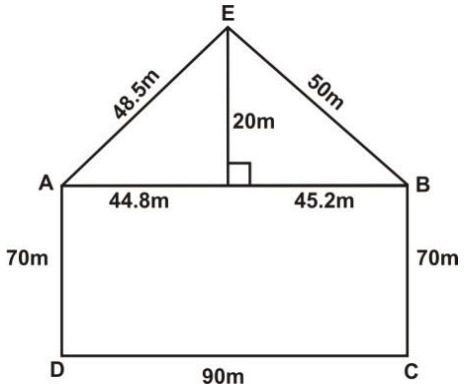
1 (c) Charge for: First 5 sheets = 28 Gp

remaining 31 sheets = 31 × 8 Gp = 248 Gp

Charge for typing 36 sheets = first 5 sheets + remaining 31 sheets

= 28Gp + 248Gp

= 276 Gp **or** GH¢2.76



2 (a) **(i)** Area of ABCD (rectangle)

= Length × width

= 90m × 70m

= 6300 m^2

(ii) Area of AEB (triangle)

= ½ × base × height

= ½ × 90m × 20m

= 900 m^2

(iii) Total area of the garden

= ABCD area + AEB area

= 6300 m^2 + 900 m^2

= 7200 m^2

(iv) Perimeter of garden (Distance around garden)

From the base in the anticlockwise direction,

Perimeter = 90m + 70m + 50m + 48.5m +70m

= 328.5 m

2 (b) If $\frac{3x-2}{5}$ is greater than $\frac{1-4x}{10}$ by 5, then

$$\Rightarrow \quad \frac{3x-2}{5} - \frac{1-4x}{10} = 5$$

Approach 1 (Simplifying the LHS first)

$$\frac{2(3x-2) - (1-4x)}{10} = 5$$

$$\Rightarrow \quad \frac{6x - 4 \quad -1 + 4x}{10} = 5$$

$$\Rightarrow \quad \frac{6x + 4x - 4 - 1}{10} = 5$$

$$\Rightarrow \quad \frac{10x - 5}{10} = 5$$

$$\Rightarrow \quad 10x - 5 = 5 \times 10$$

$$\Rightarrow \quad 10x = 50 + 5$$

$$\Rightarrow \quad \frac{10x}{10} = \frac{55}{10}$$

$$\Rightarrow \quad \underline{\underline{x = 5.5}} \quad \text{or} \quad \underline{\underline{5\tfrac{1}{2}}}$$

2 (b) **Approach 2** (Multiplying by 10 to clear fractions first)

$$10\times\left(\frac{3x-2}{5}\right) - 10\times\left(\frac{1-4x}{10}\right) = 10\times 5$$

$$\Rightarrow \quad 2\times(3x-2) - (1-4x) = 50$$

$$\Rightarrow \quad 6x - 4 - 1 + 4x = 50$$

$$\Rightarrow \quad 6x + 4x - 1 - 4 = 50$$

$$\Rightarrow \quad 10x - 5 = 50$$

$$\Rightarrow \quad 10x = 50 + 5$$

$$\Rightarrow \quad \frac{10x}{10} = \frac{55}{10}$$

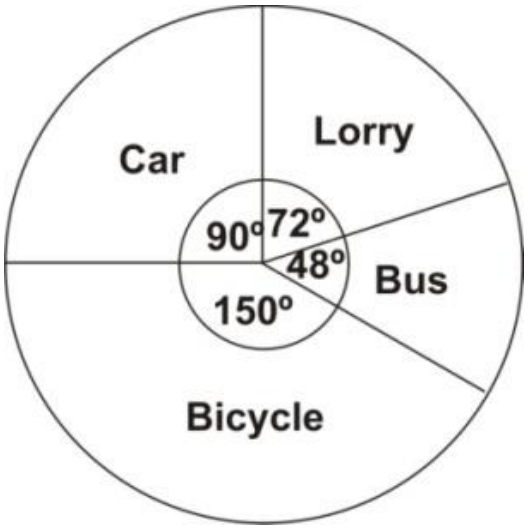
$$\Rightarrow \quad \underline{\underline{x = 5.5}} \quad \text{or} \quad \underline{\underline{5\tfrac{1}{2}}}$$

3 (a) (i)

Vehicle	Frequency	Angle of sector
Car	**15**	$\frac{15}{60} \times 360^o = 90^o$
Lorry	**12**	$\frac{12}{60} \times 360^o = 72^o$
Bus	**8**	$\frac{8}{60} \times 360^o = 48^o$

Bicycle	**25**	$\frac{25}{60} \times 360^{o} = 150^{o}$
TOTAL	**60**	**360°**

Pie chart showing results of a traffic survey



3 (a) (ii) Percentage of lorries $= \frac{\text{No. of lorries}}{\text{Total no. of vehicles}} \times 100\%$

$= \frac{12}{60} \times 100\%$

$= \underline{20\%}$

$\Rightarrow$ <u>20% of the vehicles were lorries</u>

3 (b) (i) Simple interest = Principal × Rate × Time

$= 96 \times 24\% \times 1$

$= 96 \times \frac{24}{100} \times 1 = \frac{96 \times 24}{100}$

$= \frac{2304}{100} = \underline{23.04}$

The simple interest is <u>GH¢23.04</u>

(ii) Total amount to pay = Principal + S. Interest

= GH¢96.00 + GH¢23.04

= GH¢ 119.04

(iii) Amount still owed = Total amount – amount paid

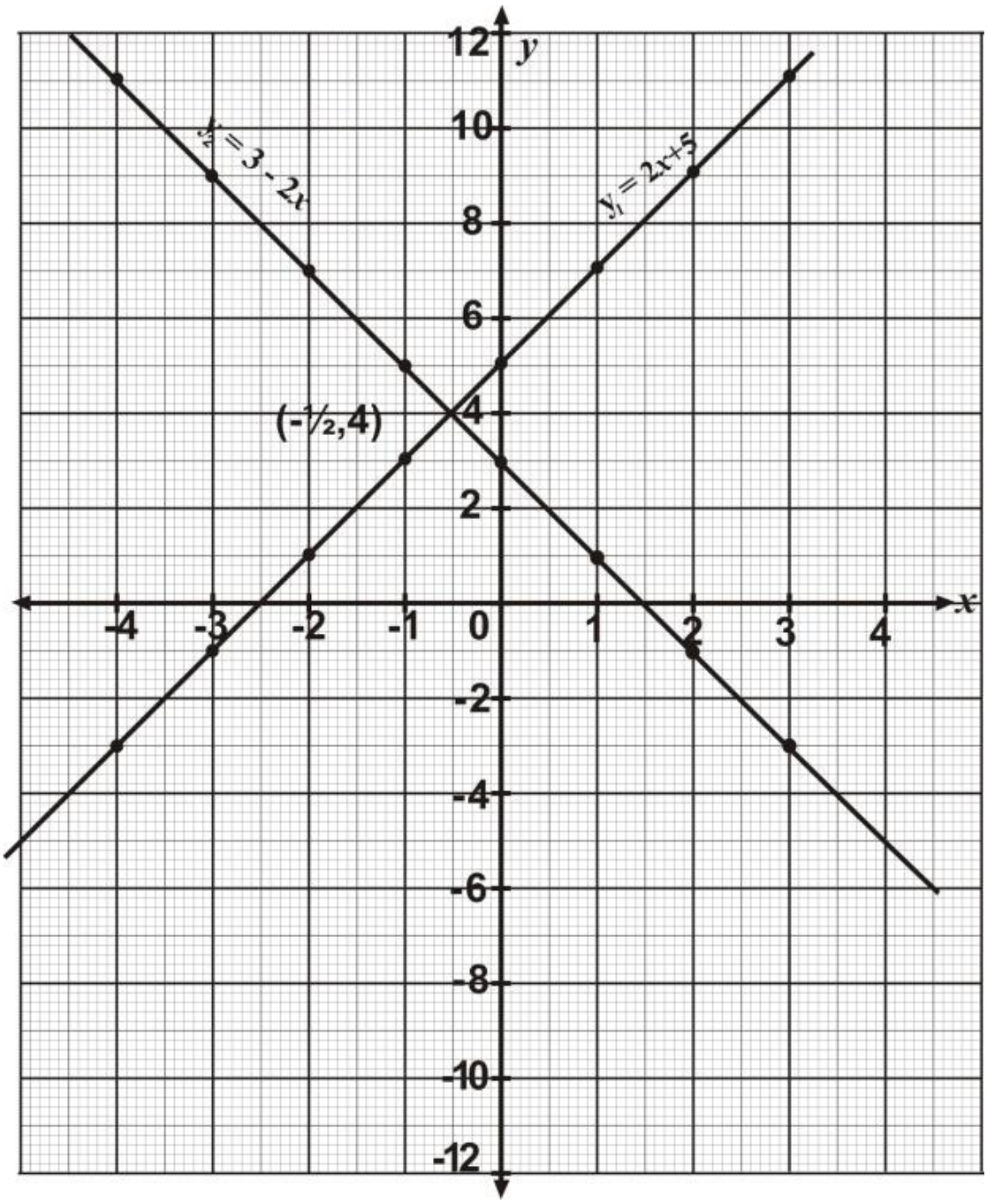
= GH¢ 119.04 – GH¢60.00

= GH¢ 59.04

4 (a) (i) By substitution of the x values in the given relations, the table is completed as shown below:

x	-4	-3	-2	-1	0	1	2	3
$y_1 = 2x+5$	-3	-1	1	3	5	7	9	11
$y_2 = 3-2x$	11	9	7	5	3	1	-1	-3

4 (b) The straight line graph is shown below



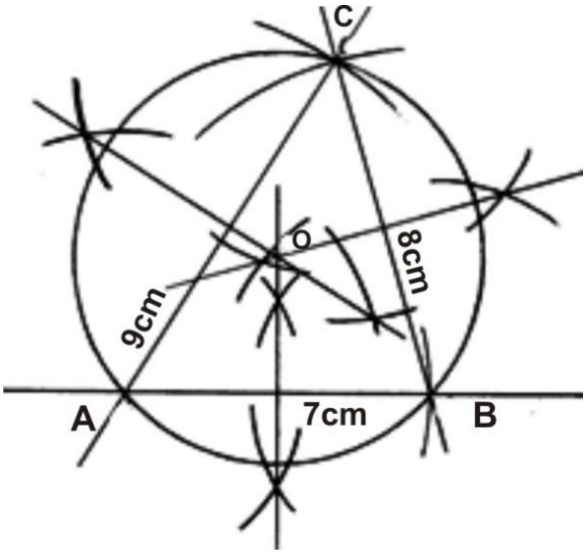
4 (c) y_1 and y_2 meet at (- 0.5, 4)

4 (d) **r** = ½ (**q** + **p**)

$$= \tfrac{1}{2}\left[\begin{pmatrix}2\\5\end{pmatrix}+\begin{pmatrix}2\\3\end{pmatrix}\right] = \tfrac{1}{2}\begin{pmatrix}2+2\\5+3\end{pmatrix}$$

$$= \tfrac{1}{2}\begin{pmatrix}4\\8\end{pmatrix} = \begin{pmatrix}\frac{1}{2}\times 4\\ \frac{1}{2}\times 8\end{pmatrix} = \underline{\underline{\begin{pmatrix}2\\4\end{pmatrix}}}$$

5 (a)



5 (b) Radius of circle ≈ 4.7 cm

5 (c) The product of (2x-3) and (x-1)

Approach 1 (Distributive property)

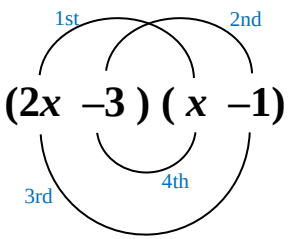
$$(2x - 3)(x - 1)$$

$$2x(x - 1) - 3(x - 1)$$

$= 2x^2 - 2x - 3x + 3$

$= 2x^2 - 5x + 3$

Approach 2 (Smiley face)



1st:	$2x$	x	x	= **$2x^2$**
2nd:	-3	x	-1	= **+3**
3rd:	$2x$	x	-1	= **- 2x**
4th:	-3	x	x	= **- 3 x**

$\Rightarrow \quad 2x^2 + 3 - 2x - 3x$

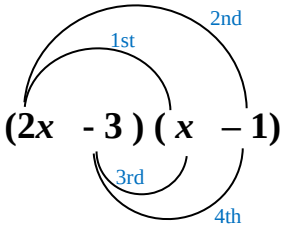
$\Rightarrow \quad 2x^2 - 2x - 3x + 3$

$\Rightarrow \quad \underline{2x^2 - 5x + 3}$

Grouping like terms

$-2x - 3x = -5x$

Approach 3 (Moon shape)



1st: $2x \times x = 2x^2$

2nd: $2x \times -1 = -2x$

3rd: $-3 \times x = -3x$

$2x^2 - 2x - 3x + 3$

$\Rightarrow \quad \underline{2x^2 - 5x + 3}$

6 (a) (i)

Mark	Tally	Frequency
1	/	1
2	//	2
3	//	2
4	////	4
5	//	2
6	//	2
7	~~////~~	5
8	//	2

6 (a) (ii)

Mode = The most occurring quantity

$\Rightarrow$ Mode = 7 marks

6 (a) (iii) **Method 1** **(Using the raw data)**

$$\text{The mean mark} = \frac{\text{The sum of marks}}{\text{The total number of pupils}}$$

$$= \frac{(1\times1)+(2\times2)+(3\times2)+(4\times4)+(5\times2)+(6\times2)+(7\times5)+(8\times2)}{1+2+2+4+2+2+5+2}$$

$$= \frac{1 + 4 + 6 + 16 + 10 + 12 + 35 + 16}{20}$$

$$= \frac{100}{20} = \underline{5}$$

6 (a) (iii) **Method 2** **(Using the frequency table)**

Mark (x)	**Tally**	**Frequency (f)**	**fx**
1	/	1	1
2	//	2	4
3	//	2	6
4	////	4	16
5	//	2	10
6	//	2	12
7	~~////~~	5	35
8	//	2	16
		$\Sigma f = 20$	**$\Sigma fx = 100$**

$$\text{The mean} = \frac{\Sigma f\ x}{\Sigma f} = \frac{100}{20} = \underline{5}$$

6 (a) (iv) Percentage of pupils who passed

$$= \frac{\text{No. of pupils who passed (obtained 6 or more)}}{\text{Total no. of pupils}} \times 100\%$$

$$= \frac{2+5+2}{20} \times 100\%$$

$= \frac{9}{20} \times 100\%$

$= \underline{\underline{45\%}}$

$\Rightarrow$ <u>45% of the pupils passed</u>

6 (a) (v) Not more than 5 marks $\Rightarrow$ 5 or less

Therefore, P (not more than 5 marks)

$$= \frac{\text{Number of pupils that obtained 5 or less marks}}{\text{Total number of pupils}}$$

$$= \frac{1+2+2+4+2}{1+2+2+4+2+2+5+2}$$

$$= \underline{\underline{\frac{11}{20}}}$$

6 (b)

Method 1 (Evaluating whole number and fractions separately)

$$7\tfrac{2}{3} - 4\tfrac{5}{6} + 2\tfrac{3}{8}$$

$$\Rightarrow 7 - 4 + 2 + \tfrac{2}{3} - \tfrac{5}{6} + \tfrac{3}{8}$$

$$\Rightarrow 5 + \frac{16 - 20 + 9}{24}$$

$$\Rightarrow 5 + \tfrac{5}{24}$$

$$= \underline{\underline{5\tfrac{5}{24}}}$$

6 (b)

Method 2 (Changing mixed fractions to improper fractions)

$$7\tfrac{2}{3} - 4\tfrac{5}{6} + 2\tfrac{3}{8}$$

$$\Rightarrow \tfrac{23}{3} - \tfrac{29}{6} + \tfrac{19}{8}$$

$$\Rightarrow \frac{184 - 116 + 57}{24}$$

$$\Rightarrow \quad \frac{125}{24}$$

$$= \quad \underline{\underline{5\frac{5}{24}}}$$

April 2007

MATHEMATICS 1

OBJECTIVE TEST
1 hour

1. List the members of the set P = {factors of 30 which are odd}

A) P = {2,3,5}

B) P = {1,2,3,5}

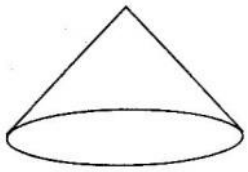
C) P = {1,3,5,15}

D) P = {2,6,10,30}

2. M = {g,o,q,s} and W = {h,p,r,t}. Find M∪W.

A. {q,r,s,t} B. {g,h,o,q,r} C. {g,h,o,p,q,r,t} D. {g,h,o,p,q,r,s,t}

3. Name the geometrical figure shown in the diagram below



A) Cuboid

B) cone

C) pyramid

D) sphere

4. Express 5/16 as a decimal fraction

A) 0.3333 B) 0.3125 C) 0.2667 D) 0.2500

5. If y = $-\frac{1}{2}x+6$, find y when x = 4

A) -2 B) 2 C) 4 D) 8

The marks obtained by 13 candidates in a test are 5, 7, 2, 9, 11, 10, 2, 12, 2, 9, 3, 18 and 2.

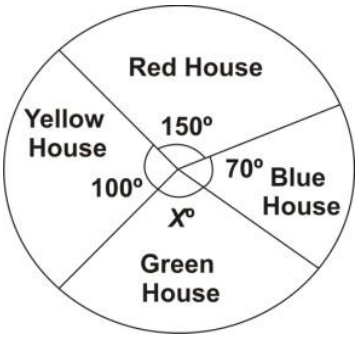
Use this information to answer Questions **6** *and* **7**

6. What is the mode?

A) 2 B) 5 C) 9 D) 18

7. Find the median

A) 9 B) 7 C) 5 D) 2



The pie chart above shows the distribution of 360 pupils to various houses in a school.

Use this information to answer Question **8** *and* **9**.

8. Find the value of the angle marked X°

A) 30° B) 40° C) 100° D) 150°

9. How many more students are in Yellow house than in Blue house?

A) 30 B) 40 C) 70 D) 100

10. Express 242 as a product of prime factors

A) $2^2 \times 11^2$ B) 2×11^2 C) $2^2 \times 11$ D) 2×11

11. Change 110011_{two} to a number in base ten

A) 51 B) 50 C) 48 D) 32

12. A boy throws a die once. What is the probability of getting the number 4?

A) $^1/_6$ B) $^1/_3$ C) ½ D) $^5/_6$

13. Arrange the following fractions from the highest to the lowest $^5/_6$, $^4/_5$ and $^4/_7$

A) $^4/_7$, $^4/_5$, $^5/_6$

B) $^4/_5$, $^5/_6$, $^4/_7$

C) $^5/_6$, $^4/_5$, $^4/_7$

D) $^4/_7$, $^5/_6$, $^4/_5$

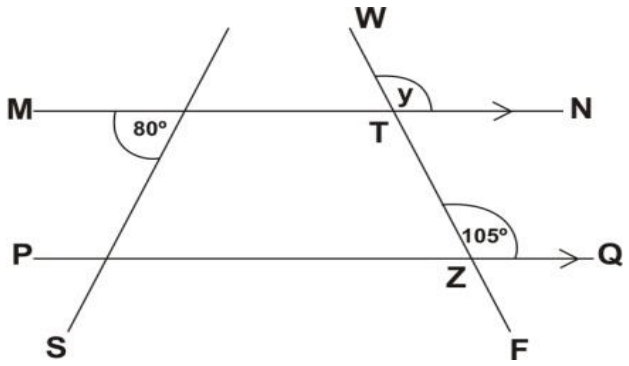
14. Solve the inequality $3x + 6 \leq 5x - 2$

A) $x \leq 2$ B) $x \geq 2$ C) $x \leq 4$ D) $x \geq 4$

15. If $8x - 3(2x - 4) = 4$, find the value of x

A) -8 B) -4 C) 4 D) 8

Use the diagram below to answer Questions **16** *and* **17**



16. Find the size of angle y

A) 75° B) 80° C) 100° D) 105°

17. Angles NTZ and QZT are …

A) alternate angles

B) corresponding angles

C) complementary angles

D) supplementary angles

18. Factorize $\frac{1}{4}px^2 + \frac{1}{8}px$

A) $\frac{1}{4}px(x+2)$

B) $\frac{1}{4}px(x+\frac{1}{2})$

C) $\frac{1}{8}px(\frac{1}{2}x+2)$

D) $\frac{1}{8}px(x+2)$

19. Jojo and Fiifi shared an amount of money in the ratio 3:4 respectively. If Fiifi had ¢140,000 how much was shared?

A) ¢200,000.00

B) ¢220,000.00

C) ¢245,000.00

D) ¢280,000.00

20. Simplify 6(7a+4) – 3(8a+9)

A) 18a–3 B) 18a+51 C) 42a–27 D) 66a–3

21. A man invested ¢500,000.00 at 15% simple interest per annum for 2 years. Calculate the amount at the end of the period

A) ¢350,000.00 B) ¢650,000.00 C) ¢750,000.00 D) ¢800,000.00

22. Make *m* the subject of $p = \frac{3m+1}{m}$

A) $m = 3p + 1$

B) $m = p - 3$

C) $m = \frac{1}{p-3}$

D) $m = \frac{3p+1}{p}$

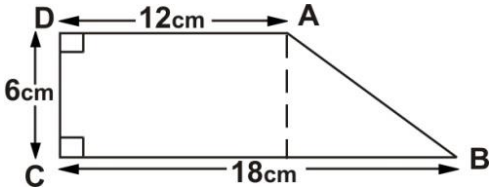
23. Simplify 200 × 0.01 × 372 leaving your answer in the standard form.

A) 74.4×10^{1} B) 7.44×10^{1} C) 7.44×10^{2} D) 7.44×10^{3}

24. What percentage of 5 is 0.25?

A) 4% B) 5% C) 20% D) 25%

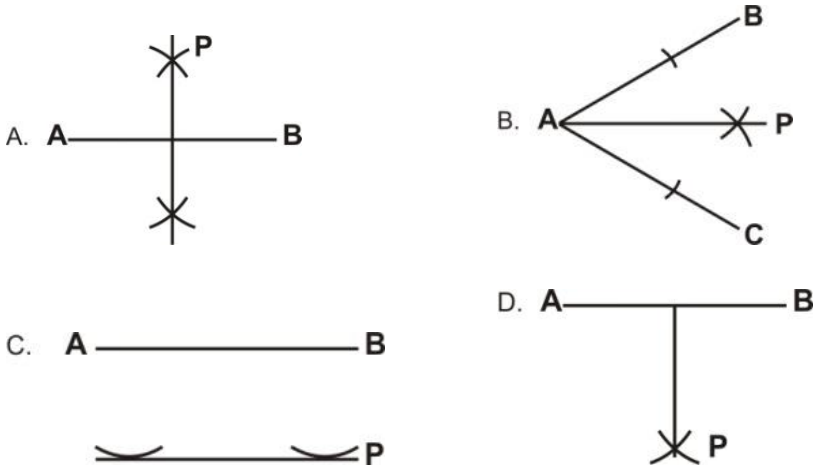
25.



Calculate the area of the figure ABCD above

A) $72cm^2$ B) $90cm^2$ C) $108cm^2$ D) $126cm^2$

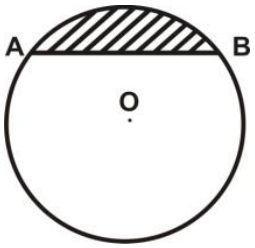
26. In which of the following constructions is P equidistant from the points A and B?



27. If a:27 = 6:18, find a

A) 4 B) 9 C) 51 D) 81

28. The diagram below shows a circle with centre O. If A and B are points on the circle, what name is given to the shaded region?



A) Chord B) Segment C) Sector D) arc

29. An amount of ¢5,400.00 is shared among three sisters in the ratio of their ages. Their ages are 10 years, 6 years and 2 years. Find the share of the youngest sister.

A) ¢300.00 B) ¢600.00 C) ¢1,200.00 D) ¢1,800.00

30. Find the Least Common Multiple (LCM) of 4,5 and 6.

A) 20 B) 24 C) 30 D) 60

31. Expand (a+2b)(a-2b)

A) $a^2 – 4ab – 4b^2$ B) $a^2 + 4ab – 4b^2$ C) $a^2 – 4b^2$ D) $a^2 + 4b^2$

32. If a woman was paid ¢180,000.00 for working for 4½ days, how much would she be paid for one day?

A) ¢40,000.00
B) ¢45,000.00
C) ¢54,000.00
D) ¢81,000.00

33. Express 0.68 as a fraction in its lowest term

A) $^7/_{25}$ B) $^{17}/_{25}$ C) $^3/_5$ D) $^7/_{15}$

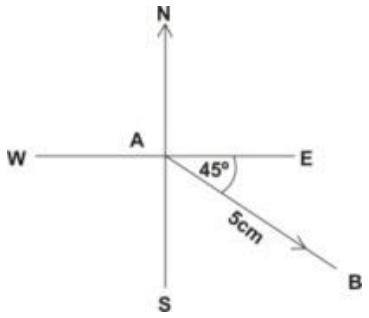
34. Write down all the integers within the interval $21 < y \leq 27$

A. {21,22,23,24,25,26,27}
B. {22,23,24,25,26,27}
C. {21,22,23,24,25,26}
D. {22,23,24,25,26}

35. Simplify $(2^6 \times 3^4) \div (2^4 \times 3^2)$

A) $2^2 \times 3^2$ B) $2^2 \times 3^6$ C) $2^{10} \times 3^2$ D) $2^{10} \times 3^6$

36.



Which of the following is represented by vector AB in the diagram above?

A. (5cm, 045°) B. (5cm, 135°) C. (5cm, 180°) D. (5cm, 225°)

37. Solve for y, if $3 + 3y = 1 - 13y$

A) - ⅛ B) - ¼ C) ⅛ D) ¼

38. Convert 37_{ten} to a base two numeral

A) 100101 B) 100111 C) 101101 D) 110101

39. The length of a rectangular playing field is 5 metres longer than its width. If the perimeter of the field is 150 metres, find its width.

A) 30 m B) 35 m C) 40 m D) 45 m

40. If the vector $\mathbf{m} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\mathbf{n} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$, find $\mathbf{m} - 2\mathbf{n}$

A. $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$ B. $\begin{pmatrix} 1 \\ -4 \end{pmatrix}$ C. $\begin{pmatrix} 5 \\ 4 \end{pmatrix}$ D. $\begin{pmatrix} 5 \\ -4 \end{pmatrix}$

April 2007

MATHEMATICS 1

OBJECTIVE TEST

SOLUTIONS

1. C) P={1,3,5,15}

2. D. {g,h,o,p,q,r,s,t}

3. B) cone

4. B) 0.3125

5. C) 4

6. A) 2

7. B) 7

8. B) 40°

9. A) 30

10. B) 2×11^{2}

11. A) 51

12. A) $^{1}/_{6}$

13. C) $^{5}/_{6}$, $^{4}/_{5}$, $^{4}/_{7}$

14. D) $x \geq 4$

15. B) -4

16. D) 105°

17. D) supplementary angles

18. B) $\frac{1}{4}px(x+\frac{1}{2})$

19. C) ¢245,000.00

20. A) 18a–3

21. B) ¢650,000.00

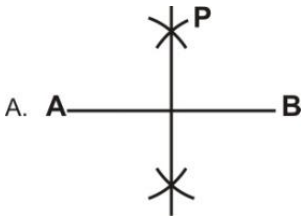
22. C) $m = \dfrac{1}{p-3}$

23. C) 7.44×10^2

24. B) 5%

25. B) 90 cm^2

26. A.



27. B) 9

28. B) Segment

29. B) ¢600.00

30. D) 60

31. C) $a^2 - 4b^2$

32. A) ¢40,000.00

33. B) $^{17}/_{25}$

34. B. {22,23,24,25,26,27}

35. A) $2^2\times3^2$

36. B. (5cm, 135º)

37. A) - ⅛

38. A) 100101

39. B) 35 m

40. D. $\begin{pmatrix} 5 \\ -4 \end{pmatrix}$

April 2007

MATHEMATICS 2

ESSAY
1 hour

[60 marks]

Answer **four** *questions* **only** *from this section*

All working must be clearly shown.

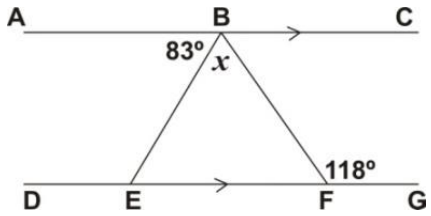
The use of calculators is **not** *allowed*

Marks will **not** *be awarded for correct answers without corresponding working.*

All questions carry equal marks

1. (a) In a class of 39 students, 19 offer French and 25 offer Ga. 5 students do not offer any of the two languages. How many students offer **only** French?

(b)



In the diagram above, AC is parallel to DG, angle BFG = 118° and angle ABE = 83°. Find the value of

(i) Angle CBF

(ii) x

(c) A fair die is thrown once,

(i) Write down the set of all the possible outcomes

(ii) Find the probability of obtaining a multiple of 2

(iii) What is the probability of obtaining a prime number?

2. (a) A water tank in the form of a cuboid of height 22cm and a rectangular base of length 7cm and width 5cm is filled with water. The water is then poured into a cylindrical container of diameter 14cm. Calculate the height of the water in the cylindrical container.

(Take $\pi = {}^{22}/_{7}$)

(b) A trader is given 15% discount on goods bought from a factory. If the original price of an item in the factory is ¢45,000,000, calculate

i) the discount on the item

ii) the amount the trader paid for the item

(c) Find the truth set of $\frac{x-3}{3} \leq \frac{1}{2} + x$ and illustrate your answer on the number line.

3. Using a ruler and a pair of compasses only,

(a) Construct triangle PQR such that the length of PQ = 10cm, angle QPR = 90° and angle PQR = 30°

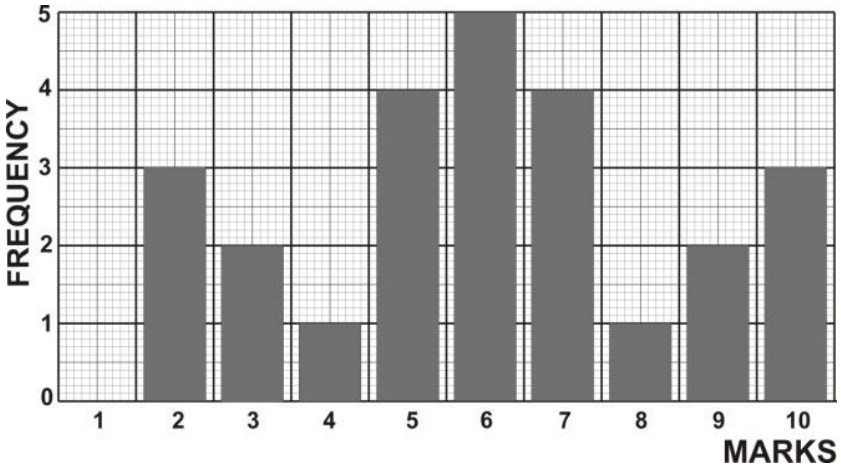
Measure the length of PR.

(b) Bisect the angle QRP to meet PQ at M.

(c) With M as centre, and radius MP draw a circle.

(d) Measure the radius of the circle

4.



The Bar Chart above is for the distribution of marks in a class test

(a) i) Write down the frequency table for the distribution

ii) Use the table to find the mean mark

(b) If the pass mark is 4, how many pupils **failed** the test?

5. (a) Mr. Jones used 173 units of electricity last month. If the charge for the first 110 units was ¢150 per unit and ¢200 per unit for the rest, calculate the total bill for Mr. Jones.

(b) Express 4 hours in seconds leaving your answer in the standard form

(c) Given the vectors $\mathbf{r} = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$ and $\mathbf{p} = \begin{pmatrix} -7 \\ -9 \end{pmatrix}$, if q = 2**p** – **r**, find **q**

6. (a) Using a scale of 2cm to 2units on both axes, draw two perpendicular axes OX and OY on a graph sheet. On the same graph sheet, mark the x-axis from -10 to 10 and the y-axis from -10 to 10.

(i) Plot A(2,2), B(6,2) and C(4,6). Join AB, BC and AC

(ii) Draw the image triangle $A_1B_1C_1$ of triangle ABC under a clockwise rotation of 90° about the origin, where $A \rightarrow A_1$, $B \rightarrow B_1$ and $C \rightarrow C_1$.

(iii) Draw the image triangle $A_2B_2C_2$ of triangle ABC under a reflection in the y-axis, where $A \rightarrow A_2$, $B \rightarrow B_2$ and $C \rightarrow C_2$

(b) When 12 is added to a certain number and the sum is multiplied by 4, the result is 60. Find the number

April 2007

MATHEMATICS 2

ESSAY

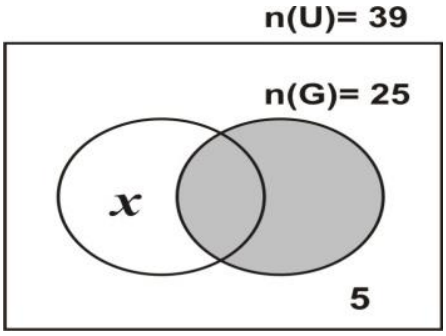
SOLUTIONS

1 (a) **Method 1**

Let n(U) = No. of students in the class

n(G) = No. of Ga students

x = No. of students that offer only French



From the Venn diagram above,

$x + 25 + 5 = 39$

$x + 30 = 39$

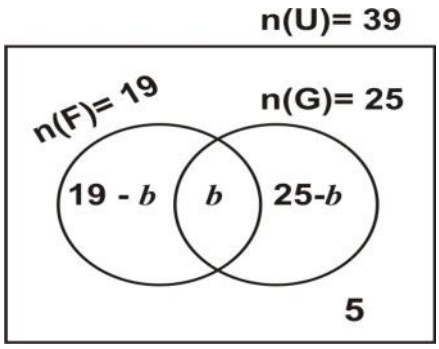
$x = 39 - 30$

$x = 9$

Number of students that offer only French = 9

1 (a) **Method 2**

Let b = No. of student that offer both French and Ga



From the above diagram,

$19 - b + b + 25 - b + 5 = 39$

$\Rightarrow \quad 19 + 0 + 25 - b + 5 = 39$

$\Rightarrow \quad 19 + 25 + 5 - b = 39$

$\Rightarrow \quad 49 - b = 39$

$\Rightarrow \quad 49 - 39 = b$

$\Rightarrow \quad b = 49 - 39$

$\Rightarrow \quad b = 10$

But French only $= n(F) - b$

$= 19 - 10 = 9$

Therefore the number of students who offer only French **= 9**

1 (b) (i) **Method 1**

Since ∠CBF and ∠BFG are Supplementary angles

$\Rightarrow \quad \angle CBF + \angle BFG = 180°$

$\Rightarrow \quad \angle CBF + 118° = 180°$

$\Rightarrow \quad \angle CBF = 180° - 118°$

$\therefore \quad$ Angle CBF $= 62°$

1 (b) (i) **Method 2**

Since angles BFE and BFG lie on a straight line

$\Rightarrow \quad \angle BFE + \angle BFG = 180°$

$\Rightarrow \quad \angle BFE + 118° = 180°$

$\Rightarrow \quad \angle BFE = 180° - 118°$

$\Rightarrow \quad \angle BFE = 62°$

But $\angle CBF = \angle BFE$ (Alternate angles)

$\therefore \quad \underline{\angle CBF = 62°}$

1 (b) (ii) Since $\angle ABE$, x and $\angle CBF$ lie on a straight line,

$\Rightarrow \quad \angle ABE + x + \angle CBF = 180°$

$\Rightarrow \quad 83° + x + 62° = 180°$

$\Rightarrow \quad x = 180° - 83° - 62°$

$\Rightarrow \quad \underline{x = 35°}$

NB: Triangle BEF could also be used to find the value of x

1 (c) (i) The set of all possible outcomes

$= \{1, 2, 3, 4, 5, 6\}$

1 (c) (ii) Multiples of 2 within the set = 2, 4, 6

$$\Rightarrow \text{P(multiple of 2)} = \frac{\text{Number of multiples of 2}}{\text{Total number of possible outcomes}}$$

$$= \frac{3}{6} = \frac{1}{2}$$

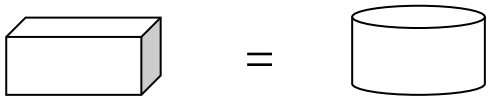
The probability of obtaining a multiple of 2 = ½

1 (c) (iii) Prime numbers within the set = 2, 3, 5

$$\Rightarrow \text{P(prime number)} = \frac{\text{Number of prime numbers}}{\text{Total number of possible outcomes}}$$

$= \quad \frac{3}{6} \quad = \quad \frac{1}{2}$

2 (a) Volume of water in cuboid = Volume of water in cylinder



Let h_{cyl} = the height of water in the cylindrical container

and h_{cub} = the height of water in the cuboid

Method 1 (Substituting values and solving for h_{cyl})

Volume of water in cuboid = Volume of water in cylinder

Area of base $_{cuboid}$ × height $_{cuboid}$ = Area of base $_{cylinder}$ × height $_{cylinder}$

$\Rightarrow \quad l \times w \times h_{cuboid} = \pi r^2 \times h_{cylinder}$

$\Rightarrow \quad [7cm \times 5cm] \times 22cm = [{}^{22}/_{7} \times (7cm)^2] \times h_{cyl}$

$\Rightarrow \quad 7cm \times 5cm \times 22cm = \frac{22}{7} \times 7cm \times 7cm \times h_{cyl}$

$\Rightarrow \quad \frac{7cm \times 5cm \times 22cm \times 7}{22 \times 7cm \times 7cm} = h_{cyl}$

$\Rightarrow \quad 5cm = h_{cyl}$

$\therefore$ The height of the water in the cylindrical container = 5cm

2 (a) **Method 2 (Making h_{cyl} the subject first, then substituting and evaluating)**

Volume of water in cuboid = Volume of water in cylinder

$l \times w \times h_{cub} = (\pi r^2) \times h_{cyl}$

$\Rightarrow \quad \frac{l \times w \times h_{cub}}{\pi r^2} = h_{cyl}$

$$\Rightarrow \quad \frac{7cm \times 5cm \times 22cm}{\frac{22}{7} \times (7cm)^2} = h_{cyl}$$

$$\Rightarrow \quad \frac{7cm \times 5cm \times 22\,cm \times 7}{22 \times 7cm \times 7cm} = h_{cyl}$$

$\Rightarrow$ 5cm = h_{cyl}

$\therefore$ The height of the water in the cylindrical container = 5cm

2 (a) **Method 3 (Finding the volume of water in cuboid first, then equating)**

Volume of water in cuboid = $l \times w \times h$ cuboid

= 7cm × 5cm × 22cm

= 770 cm^3

But Volume of water in cuboid = Volume of water in cylinder

$\Rightarrow$ 770 cm^3 = 154 $cm^2 \times h_{cyl}$

$$\Rightarrow \quad \frac{770\ cm^3}{154\ cm^2} = h_{cyl}$$

$\Rightarrow$ 5 cm = h_{cyl}

$\therefore$ The height of the water in the cylindrical container = 5cm

2 (b) (i) The discount = Discount rate × Original price

= 15% × ¢45,000,000

$$= \frac{15}{100} \times ¢45{,}000{,}000$$

= 15 × ¢450,000

$= \quad \underline{\underline{¢6{,}750{,}000}}$

(ii) Amount paid = Original amount – discount

$= ¢45{,}000{,}000 - ¢6{,}750{,}000$

$= \quad \underline{¢38{,}250{,}000}$

2 (c) $\dfrac{x-3}{3} \leq \dfrac{1}{2} + x$

Approach 1

$\Rightarrow \quad \dfrac{x-3}{3} \leq \dfrac{1}{2} + \dfrac{x}{1}$

$\Rightarrow \quad \dfrac{x-3}{3} \leq \dfrac{1+2x}{2}$

$\Rightarrow \quad 6\left(\dfrac{x-3}{3}\right) \leq 6\left(\dfrac{1+2x}{2}\right)$

$\Rightarrow \quad 2(x-3) \leq 3(1+2x)$

$\Rightarrow \quad 2x - 6 \leq 3 + 6x$

$\Rightarrow \quad 2x - 6x \leq 3 + 6$

$\Rightarrow \quad -4x \leq 9$

$\Rightarrow \quad \dfrac{-4x}{-4} \geq \dfrac{9}{-4}$

$\Rightarrow \quad \underline{\underline{x \geq -2\frac{1}{4}}}$

2 (c) **Approach 2**

$\Rightarrow \quad 6\left(\dfrac{x-3}{3}\right) \leq 6\left(\dfrac{1}{2} + x\right)$

$$\Rightarrow \quad 2(x-3) \quad \leq \quad 6(\tfrac{1}{2}) + 6x$$

$$\Rightarrow \quad 2x - 6 \quad \leq \quad 3 + 6x$$

$$\Rightarrow \quad -3 - 6 \quad \leq \quad 6x - 2x$$

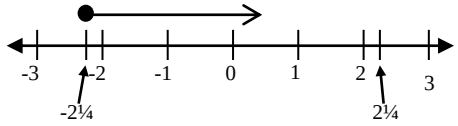
$$\Rightarrow \quad -9 \quad \leq \quad 4x$$

$$\Rightarrow \quad \frac{-9}{4} \quad \leq \quad \frac{4x}{4}$$

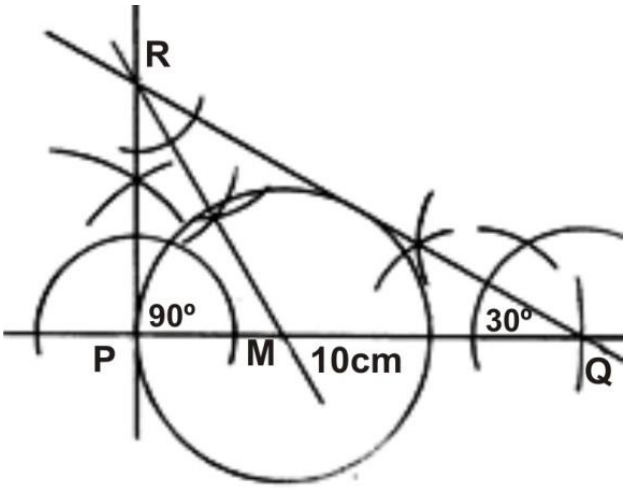
$$\Rightarrow \quad -2\tfrac{1}{4} \quad \leq \quad x$$

$$\Rightarrow \quad \underline{\underline{x \quad \geq \quad -2\tfrac{1}{4}}}$$

The number line is shown below:



3(a),(b),(c)



3(a) |PR| ≈ <u>5.8 cm</u>

3(d) Radius of circle ≈ <u>3.3 cm</u>

4(a) **(i)** The frequency table is shown below

Mark	Tally	Frequency
2	///	3
3	//	2
4	/	1
5	////	4
6	~~////~~	5
7	////	4
8	/	1
9	//	2
10	///	3

4(a) (ii)

Mark (x)	Tally	Frequency (f)	f x
2	///	3	6
3	//	2	6
4	/	1	4
5	////	4	20
6	~~////~~	5	30
7	////	4	28
8	/	1	8
9	//	2	18
10	///	3	30
		Σf = 25	Σfx = 150

The mean mark $= \frac{\Sigma f x}{\Sigma f} = \frac{150}{25} = \underline{6}$

4(b) If the pass mark is 4, ⇒ pupils who scored 2 or 3 failed

⇒ Frequency of 2 + Frequency of 3

= 3 pupils + 2 pupils

= <u>5 pupils</u>

<u>Therefore 5 pupils failed the test</u>

5(a) Charge for:

First 110 units	=	¢150 × 110	=	¢16,500
Remaining (173 – 110 = 63) units	=	¢200 × 63	=	¢12,600
∴ Total charge	=	¢12,600 + ¢16,500		
	=	¢29,100.00		

5(b) **Method 1**

4 hours	=	4 × 60 minutes
	=	240 minutes
Now, 240 min	=	240 × 60 seconds
	=	14400 seconds
	=	1.44×10^4 seconds

5(b) **Method 2**

Since 1 hour = 60 minutes and

1 minute = 60 seconds

⇒ 1 hour = 60 × 60 seconds = 3600 seconds

Now, if 1 hour = 3600 seconds

then 4 hours = ? (more)

If more, less divides

$$\Rightarrow \quad \frac{4\ hours}{1\ hour} \times 3600\ s$$

⇒ 4 × 3600 s = 14400 s

= 1.44×10^4 seconds

5(b) **Method 3**

If 1 hour = 3600 seconds

then 4 hours = x

$$\Rightarrow \quad \frac{1}{4} = \frac{3600}{x}$$

$\Rightarrow \quad x = 3600 \times 4$

$\Rightarrow \quad x = 14400$

$\Rightarrow$ 4 hours $= 14400$ seconds

$= \underline{1.44. \times 10^4 \text{ seconds}}$

5(c) $\mathbf{q} = 2\mathbf{p} - \mathbf{r}$

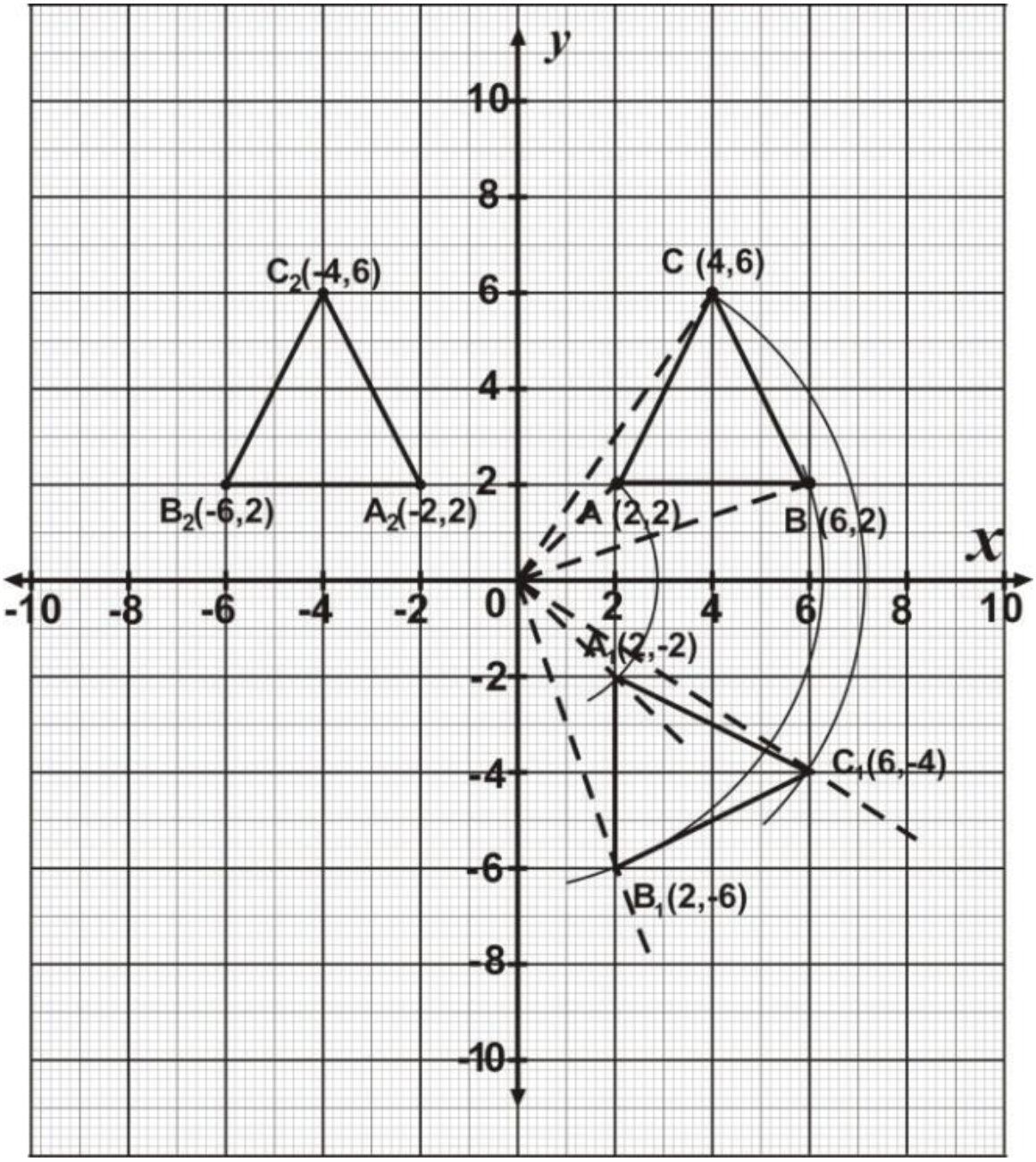
$$= 2\begin{pmatrix} -7 \\ -9 \end{pmatrix} - \begin{pmatrix} 3 \\ -5 \end{pmatrix}$$

$$= \begin{pmatrix} -14 \\ -18 \end{pmatrix} - \begin{pmatrix} 3 \\ -5 \end{pmatrix}$$

$$= \begin{pmatrix} -14 - 3 \\ -18 - (-5) \end{pmatrix}$$

$$\mathbf{q} = \begin{pmatrix} -14-3 \\ -18+5 \end{pmatrix} = \underline{\underline{\begin{pmatrix} -17 \\ -13 \end{pmatrix}}}$$

6 (a) **Approach 1** (By Inspection / Construction)



Approach 2 (Using the rule / formula)

6 (a) (ii) Rotating (x, y) through 90° clockwise about the origin

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} y \\ -x \end{pmatrix}$$

$$\Rightarrow \quad OA\begin{pmatrix} 2 \\ 2 \end{pmatrix} \rightarrow OA_1\begin{pmatrix} 2 \\ -2 \end{pmatrix}, \quad \therefore \ A_1(2,-2)$$

$$\Rightarrow \quad OB\begin{pmatrix} 6 \\ 2 \end{pmatrix} \rightarrow OB_1\begin{pmatrix} 2 \\ -6 \end{pmatrix}, \quad \therefore B_1(2,-6)$$

$$\Rightarrow \quad OC\begin{pmatrix} 4 \\ 6 \end{pmatrix} \rightarrow OC_1\begin{pmatrix} 6 \\ -4 \end{pmatrix}, \quad \therefore C_1(6,-4)$$

∴ Plot and join $A_1(2,-2)$, $B_1(2,-6)$ and $C_1(6,-4)$ as the image of triangle ABC under a rotation through 90° clockwise about the origin as shown above.

6 (a) (iii) Reflecting (x, y) in the y-axis

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} -x \\ y \end{pmatrix}$$

$$\Rightarrow \quad OA\begin{pmatrix} 2 \\ 2 \end{pmatrix} \rightarrow OA_2\begin{pmatrix} -2 \\ 2 \end{pmatrix}, \quad \therefore A_2(-2, 2)$$

$$\Rightarrow \quad OB\begin{pmatrix} 6 \\ 2 \end{pmatrix} \rightarrow OB_2\begin{pmatrix} -6 \\ 2 \end{pmatrix}, \quad \therefore B_2(-6, 2)$$

$$\Rightarrow \quad OC\begin{pmatrix} 4 \\ 6 \end{pmatrix} \rightarrow OC_2\begin{pmatrix} -4 \\ 6 \end{pmatrix}, \quad \therefore C_2(-4, 6)$$

∴ Plot and join $A_2(-2,2)$, $B_2(-6,2)$ and $C_2(-4,6)$ as the image of triangle ABC under a reflection in the y axis as shown above

6 (b) Let n = the number, then $\Rightarrow$ $(12 + n) \times 4 = 60$

Approach 1 (Expanding first)

$$4(12 + n) = 60$$

$$\Rightarrow \quad 48 + 4n = 60$$

$$\Rightarrow \quad 4n = 60 - 48$$

$$\Rightarrow \quad 4n = 12$$

$$\Rightarrow \quad n = 12 \div 4$$

$$\Rightarrow \quad \underline{n = 3}$$

6 (b) **Approach 2** (Dividing through by 4 first)

$$4(12 + n) = 60$$

$$\Rightarrow \quad \frac{4(12+n)}{4} = \frac{60}{4}$$

$$\Rightarrow \quad 12 + n = 15$$

$$\Rightarrow \quad n = 15 - 12$$

$\Rightarrow \quad n = 3$

Therefore the number is 3

April 2006

MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. If Q = {1,3,5,7,9,11,13,15} and R = {1,2,3,5,6,7,10,11,12}, find Q ∩ R.

A. {1,3,5,7,11}
B. {1,3,5,7,9,11}
C. {2,4,8,9,13,14}
D. {1, 2, 3, 5, 6, 7, 9, 10, 11, 12}

2. If x is an integer, list the members of the set, $\{2 \leq x < 10\}$

A. {3,4,5,6,7,8,9}
B. {2,3,4,5,6,7,8,9}
C. {3,4,5,6,7,8,9,10}
D. {2,3,4,5,6,7,8,910}

3. Simplify: $\frac{4}{3}x - \frac{2}{9}x$

A) $\frac{2}{9}x$ B) $\frac{2}{3}x$ C) $\frac{10}{9}x$ D) $\frac{14}{9}x$

4. The number of boys in a school is 120. If the ratio of boys to girls is 5:7, find the total number of students in the school.

A) 240 B) 288 C) 600 D) 840

5. Find the circumference of a circle with radius 3.5 cm. (Take $\pi = {}^{22}/_{7}$)

A) 11 cm B) 22 cm C) 35 cm D) 38.5 cm

6. Which of the following are the prime factors of 12?

A. {1, 3} B. {2, 3} C. {2, 4, 6, 12} D. {2, 3, 4, 6}

7. The base of an isosceles triangle is 7cm long. Each of the other two sides is x cm long. What will be the expression for its perimeter?

A) x+7 B) x+14 C) 2x-7 D) 2x+7

8. Correct 0.003858 to three significant figures

A) 0.00385 B) 0.00386 C) 0.0039 D) 386

9. Find the value of $a - 3ab$ when $a = -2$ and $b = 3$

A) -20 B) -16 C) 16 D) 20

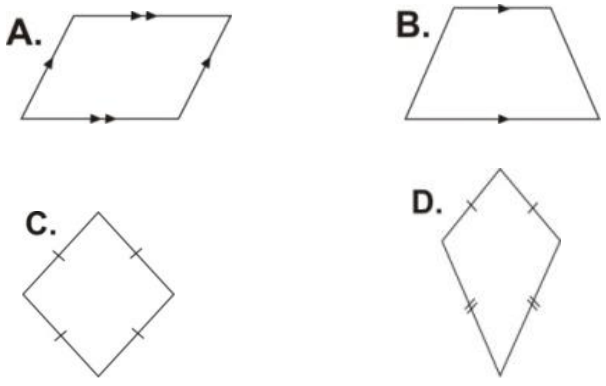
10. A boy sold some oranges at three for ¢500.00. If his total sales was ¢100,000.00, how many oranges did he sell?

A) 300 B) 400 C) 500 D) 600

11. A train travels at a speed of 80km per hour. How long will it take to travel a distance of 320km?

A) 2 hours B) 3 hours C) 4 hours D) 5 hours

12. Which of the following figures is a rhombus?



13. Three children share an amount of ¢910,800.00 in the ratio 2:3:4. What will be the highest share?

A) ¢202,400.00

B) ¢303,600.00

C) ¢404,800.00

D) ¢455,400.00

14. The next term in the sequence 3, 6, 12, 24, …

A) 27 B) 30 C) 36 D) 48

15. Express 350 as a product of prime factors

A) $2 \times 5 \times 7$ B) $2 \times 5^2 \times 7$ C) $2 \times 5 \times 7^2$ D) $2^2 \times 5 \times 7$

16. Express 962 in standard form.

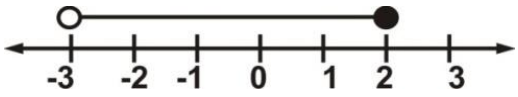
A) 96.2×10

B) 9.62×10^2

C) 0.962×10^3

D) 0.0962×10^4

17.



Which of the following inequalities is shown on the number line above, where $p \in \{\text{real numbers}\}$

A) $-2 < p < 3$ B) $-2 \geq p > 3$ C) $-2 < p \leq 3$ D) $-2 > p \geq 3$

18. Solve the equation $\frac{x+2}{3} + 2x = 10$.

A) 3 B) 4 C) 5 D) 6

19. Arrange the fractions ¾ , $^2/_3$, $^4/_5$ in ascending order of magnitude.

A) ¾, $^2/_3$, $^4/_5$

B) $^4/_5$, $^2/_3$, ¾

C) $^4/_5$, ¾, $^2/_3$

D) $^2/_3$,$^3/_4$,$^4/_5$

20. Express 87_{ten} as a base five numeral.

A) 302_{five} B) 322_{five} C) 3022_{five} D) 3202_{five}

21. A bag contains 6 blue and 5 black balls. What is the probability of picking a black ball at random?

A) $^1/_{11}$ B) $^1/_5$ C) $^5/_{11}$ D) $^6/_{11}$

22. Evaluate $\dfrac{2^3 \times 3^4 \times 3^3}{2^2 \times 2 \times 3^5}$

A) 6 B) 9 C) 12 D) 18

The table below gives the ages of members of a juvenile club.

Use it to answer Questions **23** *and* **24**

Age in years	8	9	10	11
Frequency	5	10	6	9

23. How many people are in the club?

A) 15 B) 20 C) 30 D38

24. What is the modal age of the members of the club?

A) 8 years B) 9 years C) 10 years D) 11 years

25. In an examination, 154 out of 175 candidates passed. What percentage failed?

A) 6% B) 12% C) 13% D) 18%

26. Expand $(6 – x)(6 + y)$

A) $36 – 6x + 6y – xy$

B) $36 – 6x – 6y + xy$

C) $36 – 6x – xy$

D) $36 + 6y – xy$

27. Find the highest common factor (HCF) of 20, 12 and 28.

A) 2 B) 4 C) 8 D) 12

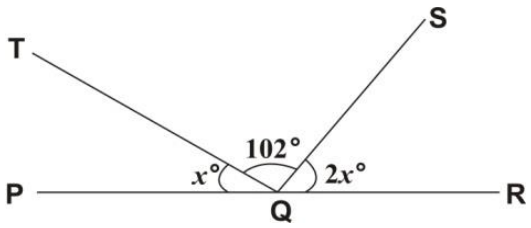
28. If $F = \frac{9}{5}c + 32$ find F when C = 40.

A) 49 B) 78.4 C) 104 D) 129.6

29. Evaluate $^2/_3\,(27 – 12) – 6$.

A) 4 B) 6 C) 14 D) 16

30.



In the figure, PQR is a straight line. Angle TQP = $x°$, angle TQS = 102° and angle SQR = $2x°$. Find the value of x.

A) 78 B) 39 C) 34 D) 26

31. Factorize 22ab – 11ac + 6rb – 3rc.

A. (2b – c) (11a + 3r)

B. (2b + c) (11a – 3r)

C. (2b – c) (11a – 3r)

D. (2b + c) (11a + 3r)

32. Find the sum of 124.3, 0.275 and 74.06. (Correct your answer to one decimal place)

A) 198.6 B) 198.7 C) 892.0 D) 892.4

33. Simplify $6p^3 \times p^2 \div 3p^4$

A) 2p B) 3p C) 18p D) $2p^2$

34. Mr. Nkrumah saved ¢75,000.00 at a simple interest rate of 20% per annum for 3 years. Calculate the interest he earned on his savings

A) ¢15,000.00 B) ¢30,000.00 C) ¢45,000.00 D) ¢60,000.00

35. Find the rule for the mapping:

1	2	3	4	5	…	n
↓	↓	↓	↓	↓		↓
10	21	32	43	54	…	-

A) n → 10n

B) n → (10n + 1)

C) n → (11n - 1)

D) n → (7n + 3)

36. What is the value of the digit 9 in the number 624.93 ?

A) 9 hundreds B) 9 tens C) 9 units D) 9 tenths

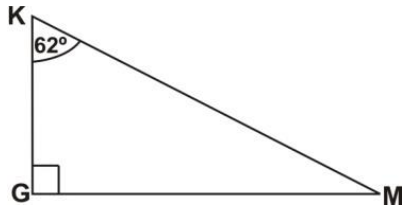
37. The marks obtained by 5 girls in a test are: 10, 15, 8, 18, 12.
Find the median mark.

A) 10 B) 11 C) 12 D) 15

38. The point P(3, 4) is translated by the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ to a new position P'. Find the coordinates of the image P'.

A. (0, 2) B. (6, -2) C. (6, 2) D. (6, 6)

39.



In the diagram above, KGM is a right-angled triangle and angle GKM = 62°. Find the angle of elevation of K from M.

A) 28° B) 62° C) 90° D) 118°

40. Given that vector **a** = $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and vector **b** = $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$, find **a** + 2**b**

A. $\begin{pmatrix} 6 \\ 13 \end{pmatrix}$ B. $\begin{pmatrix} 2 \\ -13 \end{pmatrix}$ C. $\begin{pmatrix} -2 \\ 7 \end{pmatrix}$ D. $\begin{pmatrix} 2 \\ -7 \end{pmatrix}$

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MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. A. {1,3,5,7,11}

2. B. {2,3,4,5,6,7,8,9}

3. C) $\frac{10}{9}x$

4. B) 288

5. B) 22 cm

6. B. {2, 3}

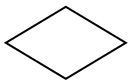
7. D) 2x+7

8. B) 0.00386

9. C) 16

10. D) 600

11. C) 4 hours

12. C. 

13. C) ¢404,800.00

14. D) 48

15. B) $2 \times 5^2 \times 7$

16. B) 9.62×10^2

17. C) $-2 < p \leq 3$

18. B) 4

19. D) $^2/_3, ^3/_4, ^4/_5$

20. B) 322_{five}

21. C) $^5/_{11}$

22. B) 9

23. C) 30

24. B) 9 years

25. B) 12%

26. A) $36 - 6x + 6y - xy$

27. B) 4

28. C) 104

29. A) 4

30. D) 26

31. A. $(2b - c)(11a + 3r)$

32. A) 198.6

33. B) 3p

34. C) ¢45,000.00

35. C) $n \rightarrow (11n - 1)$

36. D) 9 tenths

37. C) 12

38. C. (6, 2)

39. A) 28°

40. D $\begin{pmatrix} 2 \\ -7 \end{pmatrix}$.

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MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. (a) A trader sold 250 articles for ¢525,000.00 at a profit of 25%.

(i) Calculate the cost price of each article

(ii) If the trader had wanted 45% profit on the cost price, how much should he have sold each of the articles?

(b) Find the simple interest on ¢880,000.00 for 2 ½ years at 3 ¼ % per annum

2. (a) The ratio of men to women in a village is 12:25. If there are 120 men,

(i) how many women are there?

(ii) what is the total number of men and women?

(b) A bag contains 70 pencils out of which 15 are green and 30 blue.

(i) How many pencils of other colours are in the bag?

(ii) A pencil is selected from the bag at random. What is the probability that it is blue?

(c) Solve $\frac{1}{3}(x-1)-\frac{1}{2}(x-3) \leq 1\frac{1}{4}$ and illustrate your answer on the number line.

3. (a) (i) Using a pair of compasses and ruler only, construct triangle XYZ with XZ = 12cm, XY = 10cm and angle XYZ = 90°.

(ii) Measure YZ.

(iii) Calculate the area of triangle XYZ

(iv) Measure angle ZXY.

(b) An isosceles triangle has a perimeter of (9y – 15) cm.

What is the length of each of the two equal sides, if its third side is (3y – 7) cm?

4. The mapping below has the rule, $y = 2x + 3$.

x	0	1	2	3	4
↓	↓	↓	↓	↓	↓
y	-	5	7	-	-

(a) (i) Copy the mapping and fill in the missing numbers

(ii) Using a scale of 2cm to 1 unit on both axes on a graph sheet, choose the origin O and draw the perpendicular axes OX and OY.

(iii) On the same graph sheet, mark the x-axis from 0 to 5 and the y-axis from 0 to 12.

(iv) Plot on the graph sheet the ordered pairs (x, y) from the mapping and join all the points using a straight edge.

(b) From your graph, find:

(i) y when $x = 3.5$;

(ii) x when $y = 8$;

(iii) the gradient of the line $y = 2x + 3$.

5. (a) A man spent ¼ of his monthly salary on rent, $^2/_5$ on food and $^1/_6$ on books. If he still had ¢55,000 left, what was his monthly salary?

(b) The average age of a family of eight is 30 years.

The average age of the six children in the family is 19 years.

If the mother is four years younger than the father, calculate the age of the father.

6. The following table shows the distribution of grades obtained by 120 students in an examination.

Grade	A	B	C	D
No. of students	14	30	52	24

(a) Draw a pie chart for the distribution.

(b) (i) Evaluate: $5\frac{7}{15} - 2\frac{2}{3} + 1\frac{5}{12}$

(ii) Factorize: $xy - xz + 5y - 5z$

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MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1 (a) (i) **METHOD 1** [Formula-(Finding CP of all 250 articles first, then CP of 1)]

$$\text{CP} = \frac{\text{Percentage of CP}}{\text{Percentage of SP}} \times \text{SP}$$

$$\text{CP of 250 articles} = \frac{100\%}{100\% + 25\%} \times ¢525{,}000$$

$$= \frac{100}{125} \times ¢525{,}000$$

$$= ¢\ 420{,}000$$

Therefore CP of 1 article = ¢ 420,000 ÷ 250 = ¢ 1,680

1 (a) (i) **METHOD 2** [Formula –(Finding SP of 1 article first, then CP of 1)]

SP of 1 article = ¢ 525,000 ÷ 250 = ¢ 2100

$$\text{CP} = \frac{\text{Percentage of CP}}{\text{Percentage of SP}} \times \text{SP}$$

$$\text{Therefore, CP of 1 article} = \frac{100}{100 + 25} \times ¢2100$$

$$= \frac{100}{125} \times ¢2100$$

$$= ¢\ 1{,}680$$

Simple Proportion **(S. Prop)** / Equivalent Fractions **(E.Frac)** METHODS

1 (a) (i) METHOD 3 [S. Prop / E.Frac (Finding CP of all 250 articles first, then CP of 1)**]**

	Percentage	Amount (250 articles)
SP	100% + 25% = **125%**	**₵ 525,000**
CP	**100%**	x

From the table above, $\frac{125}{100} = \frac{₵\ 525{,}000}{x}$

Cross multiplying, we have $125x = ₵\ 525{,}000 \times 100$

$$\Rightarrow \quad x = \frac{₵525{,}000 \times 100}{125}$$

$$\Rightarrow \quad x = ₵\ 4200 \times 100$$

$$= ₵420{,}000$$

$\because$ CP of all 250 articles = ₵ 420,000

$\Rightarrow$ CP of 1 article = ₵ 420,000 ÷ 250 = ₵ 1,680

1 (a) (i) METHOD 4 [S. Prop / E. Frac (Finding SP of 1 article first, then CP of 1)]

	Percentage	**Amount (1 article)**
SP	100% + 25% = **125%**	₵525,000 ÷ 250 **= ₵2100**
CP	**100%**	***x***

From the table above,

$\frac{125\%}{100\%} = \frac{₵2100}{x} \quad \Rightarrow \quad \frac{125}{100} = \frac{₵2100}{x}$

Cross multiplying, we have

$$125x = ₵2100 \times 100$$

$$\Rightarrow \qquad x = \frac{¢2100 \times 100}{125} \qquad = \qquad ¢\ \underline{1{,}680}$$

Simple Proportion **(S. Prop)** / 'If more, less divides... '**(IM, LD,...)**

1 (a) (i) METHOD 5 [S. Prop / IM, LD,... (Finding CP of all 250 articles first, then CP of 1)]

	Percentage	Amount (250 articles)
SP	100% + 25% = **125%**	**¢ 525,000**
CP	**100%**	*x*

If 125% → ¢ 525,000

Then 100% → ? (less)

If less, more (i.e., 125%) divides; therefore we have

$$\frac{100\%}{125\%} \times ¢525{,}000 \qquad = \qquad \frac{100}{125} \times ¢525{,}000$$

$$= \qquad ¢\underline{420{,}000}$$

∵ CP of all 250 articles = ¢ 420,000

⇒ CP of 1 article = ¢ 420,000 ÷ 250 = ¢ 1,680

1 (a) (i) METHOD 6 [S. Prop / IM, LD,... (Finding SP of 1 article first, then CP of 1)]

	Percentage	**Amount (1 article)**
SP	100% + 25% = **125%**	¢525,000 ÷ 250 = **¢2100**
CP	**100%**	***x***

If 125% → ¢ 2,100

Then 100% → ? (less)

If less, more (i.e., 125%) divides; therefore we have

$$\frac{100\%}{125\%} \times ¢2100 = \frac{100}{125} \times ¢2100$$

$$= ¢\,\underline{1{,}680}$$

1 (a) (ii) Selling price of each article to make 45% profit

METHOD 1 **(Formula)**

SP of 1 article = Percentage of SP × CP of 1 article

= 145% × CP of 1article

$$= \frac{145}{100} \times ¢\,1680$$

$$= \frac{¢\,243600}{100} = ¢\,\underline{2436}$$

1 (a) (ii) **METHOD 2** **(Formula)**

SP of 1article = Profit on 1article + CP of 1article

= (45% × CP of 1article) + CP of 1article

$$= \left(\frac{45}{100} \times ¢\,1680\right) + ¢1680$$

= ¢756 + ¢1680

= ¢ 2436

NB:.

Simple Proportion, using the ideas of equivalent fractions and the rule 'If more, less divides; If less, more divides', could also be used [as in Methods 3 – 6 in Solution to 1 (a) (i) above] to solve the problem.

1 (b) Simple interest = Principal × Rate × time

Approach 1 (common fractions)

= 880,000 × 3¼% × 2½

$$= 880{,}000 \times \frac{13}{4}\% \times \frac{5}{2}$$

$$= \frac{880{,}000}{1} \times \frac{13}{4\times 100} \times \frac{5}{2}$$

$$= \frac{880{,}000\times 13\times 5}{4\times 100\times 2}$$

$$= 1100 \times 13 \times 5$$

$$= \underline{¢\ 71{,}500}$$

1 (b) **Approach 2a** (decimal fractions)

$$= 880{,}000 \times 3¼\% \times 2½$$

$$= 880{,}000 \times 3.25\ \% \times 2.5$$

$$= 880{,}000 \times \frac{3.25}{100} \times 2.5$$

$$= 880{,}000 \times 0.0325 \times 2.5$$

$$= \underline{¢\ 71{,}500}$$

1 (b) **Approach 2b** (decimal fractions)

$$= 880{,}000 \times 3¼\% \times 2½$$

$$= 880{,}000 \times 3.25\ \% \times 2.5$$

$$= 880{,}000 \times \frac{3.25}{100} \times 2.5$$

$$= \frac{880{,}000}{1} \times \frac{325}{10000} \times \frac{25}{10}$$

$$= \frac{88\times 325\times 25}{10} = \frac{715{,}000}{10}$$

$$= 71{,}500$$

∴ The simple interest = ¢ 71,500

2(a) (i) **Method 1** (Unitary Approach – the value of one item)

Since the men ratio, 12, corresponds to 120 persons,

⇒ ratio 1 corresponds to $\frac{120}{12} = 10$

∴ women ratio, 25, corresponds to $25 \times 10 = \underline{250\ \text{women}}$

2(a) (i) **Method 2** (Equivalent Fractions)

	Ratio	Actual no.
Men	12	120
Women	25	w

From the table above, $\frac{12}{25} = \frac{120}{w}$

Cross multiplying, we have

$$12 \times w = 120 \times 25$$

$$\Rightarrow \quad \frac{12w}{12} = \frac{120 \times 25}{12}$$

$$\Rightarrow \quad w = 10 \times 25$$

$$\Rightarrow \quad w = \underline{250}$$

NB:
Alternatively,
it can be deduced from the table (by simple logic) that
$w = 250$

2(a) (i) **Method 3** (If more, less divides …)

If 12 → 120

Then 25 → ? (more)

If more, less (i.e., 12) divides; ∴ we have

$$\frac{25}{12} \times 120 = 25 \times 10 = \underline{250 \text{ women}}$$

2(a) (i) **Method 4** (Equivalent ratios)

$$12 : 25 = 120 : w$$

$$\Rightarrow \quad \frac{12w}{12} = \frac{25 \times 120}{12}$$

$$\Rightarrow \quad \underline{\underline{w = 250}}$$

2(b) (i) Number of pencils of other colours

$$= 70 - (15+30) = 70 - 45$$

$$= \underline{\underline{25}}$$

(ii) $\text{P(Blue pencil)} = \frac{\text{Number of blue pencils}}{\text{Total number of pencils}}$

$$= \quad \frac{30}{70} \quad = \quad \underline{\underline{\frac{3}{7}}}$$

2(c) $\frac{1}{3}(x-1)-\frac{1}{2}(x-3) \leq 1\frac{1}{4}$

Method 1 (Simplifying expression on LHS first)

$$\frac{x-1}{3} - \frac{x-3}{2} \leq \frac{5}{4}$$

$$\frac{2(x-1) - 3(x-3)}{6} \leq \frac{5}{4}$$

$$\frac{2x-2-3x+9}{6} \leq \frac{5}{4}$$

$$\frac{-x+7}{6} \leq \frac{5}{4}$$

$$24\left(\frac{-x+7}{6}\right) \leq 24\left(\frac{5}{4}\right)$$

$$4(-x+7) \leq 6(5)$$

$$-4x+28 \leq 30$$

$$-4x \leq 30-28$$

$$\frac{-4x}{-4} \geq \frac{2}{-4}$$

$$\underline{\underline{x \geq -\frac{1}{2}}}$$

2(c) **Method 2** (Eliminating fractions first, by multiplying through by 12)

$$12 \times \tfrac{1}{3}(x-1) - 12 \times \tfrac{1}{2}(x-3) \leq 12(\tfrac{5}{4})$$

$$\Rightarrow \quad 4(x-1) - 6(x-3) \leq 3(5)$$

$$\Rightarrow \quad 4x-4-6x+18 \leq 15$$

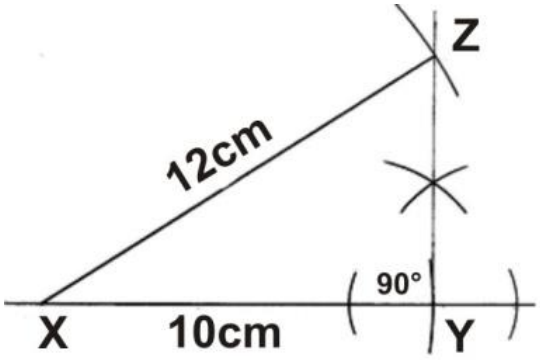
$$\Rightarrow \quad 4x-6x \leq 15+4-18$$

$$\Rightarrow \quad -2x \leq 1$$

$$\Rightarrow \quad \frac{-2x}{-2} \geq \frac{1}{-2}$$

$$\Rightarrow \quad x \geq -\frac{1}{2}$$

3(a)



(ii) |YZ| ≈ 6.6 cm

(iii) Area of triangle XYZ

= ½ bh

= ½ × |XY| × |YZ|

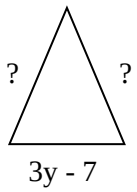
= ½ × 10cm × 6.6cm

= 10cm × 3.3cm

= $33cm^2$

(iv) Angle ZXY ≈ 34°

3(b)



Perimeter = 9y - 15

Method 1

Length of each of the two equal sides

$= [(9y - 15) - (3y - 7)] \div 2$

$= \frac{(9y - 15) - (3y - 7)}{2}$

$= \frac{9y - 15 - 3y + 7}{2}$

$= \frac{9y - 3y - 15 + 7}{2}$

$= \frac{6y - 8}{2}$

$= \frac{1}{2}(6y - 8)$

$= \underline{\underline{3y - 4}}$

$\Rightarrow$ The length of each of the two equal sides $= \underline{\underline{3y - 4}}$

3(b) **Method 2**

Let x = length of each of the other two sides

Then $x + x + 3y - 7$ = Perimeter

$\Rightarrow x + x + 3y - 7 = 9y - 15$

$\Rightarrow 2x + 3y - 7 = 9y - 15$

$\Rightarrow 2x = 9y - 15 - 3y + 7$

$\Rightarrow 2x = 9y - 3y - 15 + 7$

$\Rightarrow 2x = 6y - 8$

$\Rightarrow x = \frac{6y - 8}{2}$

$\Rightarrow x = \frac{6y}{2} - \frac{8}{2}$

$\Rightarrow x = \underline{\underline{3y - 4}}$

$\Rightarrow$ The length of each of the two equal sides

$= \underline{\underline{3y - 4}}$

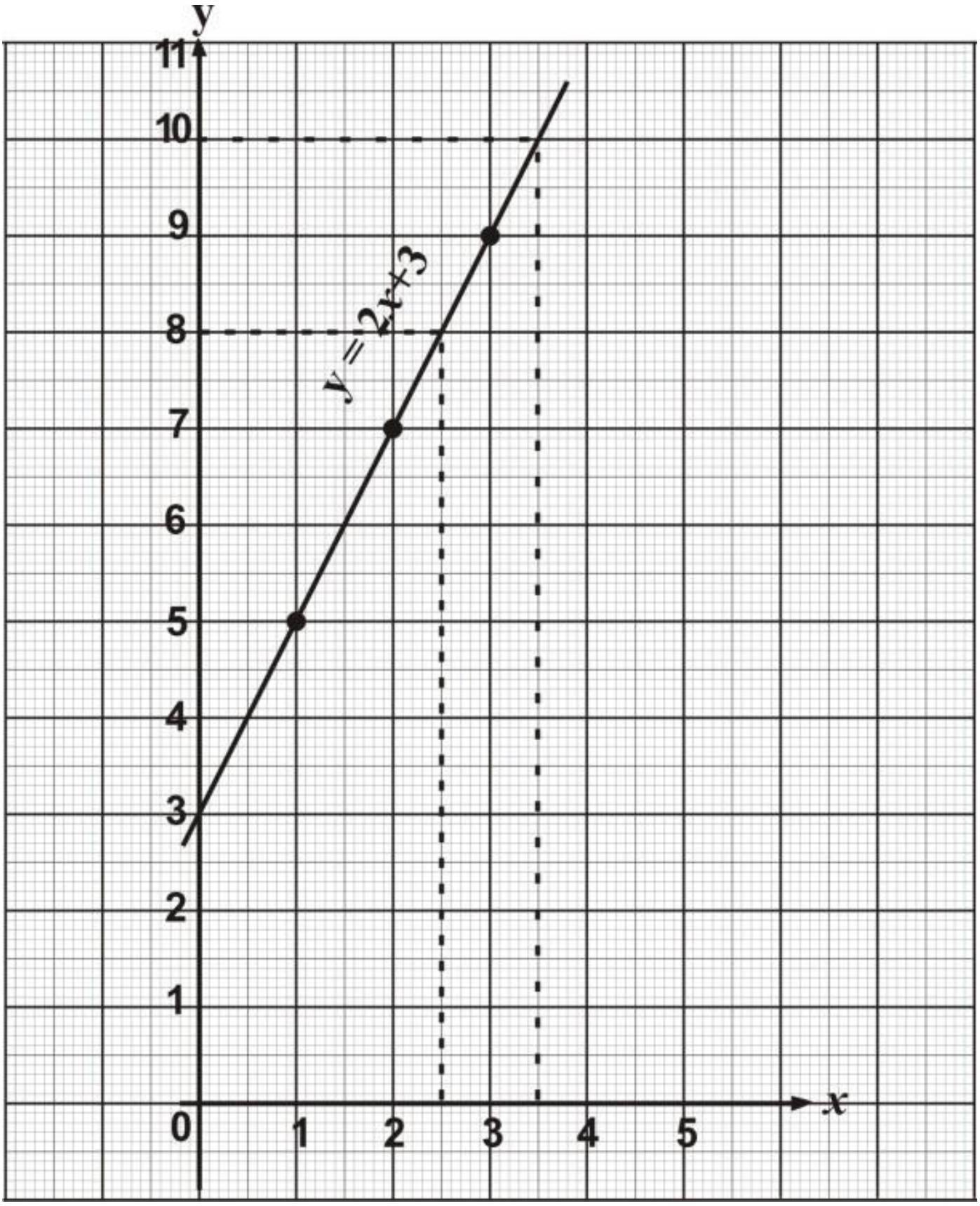
4(a)(i)

By substitution of the x values in the given relation, the mapping is completed as shown below:

x	0	1	2	3	4
	↓	↓	↓	↓	↓
y	3	5	7	9	11

(ii), (iii) The straight line graph is shown below

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4(b) From the graph,

(i) When $x = 3.5$, $\underline{y = 10}$

(ii) When $y = 8$, $\underline{x = 2.5}$

(iii) the gradient of the line:

$y = 2x + 3$ is in the form $y = mx + c$, where $m = 2$,

$\therefore$ gradient $= \underline{\underline{2}}$ (by inspection)

Or by the rule $\dfrac{y_2 - y_1}{x_2 - x_1}$, Using the points (2, 7) and (1, 5)

Gradient $= \dfrac{7-5}{2-1} = \dfrac{2}{1} = \underline{\underline{2}}$

5(a) Total fraction spent = Fractions spent on *Rent + food + books*

$$= \frac{1}{4} + \frac{2}{5} + \frac{1}{6}$$

$$\text{Fraction spent} = \frac{15 + 24 + 10}{60} = \frac{49}{60}$$

$$\Rightarrow \text{Fraction left} = 1 - \frac{49}{60} = \frac{60}{60} - \frac{49}{60} = \frac{11}{60}$$

Method 1 (Formula)

$$\text{Total amount} = \frac{\text{Fraction of Total amount}}{\text{Fraction left}} \times \text{Amount left}$$

$$\text{Total amount (M.Salary)} = \frac{60/60}{11/60} \times ¢55{,}000$$

$$= \frac{60}{60} \div \frac{11}{60} \times ¢55{,}000$$

$$= \frac{60}{60} \times \frac{60}{11} \times ¢55{,}000$$

$$= \frac{60}{11} \times ¢55{,}000$$

$$= 60 \times ¢5{,}000$$

$$= \underline{\underline{¢300{,}000}}$$

5(a) **Method 2** (Equivalent Fractions)

	Fraction	**Amount (¢)**

Spent	$\frac{49}{60}$	**?**
Left	$\frac{11}{60}$	**55,000**
Total (M. Salary)	$\frac{60}{60}$	**? (x)**

From the above table, using the 'Left' and 'Total' rows,

$$\frac{11/60}{60/60} = \frac{55{,}000}{x} \quad \Rightarrow \quad \frac{11}{60} = \frac{55{,}000}{x}$$

Cross-multiplying, we have

$$11x = 55{,}000 \times 60$$

$$\Rightarrow \quad \frac{11x}{11} = \frac{55{,}000 \times 60}{11}$$

$$= 5{,}000 \times 60$$

$$\Rightarrow \quad x = 300{,}000$$

$\therefore$ The monthly salary is ¢300,000.00

5(a) **Method 3** ('If more, less divides ...')

If $\frac{11}{60}$ → 55,000

Then $\frac{60}{60}$ → ? (more)

If more, less ($11/60$) divides, hence we have

$$\frac{60/60}{11/60} \times 55{,}000 = \frac{60}{60} \div \frac{11}{60} \times ¢55{,}000$$

$$= \frac{60}{60} \times \frac{60}{11} \times ¢55{,}000$$

$$= \frac{60}{11} \times ¢55{,}000$$

$$= 60 \times ¢5{,}000$$

$$= ¢300{,}000$$

5(a) **Method 4** (Equation)

If Monthly Salary = M, Fraction spent $= \frac{1}{4} + \frac{2}{5} + \frac{1}{6} = \frac{49}{60}$

Fraction left $= \frac{60}{60} - \frac{49}{60} = \frac{11}{60}$

Fraction left × Total amount (M) = Amount left

or $^{11}/_{60}$ of M = ¢55,000 (Left)

⇒ $^{11}/_{60}$ × M = ¢55,000

⇒ M = ¢55,000 × $^{60}/_{11}$

⇒ M = ¢300,000

5(b) **Method 1** **[Expression in terms of the father (f)]**

Let the age of the father = f,

then the mother's age = $f - 4$

Sum of children's ages = 6 × 19yrs = 114 yrs

Now, since the average age of the family = 30 years

$$\text{Average age} = \frac{\text{Ages of (children + father + mother)}}{8 \text{ people}} = 30$$

$$\Rightarrow \frac{114 + f + f - 4}{8} = 30$$

$$\Rightarrow \frac{114 - 4 + f + f}{8} = 30$$

$$\Rightarrow \frac{110 + 2f}{8} = 30$$

$$\Rightarrow 110 + 2f = 30 \times 8$$

$$\Rightarrow 2f = 240 - 110$$

$$\Rightarrow 2f = 130$$

$$\Rightarrow f = 130 \div 2$$

$$\Rightarrow f = 65$$

Therefore the father is 65 years old

5(b) **Method 2** **[Expression in terms of the mother (m)]**

Since average age of the **8** family members = **30**yrs,

⇒ Sum of ages of family members = **8** × **30** years = 240 years

⇒ Ages of children + (father) + mother = 240 years

⇒ (6 × 19) + (4 + mother) + mother = 240

Let mother's age = m

⇒ (6 × 19) + (4 +m) + m = 240

⇒ 114 + 4 + 2m = 240

⇒ 118 + 2m = 240

⇒ 2m = 240 – 118

⇒ 2m = 122

⇒ m = 122 ÷ 2

⇒ m = 61

⇒ The mother's age = 61 years

Therefore, the father's age = 61years + 4years

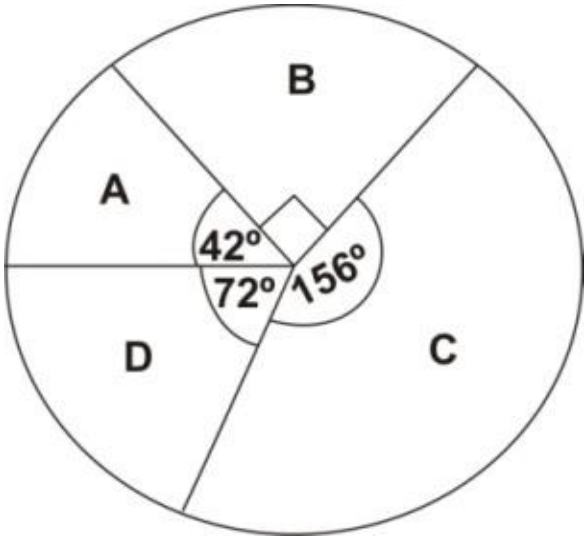
= 65 years

6 (a)

Grade	Frequency	Angle of sector
A	14	$\frac{14}{120} \times 360^{o} = 42^{o}$
B	30	$\frac{30}{120} \times 360^{o} = 90^{o}$
C	52	$\frac{52}{120} \times 360^{o} = 156^{o}$
D	24	$\frac{24}{120} \times 360^{o} = 72^{o}$

TOTAL	**120**	**= 360°**

Pie chart showing the distribution of grades obtained by 120 students in an examination



6 (a) (i) **Method 1** **(Evaluating whole numbers and fractions separately)**

$$5\frac{7}{15} - 2\frac{2}{3} + 1\frac{5}{12}$$

$$\Rightarrow \quad 5 - 2 + 1 \; + \; \frac{7}{15} - \frac{2}{3} + \frac{5}{12}$$

$$\Rightarrow \quad 4 \quad + \quad \frac{28 \; - \; 40 \; + \; 25}{60}$$

$$\Rightarrow \quad 4 \quad + \quad \frac{13}{60}$$

$$\Rightarrow \quad \underline{\underline{4\frac{13}{60}}}$$

6 (b) (i) **Method 2** **(Changing mixed fractions to improper fractions)**

$$5\frac{7}{15} - 2\frac{2}{3} + 1\frac{5}{12}$$

$$\Rightarrow \quad \frac{82}{15} - \frac{8}{3} + \frac{17}{12}$$

$$\Rightarrow \quad \frac{328 - 160 + 85}{60}$$

$$\Rightarrow \quad \frac{253}{60} \quad = \quad \underline{\underline{4\frac{13}{60}}}$$

6 (b) (ii) $xy - xz + 5y - 5z$

$= x(y - z) + 5(y - z)$

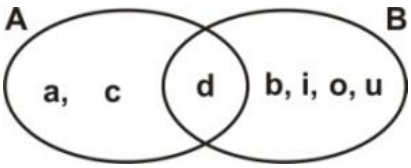
$= \underline{(y - z)(x + 5)}$

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MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1.



In the diagram above, A and B are two intersecting sets. How many members are in the set B?

A) 2 B) 3 C) 5 D) 7

2. Which of the following is not a factor of 18?

A) 3 B) 4 C) 6 D) 9

3. Multiply 0.014 by 0.2

A) 0.00028 B) 0.0028 C) 0.028 D) 0.28

4. Given that A = {b, c, i} and B = {a, e, f, i}, find $A \cap B$.

A. {a, b, c, e, f, i} B. {f, i} C. {a} D. {i}

5. If a = 64 and $b = 2^2$, find $^a/_b$.

A) 32 B) 16 C) $^1/_{16}$ D) $^1/_{32}$

6. Arrange the following numbers from the highest to the lowest: $^2/_3$, -7, 0.

A) -7, 0, $^2/_3$

B) -7, $^2/_3$, 0

C) 0, $^2/_3$, -7

D) $^2/_3$, 0, -7

7. Simplify $2^2 \times 3^2 \times 2^3 \times 3^4$

A) $2^2 \times 3^6$ B) $2^4 \times 3^7$ C) $2^5 \times 3^6$ D) $2^6 \times 3^8$

8. In an examination 40% of the students failed. The number of students that passed was 180. How many students failed?

A) 120 B) 270 C) 300 D) 450

9. Evaluate $^1/_3$ [(5 – 1) – (2 – 7)]

A) -3 B) -1 C) 1 D) 3

10. If 6n + 4 = 16, find the value of n.

A) 2 B) 3 C) 5 D) 6

11. What is the next term in the following set of numbers: {1, 3, 7, 15, 31, …} ?

A) 35 B) 39 C) 47 D) 63

12. Find the mean of the following set of numbers 10, 12, 14 and 16.

A) 12 B) 13 C) 14 D)15

13. Simplify $3a^2 \times 2ab \times 4bc$

A) $9a^3b^2c$ B) $12a^2b^2c$ C) $24a^2b^2c$ D) $24a^3b^2c$

14. Expand 3 (2a + 3b)

A) 2a +9b B) 5a + 6b C) 6a + 3b D) 6a + 9b

15. Find the area of a circle whose diameter is 28 cm.
(Take $\pi = {}^{22}/_7$)

A) 44 cm^2 B) 88 cm^2 C) 352 cm^2 D) 616 cm^2

16. In sharing 95 oranges with Dede, Fofo kept 45 of them and shared the rest equally with Dede. How many oranges did Dede get?

A) 20 B) 25 C) 40 D) 55

17. If $c = \dfrac{b^2 + 4r}{2ar}$, find c when $b = 3$, $r = 4$ and $a = 5$.

A) $^{19}/_{40}$ B) $^{11}/_{20}$ C) $^3/_8$ D) $^5/_8$

18. A woman bought a sewing machine for ¢800,000.00 and sold it for ¢920,000.00. Find the percentage profit

A) 12% B) 13% C) 15% D) 16.4%

19. What is the rule for the following mapping?

x	0	1	2	3	4
↓	↓	↓	↓	↓	↓
y	5	9	13	17	21

A) $y = x + 5$ B) $y = 4x + 5$ C) $y = 5x+4$ D) $y = 6x+1$

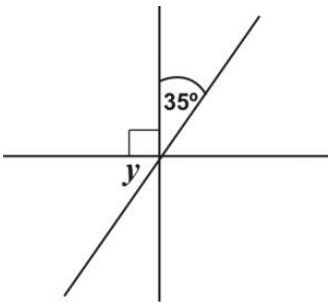
20. Find the volume of a cuboid of sides 15 m by 12 m by 3 m.

A) 36 m^3 B) 45 m^3 C) 180 m^3 D) 540 m^3

21. A bag contains 4 blue and 8 red balls. What is the probability of picking a blue ball at random from the bag?

A) $^1/_3$ B) ½ C) $^2/_3$ D) $^3/_4$

22.



Find the size of the angle marked y in the diagram.

A) 35° B) 45° C) 55° D) 90°

23. An equilateral triangle has side 16cm. A square has the same perimeter as the equilateral triangle. What is the area of the square?

A) 48 cm^2 B) 96 cm^2 C) 144 cm^2 D) 256 cm^2

24. Kofi deposited ¢500,000.00 with a bank for 2 years at a rate of 10% per annum. Find the simple interest

A) ¢10,000.00 B) ¢20,000.00 C) ¢50,000.00 D) ¢100,000.00

25. Make *T* the subject of the relation $l^2 = \frac{4\pi^2 T}{g}$

A) $T = \frac{gl}{2\pi}$ B) $T = \frac{gl^2}{2\pi}$

C) $T = \frac{l^2}{4g\pi^2}$ D) $T = \frac{gl^2}{4\pi^2}$

26. Evaluate $\frac{0.54 \times 0.7}{9}$

A) 0.0042 B) 0.042 C) 0.42 D) 4.2

27. Eric and Ebo are to share an amount of ¢800,000.00 in the ratio 5:3 respectively. What will be Ebo's share?

A. ¢30,000.00

B) ¢50,000.00

C) ¢300,000.00

D) ¢500,000.00

28. How many faces has a rectangular pyramid?

A) 4 B) 5 C) 6 D) 7

29. The perimeter of a rectangle is 24cm. If the breadth of the rectangle is 4 cm, find the area of the rectangle.

A) 32 cm^2 B) 48 cm^2 C) 64 cm^2 D) 144 cm^2

30. Factorize $4ab^2 - 20ba^2$.

A. $4a(b^2 - 5b)$

B. $4b(b - 5a)$

C. 4ab(b-5a)

D. 4ab(a-5b)

31. Solve the equation $\frac{2}{3}(x-3) = \frac{5}{6}(x+6)$

A) -42 B) -12 C) 18 D) 42

32. Kofi had 100 mangoes and sold 80 of them. What is the percentage of mangoes left?

A) 12.5% B) 18% C) 20% D) 25%

33. Find the median of the following marks: 2, 4, 10, 3, 6, 12.

A) 4 B) 5 C) 6 D) 10

In a class of 20 pupils, 8 pupils read Mathematics, 13 read English and 3 read both Mathematics and English.

Use this information to answer Questions **34** *and* **35**

34. How many pupils read English only?

A) 2 B) 3 C) 5 D) 10

35. How many pupils do not read either Mathematics or English?

A) 2 B) 3 C) 5 D) 7`

36. Calculate 82.5 ÷ 0.25, expressing the answer in the standard form

A) 3.3×10^{-3} B) 3.3×10 C) 3.3×10^{2} D) 3.3×10^{3}

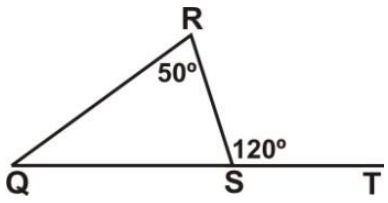
37. The price of a gas cooker costing ¢450,000.00 was increased by 10%. Find its new price.

A) ¢45,000.00 B) ¢49,000.00 C) ¢405,000.00 D) ¢495,000.00

38. What is the number of all possible outcomes of tossing a die?

A) 2 B) 3 C) 4 D) 6

39. In the diagram, QRS is a triangle. Angle QRS = 50° and angle RST = 120°. Find angle RQS.



A) 60° B) 65° C) 70° D) 80°

40.

$$If \quad r = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \; and \; s = \begin{pmatrix} -4 \\ -1 \end{pmatrix}, \; find \; r + 2s$$

$$A.\begin{pmatrix} -10 \\ 1 \end{pmatrix} \quad B.\begin{pmatrix} -10 \\ -1 \end{pmatrix} \quad C.\begin{pmatrix} -6 \\ 1 \end{pmatrix} \quad D.\begin{pmatrix} -6 \\ 2 \end{pmatrix}$$

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MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. C) 5

2. B) 4

3. B) 0.0028

4. D. {i}

5. B) 16

6. D) $^{2}/_{3}$, 0, -7

7. C) $2^5 \times 3^6$

8. A) 120

9. D) 3

10. A) 2

11. D) 63

12. B) 13

13. D) $24a^3b^2c$

14. D) 6a + 9b

15. D) 616 cm^2

16. B) 25

17. D) $^{5}/_{8}$

18. C) 15%

19. B) $y = 4x + 5$

20. D) $540m^3$

21. A) $^{1}/_{3}$

22. C) 55°

23. C) $144cm^2$

24. D) ¢100,000.00

25. D) $T = \frac{gl^2}{4\pi^2}$

26. B) 0.042

27. C) ¢300,000.00

28. B) 5

29. A) 32 cm^2

30. C. 4ab (b – 5a)

31. A) -42

32. C) 20%

33. B) 5

34. D) 10

35. A) 2

36. C) 3.3×10^2

37. D. ¢495,000.00

38. D) 6

39. C) 70°

40. A) $\begin{pmatrix} -10 \\ 1 \end{pmatrix}$

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MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. (a) A boy has enough money to buy 14 pencils at ¢450.00 each. How many erasers costing ¢300.00 each can he buy with the same amount of money?

(b) A circle has a circumference of 44cm. Calculate its radius if $\pi = {}^{22}/_{7}$.

(c) Simplify $\left(2\frac{1}{4} - 1\frac{5}{8}\right) \div 3\frac{7}{16}$

(d) If $\frac{5(n+6)}{n} = 1$ find n.

2. (a) Copy and complete the table below for the relation; $x + y = 180$

X	0	30	60	90	120	150	180
Y	180			90			

(b) (i) Using a scale of 2cm to 20 units on both axes, draw two perpendicular axes OX and OY.

(ii) Mark both axes from 0 to 180.

(iii) Plot all the seven points. Use a ruler to join all the points.

(c) Using your graph, find

(i) y when $x = 100$;

(ii) x when $y = 70$.

3. (a) There are 30 boys in a sporting club. 20 of them play hockey and 15 play volley-ball. Each boy plays at least one of the two games.

(i) Illustrate the information on a Venn diagram

(ii) How many boys play volley-ball only?

(b) Factorize $xy + 3x + 6y + 18$

(c) Multiply $(3 + x)$ by $(5 – 2x)$

4. (a) A woman borrowed ¢2,000,000.00 from a bank at a rate of 15% per annum simple interest for 2 years.

Find

(i) the interest for the 2 years

(ii) how much she paid in all to the bank after the 2 years.

(b) A woman borrowed ¢2,000,000.00 from a bank. She used the ¢2,000,000.00 to purchase a fridge and sold it at a profit of 45%. Find the selling price of the fridge.

5. (a) Using a pair of compasses and ruler only, construct

(i) triangle ABC with $|AB| = 10$cm, angle ABC = 30° and angle CAB = 60° ;

(ii) a perpendicular from the point C to meet the line AB at P.

(b) (i) Extend line CP to the point D such that $|BC| = |BD|$.

(ii) Join A to D and B to D.

(iii)Measure $|AC|$ and $|AD|$.

(c) What type of quadrilateral is ADBC?

6. The following marks were obtained by pupils in a test.

6	4	8	2	8
6	8	8	8	10
8	9	8	6	10
2	2	6	6	6

(a) Construct a frequency distribution table for the data.

(b) What is the modal mark?

(c) Calculate the mean mark.

(d) How many pupils scored more than 6 marks?

(e) What is the probability that a student chosen at random obtained 2 marks?

April 2005

MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1(a) **Method 1** **(Simple logic)**

Total amount = 14 × ¢450.00 = ¢6300.00

Now, ¢ 6300 ÷ ¢300 = 21

∴ He can buy 21 erasers with the same amount

1(a) **Method 2** **[Simple Proportion (indirect)]**

If @ ¢450 each, he can buy 14 pencils

Then @ ¢300 each, he can buy ? (more) erasers

If more, less (i.e., ¢300) divides,

$$\Rightarrow \quad \frac{¢450}{¢300} \times 14 = 21$$

∴ He can buy 21 erasers with the same amount

Circumference (C) = 2 × pi (π) × radius (r)

1(b) **Method 1** (Substituting first)

$$C = 2\ \pi\ r$$

$$\Rightarrow \quad 44 = 2 \times {}^{22}/_{7} \times r$$

$$\Rightarrow \quad 44 = {}^{44}/_{7} \times r$$

$$\Rightarrow \quad 44 \times \frac{7}{44} = r$$

$$\Rightarrow \quad 7 = r$$

$$\Rightarrow \quad \underline{\text{Radius} = 7 \text{ cm}}$$

1(b) **Method 2** (Making r the subject first)

$$C = 2\,\pi\, r$$

$$\Rightarrow \quad \frac{C}{2\ \pi} = r$$

$$\Rightarrow \quad \frac{44}{2 \times {}^{22}\!/_{7}} = r$$

$$\Rightarrow \quad \frac{44}{{}^{44}\!/_{7}} = r$$

$$\Rightarrow \quad 44 \div \frac{44}{7} = r$$

$$\Rightarrow \quad 44 \times \frac{7}{44} = r$$

$$\Rightarrow \quad 7 = r$$

$$\Rightarrow \quad \underline{\text{Radius} = 7 \text{ cm}}$$

1(c) $\left(2\frac{1}{4} - 1\frac{5}{8}\right) \div 3\frac{7}{16}$

$$\Rightarrow \quad \left(\frac{9}{4} - \frac{13}{8}\right) \div \frac{55}{16}$$

$\Rightarrow \quad \left(\frac{18 - 13}{8}\right) \div \frac{55}{16}$

$\Rightarrow \quad \frac{5}{8} \times \frac{16}{55}$

$\Rightarrow \quad \underline{\underline{2/11}}$

1(d) $\frac{5(n+6)}{n} = 1$

Multiplying both sides by n

$$n \times \frac{5(n+6)}{n} = 1 \times n$$

$\Rightarrow \quad 5(n + 6) = n$

$\Rightarrow \quad 5n + 30 = n$

$\Rightarrow \quad 5n - n = - 30$

$\Rightarrow \quad 4n = - 30$

$\Rightarrow \quad \frac{4n}{4} = \frac{-30}{4}$

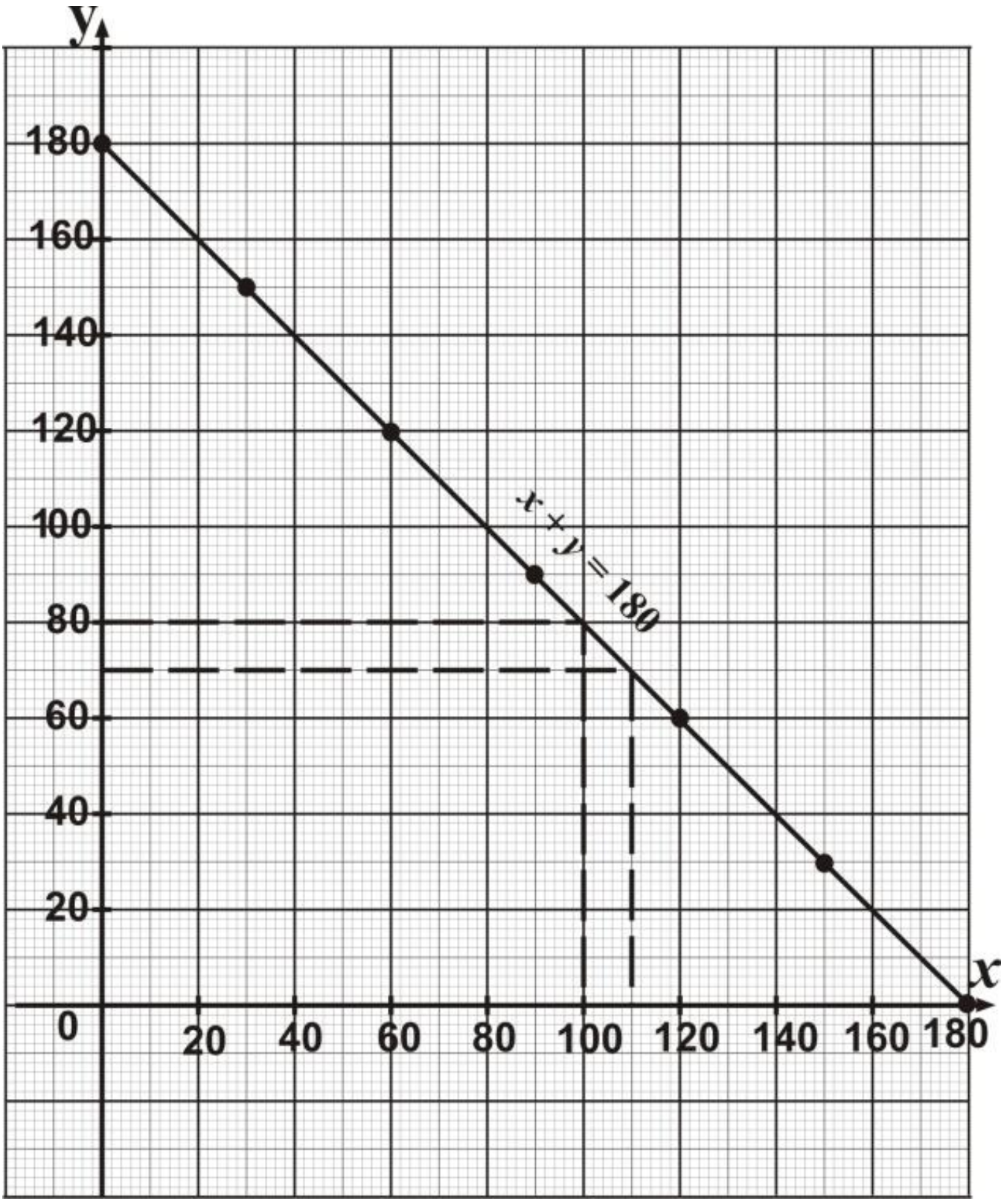
$\Rightarrow \quad n = \frac{-15}{2}$

$\Rightarrow \quad n = \underline{\underline{-7\frac{1}{2}}}$

2 (a) By substitution of the x values in the given relation, the table is completed as shown below:

X	0	30	60	90	120	150	180
Y	180	150	120	90	60	30	0

2(b) The straight line graph is shown below



2(c) From the graph,

(i) When $x = 100$, $\underline{y = 80}$

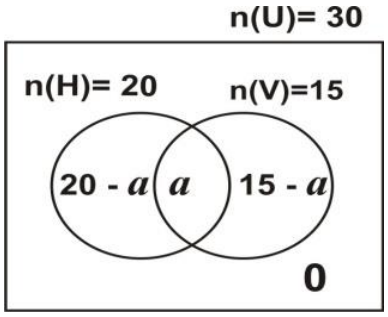
(ii) When $y = 70$, $\underline{x = 110}$

3(a) (i) Let n(U) = No. of boys in the club

n(H) = No. of boys that play hockey

n(V) = No. of boys that play volleyball

a = No. of boys that play both hockey and volleyball



3(a) (ii) From the above diagram,

$$20 - a + a + 15 - a + 0 = 30$$

$$\Rightarrow \quad 20 + 0 + 15 - a + 0 = 30$$

$$\Rightarrow \quad 20 + 15 - a = 30$$

$$\Rightarrow \quad 35 - a = 30$$

$$\Rightarrow \quad \underline{a = 5}$$

No. that play volleyball only

$$= \quad n(V) - a$$

$$= \quad 15 - 5$$

$$= \quad \underline{10}$$

<u>10 boys play volleyball only</u>

3(b) $xy + 3x + 6y + 18$

$$= \quad x(y + 3) + 6(y + 3)$$

$$= \quad \underline{(y + 3)(x + 6)}$$

3(c) $(3 + x)(5 - 2x)$

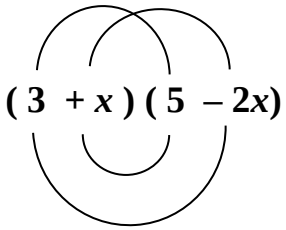
<u>Approach 1 (Distributive property idea)</u>

$$3(5 - 2x) + x(5 - 2x)$$

$\Rightarrow \quad 15 - 6x + 5x - 2x^2$

$\Rightarrow \quad \underline{\underline{15 - x - 2x^2}}$

3(c) **Approach 2 (Smiley face)**

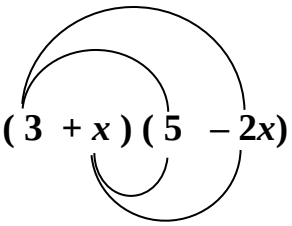


$15 - 2x^2 + 5x - 6x$

$\Rightarrow \quad 15 + 5x - 6x - 2x^2$

$\Rightarrow \quad \underline{\underline{15 - x - 2x^2}}$

3(c) **Approach 3 (Moon shape)**



(3 + x) (5 – 2x)

$\Rightarrow \quad 15 - 6x + 5x - 2x^2$

$\Rightarrow \quad \underline{\underline{15 - x - 2x^2}}$

4(a) (i) Simple interest = Principal × Rate × Time

$= 2{,}000{,}000 \times 15\% \times 2$

$= 2{,}000{,}000 \times \frac{15}{100} \times 2$

$= 20{,}000 \times 15 \times 2$

$= 600{,}000$

The simple interest is ¢600,000.00

(ii) Amount paid to bank = Principal + S. interest

= ¢2,000,000 + ¢600,000

= ¢2,600,000

She paid ¢2,600,000 to the bank after the 2 years

4(b) **Method 1** (Formula 1)

SP = CP + Profit

= ¢2,000,000 + (45% of ¢2,000,000)

$= ¢2{,}000{,}000 + \left(\frac{45}{100} \times ¢2{,}000{,}000\right)$

= ¢2,000,000 + 45 × ¢20,000

= ¢2,000,000 + ¢900,000

= ¢2,900,000

4(b) **Method 2** (Formula 2)

$$SP = \frac{\text{SP percentage}}{\text{CP percentage}} \times CP$$

$$= \frac{145\%}{100\%} \times 2{,}000{,}000$$

$$= \frac{145}{100} \times 2{,}000{,}000$$

$$= 145 \times 20{,}000$$

= ¢2,900,000

NB
The Percentage of the Original (Cost) Price is always 100%

4(b) **Method 3** (Equivalent Fractions)

	Percentage	Amount
CP	**100%**	**2,000,000**
SP	100% + 45% = **145%**	***x***

$$\frac{100\%}{145\%} = \frac{2{,}000{,}000}{x}$$

← From the above table

Cross multiplying,

$$100 \times x = 2{,}000{,}000 \times 145$$

Dividing both sides by 100

$$\Rightarrow \quad x = \frac{2{,}000{,}000 \times 145}{100}$$

$$\Rightarrow \quad x = 20{,}000 \times 145$$

$$\Rightarrow \quad \underline{\underline{x = 2{,}900{,}000}}$$

4(b) **Method 4** (If more, less divides …)

If 100% → 2,000,000

Then 145% → ? (more)

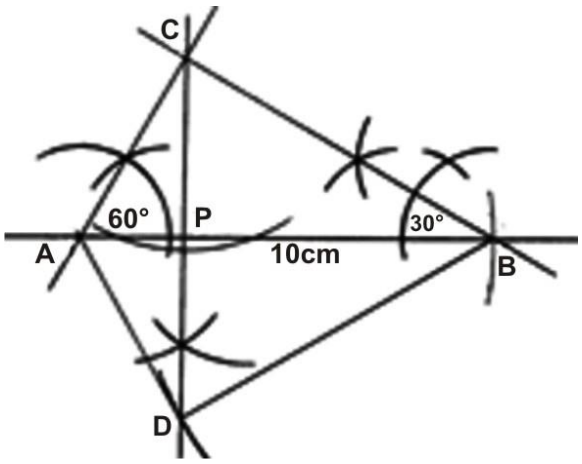
If more, less (i.e., 100) divides; therefore we have

$$\frac{145}{100} \times 2{,}000{,}000$$

$$= 145 \times 20{,}000$$

$$= ¢\ \underline{2{,}900{,}000}$$

5 (a), (b)



(b) (iii) |AC| ≈ 5 cm, |AD| ≈ 5 cm

(c) ADBC is a kite.

6 (a) Frequency Table

Mark	Tally	Frequency
2	///	3
4	/	1
6	~~////~~ /	6
8	~~////~~ //	7
9	/	1
10	//	2

6(b) The modal mark is the most occurring mark

⇒ The modal mark = 8

6(c) **Method 1** **(Using raw data)**

$$\text{The mean mark} = \frac{\text{The sum of marks}}{\text{The total number of pupils}}$$

$$= \frac{(2\times3)+(4\times1)+(6\times6)+(8\times7)+(9\times1)+(10\times2)}{3+1+6+7+1+2}$$

$$= \frac{6+4+36+56+9+20}{20}$$

$$= \frac{131}{20} = \underline{\underline{6\frac{11}{20}}} \quad \text{or} \quad \underline{\underline{6.55}}$$

6(c) **Method 2** **(Using frequency table)**

Mark (x)	**Tally**	**Frequency** (f)	fx
2	///	3	6
4	/	1	4
6	//// /	6	36
8	//// //	7	56
9	/	1	9
10	//	2	20
		$\Sigma f = 20$	**$\Sigma fx = 131$**

The mean mark $= \frac{\Sigma f x}{\Sigma f} = \frac{131}{20}$

$$= \underline{\underline{6\frac{11}{20}}} \quad \text{or} \quad \underline{\underline{6.55}}$$

6 (d) Sum of frequencies of 8, 9 and 10

$\Rightarrow$ 7 + 1 + 2 pupils = <u>10 pupils</u>

6 (e) $P(2 \text{ marks}) = \frac{\text{Number of pupils that obtained 2 marks}}{\text{Total number of pupils}}$

$$= \underline{\underline{\frac{3}{20}}}$$

April 2004

MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. If the set A = {1, 2, 3, 4, 5} and the set B = {2, 4, 6}, find the number of members in the set A B.

A) 8 B) 6 C) 5 D) 2

2. How many faces has a cuboid ?

A) 3 B) 4 C) 5 D) 6

3. Evaluate $\frac{0.036}{0.02}$

A) 0.018 B) 0.18 C) 1.8 D) 18.0

4. The angle formed by one complete revolution is equivalent to

A) one right angle
B) two right angles
C) three right angles
D) four right angles

5. If the set P = {1, 2, 3, 4, 5} which of the following statements best describes P?

A) Set of whole numbers up to 6.
B) Set of counting numbers less than 6
C) Set of counting numbers greater than 6
D) Set of integers less than 6.

6. Which of the following numbers is the next prime number greater than 23 ?

A) 17 B) 24 C) 25 D) 29

7. Simplify -35 – (-15) + (-30)

A) –10 B) –20 C) –50 D) -80

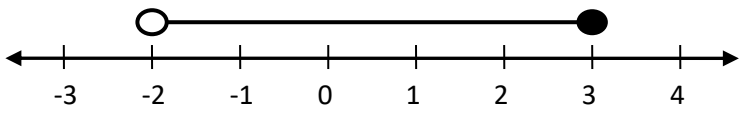
8. Write 78910 correct to the **nearest** thousand

A) 70,000 B) 78,000 C) 79,000 D) 80,000

9. Simplify $\dfrac{2^{10} \times 3^2}{3^6 \times 2^8}$

A) $\dfrac{2^2}{3^4}$ B) $\dfrac{3^4}{2^2}$ C) $\dfrac{3^2}{2^4}$ D) $\dfrac{2^4}{3^4}$

10. Which of the following inequalities is represented on the number line below?



A. $-2 < x < 3$

B. $-2 \leq x < 3$

C. $-2 \leq x \leq 3$

D. $-2 < x \leq 3$

11. Factorize completely the expression $4xy - 16x + 10y - 40$.

A.. $(y + 4)(4x - 10)$

B. $(y - 4)(4x + 10)$

C. $(4 - y)(10 - 4x)$

D. $(y + 4)(4x + 10)$

12. What is the image of 2 in the mapping $x \rightarrow 2x + 5$?

A) 7 B) 9 C) 12 D) 20

13. Simplify $3a^2b^3 \times 4a^3b$

A) $12a^5b^4$ B) $12a^4b^5$ C) $7a^5b^4$ D) $7a^4b^5$

14. Find the solution set of $2x + 1 < 5$ in the domain $\{-1, 0, 1, 2, 3\}$.

A. {-1, 1, 3,} B. {-1, 0, 1} C. {-1, 1, 2} D. {0, 1, 2}

15. Find the highest common factor (HCF) of 48, 60 and 96.

A) 12 B) 24 C) 36 D) 48

16. P = {odd numbers between 20 and 30} and Q = {23, 29}. Which of the following is true?

A) $P \subset Q$ B) $Q \subset P$ C) $P = Q$ D) $P \cap Q = \Phi$

17. Find the rule for the following mapping

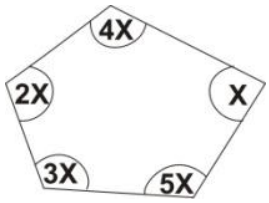
x	0	1	2	3	4
↓	↓	↓	↓	↓	↓
y	-1	1	3	5	7

A) $y = 2x + 1$ B) $y = 2x - 1$ C) $y = x^2-1$ D) $y=x^2+1$

18. If $\mathbf{a} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$ find $2\mathbf{a} - \mathbf{b}$.

A. $\begin{pmatrix} 1 \\ 6 \end{pmatrix}$ B. $\begin{pmatrix} 7 \\ 6 \end{pmatrix}$ C. $\begin{pmatrix} 1 \\ 10 \end{pmatrix}$ D. $\begin{pmatrix} 7 \\ 10 \end{pmatrix}$

19. Find the value of x in the polygon below.



A) 12° B) 36° C) 60° D) 72°

20. An article costs ¢60,000.00. The price was increased by 10%. Find the new price.

A) ¢54,000.00 B) ¢61,000.00 C) ¢66,000.00 D) ¢70,000.00

21. Solve for x in the equation $15 - 2x = 6$

A) –10.5 B) –4.5 C) 4.5 D) 10.5

22. The population of Ghana was 5,000,000 in 1957. The population in 1998 was estimated to be 17,000,000. Find the percentage increase in population from 1957 to 1998.

A) 2.4% B) 24% C) 240% D) 2400%

23. The angles of a triangle are in the ratio 3:2:1. Find the value of the smallest angle.

A) 30° B) 45° C) 60° D) 90°

24. Find the least common multiple (LCM) of 4, 6 and 10

A) 10 B) 24 C) 30 D) 60

25. Convert 134_{five} to a base ten numeral

A) 16 B) 40 C) 44 D) 220

26. A circle has radius 7cm; find its area.

(Take $\pi = {}^{22}/_{7}$)

A) 22 cm^2 B) 44 cm^2 C) 145 cm^2 D) 154 cm^2

27. Simplify: ½ - $^{2}/_{3}$ + ¾

A) $^{11}/_{12}$ B) $^{7}/_{12}$ C) $^{2}/_{3}$ D) $^{5}/_{12}$

28. Solve for x, if $\frac{1}{2}x - 4x > 20$

A) x < -13 $^{1}/_{3}$ B) x < -5 $^{5}/_{7}$ C) x > 5 $^{5}/_{7}$ D) x > 13 $^{1}/_{3}$

29. Which of the following is equivalent to $2^2 \times 6^3$?

A) $2^6 \times 3^3$ B) $2^5 \times 3^3$ C) $2^4 \times 3$ D) $2^3 \times 3^3$

30. The volume of water in a rectangular tank is 30 cm^3. The length of the tank is 5 cm and its breadth is 2 cm. Calculate the depth of water in the tank.

A) 4.0 cm B) 3.0 cm C) 5.0 cm D) 6.0 cm

31. The following temperatures in ^{o}C were recorded in 10 cities in Europe;

-4, 5, 2, 0, -6, -4, 3, -6, -4, 7.

Find the modal temperature.

A) –2°C B) –4°C C) 0°C D) –6°C

32. The ratio of the ages of two sisters is 4:3. The elder sister is 3 years older than the younger one. How old is the younger sister?

A) 9 years B) 12 years C) 15 years D) 18 years

33. If $n^2 + 4 = 40$, find n

A) 6 B) 18 C) 22 D) 44

34. An angle which is more than 90° but less than 180° is

A) an acute angle
B) a right angle
C) an obtuse angle
D) a reflex angle

35. Arrange the following fractions in ascending order: 4 ¼ , 4 ½ , 4 $^1/_3$

A) 4 ¼ , 4 $^1/_3$, 4 ½
B) 4 $^1/_3$, 4 ½ , 4 ¼
C) 4 ½ , 4 ¼ , 4 $^1/_3$
D) 4 ½ , 4 $^1/_3$, 4 ¼

36. Prices of items in a shop were reduced by 10% during a reduction sale. If a customer bought an electric fan for ¢81,000.00, what was its original price?

A) ¢72,900.00
B) ¢73,900.00
C) ¢89,100.00
D) ¢90,000.00

37. The marks scored by 8 pupils in a science test are: 3, 7, 8, 8, 5, 4, 8, 4.
What is the median mark?

A) 5 B) 6 C) 7 D) 8

38. The marks scored by 8 pupils in a science test are: 3, 7, 8, 8, 5, 4, 8, 4.
What is the probability that a pupil chosen at random scored 4 marks?

A) ½ B) ¼ C) $^1/_8$ D) $^2/_{47}$

39. If $y = k + ax^2$, find y when $k = \frac{14}{5}$, $a = \frac{4}{5}$ and $x = 2$

A) 30 B) $7\frac{3}{5}$ C) 6 D) 4

40. Which of the following dimensions of a triangle form the sides of a right –angled triangle ?

A) 3cm, 4cm, 6cm

B) 3cm, 5cm, 7cm

C) 5cm, 12cm, 13cm

D) 5cm, 13cm, 17cm

April 2004

MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. B) 6

2. D) 6

3. C) 1.8

4. D) four right angles

5. B) Set of counting numbers less than 6

6. D) 29

7. C) –50

8. C) 79,000

9. A) $\frac{2^2}{3^4}$

10. D. $-2 < x \leq 3$

11. B. $(y - 4)(4x + 10)$

12. B) 9

13. A) $12a^5b^4$

14. B. {-1, 0, 1}

15. A) 12

16. B) $Q \subset P$

17. B) $y = 2x - 1$

18. B. $\binom{7}{6}$

19. B) 36°

20. C) ¢66,000.00

21. C) 4.5

22. C) 240%

23. A) 30°

24. D) 60

25. C) 44

26. D) 154 cm^2

27. B) $^{7}/_{12}$

28. B) $x < -5\ ^{5}/_{7}$

29. B) $2^5 \times 3^3$

30. B) 3.0cm

31. B) $-4^{o}C$

32. A) 9 years

33. A) 6

34. C) an obtuse angle

35. A) 4 ¼ , $4\ ^{1}/_{3}$, 4 ½

36. D) ¢90,000.00

37. B) 6

38. B) ¼

39. C) 6

40. C) 5cm, 12cm, 13cm

April 2004

MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. (a) (i) Factorize completely the expression $2xy - 8x + 3y - 12$

(ii) Evaluate the expression in (i) if $x = 5$ and $y = 7$

(b) Make q the subject of the equation $t = \frac{1}{p} + \frac{1}{q}$

(c) Given that $\mathbf{U} = \begin{pmatrix} -5 \\ 9 \end{pmatrix}$ and $\mathbf{V} = \begin{pmatrix} 8 \\ 12 \end{pmatrix}$, find $3(\mathbf{U} + \frac{1}{2}\mathbf{V})$

2. (a) Ama and Kofi shared the profit earned from their business in the ratio 3:4. The profit was ¢1,743,000.00

(i) Find how much of the profit each person received.

(ii) Kofi lent out his share of the profit at a rate of 20% per annum for 2 years. Find the interest on his share.

(iii) What will be Kofi's total amount at the end of the 2 years?

.

(b) Change 243_{five} to a base ten numeral.

3. (a) Using a ruler and a pair of compasses only, construct,

(i) triangle PQR such that | PQ | = 8cm, angle QPR = 60° and angle PQR = 45°.

(ii) Measure | QR |.

(b) A rectangular water tank has length 60cm, width 45cm and height 50cm.

Find

(i) the total surface area of the tank when closed

(ii) the volume of the tank

(iii) the height of the water in the tank, if the tank contains 81,000cm^3 of water.

4. (a) A car runs on the average at 45 km to 5litres of fuel. Calculate how many litres of fuel are required for a journey of 117km.

(b) (i) Solve for x in the inequality $\frac{2}{3}(2x+5) \leq 8\frac{2}{3}$

(ii) Illustrate the solution on the number line

(c) A factory increased its production by 22 ½% and produced 49,000 tonnes. How many tonnes was it producing before?

5. (a) (i) Using a scale of 2cm to 1 unit on both axes, draw two perpendicular axes OX and OY on a graph sheet.

(ii) On the same graph sheet, mark the x-axis from –5 to 5 and the y-axis from –6 to 6.

(b) Plot the points,

(i) P (1, -2) and Q (4, 5)

(ii) P′ the image of P under a translation by the vector $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$ and Q′, the image of Q by the same vector.

(NB: The original *given vector* $\begin{pmatrix} 5 \\ 0 \end{pmatrix}$*, which was an error, is corrected to* $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$

(c) (i) Join PQQ′P′.

(ii) Measure angles PQQ′ and PP′Q′.

(d) (i) Find the vectors $\overrightarrow{PQ}$ and $\overrightarrow{P'Q'}$

(ii) What is the shape of PQQ′P′ ?

6. (a) The following are the ages in years of members of a group:
8, 11, 8, 10, 6, 7, 3x, 11, 11. If the mean age is 9 years, find

(i) x

(ii) the modal age

(iii) the median age.

(b) Draw a bar chart for the distribution

April 2004

MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1(a) (i) $2xy - 8x + 3y - 12$

$\Rightarrow \quad 2x(y - 4) + 3(y - 4)$

$\Rightarrow \quad \underline{(y - 4)\,(2x + 3)}$

1. Factorize $2xy - 8x$ and $3y - 12$ separately
2. Factorize $(y - 4)$ out

(ii) **Approach 1** (Using $2xy - 8x + 3y - 12$)

since $x = 5$ and $y = 7$, we have

$2(5)(7) - 8(5) + 3(7) - 12$

$\Rightarrow \quad 70 - 40 + 21 - 12$

$= \quad \underline{39}$

(ii) **Approach 2** [Using $(y - 4)\,(2x + 3)$]

since $x = 5$ and $y = 7$, we have

$[7 - 4]\ [2(5) + 3]$

$[3]\ [10 + 3]$

$\Rightarrow \quad 3 \times 13$

$= \quad \underline{39}$

1 (b)

$t = \frac{1}{p} + \frac{1}{q}$, making q the subject

Method 1 (Removing fractions first)

$pq(t) = pq(\frac{1}{p}) + pq(\frac{1}{q})$

Multiplying through by pq (LCM of the denominators) to remove fractions

$\Rightarrow \quad pqt = q + p$

$\Rightarrow \quad pqt - q = p$

Grouping all terms containing q on one side

$\Rightarrow \quad q(pt-1) = p$

Factorizing q out

$\Rightarrow \quad \frac{q(pt-1)}{(pt-1)} = \frac{p}{(pt-1)}$

Dividing both sides by *(pt–1)*

$\Rightarrow \quad \underline{\underline{q = \frac{p}{pt-1}}}$

1 (b) **Method** 2 (Separating term(s) containing q on one side first)

$t = \frac{1}{p} + \frac{1}{q}$

Separating term(s) containing q on one side and simplifying both sides

$t - \frac{1}{p} = \frac{1}{q}$

$\Rightarrow \quad \frac{t}{1} - \frac{1}{p} = \frac{1}{q}$

$\Rightarrow \quad \frac{pt-1}{p} = \frac{1}{q}$

$q(pt-1) = p$

Cross multiplying

$\Rightarrow \quad \underline{\underline{q = \frac{p}{pt-1}}}$

1 (c) $3\ (\mathbf{U} + \frac{1}{2}\ \mathbf{V})$

Method 1 (Evaluating within the bracket first)

$$= 3\left[\begin{pmatrix}-5\\9\end{pmatrix} + \frac{1}{2}\begin{pmatrix}8\\12\end{pmatrix}\right]$$

$$= 3\left[\begin{pmatrix}-5\\9\end{pmatrix} + \begin{pmatrix}\frac{1}{2}\times 8\\\frac{1}{2}\times 12\end{pmatrix}\right]$$

$$= 3\left[\begin{pmatrix}-5\\9\end{pmatrix} + \begin{pmatrix}4\\6\end{pmatrix}\right]$$

$$= 3\begin{pmatrix}-5+4\\9+6\end{pmatrix}$$

$$= 3\begin{pmatrix}-1\\15\end{pmatrix} \qquad = \qquad \underline{\underline{\begin{pmatrix}-3\\45\end{pmatrix}}}$$

1 (c) **Method 2** (Expanding / Removing brackets first)

$$3\ (\mathbf{U} \quad + \quad \tfrac{1}{2}\ \mathbf{V})$$

$$= \quad 3\ \mathbf{U} \quad + \quad \tfrac{3}{2}\ \mathbf{V}$$

$$= \quad 3\begin{pmatrix}-5\\9\end{pmatrix} \quad + \quad \frac{3}{2}\begin{pmatrix}8\\12\end{pmatrix}$$

$$= \quad \begin{pmatrix}3\times -5\\3\times\ 9\end{pmatrix} \quad + \quad \begin{pmatrix}\frac{3}{2}\times 8\\\frac{3}{2}\times 12\end{pmatrix}$$

$$= \quad \begin{pmatrix}-15\\27\end{pmatrix} \quad + \quad \begin{pmatrix}12\\18\end{pmatrix}$$

$$= \quad \underline{\underline{\begin{pmatrix}-3\\45\end{pmatrix}}}$$

2 (a) (i) **Method 1** (Formula)

$$\text{Person's share} = \frac{\text{Person's ratio}}{\text{Total ratio}} \times \text{Total amount}$$

Ama's share:

$$= \frac{3}{7} \times ₵1{,}743{,}000$$

$$= 3 \times ₵249{,}000 = ₵747{,}000$$

Kofi's share:

Approach 1

$$= ₵1{,}743{,}000 - ₵747{,}000$$

$$= ₵996{,}000$$

Approach 2

$$= \frac{4}{7} \times 1{,}743{,}000$$

$$= 4 \times 249{,}000 = ₵996{,}000$$

2 (a) (i) **Method 2** (Equivalent Fractions)

	Ratio	**Amount**
Total	**3 + 4 = 7**	**1,743,000**
Ama	3	***a***
Kofi	4	**k**

$$\frac{7}{3} = \frac{1{,}743{,}000}{a}$$

From the table above - using the 'Total' and 'Ama' rows,

$$7 \times a = 1{,}743{,}000 \times 3$$

Cross multiplying

$$\Rightarrow a = \frac{1{,}743{,}000 \times 3}{7}$$

$$\Rightarrow a = 249{,}000 \times 3$$

Dividing both sides by 7 to find '*a*'

$\Rightarrow \quad a = 747{,}000$

$\Rightarrow$ Ama's share = ¢747,000

Hence, Kofi's share = ¢1,743,000 – ¢747,000

= ¢996,000

NB: Other approaches using the idea of Simple Proportion could also be used to solve the problem

2(a) (ii) Simple interest on Kofi's share

= Principal × Rate × Time

= 996,000 × 20% × 2

= $996{,}000 \times \frac{20}{100} \times 2$

= 9,960 × 20 × 2

= GH¢398,400.00

(iii) Kofi's total amount = Principal + S. interest

= 996,000 + 398,400

= GH¢1,394,400

2 (b) Changing $2\ 4\ 3_{\text{five}}$ to a base ten numeral.

2	4	3
5^2	5^1	5^0

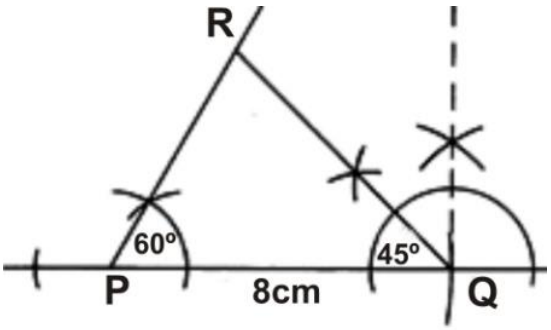
$\Rightarrow \quad (2 \times 5^2) + (4 \times 5^1) + (3 \times 5^0)$

$= \quad (2 \times 25) + (4 \times 5) + (3 \times 1)$

$= \quad 50 + 20 + 3$

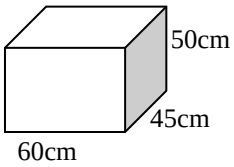
$= \quad \underline{73}$

3 (a) (i)



(ii) $|QR| \approx \underline{7.1 \text{ cm}}$

3 (b) (i)



Total surface area of a cuboid of length l, width w and height h

$=$ Sum of all 6 faces areas of the cuboid

$= 2(l \times w) + 2(l \times h) + 2(w + h)$

$= 2(60\text{cm} \times 45\text{cm}) + 2(60\text{cm} \times 50\text{cm}) + 2(45\text{cm} \times 50\text{cm})$

$= 2\,(2700\text{cm}^2) + 2(3000\text{cm}^2) + 2(2250\text{cm}^2)$

$= 5400\text{cm}^2 + 6000\text{cm}^2 + 4500\text{cm}^2$

$= \underline{15900\ \text{cm}^2}$

3 (b) (ii) The volume = area of base × height

= (length × width) × height

$= (60\text{cm} \times 45\text{cm}) \times 50\text{cm}$

$= 2700\ \text{cm}^2 \times 50\text{cm}$

$= \underline{135{,}000\ \text{cm}^3}$

3 (b) (iii) **Method 1** **(Substituting first, then finding height)**

If the tank contains $81{,}000\text{cm}^3$ then

$\Rightarrow$ Volume of water in the tank $= 81{,}000\text{cm}^3$

Volume .= Area of base × height

$\Rightarrow 81{,}000\ \text{cm}^3 = 2700\ \text{cm}^2 \times \text{height}$

$\Rightarrow \dfrac{81{,}000\ cm^3}{2700\ cm^2} = \text{height}$

$\Rightarrow$ 30 cm = height (of water in the tank)

$\Rightarrow$ The height of water in the tank = $\underline{30\text{ cm}}$

3 (b) (iii) **Method 2** **(Making height the subject first, then substituting)**

Length × width × height = Volume (of water)

$\Rightarrow \text{height} = \dfrac{\text{Volume}}{\text{length} \times \text{width}}$

$= \dfrac{81{,}000\ \text{cm}^3}{60\text{cm} \times 45\text{cm}}$

$= 30\text{ cm}$

$\Rightarrow$ The height of water in the tank = $\underline{30\text{ cm}}$

4 (a) **Method 1** (Unitary Approach – the value of one item)

Since 45km requires 5 litres

$\Rightarrow$ 1 km requires $\frac{5}{45}$ litre $= \frac{1}{9}$ litre

$\therefore$ 117 km requires $\frac{1}{9}\text{ litre} \times 117 = \underline{13 \text{ litres}}$

4 (a) **Method 2** (Equivalent Fractions)

Distance (km)	Fuel (litres)
45	5
117	f

$\frac{45}{117} = \frac{5}{f}$ From the table above,

$45 \times f = 5 \times 117$ Cross multiplying,

$\Rightarrow \quad \frac{45\ f}{45} = \frac{5 \times 117}{45} = \frac{117}{9}$

$\Rightarrow \quad f = 13$

$\therefore$ A journey of 117 km requires 13 litres of fuel

4 (a) **Method 3** (If more, less divides …)

If 45 km $\rightarrow$ 5 litres

Then 117 km $\rightarrow$? (more)

If more, less (i.e., 45km) divides; therefore we have

$\frac{117\,\text{km}}{45\,\text{km}} \times 5\,\text{litres} = \frac{117}{45} \times 5\,\text{litres}$

$= \underline{13\,\text{litres}}$

4 (b) (i) $\frac{2}{3}(2x+5) \leq 8\frac{2}{3}$

$\Rightarrow \frac{2}{3}(2x+5) \leq \frac{26}{3}$

$\Rightarrow 3 \times \frac{2}{3}(2x+5) \leq 3 \times \frac{26}{3}$

$\Rightarrow 2(2x+5) \leq 26$

$\Rightarrow 4x + 10 \leq 26$

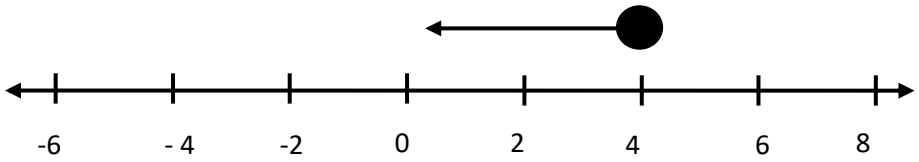
$\Rightarrow 4x \leq 26 - 10$

$\Rightarrow 4x \leq 16$

$\Rightarrow x \leq 16 \div 4$

$\Rightarrow \underline{\underline{x \leq 4}}$

4 (b)(ii)



4 (c) **Method 1** (Equivalent Fractions)

	Percentage	**Production**
After	100% + 22½% = **122½%**	**49,000**
Before	**100%**	***x***

From the table above, $\frac{122\frac{1}{2}\%}{100\ \%} = \frac{49{,}000}{x}$

$$\Rightarrow \quad \frac{122\frac{1}{2}}{100} = \frac{49{,}000}{x} \qquad \Rightarrow \quad \frac{122.5}{100} = \frac{49{,}000}{x}$$

Cross-multiplying, we have: $122.5 \times x = 49000 \times 100$

$$\Rightarrow \quad \frac{122.5 \times x}{122.5} = \frac{49000 \times 100}{122.5}$$

Dividing both sides by 122.5 to find x

$$x = \frac{49000 \times 100}{122.5} \times \frac{10}{10}$$

Multiplying both numerator and denominator by 10 (to convert 122.5 to 1225)

$$x = \frac{49000 \times 1000}{1225} = 40{,}000$$

⇒ The factory was producing 40,000 tonnes before.

4 (c) **Method 2** (If more, less divides …)

If 122½% → 49,000 tonnes

Then 100 % → ? (less)

If less, more (i.e., 122½%) divides; therefore we have

$$\frac{100\%}{122\frac{1}{2}\%} \times 49{,}000 \text{ tonnes} = \frac{100}{122.5} \times 49{,}000 \text{ tonnes}$$

$$= 40{,}000 \text{ tonnes}$$

4 (c) **Method 3** (Equation)

Increased q'ty = 122½ % of Original q'ty = 49,000 tonnes

$$\Rightarrow \quad \frac{122\frac{1}{2}}{100} \times \text{Original Quantity} = 49{,}000 \text{ tonnes}$$

$$\Rightarrow \quad \frac{122.5}{100} \times \text{Original q' ty} = 49{,}000$$

Multiplying both sides by $\frac{100}{122.5}$ (to remove $\frac{122.5}{100}$)

To find original quantity

$$\Rightarrow \quad \text{Original q' ty} = 49{,}000 \times \frac{100}{122.5}$$

Multiplying both numerator and denominator by 10 (to

$$= \frac{49{,}000 \times 1000}{1225}$$

$$= 40{,}000$$

$\Rightarrow$ It was producing 40,000 tonnes before

5 (a), (b) Approach 1 (Inspection / Construction)

[NB: Translating PQ by vector $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$ simply means moving PQ 5 units to the left (-5),

and 0 units downwards [i.e., no movement downwards (0).

So, e.g., the point Q(4,5) moves to Q′(-1,5)

NB: **Horizontal direction movement first; Vertical direction movement second (always)**

5 (a), (b) Approach 2 (The rule / formula)

Translating (x, y) by vector $\begin{pmatrix} a \\ b \end{pmatrix}$

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} x+a \\ y+b \end{pmatrix}$$

$\Rightarrow$

$$OP\begin{pmatrix} 1 \\ -2 \end{pmatrix} \rightarrow OP'\begin{pmatrix} 1+-5 \\ -2+0 \end{pmatrix}$$

$$OP\begin{pmatrix} 1 \\ -2 \end{pmatrix} \rightarrow OP'\begin{pmatrix} -4 \\ -2 \end{pmatrix}, \quad \therefore P'(-4,-2)$$

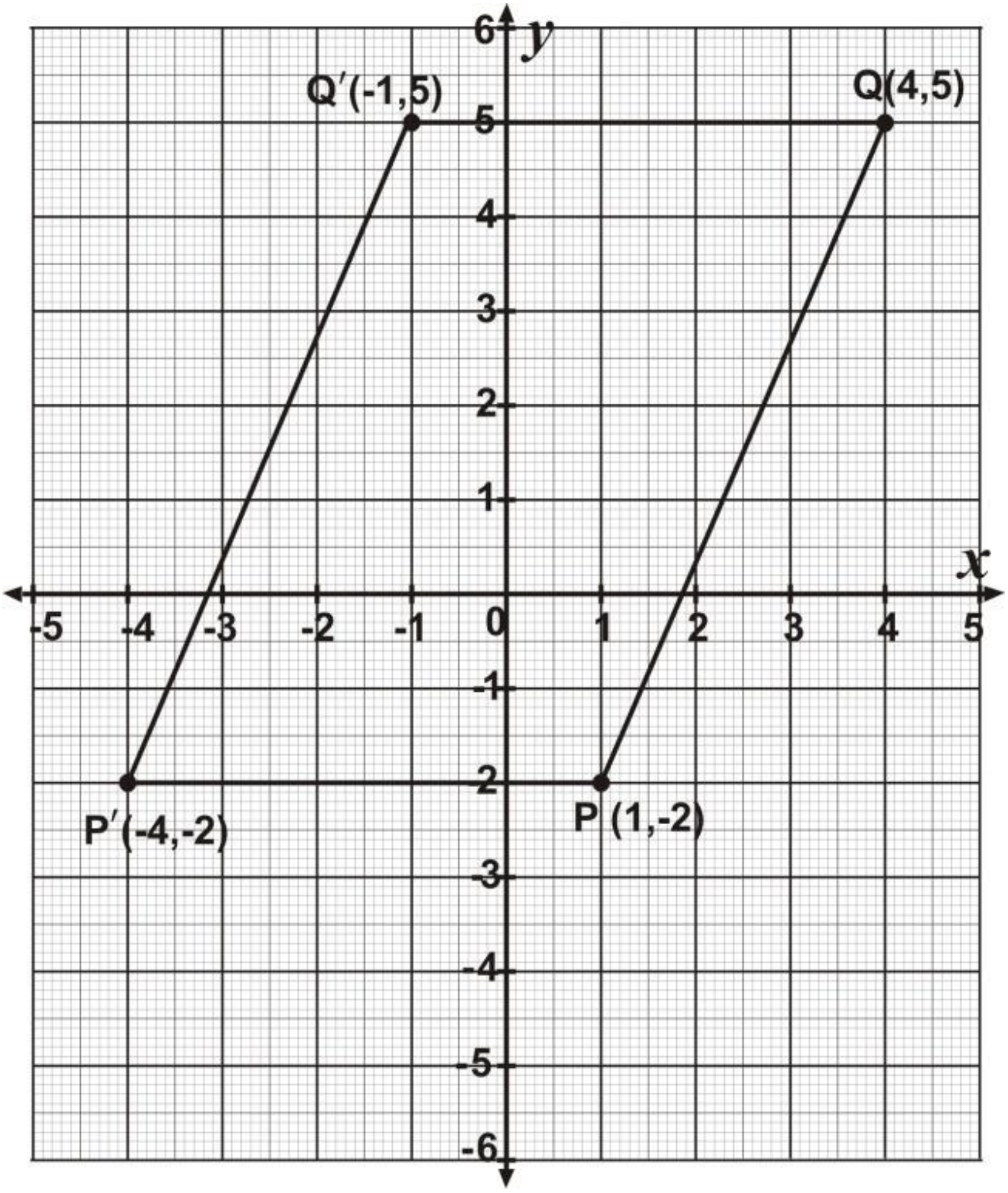
$\Rightarrow$

$$OQ\begin{pmatrix} 4 \\ 5 \end{pmatrix} \rightarrow OQ'\begin{pmatrix} 4+-5 \\ 5+0 \end{pmatrix}$$

$$OQ\begin{pmatrix} 4 \\ 5 \end{pmatrix} \rightarrow OQ'\begin{pmatrix} -1 \\ 5 \end{pmatrix}, \quad \therefore Q'(-1,5)$$

Plot and join P′(-4,-2) and Q′(-1,5) as the image of PQ under a translation by

the vector $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$



5 (c) (ii) Angle PQQ′ ≈ 67°

Angle PP′Q′ ≈ 67°

5 (d) (i) Vectors PQ and P′Q′

Method 1 (Inspection)

From P to Q, we move 3 units to the right (horizontal),

followed by 7 units upwards (vertical)

$\Rightarrow$ Vector PQ $(\overrightarrow{PQ})$ $= \begin{pmatrix} 3 \\ 7 \end{pmatrix}$

From P′ to Q′, we move 3 units to the right (horizontal),

followed by 7 units upwards (vertical)

$\Rightarrow$ Vector P′Q′ $(\overrightarrow{P'Q'})$ $= \begin{pmatrix} 3 \\ 7 \end{pmatrix}$

NB:
Horizontal direction movement first; Vertical direction movement second (always)

5 (d) (i) **Method 2** (Calculation / Rule)

Vector PQ $= \overrightarrow{OQ} - \overrightarrow{OP}$

$= \begin{pmatrix} 4 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 4 - 1 \\ 5 - (-2) \end{pmatrix}$

$= \begin{pmatrix} 4-1 \\ 5+2 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$

Vector P′Q′ $= \overrightarrow{OQ'} - \overrightarrow{OP'}$

$= \begin{pmatrix} -1 \\ 5 \end{pmatrix} - \begin{pmatrix} -4 \\ -2 \end{pmatrix} = \begin{pmatrix} -1-(-4) \\ 5-(-2) \end{pmatrix}$

$= \begin{pmatrix} -1+4 \\ 5+2 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$

(ii) PQQ′P′ is a parallelogram

6 (a) (i) If the mean age = 9 years, then

$$\frac{8 + 11 + 8 + 10 + 6 + 7 + 3x + 11 + 11}{9} = 9$$

$$\Rightarrow \quad \frac{72 + 3x}{9} = 9$$

$$\Rightarrow \quad 72 + 3x = 9 \times 9$$

$$\Rightarrow \quad 72 + 3x = 81$$

$$\Rightarrow \quad 3x = 81 - 72$$

$$\Rightarrow \quad 3x = 9$$

$$\Rightarrow \quad x = \frac{9}{3}$$

$$\Rightarrow \quad x = 3$$

6 (a) (ii) The modal age is 11 years

6 (a) (iii) Since x = 3, $\Rightarrow 3x = 3(3) = 9$

Therefore, the ages in years of the members are

8, 11, 8, 10, 6, 7, 9, 11, 11.

Hence re-arranging in ascending order, we have

6, 7, 8, 8, 9, 10, 11, 11, 11

The middle ages (in years) are 9 and 10; therefore

The median age $= \frac{9+10}{2}$

or ½ (9+10) = 4.5 or 4½ years

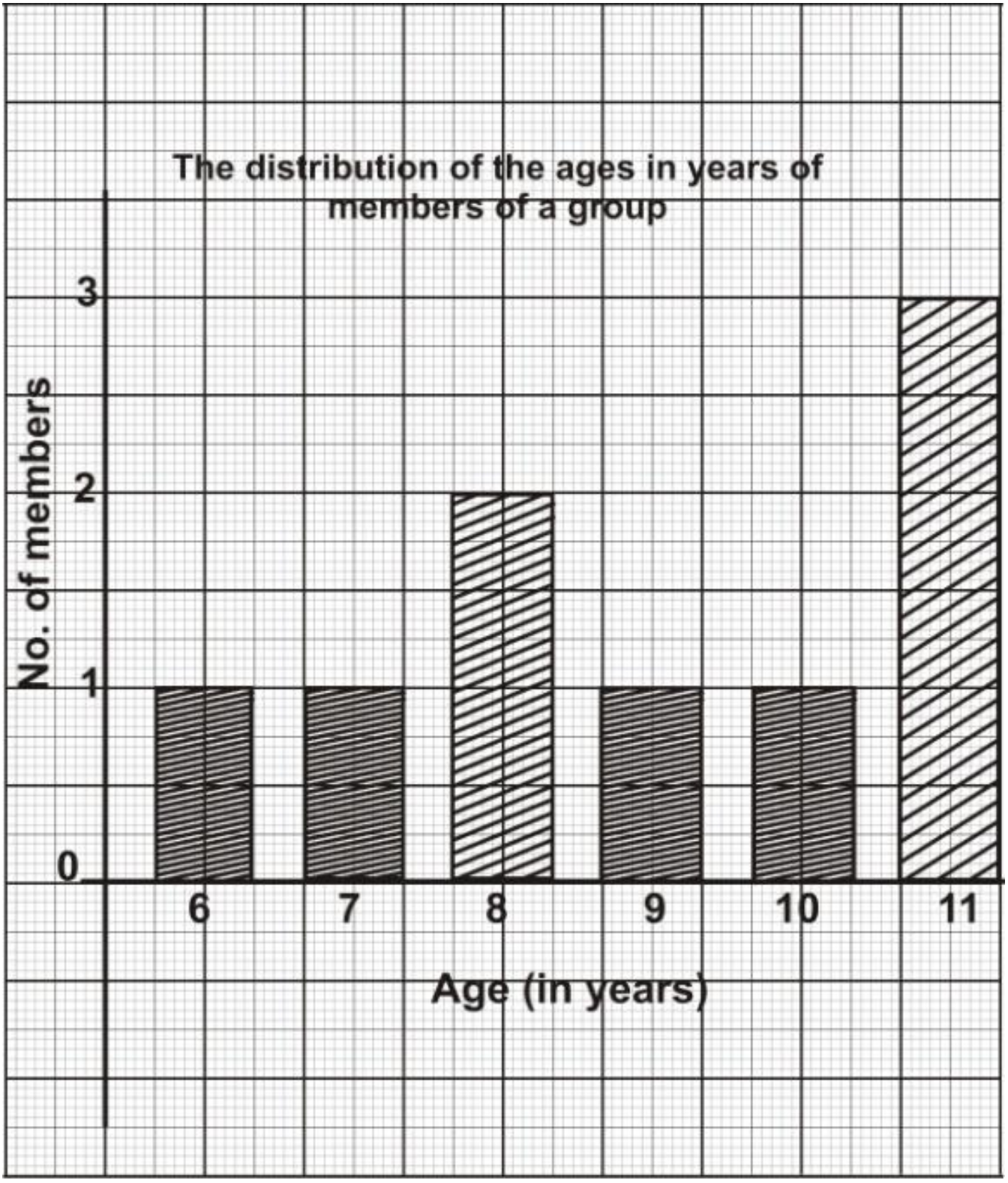
6 (b) Hence the frequency table and bar chart are shown below

Age (in yrs)	6	7	8	9	10	11

No. of members	1	1	2	1	1	3

0201054428.

Vertical AxisScale:4 cm to 1 member



April 2003

MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. P = {3, 6, 9, 12, 15}. Which of the following best describes the set P?

A) The set of multiples of 3 less than 18

B) The set of multiples of 3

C) The set of odd numbers

D) The set of odd numbers less than 16

2. If Q = {2, 4, 6, 7, 8, 10} and R = {3, 5, 7, 9, 10, 11}, find Q ∩ R.

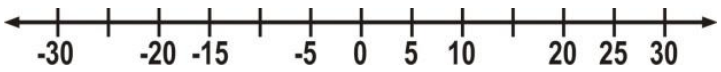
A. {2, 3, 4, 5, 6, 7, 8, 9, 10, 11}

B. {7, 10}

C. {10}

D. {7}

3.



Find the set of missing numbers on the number line above.

A. {-21, -6, 11} B. {-21, -10, 11} C. {-25, -6, 15} D. {-25, -10, 15}

4. If 7.2 and 7.9 are two points on a number line, find the number in the middle of these points.

A) 7.35 B) 7.45 C) 7.55 D) 7.65

5. Find the least common multiple (LCM) of 12 and 20.

A) 24 B) 48 C) 60 D) 80

6. Express 72 as a product of prime factors.

A) 2×3 B) $2^2\times3^2$ C) $2^2\times3^3$ D) $2^3\times3^2$

7. If $3t - 2(t + 12) = 11$, find the value of t.

A) –35 B) –13 C) 13 D) 35

8. What is the value of the digit 8 in the number 78000 ?

A) 8 ten thousands B) 8 thousands C) 8 hundreds D) 8 tens

9. Find the product of 17 and 121.

A) 968 B) 1,751 C) 2,057 D) 8,591

10. Araba owes ¢550,000.00 at the bank. She goes to pay ¢150,000.00. How much does Araba owe the bank now?

A) ¢700,000.00 B) ¢600,000.00 C) ¢500,000.00 D) ¢400,000.00

11. Evaluate $\frac{37}{100} \times \frac{7}{10}$

A) 0.259 B) 2.590 C) 25.900 D) 259.000

12. Express 1.25 as a mixed fraction in its lowest form.

A) $1\frac{1}{25}$ B) $1\frac{4}{25}$ C) 1 ¼ D) 1 ¾

13. The total number of match sticks in 6 match boxes was 270. Find the total number of sticks in 20 similar boxes.

A) 710 B) 800 C) 810 D) 900

14. A car travels at an average speed of 45 km per hour. What distance does it cover in 12 hours?

A) 450km B) 480km C) 500km D) 540km

15. Calculate the simple interest on ¢130,000.00 for 2 ½ years at 12% per annum.

A) ¢78,000.00 B) ¢39,000.00 C) ¢36,000.00 D) ¢31,200.00

16. Express $2/5$ as a percentage

A) 20% B) 25% C) 40% D) 80%

17. Kwame gets a commission of 20% on bread sold. In one week, Kwame's commission was ¢45,000.00. How much bread did he sell during that week?

A) ¢205,000.00

B) ¢220,000.00

C) ¢225,000.00

D) ¢235,000.00

18. Write 83000 in standard form.

A) 8.3×10^{-4}

B) 8.3×10^{-3}

C) 8.3×10^{3}

D) 8.3×10^{4}

19. The following marks are the marks obtained by pupils in a test:

2, 3, 5, 2, 3, 4, 2, 3, 5, 3.

What is the mode?

A) 2 B) 3 C) 4 D)5

20. The following marks are the marks obtained by pupils in a test:

2, 3, 5, 2, 3, 4, 2, 3, 5, 3. Calculate the mean

A) 3.0 B) 3.2 C) 4.0 D) 4.2

21. There are 12 red and 8 blue balls in a bag. If a ball is selected at random from the bag, what is the probability that it is red?

A) $^{2}/_{5}$ B) $^{2}/_{3}$ C) $^{3}/_{5}$ D) $^{4}/_{5}$

22. If $y = \frac{1}{3}(x-2)$ express x in terms of y.

A) $x = 3y - 2$

B) $x = 3y + 2$

C) $x = \frac{3}{2}y$

D) $x = -\frac{3}{2}y$

23. Simplify $6a^2 \times 4a^2b^2$

A) $10a^4b^2$ B) $24a^2b^2$ C) $24a^4b^2$ D) $24a^2b^4$

24. Change 17_{ten} to a base two numeral.

A) 101 B) 1001 C) 1000 D) 10001

25. Which of the following statements is true?

A) 8+4 < 10 B) 7+4 < 10 C) 6+4 < 10 D) 5+4 < 10

26. Find the set of integers within the interval $-2 < x < 2$

A. {-2,-1,2} B. {-2,-1,0} C. {-1,0,1} D. {-1,1,2}

27. Use the mapping below to answer question **27**

x	1	2	3	4	5
↓	↓	↓	↓	↓	↓
y	-4	-2	0	2	m

What is the rule for this mapping ?

A) x→ 2(x-3) B) x→ x-5 C) x→ 2(x-2) D) x→ 2x-3

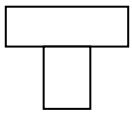
28.

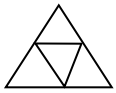
x	1	2	3	4	5
↓	↓	↓	↓	↓	↓
y	-4	-2	0	2	m

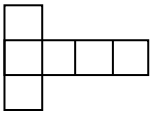
Find m

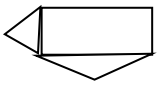
A) –4 B) 4 C) 6 D) 8

29. Which of the following represents the net of a pyramid?

A 

B 

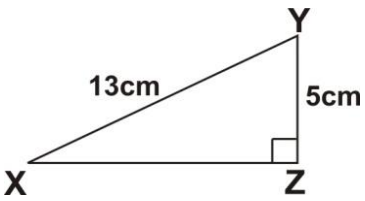
C 

D 

30. A rectangular tank has dimensions 2.5 m by 4 m by 5 m. It is filled with water to the brim. If 35 m^3 of the water is used, how much water is left in the tank?

A) 50 m^3 B) 35 m^3 C) 25 m^3 D) 15 m^3

31.



In the triangle XYZ, angle XZY = 90º, |XY| = 13cm and

|YZ| = 5cm. What is the length of XZ ?

A) 4cm B) 8cm C) 12cm D) 65cm

32. How many lines of symmetry has an equilateral triangle ?

A) 1 B) 2 C) 3 D) 4

33. The value of an obtuse angle lies between

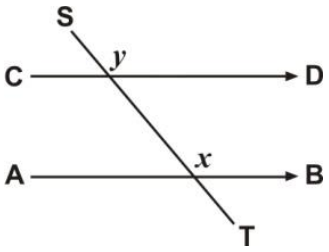
A) 0° and 90°

B) 90° and 180°

C) 90° and 270°

D) 180° and 360°

34.



In the diagram above, AB is parallel to CD. Angles x and y are

A) alternate angles

B) corresponding angles

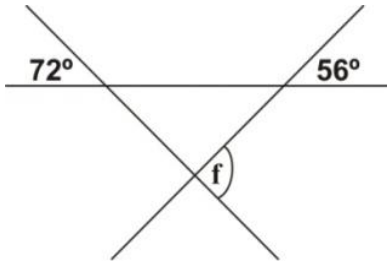
C) vertically opposite angles

D) co-interior angles

35. If $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$, find $2\mathbf{a} + \mathbf{b}$.

A. $\begin{pmatrix} 7 \\ -2 \end{pmatrix}$ B. $\begin{pmatrix} 5 \\ -3 \end{pmatrix}$ C. $\begin{pmatrix} 8 \\ -7 \end{pmatrix}$ D. $\begin{pmatrix} 1 \\ 6 \end{pmatrix}$

36.



Find the size of the angle marked f, in the diagram above.

A) 56° B) 72° C) 108° D) 128°

37. The dimensions of a cuboid are 2cm, P cm and 5cm. Which of the following is an expression for the volume of the cuboid?

A. $7P\ cm^3$

B. $(7+P)\ cm^3$

C. $10P\ cm^3$

D. $(10+P)\ cm^3$

38. Simplify $5(3t + 1) – 6(t – 1)$.

A) $9t + 11$ B) $9t + 7$ C) $9t + 1$ D) $9t – 5$

39. Simplify: $({}^{2}/_{3} – ½) ÷ {}^{1}/_{6}$

A) ${}^{1}/_{36}$ B) ${}^{1}/_{12}$ C) 1 D) 6

40. The point P moves in a plane such that it is always at equal distance from two fixed points, A and B in the same plane. Which of these is the locus of the point P?

A) The bisector of angle PAB

B) A circle centre B and radius AB

C) A circle with AB as the diameter

D) The perpendicular bisector of line AB.

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MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. A) The set of multiples of 3 less than 18

2. B. {7, 10}

3. D. {-25, -10, 15}

4. C) 7.55

5. C) 60

6. D) $2^3 \times 3^2$

7. C) 2,057

8. D) 35

9. B) 8 thousands

10. D) ¢400,000.00

11. A) 0.259

12. C) 1 ¼

13. D) 900

14. D) 540km

15. B) ¢39,000.00

16. C) 40%

17. C) ¢225,000.00

18. D) 8.3×10^4

19. B) 3

20. B) 3.2

21. C) $^3/_5$

22. B) $x = 3y + 2$

23. C) $24a^4b^2$

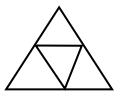
24. D) 10001

25. D) $5+4 < 10$

26. C. {-1,0,1}

27. A) $x \rightarrow 2(x-3)$

28. B) 4

29. B 

30. D) $15m^3$

31. C) 12cm

32. C) 3

33. B) 90° and 180°

34. B) corresponding angles

35. A. $\begin{pmatrix} 7 \\ -2 \end{pmatrix}$

36. D) 128°

37. C. $10Pcm^3$

38. A) $9t + 11$

39. C) 1

40. D) The perpendicular bisector of line AB.

April 2003

MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. $\varepsilon = \{1, 2, 3, 4, \ldots, 18\}$

$A = \{\text{Prime numbers}\}$

$B = \{\text{Odd numbers greater than 3}\}$

(a) If A and B are subsets of the Universal set, ε, list the members of A and B

(b) Find the set

(i) $A \cap B$;

(ii) $A \cup B$

(c) (i) Illustrate ε, A and B on a Venn diagram.

(ii) Shade the region for prime factors of 18 on the Venn diagram

2. (a) If $2n - 5m + 10 = 0$, find

(i) m, when n = 2;

(ii) n, when m = 5.

(b) Simplify $\frac{6.4\times0.25\times16}{0.08\times0.5}$ leaving your answer in standard form

(c) A number of sweets were shared among 8 children and each child received 30. If 12 children shared the same number of sweets, how many will each receive?

3. The table below shows the distribution of the ages (in years) of children in a nursery school.

Age (years)	1	2	3	4	5
Number of children	6	4	2	3	5

(a) Find

(i) the modal age

(ii) the mean age

(b) Draw a bar chart for the distribution.

(c) What is the probability that a child chosen at random from the school is 4 years old?

4. (a) If $\mathbf{p} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}$ and $\mathbf{r} = \begin{pmatrix} -3 \\ 7 \end{pmatrix}$, find $\mathbf{p} + 2\mathbf{q} + \mathbf{r}$

(b) Find the solution set of the inequality $x - \frac{4}{5} \leq \frac{1}{5}$, if the domain is the set {-2, -1, 0, 1, 2}.

(c) A rectangular sheet of metal has length 44cm and breadth 10cm. It is folded to form a cylinder with the breadth becoming the height.

Calculate

(i) the radius of the cylinder formed;

(ii) the volume of the cylinder.

(Take $\pi = {}^{22}/_{7}$)

5. (a) Using a pair of compasses and ruler only,

(i) construct the triangle ABC with |AB| = 8 cm, |BC| = 8cm and |AC| = 7cm.

(ii) Bisect angle ABC and let the bisector meet AC at D. Produce |BD| to P such that |BD| = |DP|. Join AP and CP.

(b) Measure (i) angle ADB; (ii) |AP|.

(c) What kind of quadrilateral is ABCP?

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MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

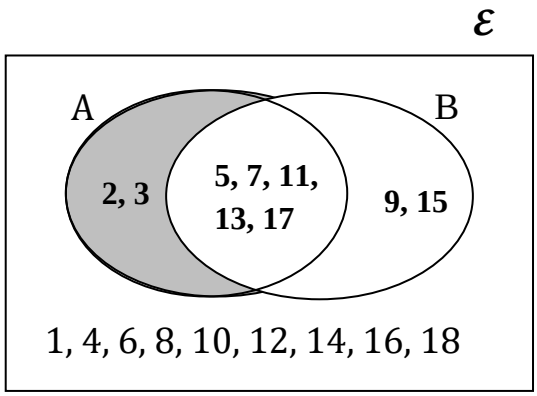
1 (a) A = {2, 3, 5, 7, 11, 13, 17}

B = {5, 7, 9, 11, 13, 15, 17}

(b) (i) $A \cap B$ = {5, 7, 11, 13, 17}

(ii) $A \cup B$ = {2, 3, 5, 7, 9, 11, 13, 15, 17}

(c)(i), (ii)



2 (a) (i) When $n = 2$, we have

$$2(2) - 5m + 10 = 0$$

$$\Rightarrow \quad 4 - 5m + 10 = 0$$

$$\Rightarrow \quad 4 + 10 = 0 + 5m$$

$$\Rightarrow \quad 14 = 5m$$

$$\Rightarrow \quad \frac{14}{5} = m$$

$$\Rightarrow \quad m = \frac{14}{5}$$

$$\Rightarrow \quad m = \underline{\underline{2\frac{4}{5}}}$$

(ii) When $m = 5$, we have

$$2n - 5(5) + 10 = 0$$

$$\Rightarrow \quad 2n - 25 + 10 = 0$$

$$\Rightarrow \quad 2n - 15 = 0$$

$$\Rightarrow \quad 2n = 15$$

$$\Rightarrow \quad n = \frac{15}{2}$$

$$\Rightarrow \quad n = \underline{\underline{7\frac{1}{2}}}$$

2 (b) $\dfrac{6.4 \times 0.25 \times 16}{0.08 \times 0.5}$

Method 1 **(Converting Decimals to Common Fractions)**

$$= \quad (6.4 \times 0.25 \times 16) \div (0.08 \times 0.5)$$

$$= \quad \left(\frac{64}{10} \times \frac{25}{100} \times \frac{16}{1}\right) \div \left(\frac{8}{100} \times \frac{5}{10}\right)$$

$$= \quad \left(\frac{64}{10} \times \frac{25}{100} \times \frac{16}{1}\right) \times \left(\frac{100}{8} \times \frac{10}{5}\right)$$

$$= \quad \frac{64 \times 25 \times 16}{8 \times 5}$$

$$= \quad \frac{8 \times 5 \times 16}{1 \times 1}$$

$$= \quad 640 = \underline{6.4 \times 10^2}$$

2 (b) **Method 2 (Converting Decimals to Whole Numbers)**

Multiplying both numerator and denominator by 1000

(to convert to whole numbers)

$$\frac{6.4\times 0.25\times 16}{0.08\times 0.5}\times\frac{1000}{1000}$$

$$= \frac{64\times 25\times 16}{8\times 5}$$

$$= \frac{8\times 5\times 16}{1\times 1}$$

$$= 640 = 6.4\times 10^2$$

2 (c) **Method 1 (Simple Sharing)**

Total no. of sweets = 8 × 30 sweets = 240 sweets

Now, 240 ÷ 12 = 20

∴ If 12 children shared 240 sweets, each gets 20 sweets

2 (c) **Method 2 (Simple Proportion)**

In sharing the same number of sweets,

If for 8 children, each receives 30 sweets,

then for 12 children, each receives ?(less) sweets

If less, more (i.e., 12 children) divides,

$$\Rightarrow \frac{8\,\text{children}}{12\,\text{children}}\times 30\,\text{sweets} = 20\,\text{sweets}$$

3 (a) (i) The modal age is the age that occurs most

⇒ The modal age = 1 year

3 (a) (i) **Approach 1**

$$\text{The mean age} = \frac{\text{The sum of ages}}{\text{The total number of children}}$$

$$= \frac{(1\times 6)+(2\times 4)+(3\times 2)+(4\times 3)+(5\times 5)}{6+4+2+3+5}$$

$$= \frac{6+8+6+12+25}{6+4+2+3+5}$$

$$= \frac{57}{20}$$

$$= \underline{\underline{2\tfrac{17}{20}\ years}} \quad \text{or} \quad \underline{2.85 \text{ years}}$$

3 (a)(i) **Approach 2**

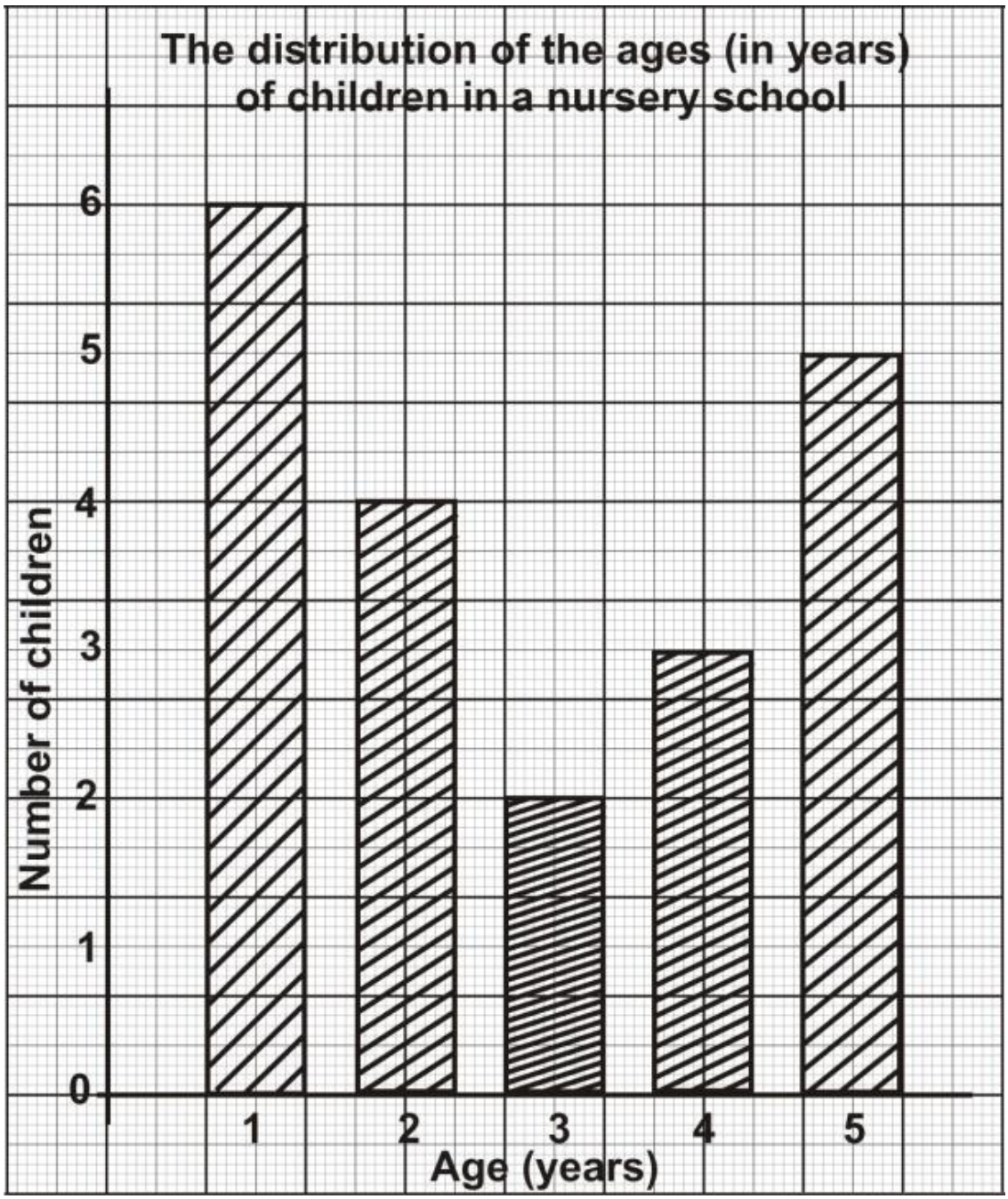
Age (years) (*x*)	**No. of Children (f)**	**f *x***
1	6	6
2	4	8
3	2	6
4	3	12
5	5	25
	Σf = 20	**Σfx = 57**

The mean age $= \frac{\Sigma f x}{\Sigma f} = \frac{57}{20}$

$$= \underline{\underline{2\tfrac{17}{20}\ years}} \quad \text{or} \quad \underline{2.85 \text{ years}}$$

3 (b)

Vertical AxisScale: 3 cm to 1 child



3 (c) P(4 years old) $= \dfrac{\text{Number of children aged 4 years}}{\text{Total number of children}}$

$= \dfrac{3}{20}$

4 (a) $\mathbf{p} + 2\mathbf{q} + \mathbf{r}$

$= \begin{pmatrix} 4 \\ 5 \end{pmatrix} + 2\begin{pmatrix} 0 \\ -2 \end{pmatrix} + \begin{pmatrix} -3 \\ 7 \end{pmatrix}$

$$= \begin{pmatrix} 4 \\ 5 \end{pmatrix} + \begin{pmatrix} 2 \times 0 \\ 2 \times -2 \end{pmatrix} + \begin{pmatrix} -3 \\ 7 \end{pmatrix}$$

$$= \begin{pmatrix} 4 + 0 + (-3) \\ 5 + (-4) + 7 \end{pmatrix}$$

$$= \begin{pmatrix} 4 + 0 - 3 \\ 5 - 4 + 7 \end{pmatrix}$$

$$= \underline{\underline{\begin{pmatrix} 1 \\ 8 \end{pmatrix}}}$$

4 (b) **Method 1**

$x - \frac{4}{5} \leq \frac{1}{5}$, domain = {-2, -1, 0, 1, 2}

$\Rightarrow \quad x \leq \frac{1}{5} + \frac{4}{5}$ Grouping constants

$\Rightarrow \quad x \leq 1$

$\Rightarrow$ Truth set = {-2, -1, 0, 1}

4 (b) **Method 2**

$x - \frac{4}{5} \leq \frac{1}{5}$ domain = {-2, -1, 0, 1, 2}

$5(x) - 5\left(\frac{4}{5}\right) \leq 5\left(\frac{1}{5}\right)$ Multiplying through by 5 to remove fractions

$\Rightarrow \quad 5x - 4 \leq 1$

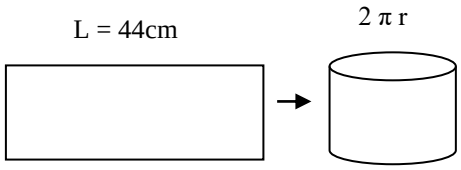
$\Rightarrow \quad 5x \leq 1 + 4$

$\Rightarrow \quad \frac{5x}{5} \leq \frac{5}{5}$

$\Rightarrow \quad x \leq 1$

$\Rightarrow$ Truth set = {-2, -1, 0, 1}

4 (c) (i) **METHOD 1** **(Length of rectangle = Circumference of circle)**



Approach 1 **(Substituting first, then finding r)**

L = 2 π r

⇒ 44cm = $2 \times {}^{22}/_{7} \times r$

⇒ $\frac{44\ cm}{2 \times \frac{22}{7}} = r$

Dividing both sides by $2 \times \frac{22}{7}$ to find r

⇒ $\frac{44cm \times 7}{2 \times 22} = r$

⇒ 7cm = r

⇒ Radius of the cylinder formed = 7 cm

4 (c) (i) **Approach 2** **(Making r the subject first, then substituting)**

L = 2 π r

⇒ $\frac{L}{2\ \pi} = r$

⇒ $\frac{44\,cm}{2 \times {}^{22}/_{7}} = r$

⇒ $\frac{44\,cm}{{}^{44}/_{7}} = r$

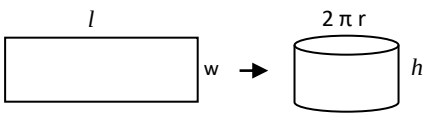
⇒ $44\,cm \div {}^{44}/_{7} = r$

⇒ $44\ \text{cm} \times \frac{7}{44} = r$

⇒ 7cm = r

⇒ Radius of the cylinder formed = 7 cm

4 (c) (i) **METHOD 2** **(Area of rectangle = Curved surface area of cylinder)**



Approach 1 (Substituting first)

$$l \quad w \quad = \quad 2\,\pi\, r\, h$$

$$\Rightarrow \quad 44\text{cm} \times 10\text{cm} \quad = \quad 2 \times \tfrac{22}{7} \times r \times 10cm$$

$$\Rightarrow \quad \frac{44cm \times 10cm}{2 \times \frac{22}{7} \times 10cm} \quad = \quad r$$

$$\Rightarrow \quad \frac{44cm \times 10cm \times 7}{2 \times 22 \times 10cm} = r$$

$$\Rightarrow \quad 7\text{cm} \quad = \quad r$$

$$\Rightarrow \quad \text{Radius of the cylinder formed} \quad = \quad 7 \text{ cm}$$

4 (c) (i)

Approach 2 (Making *r* the subject first)

$$l \quad w \quad = \quad 2\,\pi\, r\, h$$

$$\Rightarrow \quad \frac{l\ w}{2\ \pi\ h} \quad = \quad r$$

$$\Rightarrow \quad \frac{44cm \times 10cm}{2 \times \frac{22}{7} \times 10cm} = r$$

$$\Rightarrow \quad \frac{44cm}{44/7} \quad = \quad r$$

$$\Rightarrow \quad 44cm \div 44/7 \quad = \quad r$$

$$\Rightarrow \quad 44cm \times 7/44 \quad = \quad r$$

$$\Rightarrow \quad 7\text{cm} \quad = \quad r$$

$$\Rightarrow \quad \text{Radius of the cylinder formed} \quad = \quad 7 \text{ cm}$$

4 (c) (ii)

Volume of cylinder = Area of base × height

$$= \quad \pi\, r^2 \times h$$

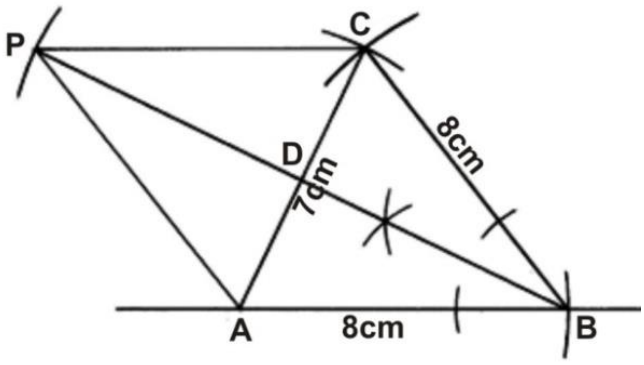
$$= \quad 22/7 \times (7\text{cm})^2 \times 10\text{cm}$$

$$= \quad \tfrac{22}{7} \times 7\text{cm} \times 7\text{cm} \times 10\text{cm}$$

$$= \quad 22 \times 7\text{cm}^2 \times 10\text{cm}$$

$$= \quad 1540\ \text{cm}^3$$

5 (a)



(b) (i) Angle ADB ≈ 90°

(ii) |AP| ≈ 8 cm

(c) ABCP is a rhombus

July 2002

MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. State the property used in the operation a(b+c) = ab + ac.

A) Associative B) Distributive C) Commutative D) Universal

2. Simplify 0.1 x 0.02 x 0.003

*(*leaving *your* answer *in standard form)*

A) $6x10^{-7}$ B) $6x10^{-6}$ C) $6x10^{5}$ D) $6x10^{6}$

3. Express $^{1}/_{25}$ as a decimal fraction.

A) 0.4 B) 0.04 C) 0.004 D) 0.0004

4. In an examination, 25% of the candidates failed to obtain the pass mark. The number of candidates who passed was 150. How many candidates failed?

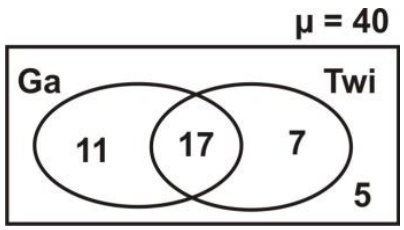
A) 113 B) 100 C) 50 D) 38

5. How many faces has a cube ?

A) 8 B) 6 C) 5 D) 4

6. Simplify $2^{8} \div 2^{3}$

A) 2^{24} B) 2^{10} C) 2^{5} D) 2^{3}



The Venn diagram above shows the number of pupils in a class of 40, who speak Twi or Ga or neither.

7. How many pupils speak both Twi and Ga?

A) 17 B) 11 C) 7 D) 5

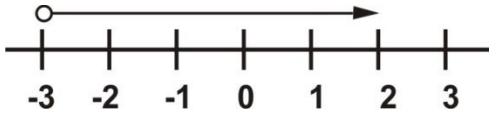
8. How many pupils speak neither Twi nor Ga?

A) 17 B) 11 C) 7 D) 5

9. Find the least common multiple (LCM) of the numbers 10, 15 and 25.

A) 30 B) 60 C) 120 D) 150

10. Which of the following expressions is illustrated on the number line below?



A) $x < -3$ B) $x \leq -3$ C) $x > -3$ D) $x \geq -3$

11. Simplify $\frac{2a}{3} - \frac{a-b}{2}$

A) $\frac{a+3b}{6}$ B) $\frac{a-3b}{6}$ C) $\frac{a-b}{6}$ D) $\frac{7a-3b}{6}$

12. Correct 0.00025 to one significant figure.

A) 0.2 B) 0.003 C) 0.0002 D) 0.0003

13. If $y : 28 = 5 : 7$, find y.

A) 5 B) 20 C) 31.2 D) 37.5

14. Convert 121_{five} to a base ten numeral.

A) 21 B) 35 C) 36 D) 40

15. In the mapping

x	=	1	2	3	4	5
y	=	4	7	10	k	16

Find the image of 4 under the mapping.

A) 11 B) 12 C) 13 D) 14

16. The point K (3,4) is rotated through 180° about the origin. Find its image.

A. (-3, 4) B. (-4, 3) C. (-3, -4) D. (3, -4)

17. Arrange the following fractions in descending order of magnitude: ½ , $^{17}/_{20}$, ¾

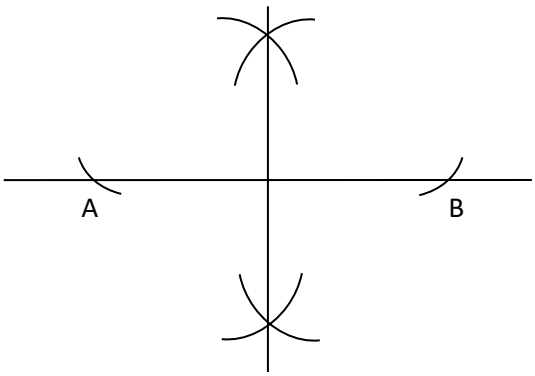
A) ¾ , $^{17}/_{20}$, ½

B) $^{17}/_{20}$, ¾ , ½

C) ½ , ¾ , $^{17}/20$

D) ½ , $^{17}/_{20}$, ¾

18. Which of the following best describes the construction below?



A. Constructing an angle of 60°

B. Constructing the perpendicular bisector of AB

C. Constructing a circle about the midpoint of AB

D. Constructing an angle of 180°

19. Find the image of the point K (3, 5) when it is reflected in the x-axis.

A. (3, 5) B. (5, 3) C. (3, -5) D. (-3, -5)

20. A set of furniture was sold for ¢300,000.00 at a profit of 20%. Find the cost price.

A) ¢250,000

B) ¢240,000

C) ¢200,000

D) ¢180,000

21. Make x the subject of the relation, $v^2 = u^2 + 2ax$.

A) $x = u^2 + 2av^2$

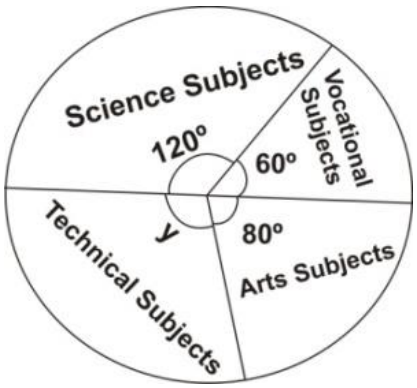
B) $x = \frac{v^2 - u^2}{a^2}$

C) $x = \frac{u^2 + v^2}{2a}$

D) $x = \frac{v^2 - u^2}{2a}$

22. Three baskets contain 95 oranges, x oranges and 2x oranges. How many oranges are in the three baskets?

A) 96x B) 95 + 3x C) 95 + $2x^2$ D) 98x



The pie chart shows the distribution of programmes offered by 720 students at Kofikrom.

Use this information to answer Questions **23** *and* **24**

23. Find the value of the angle marked y.

A) 90° B) 100° C) 110° D) 120°

24. How many more students offered Science subjects than Arts subjects?

A) 40 B) 80 C) 160 D) 240

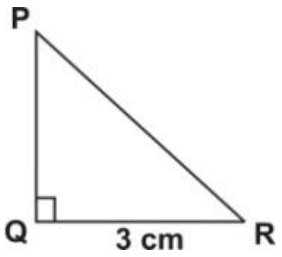
25. Kofi is two years older than Ama. If the sum of their ages is 16, find Ama's age.

A) 7 yrs B) 9 yrs C) 14 yrs D) 18 yrs

26. $J = \frac{1}{2} mv^2$. If v = 4 and J = 12, find m.

A) 1.5 B) 3.0 C) 3.6 D) 6.3

27. PQR is a right-angled triangle. The area of the triangle is $6cm^2$ and |QR| = 3cm. Find |PQ|.



A) 2cm B) 3cm C) 4cm D) 6cm

28. A rectangle has a length of 8cm and a breadth of 6cm. How long is its diagonal?

A) 10 B) 14 C) 28 D) 50

29. Simplify $\frac{1}{3}(2\frac{2}{3} + \frac{5}{6})$.

A) $\frac{13}{4}$ B) $\frac{7}{6}$ C) $\frac{13}{18}$ D) $\frac{31}{18}$

30. A man divided his 360 cattle among his three sons in the ratio 7:6:5. How many cattle did the one who had the smallest share receive?

A) 100 B) 120 C) 140 D) 160

31. The marks obtained by 9 students in a test are 3, 3, 4, 5, 6, 7, 7, 7, 8.
Find the median

A) 4 B) 5 C) 6 D) 7

32. The marks obtained by 9 students in a test are 3, 3, 4, 5, 6, 7, 7, 7, 8.
What is the mode?

A) 3 B) 4 C) 6 D) 7

33. What is the fifth term of the sequence $\frac{1}{3}, \frac{2}{5}, \frac{3}{7}, \ldots$?

A) $\frac{4}{9}$ B) $\frac{5}{11}$ C) $\frac{6}{13}$ D) $\frac{7}{15}$

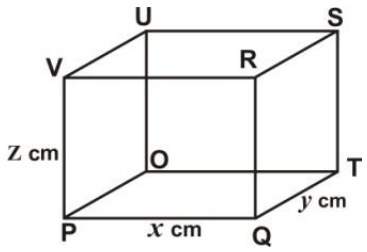
34. 30 men dig a pit in 21 days. How many days will 14 men take to dig the pit, working at the same rate?

A) 20 B) 25 C)30 D) 45

35. A bag contains 6 white and 8 red balls. What is the probability that a ball picked at random will be a white ball?

A) $^1/_7$ B) $^3/_7$ C) $^4/_7$ D) $^6/_7$

36. The figure below is a cuboid with dimensions *x* cm by *y* cm by *z* cm. Find the area of the face PQRV.

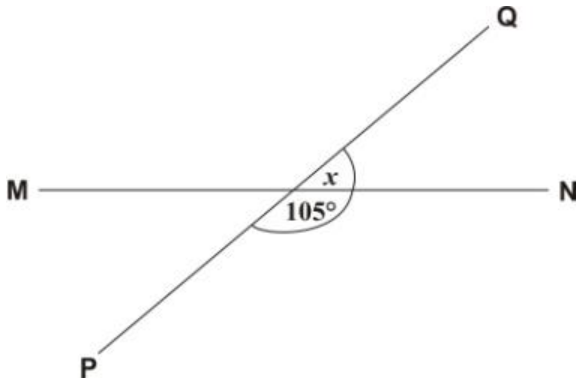


A) xy cm^2 B) yz cm^2 C) x^2 cm^2 D) xz cm^2

37. A rod 200cm long is broken into two parts. The shorter part is one-quarter of the length of the rod. Express the shorter part as a percentage of the longer part.

A) 25% B) 30% C) 33.33% D) 66.67%

38. In the figure below, MN and PQ are straight lines. Find the value of the angle marked x in the figure.



A) 65° B) 75° C) 85° D) 155°

39. **F**ind the value of x in the equation $^x/_4 = 2$

A) 2 B) 4 C) 6 D) 8

40. The circumference of a circular track is 15.4m. Find the diameter of the track. (Take $\pi = {}^{22}/_{7}$)

A) 4.9 m B) 22 m C) 24 m D) 24.5 m

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MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1.	B)	Distributive
2.	B)	$6x10^{-6}$
3.	B)	0.04
4.	C)	50
5.	B)	6
6.	C)	2^5
7.	A)	17
8.	D)	5
9.	D)	1311_{five}
10.	D)	150
11.	C)	$x > -3$
12.	A)	$\frac{a+3b}{6}$
13.	D)	0.0003
14.	B)	20
15.	C)	36

16. C. (-3, -4)

17. B) $^{17}/_{20}$, ¾ , ½

18. C. (3, -5)

19. A) ¢250,000

20. D) $x = \frac{v^2 - u^2}{2a}$

21. B) $95 + 3x$

22. B) 100°

23. B) 80

24. A) 7 yrs

25. A) 1.5

26. C) 4cm

27. C) 13

28. A) 10

29. B) $^{7}/_{6}$

30. A) 100

31. C) 6

32. D) 7

33. B) $\frac{5}{11}$

34. D) 45

35. B) $^{3}/_{7}$

36. D) $x\ z\ \text{cm}^2$

37. C) 33.33%

38. B) 75°

39. D) 8

40. A) 4.9 m

July 2002

MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. (a) Simplify $2\frac{3}{4} \div (3\frac{3}{8} - 1\frac{1}{2})$

(b) There are 50 pupils in a class. Out of this number, $\frac{1}{10}$ speaks French only and $\frac{4}{5}$ of the remainder speak both French and English. If the rest speak English only,

(i) find the number of students who speak

(α) both French and English

(β) only English

(ii) draw a Venn diagram to illustrate the above information

2. A shop sells a pencil at ¢500.00 and a pen at ¢2,000.00

a) If Afua bought 8 pencils and 5 pens, how much did she pay together for them?

b) The price of a pencil is increased by 20% and a pen by 10%. Find how much Otu will pay for 10 pencils and 8 pens.

3. a) Using a ruler and a pair of compasses only, construct

(i) triangle ABC such that AB = 12cm, AC = 8cm and BAC = 30°;

(ii) a perpendicular from C to meet AB at M.

b) Measure

(i) angle ABC; (ii) |CM|.

c) Calculate the area of triangle ABC.

4. (a) Solve the inequality $\frac{5x-3}{6} - \frac{2x-4}{4} < 2$

(b) (i) Copy and complete the table of values for the relation, $y = 2x + 1$.

x	-3	-2	-1	0	1	2	3	4
y	-5	-3		1			7	

(ii) Using a scale of 2cm to 1 unit on the x-axis and 2cm to 2units on the y-axis, plot the ordered pairs (x, y) on a graph sheet.

(iii) Use a ruler to join the points plotted.

(iv) Use your graph to find

α) x, when $y = 4$;

β) y, when $x = -2.5$

5. The table shows the number of students in a JSS class who prepared various dishes for their practical.

Dishes	**No. of students**
Fufu and light soup	5
Banku and Okro stew	20
Fried Rice	30
Fried Plantain and Beans	25
Boiled yam and palaver sauce	10

(a) (i) Draw a pie chart for the distribution above.

(ii) What dish was prepared most?

(iii) What percentage of students prepared Fried Rice?

(b) What is the probability that a student chosen at random cooked Fried Plantain and Beans?

July 2002

MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1 (a) $2\frac{3}{4} \div (3\frac{3}{8} - 1\frac{1}{2})$

$\Rightarrow \quad \frac{11}{4} \div (\frac{27}{8} - \frac{3}{2})$

$\Rightarrow \quad \frac{11}{4} \div (\frac{27 - 12}{8})$

$\Rightarrow \quad \frac{11}{4} \div \frac{15}{8}$

$\Rightarrow \quad \frac{11}{4} \times \frac{8}{15}$

$\Rightarrow \quad \frac{22}{15} = \underline{\underline{1\frac{7}{15}}}$

1 (b) (i)

α) If $(^1/_{10}$ of 50) pupils speak French only

$\Rightarrow \quad \frac{1}{10} \times 50 = 5$ speak French only

$\therefore$ The remainder $= 50 - 5 = 45$

But $^4/_5$ of the remainder speak both French and English

$\Rightarrow \quad \frac{4}{5}$ *of* 45 speak both French and English

$\Rightarrow \quad \frac{4}{5} \times 45 = \underline{\mathbf{36}}$

$\Rightarrow$ **36** pupils speak both French and English

β) If the rest speak English only

⇒ Total no. – (F only + both F and E)

⇒ 50 – (5 + 36)

⇒ 50 – 41

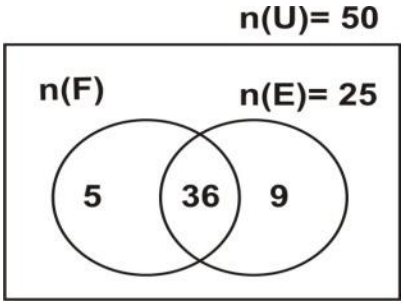
⇒ **9**

∴ **9** students speak only English

(ii) Let n(U) = No. of pupils in the class

n(F) = No. of French speakers

n(E) = No. of English speakers



2 (a) Cost of 8 pencils = 8 × ¢500 = ¢4,000

Cost of 5 pens = 5 × ¢2,000 = ¢10,000

∴ Total cost = ¢4,000 + ¢10,000

= ¢ 14,000

(b) New (increased) price of pen

= Original price + Increase

= ¢2,000 + (20% × ¢2,000)

= $¢2000 + \left(\frac{20}{100} \times ¢2000\right)$

= ¢2,000 + ¢400

= ¢2,400

NB:

We also could have used the idea of Simple Proportion (Equivalent fractions / 'If more, less divides …')

New (increased) price of a pencil

= Original price + Increase

= ¢500 + (10% × ¢500)

= ¢500 + ¢50

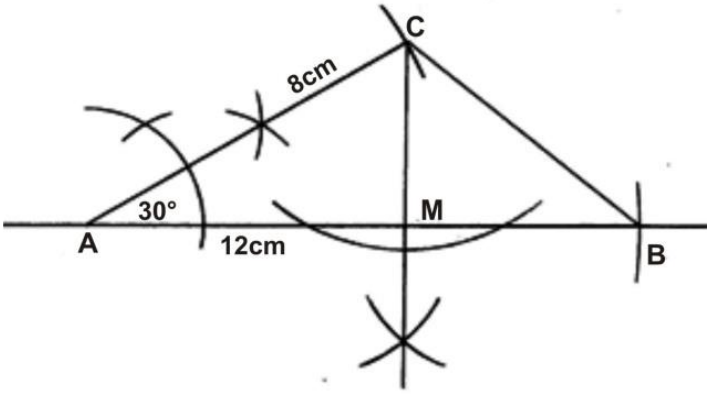
= ¢550

New price of 10 pencils = 10 × ¢550 = ¢5,500

New price of 8 pens = 8 × ¢2,400 = ¢19,200

∴ The total price = ¢5,500 + ¢19,200 = ¢24,700

3 (a)



(b) **(i)** angle ABC ≈ 39°

(ii) |CM| ≈ 4.0 cm

(c) Area of triangle = ½ bh

Area of ABC = ½ × |AB| × |CM|

= ½ × 12cm × 4cm

= 6cm × 4cm

= $24cm^2$

4 (a)

$$\frac{5x-3}{6} - \frac{2x-4}{4} < 2$$

Approach 1

$$\frac{2(5x-3)-3(2x-4)}{12} < 2$$

$$\Rightarrow \quad \frac{2(5x-3)-3(2x-4)}{12} < 2$$

$$\Rightarrow \quad \frac{10x-6-6x+12}{12} < 2$$

$$\Rightarrow \quad \frac{10x - 6x - 6 + 12}{12} < 2$$

$$\Rightarrow \quad \frac{4x + 6}{12} < 2$$

$$\Rightarrow \quad 4x + 6 < 12 \times 2$$

$$\Rightarrow \quad 4x < 24 - 6$$

$$\Rightarrow \quad \frac{4x}{4} < \frac{18}{4}$$

$$\Rightarrow \quad \underline{\underline{x < 4\tfrac{1}{2}}} \quad \text{or} \quad \underline{\underline{x < 4.5}}$$

4 (a)

Approach 2

$$\frac{5x-3}{6} - \frac{2x-4}{4} < 2$$

$$\Rightarrow \quad 12\left(\frac{5x-3}{6}\right) - 12\left(\frac{2x-4}{4}\right) < 12 \times 2$$

$$\Rightarrow \quad 2(5x-3) - 3(2x-4) < 12 \times 2$$

$$\Rightarrow \quad 10x - 6 - 6x + 12 < 24$$

$$\Rightarrow \quad 10x - 6x - 6 + 12 < 24$$

$$\Rightarrow \quad 4x + 6 < 24$$

$$\Rightarrow \quad 4x < 24 - 6$$

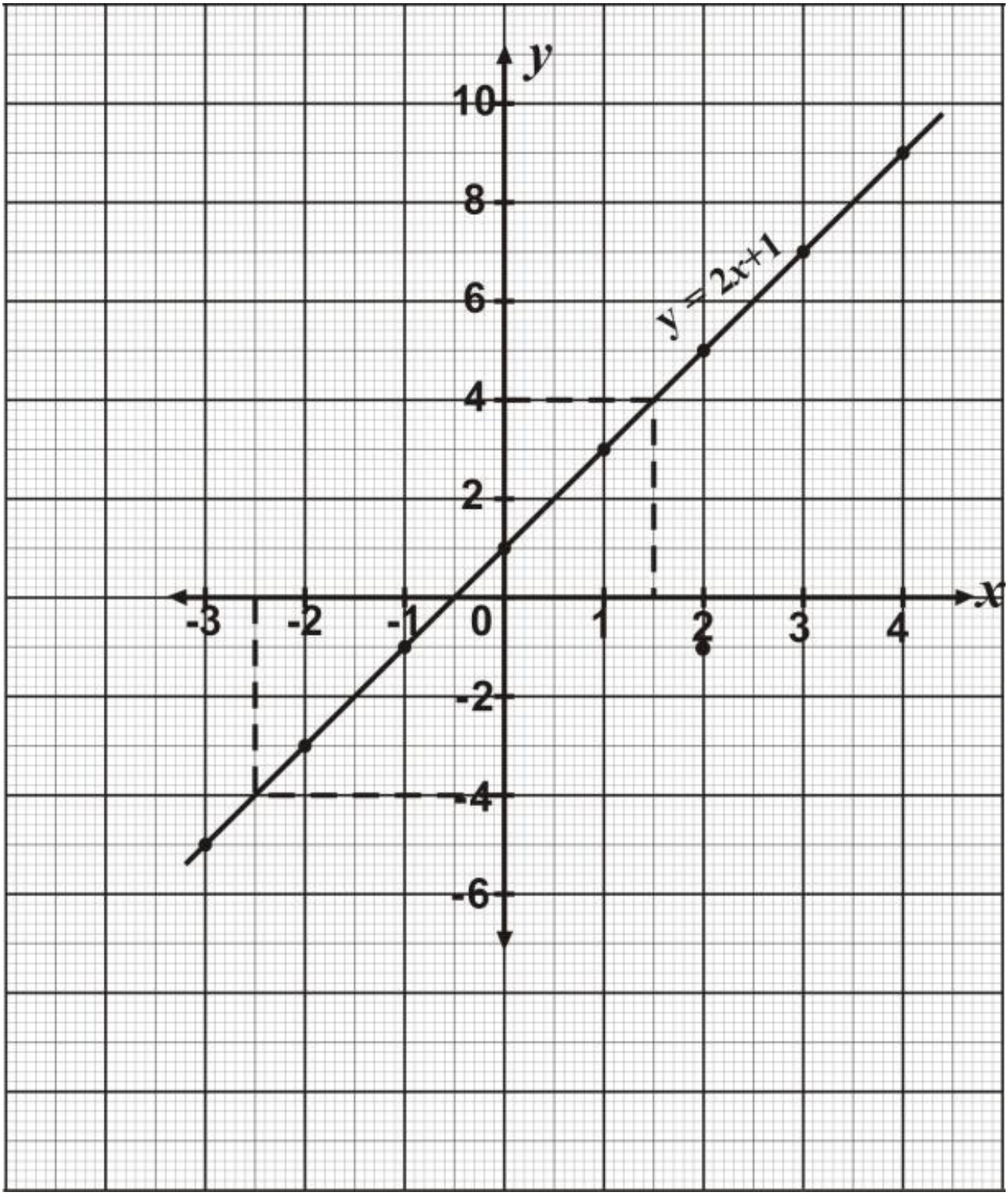
$$\Rightarrow \quad \frac{4x}{4} < \frac{18}{4}$$

$$\Rightarrow \quad \underline{\underline{x < 4\tfrac{1}{2}}} \quad \text{or} \quad \underline{\underline{x < 4.5}}$$

4 (b) (i) By substitution of the x values in the given relation, the table is completed as shown below:

x	-3	-2	-1	0	1	2	3	4
y	-5	-3	-1	1	3	5	7	9

(ii), (iii) The straight line graph is shown below



(iv) From the graph,

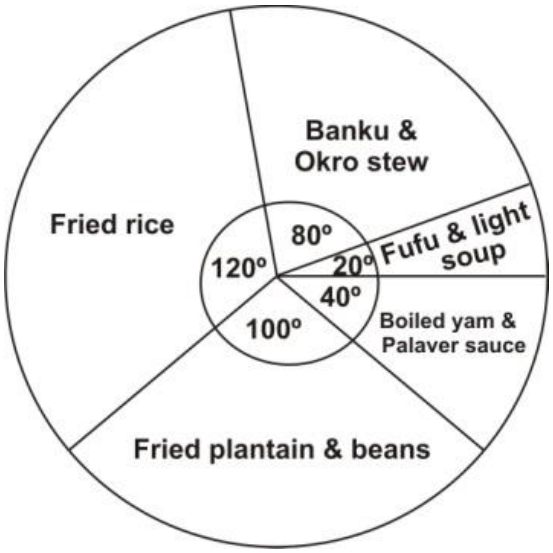
α) When $y = 4$, $\underline{x = 1.5}$

β) When $x = -2.5$, $\underline{y = -4}$

5 (a) (i)

Grade	Frequency	Angle of sector
Fufu and light soup	5	$\frac{5}{90} \times 360^{o} = 20^{o}$
Banku and Okro stew	20	$\frac{20}{90} \times 360^{o} = 80^{o}$
Fried Rice	30	$\frac{30}{90} \times 360^{o} = 120^{o}$
Fried Plantain and Beans	25	$\frac{25}{90} \times 360^{o} = 100^{o}$
Boiled yam and palaver sauce	10	$\frac{10}{90} \times 360^{o} = 40^{o}$
TOTAL	**90**	**360°**

Pie chart showing the distribution of students in a JSS class who prepared various dishes



(a) (ii) Fried Rice was prepared most

(iii) $$\frac{\text{No. of students who prepared fried rice}}{\text{Total no. of students}} \times 100\%$$

$$= \frac{30}{90} \times 100\%$$

$$= \underline{33\tfrac{1}{3}\%}$$

(b) P(Fried Plantain and Beans)

$$= \frac{\text{No. of students for Fried Plantain and Beans}}{\text{Total number of students}}$$

$$= \frac{25}{90} = \underline{\underline{\frac{5}{18}}}$$

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MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. If P = {multiples of 4 less than 16}, find P.

A. {4,8,10} B. {4,8,12} C. {1,4,8,12} D. {4,8,12,16}

2. The addition below was carried out in base x. Find x.

$$\begin{array}{r} 243 \\ +\ 221 \\ \hline 1014_x \end{array}$$

A) four B) five C) six D) seven

3. A farmer left home at 4:35 am and arrived on his farm at 6:18 am. How long did he take to get to his farm?

A) 1 hour 23 minutes
B) 1 hour 43 minutes
C) 2 hours 43 minutes
D) 10 hours 53 minute

4. Express 2474.5 in standard form

A) 2.4745×10^2
B) 2.4745×10^3
C) 2.4745×10^{-2}
D) 2.4745×10^{-3}

5. Simplify $1\frac{1}{2} + 2\frac{1}{4} - 3\frac{5}{8}$

A) $\frac{1}{8}$ B) $\frac{3}{8}$ C) $\frac{3}{16}$ D) $\frac{5}{16}$

6. Find the next two numbers in the sequence 2, 5, 9, 14, 20, _ , _ .

A) 26, 34 B) 26,35 C) 27,34 D) 27,35

7. The sum of three numbers is 28,542. Two of the numbers are 10,250 and 9,750. Find the third number.

A) 8,452 B) 8,542 C) 9,452 D) 9,542

8. 135 pencils were to be packed into boxes. Each box could take 12 pencils. Find the number of boxes that were fully packed.

A) 10 boxes B) 11 boxes C) 12 boxes D) 13 boxes

9. Which of the fractions $^{13}/_{20}$, $^{3}/_{5}$, ¾ and $^{7}/_{10}$ is the greatest ?

A) $^{3}/_{5}$ B) ¾ C) $^{7}/_{10}$ D) $^{13}/_{20}$

10. Out of ¢550,000.00 given to a school, an amount of ¢325,000.00 was used. What fraction of the total amount was used?

A) $^{4}/_{13}$ B) $^{9}/_{13}$ C) $^{9}/_{22}$ D) $^{13}/_{22}$

11. Kofi bought four pencils at ¢200.00 each and five pens at ¢350.00 each. How much did he pay altogether?

A) ¢2,400.00 B) ¢2,450.00 C) ¢2,550.00 D) ¢2,650.00

12. John's uncle sent him ¢120,000.00 through a bank which charges 5% commission. How much commission was paid ?

A) ¢5,000.00 B) ¢6,000.00 C) ¢7,000.00 D) ¢7,200.00

13. 8 girls can weed a plot of land in 10 days. How many days will 5 girls take to weed the same plot of land, working at the same rate?

A) 6 days B) 8 days C) 12 days D) 16 days

14. A mapping is defined by $x \rightarrow x^2 - 1$. What is the image of 3 under the mapping?

A) 5 B) 6 C) 7 D) 8

15. A typist charges ¢2,000.00 for the first 5 sheets typed and ¢600.00 for any additional sheet. How much will Barbara pay, if she presents 20 sheets for typing ?

A) ¢8,000.00 B) ¢11,000.00 C) ¢12,000.00 D) ¢19,000.00

16. Ama covered a distance of 100m in 12 seconds. Express her speed in kilometres per hour.

A) 5km/h B) 10 km/h C) 20 km/h D) 30 km/h

17. Yakubu, Seidu and Amina shared an amount of money in the ratio 3:4:5. Find how much was shared, if Amina received ¢40,000.00

A) ¢72,000.00

B) ¢80,000.00

C) ¢96,000.00

D) ¢120,000.00

18. Abban paid ¢630,000.00 for a bicycle at a discount of 10%. Find the actual selling price of the bicycle

A) ¢567,000.00

B) ¢573,000.00

C) ¢693,000.00

D) ¢700,000.00

19. The population of a town in 1990 was 88,000. The population increased by 20% in 1998. Find the population in 1998.

A) 70,400 B) 94,600 C) 96,800 D) 105,600

20. If $a * b = 2a - b$, find $3 * 2$

A) 1 B) 4 C) 5 D) 6

21. If $r = \frac{m+1}{m-2}$, express m in terms of r.

A) $m = \frac{2r+1}{r-1}$

B) $m = \frac{2r+1}{r+1}$

C) $m = \frac{2r-1}{r-1}$

D) $m = \frac{2r-1}{r+1}$

Use the information below to answer questions **22** *and* **23**

Forty per cent of students in a class speak Ga and seventy-five per cent speak Twi. Each student speaks at least one of the two languages.

22. What percentage of the class speaks both Ga and Twi?

A) 5% B) 10% C) 15% D) 30%

23. If there are 40 students in the class, how many of them speak Twi?

A) 10 B) 16 C) 24 D) 30

24. A point (2, 1) is reflected in the y-axis. Find its image.

A. (-1, 2) B. (1, -2) C. (-2, 1) D. (2, -1)

25. Expand (2a + b)(a – b).

A) $2a^2 – 3ab – b^2$

B) $2a^2 – ab – b^2$

C) $2a^2 + ab + b^2$

D) $2a^2 + 3ab + b^2$

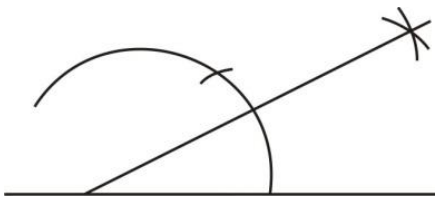
26. If **s** = $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$ and **r** = $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$, find 3**s** + **r**.

A. $\begin{pmatrix} 0 \\ 14 \end{pmatrix}$ B. $\begin{pmatrix} 0 \\ 6 \end{pmatrix}$ C. $\begin{pmatrix} 6 \\ 6 \end{pmatrix}$ D. $\begin{pmatrix} 6 \\ 14 \end{pmatrix}$

27. Solve for y in the equation $\frac{1}{3}y + \frac{1}{5}y = 8$

A) 15 B) 30 C) 45 D) 60

28.



Which of the following statements best describes the construction above?

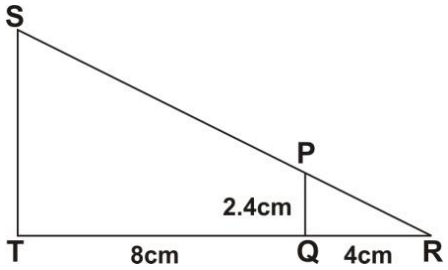
A) Constructing 30°

B) Constructing 60°

C) Constructing 120°

D) Constructing 135°

29.



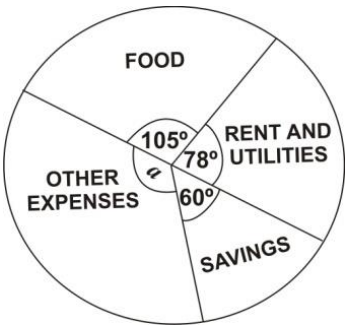
In the diagram, ΔSTR is the enlargement of ΔPQR.

If |TQ| = 8cm, |QR| = 4cm and |PQ| = 2.4cm, find |ST|.

A) 4.8cm B) 7.2cm C) 8.4cm D) 9.6cm

Use the following information to answer questions **30** *to* **32**

The pie chart below shows how a man spends his monthly salary.



30. Find the value of *a*

A) 75° B) 78° C) 117° D) 120°

31. If the man earns ¢720,000.00 monthly, find how much he spends on food.

A) ¢120,000.00

B) ¢180,000.00

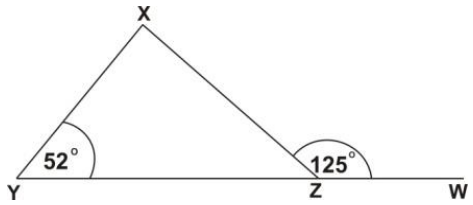
C) ¢210,000.00

D) ¢240,000.00

32. What percentage of his salary does he save?

A) 16.7% B) 21.7% C) 25.0% D) 29.2%

33.



In the diagram, XYZ is a triangle. YW is a straight line. Angle XYZ = 52° and angle XZW = 125°. Find angle YXZ.

A) 35° B) 55° C) 73° D) 107°

34. There are 18 girls and 22 boys in a class. A prefect is to be chosen at random from the class. What is the probability that the prefect will be a girl?

A) $^{1}/_{18}$ B) $^{9}/_{20}$ C) $^{11}/_{20}$ D) $^{9}/_{11}$

35. The length of a rectangle is three times its width. If its perimeter is 24cm, find its width.

A) 3 cm B) 4 cm C) 6 cm D) 8 cm

36. The interior angle of a regular polygon is 135°. How many sides has the polygon ?

A) 6 B) 8 C) 9 D) 12

37. A square of side 4cm is enlarged by a scale factor of 3. Calculate the area of the enlarged square.

A) 36cm^2 B) 72cm^2 C) 96cm^2 D)144cm^2

38. A rectangular tank has length 3m, width 2m and height 1.5m.

If the tank is filled with water to ⅔ of its capacity, calculate the volume of water in the tank.

A) 4.5m^3 B) 6.0m^3 C) 7.5m^3 D) 13.5m^3

39. A rectangular tank has length 3m, width 2m and height 1.5m.

The Alpha Water Company charges ¢40,000.00 for every 1 m^3 of water. Find how much it will cost to fill the tank completely.

A) ¢180,000.00

B) ¢240,000.00

C) ¢300,000.00

D) ¢360,000.00

40. Express 34m 5cm 6mm in millimetres.

A) 3,456 mm B) 34,056 mm C) 34,506 mm D) 340,506 mm

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MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. B. {4,8,12}

2. B) five

3. B) 1 hour 43 minutes

4. B) 2.4745×10^3

5. A) $^1/_8$

6. D) 27,35

7. B) 8,542

8. B) 11 boxes

9. B) ¾

10. D) $^{13}/_{22}$

11. C) ¢2,550.00

12. B) ¢6,000.00

13. D) 16 days

14. D) 8

15. B) ¢11,000.00

16. D) 30 km/h

17. C) ¢96,000.00

18. D) ¢700,000.00

19. D) ¢105,600

20. B) 4

21. A) $m = \frac{2r+1}{r-1}$

22. C) 15%

23. D) 30

24. C. (-2, 1)

25. B) $2a^2 - ab - b^2$

26. A. $\begin{pmatrix} 0 \\ 14 \end{pmatrix}$

27. A) 15

28. A) Constructing 30°

29. B) 7.2cm

30. 30. C) 117°

31. 31. C) ¢210,000.00

32. 32. A) 16.7%

33. C) 73°

34. B) 9/20

35. A) 3 cm

36. B) 8

37. B) $6.0m^3$

38. D) $144cm^2$

39. D) ¢360,000.00

40. B) 34,056 mm

April 2002

MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. (a) Simplify $2\frac{3}{4} \div (3\frac{3}{8} - 1\frac{1}{12})$

(b) A ribbon of length 3m is to be cut into strips of length 14cm.

(i) How many strips of length 14cm were cut?

(ii) What length of ribbon remained?

(c) One-fourth of a number is added to one-fifth of the same number. If the result is 18, find the number.

2. A trader took a loan of ¢1,800,000.00 at an interest of 12 ½ per cent per annum. It was agreed that the loan and the interest must be paid in one year in equal monthly instalments.

(a) Calculate

(i) the interest on the loan

(ii) the amount to be paid at the end of the year;

(iii) the monthly instalment.

(b) Another trader took a loan at the same rate of interest and conditions. If she had to pay ¢200,000.00 monthly instalment, find to the nearest cedi how much loan she took.

3. A teacher conducted a class test and the result is displayed in a frequency table below.

Marks	1	2	3	4	5	6	7	8	9
Frequency	2	4	2	2	10	5	6	7	2

(a) Using the frequency table, find

(i) the modal mark for the class

(ii) the number of candidates who wrote the test

(iii) the mean mark for the test.

(b) If the teacher decides that the pass mark is 4, what is the probability that a student chosen at random from the class failed the test?

(c) Draw a bar chart for the distribution

4. (a) Three numbers are in the ratio 2:3:4 and their average is 36.

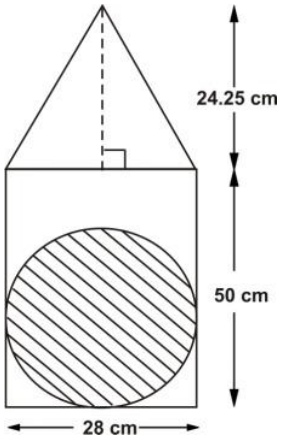
(i) Calculate the sum of the three numbers

(ii) Find the greatest number.

(b) If costs ¢875,000.00 to tile a rectangular floor measuring 7m by 5m.
How much does it cost to tile a floor measuring 8m by 4m if the same types of tiles are used?

(c) It is estimated that the population of a village increases by 10 per cent every five years. If the population of the village is 150,000 this year, what would be the estimated population ten years from now?

5.



The above diagram is a plane figure made up of a rectangle of sides 50cm by 28cm and an equilateral triangle of height 24.25cm. A circle is cut out of the rectangle as shown. If the circle touches three sides of the triangle,

Calculate

(a) the perimeter of the figure

(b) the area of the remaining portion of the figure.

[Take $\pi = {}^{22}/_{7}$]

April 2002

MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1 (a) $2\frac{3}{4} \div (3\frac{3}{8} - 1\frac{1}{12})$

$\Rightarrow \quad \frac{11}{4} \div (\frac{27}{8} - \frac{13}{12})$

$\Rightarrow \quad \frac{11}{4} \div (\frac{81 - 26}{24})$

$\Rightarrow \quad \frac{11}{4} \div \frac{55}{24}$

$\Rightarrow \quad \frac{11}{4} \times \frac{24}{55}$

$\Rightarrow \quad \frac{6}{5} = \underline{\underline{1\frac{1}{5}}}$

1 (b) (i) No. of strips = 3m ÷ 14cm

= 300 cm ÷ 14cm

$= 21\frac{3}{7}$

$\Rightarrow$ <u>21 strips of length 14 cm were cut</u>

(ii) **<u>Method 1</u>**

$^3/_7$ of 14cm remained

$\Rightarrow \quad \frac{3}{7} \times 14cm$

$\Rightarrow$ **6 cm**

⇒ Remaining length = 6 cm

(ii) **Method 2**

Total length cut = 21 × 14cm = 294cm

∴ Remaining length = 300 cm – 294cm

= 6 cm

1 (c) Let the number = n, then

⇒ $\frac{1}{4}n + \frac{1}{5}n = 18$

Approach 1 (Simplifying the LHS first)

⇒ $\frac{n}{4} + \frac{n}{5} = 18$

⇒ $\frac{5n + 4n}{20} = 18$

⇒ $\frac{9n}{20} = 18$

⇒ $9n = 18 \times 20$

⇒ $9n = 360$

⇒ $n = 360 \div 9$

⇒ $n = 40$

1 (c) **Approach 2** (Multiplying by 20 first – to clear fractions)

$20\times(\frac{1}{4}n) + 20\times(\frac{1}{5}n) = 20\times18$

⇒ $5n + 4n = 20 \times 18$

⇒ $9n = 360$

⇒ $\frac{9n}{9} = \frac{360}{9}$

⇒ $n = 40$

2 (a) (i) Simple interest = Principal × Rate × Time

= 1,800,000 × 12½% × 1

= $1{,}800{,}000 \times \frac{25}{2}\% \times 1$

$$= 1{,}800{,}000 \times \frac{25}{2 \times 100} \times 1$$

$$= 9000 \times 25 \times 1$$

$$= \underline{225{,}000}$$

(ii) The amount to be paid = Principal + S. Interest

$$= 1{,}800{,}000 + 225{,}000$$

$$= \underline{2{,}025{,}000}$$

(iii) The monthly instalment $= \dfrac{\text{Amount to be paid}}{\text{No. of months}}$

$$= \frac{2{,}025{,}000}{12}$$

$$= \underline{168{,}750}$$

2 (b) Monthly instalment $= \dfrac{\text{Total amount}}{\text{No. of months}}$

$$= \frac{\text{Principal + Simple interest}}{\text{No. of months}}$$

$$= \frac{P + PRT}{\text{No. of months}}$$

$$\Rightarrow \quad 200{,}000 = \frac{P + P \times 12.5\% \times 1}{12}$$

Solving for P by cross multiplying, we have:

$$\Rightarrow \quad 200{,}000 \times 12 = P + P \times 12.5\%$$

$$\Rightarrow \quad 200{,}000 \times 12 = P + \frac{12.5P}{100}$$

$$\Rightarrow \quad 200{,}000 \times 12 = P + 0.125P$$

$$\Rightarrow \quad 2{,}400{,}000 = P(1 + 0.125)$$

$$\Rightarrow \quad 2{,}400{,}000 = P(1.125)$$

$$\Rightarrow \quad \frac{2{,}400{,}000}{1.125} = P$$

$$\Rightarrow \quad ¢2{,}133{,}333.33 = P$$

$\therefore$ The loan she took $\approx$ $\underline{¢2{,}133{,}333}$ (to the nearest cedi)

3 (a) (i) The modal mark

= the mark with the highest frequency

⇒ The modal mark = 5

(ii) Total number of candidates

= 2 + 4 + 2 + 2 + 10 + 5 + 6 + 7 + 2

= 40 pupils

3 (a) (iii) Using the frequency table,

Approach 1

$$\text{The mean} = \frac{\text{The sum of marks}}{\text{The total number of candidates}}$$

$$= \frac{(1\times2)+(2\times4)+(3\times2)+(4\times2)+(5\times10)+(6\times5)+(7\times6)+(8\times7)+(9\times2)}{2+4+2+2+10+5+6+7+2}$$

$$= \frac{2+8+6+8+50+30+42+56+18}{40}$$

$$= \frac{220}{40} = 5\tfrac{1}{2} \text{ or } 5.5$$

3 (a) (iii) **Approach 2**

Marks (x)	**Frequency (f)**	**f x**
1	2	2
2	4	8
3	2	6
4	2	8
5	10	50
6	5	30
7	6	42
8	7	56
9	2	18
	$\Sigma f = 40$	**$\Sigma fx = 220$**

$$\text{The mean mark} = \frac{\Sigma f x}{\Sigma f} = \frac{220}{40}$$

$= \underline{\underline{5\frac{1}{2}}}$ or $\underline{\underline{5.5}}$

3 (b) Probability that a student chosen at random failed the test

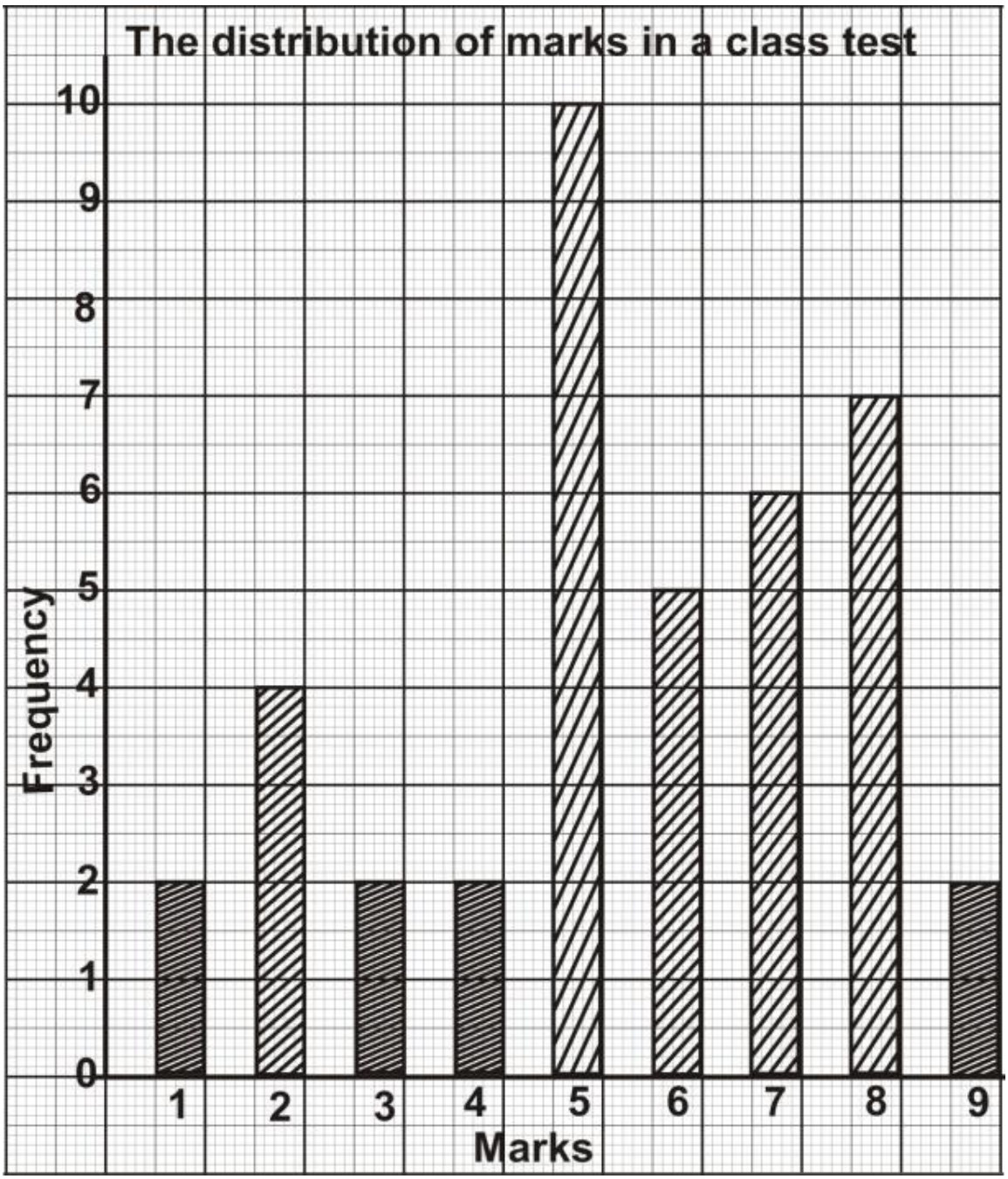
$$= \frac{\text{Number of students that failed}}{\text{Total number of students}} = \frac{2+4+2}{40}$$

$$= \frac{8}{40} = \frac{1}{5}$$

Probability that a student chosen at random failed the test $= \underline{\underline{1/5}}$

3 (c)

Vertical AxisScale: 2cm to 1 student / pupil



4 (a)

Method 1 (Unitary Approach – the value of one(1) item)

The average ratio $= \frac{2+3+4}{3} = \frac{9}{3} = 3$

Now, if the average ratio, 3, $\rightarrow$ 36,

Then ratio 1 $\rightarrow$ $36 \div 3 = 12$,

$\therefore$ The sum of ratios, 9, (i.e., 2+3+4) $\rightarrow 9 \times 12 = 108$

$\Rightarrow$ The sum of the three numbers = 108

4 (a) **Method 2** (Equivalent Fractions)

	Ratio	Actual no.
Average	$\frac{2+3+4}{3} = 3$	**36**
Sum	2+3+4 = **9**	***x***

From the table above, $\frac{3}{9} = \frac{36}{x}$

Cross multiplying, we have $\Rightarrow \quad 3 \times x = 36 \times 9$

$$\Rightarrow \quad \frac{3x}{3} = \frac{36 \times 9}{3}$$

$$\Rightarrow \quad x = 108$$

$\therefore$ The sum of the three numbers is 108

4 (a) **Method 3** (If more, less divides …)

If the average ratio, 3 → 36

Then sum of ratios, 9 → ? (more)

If more, less (i.e., 3) divides;

$$\Rightarrow \quad \frac{9}{3} \times 36 = 108$$

4 (b) **Approach 1** (Equivalent Fractions)

	Area	Cost
1st floor	7m by 5m = 7m × 5m = **35 m²**	**¢875, 000**
2nd floor	8m by 4m = 8m × 4m = **32 m²**	***x***

From the table above, $\frac{35}{32} = \frac{875{,}000}{x}$

Cross multiplying, we have

$$35 \times x = 875{,}000 \times 32$$

$$\Rightarrow \quad \frac{35x}{35} = \frac{875{,}000 \times 32}{35}$$

$$\Rightarrow \quad x = 25{,}000 \times 32$$

$$\Rightarrow \quad x = 800{,}000$$

$\therefore$ The cost of tiling a 8m by 4m floor = ¢800,000.00

4 (b) **Approach 2** (If more, less divides …)

	Area	Cost
1st floor	7m by 5m = 7m × 5m = **35 m²**	**¢875, 000**
2nd floor	8m by 4m = 8m × 4m = **32 m²**	**?**

If 35 m^2 area costs ¢875,000

Then 32 m^2 area costs ? (less)

If less, more (i.e., 35m^2) divides, therefore we have

$$\frac{32 \times ¢875{,}000}{35} = ¢800{,}000$$

$\therefore$ The cost of tiling a 8m by 4m floor = ¢800,000.00

4 (c) **METHOD 1** (Formula 1)

$$\text{New population} = \frac{\text{New population percentage}}{\text{Old population percentage}} \times \text{Old population}$$

$$\text{Population in 5 yrs} = \frac{110\%}{100\%} \times 150{,}000$$

$$= 110 \times 1{,}500 = 165{,}000$$

$$\therefore \text{ Population in 10 yrs} = \frac{110\%}{100\%} \times 165{,}000$$

$$= 110 \times 1{,}650 = 181{,}500$$

4 (c) **METHOD 2** (Formula 2)

Estimated population in 5 years' time

$$= \text{Current population} + \text{Increase}$$

$$= 150{,}000 + (10\% \times 150{,}000)$$

$$= 150{,}000 + \left(\frac{10}{100} \times 150{,}000\right)$$

$$= 150{,}000 + 15{,}000$$

$$= \underline{165{,}000}$$

Estimated population in 10 years' time

$$= \text{Current population} + \text{Increase}$$

$$= 165{,}000 + (10\% \times 165{,}000)$$

$$= 165{,}000 + 16{,}500$$

$$= \underline{\underline{181{,}500}}$$

4 (c) **METHOD 3** **(Equivalent Fractions)**

[*NB: The percentage of the original quantity is always 100%*]

	Percentage	**Population**
This year (original)	**100%**	**150,000**
5 years' time	100% + 10% = **110%**	***x***

From the table above, $\frac{100\ \%}{110\ \%} = \frac{150{,}000}{x}$

Cross multiplying, we have $100 \times x = 150{,}000 \times 110$

$\Rightarrow \quad x = \frac{150{,}000 \times 110}{100} \quad = 1{,}500 \times 110 \quad \underline{\underline{= 165{,}000}}$

$\Rightarrow$ The population in 5 years' time is 165,000

$\therefore$ Similarly,

	Percentage	**Population**
5 years' time (original)	**100%**	**165,000**
10 years'	100% + 10%	

time	= **110%**	**y**

From the table above, $\frac{100\ \%}{110\ \%} = \frac{165{,}000}{y}$

Cross-multiplying, we have $100 \times y = 165{,}000 \times 110$

$\Rightarrow \quad y = \frac{165{,}000 \times 110}{100} = 1{,}650 \times 110$

$\Rightarrow \quad y = 181{,}500$

$\therefore$ The estimated population 10 years from now = 181,500

4 (c)

METHOD 4 ('If more, less divides ...')

If this year: 100% → 150,000

Then in 5 years: 110% → ? (more)

If more, less (i.e., 100%) divides; $\therefore$ we have

$\frac{110\ \%}{100\ \%} \times 150{,}000 = 110 \times 1{,}500 = 165{,}000$

If in 5 years: 100% → 165,000

Then in 10 yrs 110% → ? (more)

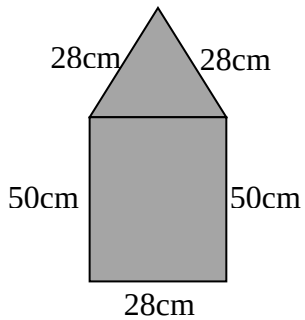
If more, less (i.e., 100%) divides; $\therefore$we have

$\frac{110}{100} \times 165{,}000 = 110 \times 1{,}650 = 181{,}500$

$\therefore$ The estimated population 10 years from now = 181,500

5 (a)

Perimeter of the figure (distance around the figure)



NB:

The triangle is equilateral

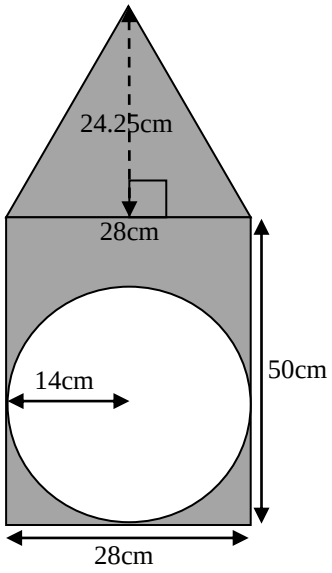
Taken from the base, in the anticlockwise direction,

Perimeter = 28cm + 50cm + 28cm + 28cm + 50cm

= <u>184 cm</u>

5 (b)

Area of remaining portion (after circle is cut out)



<u>Approach 1</u>

= Area of entire figure – Area of circle

= [rectangle + triangle] – [circle]

= [($l \times w$) + (½ × b × h)] – [πr^2]

= [(50×28) + (½×28×24.25)] – [$^{22}/_{7} \times (14)^2$]

= [(50×28) + (14 × 24.25)] – [$^{22}/_{7} \times 14 \times 14$]

= [1400+ 339.5] – [616]

= 1739.5 – 616

= 1123.5

⇒ Area of the remaining portion = <u>1123. 5 cm^2</u>

5 (b)

<u>Approach 2</u>

= Area of triangle + Area of [rectangle – circle]

= (½ × b × h) + [($l \times w$) – πr^2]

= (½ × 28 × 24.25) + [(50×28) – $^{22}/_{7} \times (14)^2$]

= 339.5 + (1400 – 616)

= 339.5 + 784

= 1123.5

⇒ Area of the remaining portion = 1123. 5 cm^2

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MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. R = {1, 3, 5, 7} and S = {2, 4, 6, 8}. Find R ∪ S.

A. { }
B. {1,2,3,5,6,8}
C. {1,2,3,4,5,7,8}
D. {1,2,3,4,5,6,7,8}
E. {1,2,3,4,5,6,7,8,9,10}

2. Find the missing members in the set {5, 10, 15, _ , 25, _ , _ ,40}

A) 20 and 30
B) 30 and 35
C) 20 and35
D) 20, 30 and 35
E) 30, 35 and 45

3. What property of addition is defined by (a+b)+c = a+(b+c) ?

A) Union
B) Inverse
C) Commutative
D) Distributive
E) Associative

4. Find the highest (greatest) common factor of 63 and 81.

A) 3 B) 7 C) 9 D) 21 E) 27

5. Change 124_{five} to a base ten numeral.

A) 24 B) 35 C) 39 D) 42 E) 55

6. Kofi bought 4 books at an average price of ¢2,500.00. If the total cost of 3 of the books was ¢6,500.00, find the cost of the fourth book.

A) ¢3,500.00
B) ¢4,000.00
C) ¢4,500.00
D) ¢6,500.00
E) ¢10,000.00

7. Find the sum of all even numbers between 70 and 80.

A) 200 B) 223 C) 280 D) 300 E) 375

8. By how much is $^5/_6$ greater than $^3/_4$?

A) $^1/_{12}$ B) $^1/_6$ C) $^5/_{12}$ D) $^2/_3$ E) $^4/_5$

9. Evaluate 0.25 x 0.006, correct to three decimal places.

A) 0.001 B) 0.002 C) 0.015 D) 0.075 E) 0.105

10. Express $^5/_8$ as a decimal fraction

A) 0.125 B) 0.375 C) 0.625 D) 0.750 E) 0.875

11. The following are the angles formed at the centre of a circle:
30°, 70°, 120°, 2*x* and 5*x*. Find the value of *x*.

A) 20 B) 40 C) 70 D) 100 E) 140

12. On a map 1cm represents 4.5km. What is the actual distance between two towns which are 4cm apart on the map?

A) 9km B) 16km C) 18km D) 19km E) 21km

13. If $2y = 6 - 3x$, find y when $x = 0$

A) –3 B) –2 C) 0 D) 2 E) 3

14. How many lines of symmetry has an isosceles triangle ?

A) 1 B) 2 C) 3 D) 4 E) 5

15. Kofi's age in the next ten years will be four times his age five years ago. How old is Kofi now?

A) 5 B) 6 C) 10 D) 15 E) 20

16. A point P (3, 4) is rotated through an angle of 90° anticlockwise about the origin O. Find the image P_1 of rotation

A. (3, -4) B. (4, -3) C. (-3, 4) D. (-4, 3) E. (-4, -3)

17. Which of the following is not quadrilateral?

A) square

B) rectangle

C) rhombus

D) triangle

E) parallelogram

18. Express 2345 in standard form.

A) 2.345×10^1

B) 2.345×10^2

C) 2.345×10^3

D) 2.345×10^4

E) 2.345×10^5

19. Solve the inequality $3(x - 1) \leq 12$

A) $x \leq -5$ B) $x \leq -3$ C) $x \leq 3$ D) $x \leq 5$ E) $x \leq {}^{13}/_{3}$

20. Kofi and Ama shared an amount of money in the ratio 3:2 respectively. If Kofi had ¢60,000.00, how much was shared?

A) ¢36,000.00

B) ¢40,000.00

C) ¢90,000.00

D) ¢100,000.00

E) ¢120,000.00

21. Workers are required to pay 4 ½ % of their salaries into an educational fund. A worker's salary is ¢120,000.00. How much does he pay into the educational fund?

A) ¢5,332.00

B) ¢5,400.00

C) ¢6,000.00

D) ¢10,800.00

E) ¢11,500.00

22. Araba bought an electric cooker for ¢540,000.00 at a discount of 10%. Find the actual price of the electric cooker.

A) ¢469,000.00

B) ¢496,000.00

C) ¢594,000.00

D) ¢600,000.00

E) ¢605,000.00

23. Given that $\sqrt{p^2 \times q^2} = p \times q$, find the value of a, if $a = \sqrt{13^2 \times 15^2}$

A) 28 B) 175 C) 195 D) 247 E) 494

24. A mapping is defined by $n \rightarrow 2n - 3$. What is the image of –2 under the mapping?

A) –1 B) –5 C) –7 D) 7 E) 1

25. Simplify $4p + 6p^2 - 2p + 2p^2$

A) $2p + 8p^2$

B) $6p + 8p^2$

C) $2p - 8p^2$

D) $6p - 8p^2$

E) $-4 + 10p^2$

26. Make q the subject of the relation $W = \frac{n-q}{q}$

A) $q = \frac{n-1}{w}$ B) $q = \frac{n+1}{w}$ C) $q = \frac{n+2}{1+w}$

D) $q = \frac{n}{1+w}$ E) $q = \frac{n}{w-1}$

27. Given that $E = \frac{1}{2} mc^2$, find E when m = 10 and c = 4

A) 20 B) 40 C) 60 D) 80 E) 160

28. The scores obtained by 8 pupils in a test are 2, 3, 3, 5, 7, 8, 8 and10.

What is the median score?

A) 3 B) 5 C) 6 D) 7 E) 8

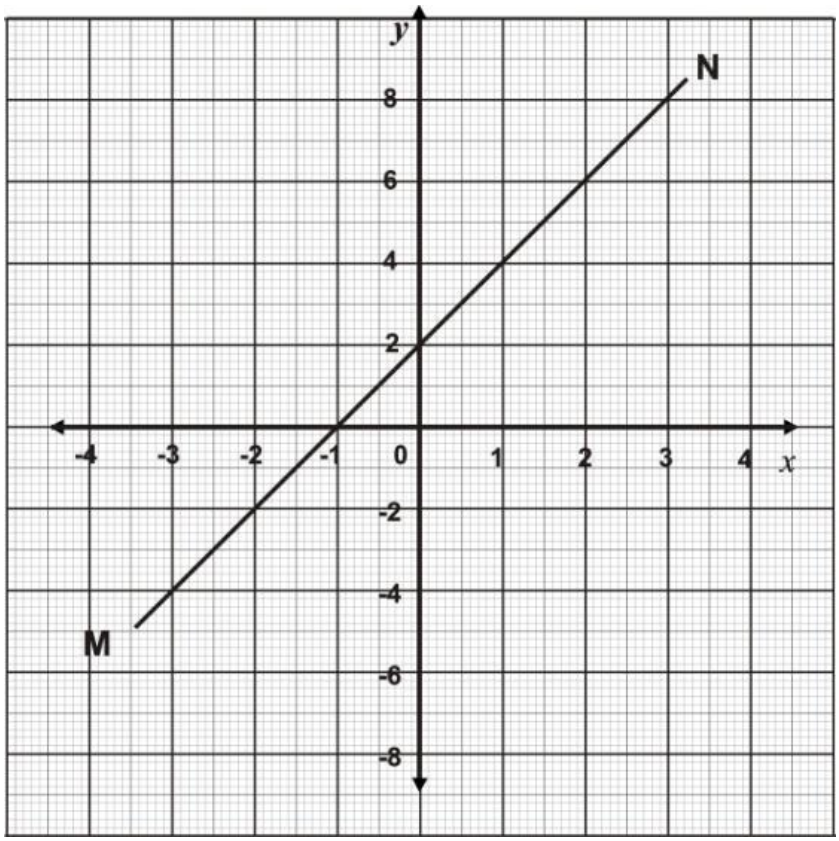
29. Find the mean score.

A) 4.50 B) 5.75 C) 6.00 D) 8.75 E) 10.00

30. The scores obtained by 8 pupils in a test are: 2, 3, 3, 5, 7, 8, 8 and10.
What is the probability that a pupil chosen at random scores 8 marks?

A) $^{1}/_{8}$ B) $^{4}/_{23}$ C) ¼ D) $^{1}/_{3}$ E) ½

The graph of the straight line MN is shown below. Use it to answer questions **31** *and* **32**



31. Determine the value of x when y is 3

A) ½ B) 1 ½ C) 2 D) 3 E) 8

32. Find the gradient of line MN.

A) –2 B) –1 C) 1 D) $^{3}/_{2}$ E) 2

33. State the rule for the mapping

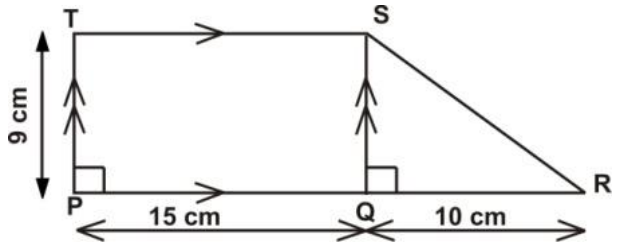
x	1	2	3	4	5
↓	↓	↓	↓	↓	↓
y	1	4	9	16	25

A) $x \rightarrow x$ B) $x \rightarrow 2x$ C) $x \rightarrow 3x$ D) $x \rightarrow x^2$ E) $x \rightarrow x^3$

34. The area of a rectangle is $18cm^2$. One of its sides is 9cm long. Find its perimeter.

A) 11cm B) 18cm C) 22cm D) 36cm E) 54cm

35. Calculate the area of figure PQRST



A) 112.5 cm^2

B) 135.0 cm^2

C) 180.0 cm^2

D) 215.0 cm^2

E) 315.0 cm^2

36. In an enlargement with scale factor k, which of the following statements is not true?

A) Each length is multiplied by k

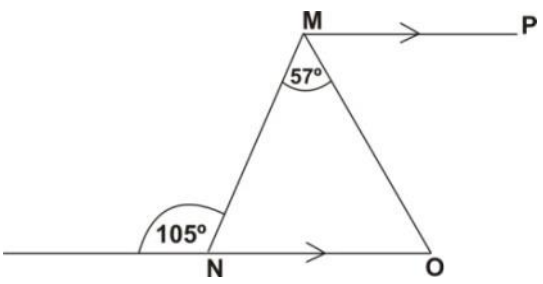
B) Each angle remains the same

C) The shape of the figure does not change

D) The size of the figure does not change

E) Corresponding lines are parallel

Use the diagram below to answer questions **37** *and* **38**



Line MP is parallel to line QNO. Angle NMO = 57° and angle MNQ = 105°

37. Find the angle MON

A) 48° B) 52° C) 58° D) 65° E) 75°

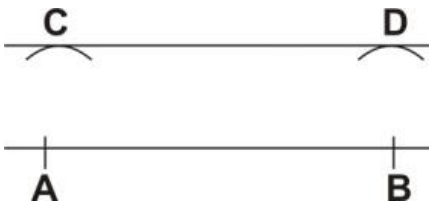
38. Angle MON and PMO are

A) Alternate angles
B) vertically opposite angles
C) Corresponding angles
D) adjacent angles
E) Complementary angles

39. If $\mathbf{u} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$, find $2\mathbf{u}+3\mathbf{v}$.

A. $\begin{pmatrix} 5 \\ 1 \end{pmatrix}$ B. $\begin{pmatrix} 11 \\ 5 \end{pmatrix}$ C. $\begin{pmatrix} 11 \\ 13 \end{pmatrix}$ D. $\begin{pmatrix} 14 \\ 0 \end{pmatrix}$ E. $\begin{pmatrix} 14 \\ 12 \end{pmatrix}$

40.



Which of the following statements best describes the construction above?

A) Construction of a horizontal line CD
B) Construction of a line parallel to AB
C) Construction of the bisector of AB.
D) Construction of a top line
E) Construction of a vertical line.

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MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. D. {1,2,3,4,5,6,7,8}

2. D) 20, 30 and 35

3. E) Associative

4. C) 9

5. C) 39

6. A) ¢3,500.00

7. D) 300

8. A) $^{1}/_{12}$

9. B) 0.002

10. C) 0.625

11. A) 20

12. C) 18km

13. E) 3

14. A) 1

15. C) 10

16. D. (-4, 3)

17. D) triangle

18. C) 2.345×10^3

19. D) $x \leq 5$

20. D) ¢100,000.00

21. B) ¢5,400.00

22. D) ¢600,000.00

23. C) 195

24. C) –7

25. A) $2p + 8p^2$

26. D) $q = \frac{n}{1+w}$

27. D) 80

28. C) 6

29. B) 5.75

30. C) ¼

31. A) ½

32. E) 2

33. D) $x \rightarrow x^2$

34. C) 22 cm

35. C) 180.0 cm^2

36. D) The size of the figure does not change

37. A) 48°

38. 38. Alternate angles

39. D. $\begin{pmatrix} 14 \\ 0 \end{pmatrix}$

40. B)construction of a line parallel to AB

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MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal marks

1. (a) M is a set consisting of all positive integers between 1 and 10. P and Q are subsets of M such that

P = {factors of 6}, Q = {multiples of 2}

(i) List the elements of M, P and Q

(ii) Represent M, P and Q on a Venn diagram

(iii) Find $P \cap Q$

(b) (i) Solve the inequality: $\frac{2x-2}{4} - \frac{2x-1}{3} \leq 1$

(ii) Illustrate your answer on the number line

2. **(a)** Express 131_{five} as a binary numeral

(b) Three children, Kwabena, Esi and Yaw were given 160 oranges to share. Kwabena gets ¼ of the oranges. Esi and Yaw share the remainder in the ratio 3:2 respectively.

(i) Find how many oranges Esi received

(ii) How many more oranges did Yaw receive than Kwabena?

3. (a) Using a ruler and a pair of compasses only, construct triangle XYZ, such that |XY| = 6cm, |XZ| = 8cm and |YZ| = 10cm.

(b) (i) Construct the mediator of line YZ

(ii) construct the mediator of line XZ

(iii) Locate O, the point of intersection of the mediators of lines YZ and XZ.

(iv) With centre O and radius OY, draw a circle.

(c) Measure the radius of the circle you have drawn in (b) (iv) above and hence calculate the circumference of the circle. [Take $\pi = 3.14$]

.

4. a) (i) Using a scale of 2cm to 1 unit on both axes, draw two perpendicular axes OX and OY on a graph sheet.

(ii) On the same graph sheet, mark the x-axes from –5 to 5 and the y-axis from –6 to 6.

b) On the same graph sheet, plot the points A (2, 5), B(4, 3) and C (1, 1). Join the points A,B and C to form a triangle.

c) Reflect triangle ABC in the y-axis such that $A \rightarrow A_1$, $B \rightarrow B_1$ and $C \rightarrow C_1$. Label the vertices of triangle $A_1B_1C_1$

d) Translate triangle $A_1B_1C_1$ by the vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ such that $A_1 \rightarrow A_2$, $B_1 \rightarrow B_2$, and $C_1 \rightarrow C_2$. Label the vertices of triangle $A_2B_2C_2$

e) Join the vertices A_1 B_1 B_2 and C. Name the figure formed.

f) Find $\overrightarrow{A_1B_1}$

5. (a) A cylinder closed at one end has radius 7 cm and height 20 cm.

(i) Find its total surface area.

(ii) If the cylinder is filled with water to a depth of 5cm, calculate the volume of the water in it.

[Take $\pi = {}^{22}/_{7}$]

(b) Evaluate $\frac{0.07 \times 0.6}{0.014 \times 0.03}$, leaving your answer in standard form

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MATHEMATICS

SECTION B
ESSAY

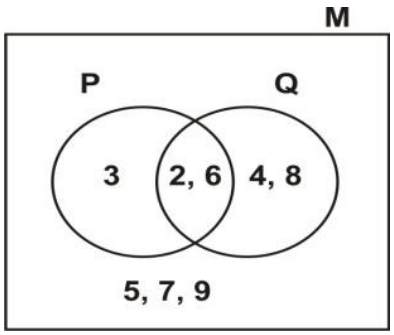
SOLUTIONS

1 (a) (i) M = {2, 3, 4, 5, 6, 7, 8, 9}

P = {2, 3, 6}

Q = {2, 4, 6, 8}

(ii)



(iii) $P \cap Q$ = {2, 6}

1 (b) (i) $\frac{2x-2}{4} - \frac{2x-1}{3} \leq 1$

Approach 1 (Simplifying the LHS first)

$$\frac{3(2x-2)-4(2x-1)}{12} \leq 1$$

$$\Rightarrow \quad \frac{6x-6-8x+4}{12} \leq 1$$

$$\Rightarrow \quad \frac{6x-8x-6+4}{12} \leq 1$$

$$\Rightarrow \quad \frac{-2x-2}{12} \leq 1$$

$$\Rightarrow \quad -2x-2 \leq 12 \times 1$$

$$\Rightarrow \quad -2x \leq 12+2$$

$$\Rightarrow \quad \frac{-2x}{-2} \geq \frac{14}{-2}$$

$$\Rightarrow \quad \underline{\underline{x \geq -7}}$$

1 (b) (i) **Approach 2** (Multiplying by 12 to clear fractions)

$$12\left(\frac{2x-2}{4}\right)-12\left(\frac{2x-1}{3}\right) \leq 12 \times 1$$

$$\Rightarrow \quad 3(2x-2)-4(2x-1) \leq 12$$

$$\Rightarrow \quad 6x-6-8x+4 \leq 12$$

$$\Rightarrow \quad 6x-8x-6+4 \leq 12$$

$$\Rightarrow \quad -2x-2 \leq 12$$

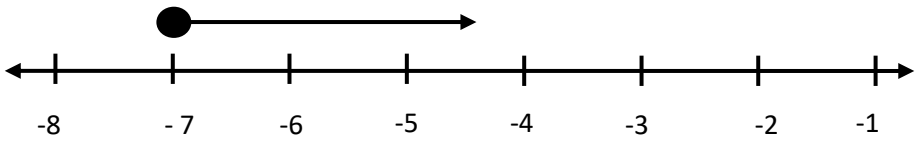
$$\Rightarrow \quad -12-2 \leq 2x$$

$$\Rightarrow \quad \frac{-14}{2} \leq \frac{2x}{2}$$

$$\Rightarrow \quad -7 \leq x$$

$$\Rightarrow \quad \underline{\underline{x \geq -7}}$$

1 (b) (ii)



2 (a) *First change to base ten, then change from base ten to base two (binary numeral)*

Changing $1\ 3\ 1_{\text{five}}$ to a base ten numeral.

1	3	1
5^2	5^1	5^0

$$\Rightarrow \quad (1 \times 5^2) + (3 \times 5^1) + (1 \times 5^0)$$
$$= \quad (1 \times 25) + (3 \times 5) + (1 \times 1)$$
$$= \quad 25 + 15 + 1$$
$$= \quad 41_{\text{ten}}$$

Changing 41_{ten} to base two (binary numeral)

Base	Number	Remainder
2	41	
	20	1
	10	0
	5	0
	2	1
	1	0
	0	1



Reading the remainder column upwards, we have 101001_{two}

131_{five} expressed as a binary numeral = $\underline{101001_{\text{two}}}$

2 (b) (i) Kwabena gets ¼ of 160 oranges

⇒ he gets ¼ × 160 oranges = 40 oranges

⇒ The remainder = 160 – 40 = 120 oranges

Since Esi and Yaw shared the remainder in the ratio 3:2 respectively,

⇒ Esi received $\frac{3}{3+2} \times 120 \text{ oranges} = \frac{3}{5} \times 120 \text{ oranges}$

$= 3 \times 24$ oranges

⇒ <u>Esi received 72 oranges</u>

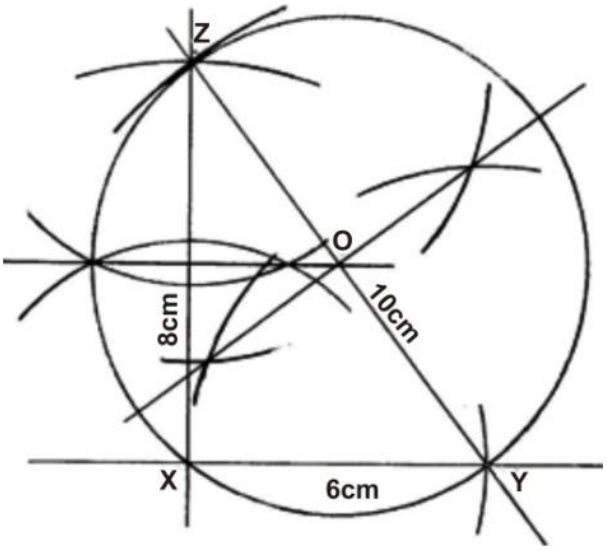
(ii) Yaw received (120 – 72) oranges = 48 oranges

Yaw's share – Kwabena's share = 48 oranges – 40 oranges

= <u>8 oranges</u>

⇒ Yaw received <u>8 more oranges</u> than Kwabena

3 (a), (b)



(c) Radius of circle = <u>5 cm</u>

Circumference = π d

= 3.14 × 10 cm

= 31.4 cm

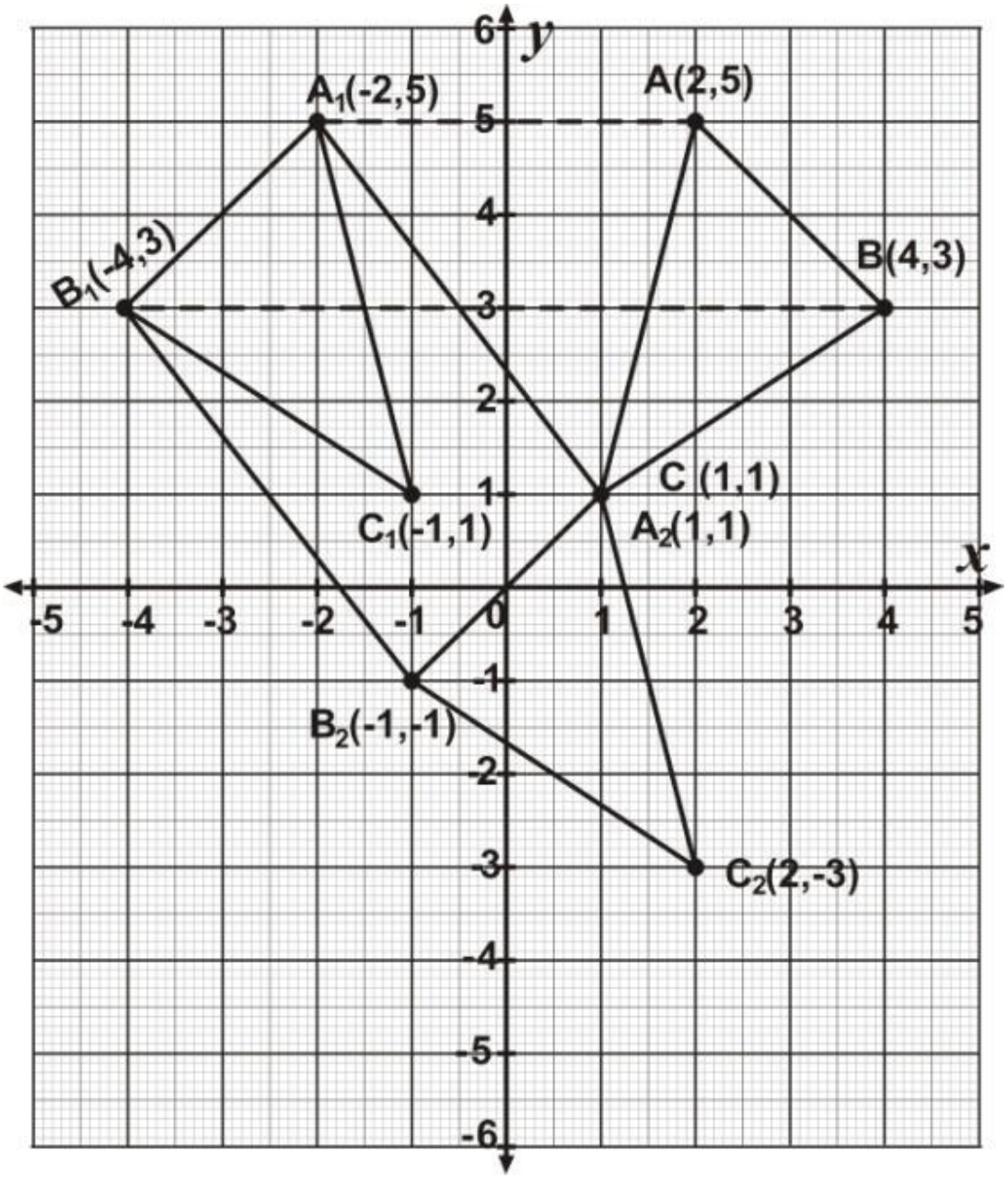
4 (a) – (e) **Approach 1** (Inspection / Construction)

[**NB**: Translating $\Delta A_1B_1C_1$ by vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ simply means

moving $\Delta A_1B_1C_1$ 3 units to the right (3), and then 4 units downwards (-4).

So, e.g., the point $A_1(-2,5)$ moves to $A_2(1,1)$

NB: **Horizontal direction movement first; Vertical direction movement second (always)**



4 (c), (d) **Approach 2** (The rule / formula)

c) Reflecting (x, y) in the y-axis

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} -x \\ y \end{pmatrix}$$

$$\Rightarrow \quad OA\begin{pmatrix} 2 \\ 5 \end{pmatrix} \rightarrow OA_1\begin{pmatrix} -2 \\ 5 \end{pmatrix}, \ \therefore A_1(-2,5)$$

$$\Rightarrow \quad OB\begin{pmatrix} 4 \\ 3 \end{pmatrix} \rightarrow OB_1\begin{pmatrix} -4 \\ 3 \end{pmatrix}, \therefore B_1(-4,3)$$

$$\Rightarrow \quad OC\begin{pmatrix}1\\1\end{pmatrix} \rightarrow OC_1\begin{pmatrix}-1\\1\end{pmatrix}, \therefore C_1(-1,1)$$

∴ Plot and join A_1(-2,5), B_1(-4,3) and C_1(-1,1) as the image of triangle ABC under a reflection in the y axis.

d) Translating (x, y) by vector $\begin{pmatrix}a\\b\end{pmatrix}$

$$\begin{pmatrix}x\\y\end{pmatrix} \rightarrow \begin{pmatrix}x+a\\y+b\end{pmatrix}$$

$$OA_1\begin{pmatrix}-2\\5\end{pmatrix} \rightarrow OA_2\begin{pmatrix}-2+3\\5+-4\end{pmatrix} = \begin{pmatrix}1\\1\end{pmatrix}, \quad \therefore A_2(1, 1)$$

$$OB_1\begin{pmatrix}-4\\3\end{pmatrix} \rightarrow OB_2\begin{pmatrix}-4+3\\3+-4\end{pmatrix} = \begin{pmatrix}-1\\-1\end{pmatrix}, \quad \therefore B_2(-1,-1)$$

$$OC_1\begin{pmatrix}-1\\1\end{pmatrix} \rightarrow OC_2\begin{pmatrix}-1+3\\1+-4\end{pmatrix} = \begin{pmatrix}2\\-3\end{pmatrix}, \quad \therefore C_2(2,-3)$$

∴ Plot and join A_2(1,1), B_2(-1,-1) and C_2(2,-3) as the image of triangle ABC under a translation by vector $\begin{pmatrix}3\\-4\end{pmatrix}$

4 (f) $\overrightarrow{A_1B_1}$

Method 1 (By inspection)

From A_1 to B_1, we move 2 units to the left [horizontal] (-2), then 2 units downwards [vertical] (-2), hence

Vector A_1B_1 $(\overrightarrow{A_1B_1})$ = $\begin{pmatrix}-2\\-2\end{pmatrix}$

NB: Horizontal direction movement first;
Vertical direction movement second (always)

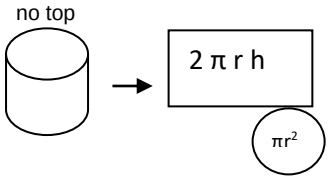
4 (f) **Method 2** (By calculation / the rule)

Vector A_1B_1 $(\overrightarrow{A_1B_1})$ = $\overrightarrow{OB_1}$ - $\overrightarrow{OA_1}$

$$= \begin{pmatrix} -4 \\ 3 \end{pmatrix} - \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

$$= \begin{pmatrix} -4+2 \\ 3-5 \end{pmatrix} = \underline{\underline{\begin{pmatrix} -2 \\ -2 \end{pmatrix}}}$$

5 (a)



(i) Total surface area of cylinder closed at one end

= Areas of (Curved surface+ bottom)

= $2\pi r h + \pi r^2$

= $2 \times {}^{22}/_{7} \times 7\text{cm} \times 20\text{cm} + {}^{22}/_{7} \times (7\text{cm})^2$

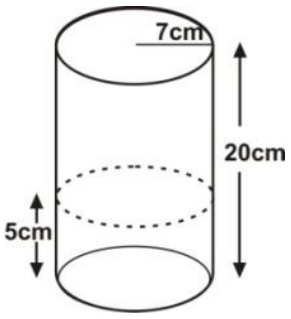
= $2 \times {}^{22}/_{7} \times 7\text{cm} \times 20\text{cm} + {}^{22}/_{7} \times 7\text{cm} \times 7\text{cm}$

= $2 \times 22 \times 20\text{cm}^2 + 22 \times 7\text{cm}^2$

= $880\ \text{cm}^2 + 154\text{cm}^2$

= $\underline{\underline{1034\text{cm}^2}}$

(ii)



Not drawn to scale

Volume of water = Area of base × depth

$= \quad \pi r^2 \quad \times \quad d$

$= \quad {}^{22}/_{7} \times (7\text{cm})^2 \times 5\text{cm}$

$= \quad {}^{22}/_{\cancel{7}} \times \cancel{7}\text{cm} \times 7\text{cm} \times 5\text{cm}$

$= \quad 22 \times 7\text{cm}^2 \times 5\text{cm}$

$= \quad \underline{\underline{770\ \text{cm}^3}}$

The volume of water in the cylinder $= \underline{\underline{770\text{cm}^3}}$

5 (b) $\dfrac{0.07 \times 0.6}{0.014 \times 0.03}$

Method 1 **(Converting to common fractions)**

$= (0.07 \times 0.6) \div (0.014 \times 0.03)$

$= \left(\frac{7}{100} \times \frac{6}{10}\right) \div \left(\frac{14}{1000} \times \frac{3}{100}\right)$

$= \left(\frac{7}{100} \times \frac{6}{10}\right) \times \left(\frac{1000}{14} \times \frac{100}{3}\right)$

$= \quad \dfrac{7 \times 6 \times 100}{14 \times 3}$

$= \quad 100 \quad = \quad \underline{\underline{1 \times 10^2}}$

5 (b) **Method 2** **(Converting to whole numbers)**

Multiplying both numerator and denominator by 100,000

(to convert to whole numbers)

$$\frac{0.07 \times 0.6}{0.014 \times 0.03} \times \frac{100{,}000}{100{,}000}$$

$$= \quad \frac{7 \times 600}{14 \times 3}$$

$= \quad 100 \quad = \quad \underline{\underline{1 \times 10^2}}$

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MATHEMATICS

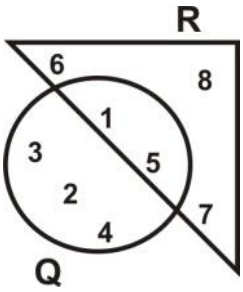
SECTION A
OBJECTIVE TEST
1 hour

1. M = {1, 2, 3, 8, 10} and N = {8, 1, x, 3, 2}.

If M is equal to N, what is the value of x?

A) 1 B) 2 C) 3 D) 8 E) 10

2. In the Venn diagram Q is the set of numbers inside the circle and R is the set of numbers inside the triangle.



Find Q ∩ R.

A. {1, 5}

B. {2, 3, 4}

C. {6, 7, 8}

D. {1,2,3,4,5}

E. {1,2,3,4,5,6,7,8}

3. Simplify: 11 – (11 – 4) + 13.

A) –7 B) –17 C) 9 D) 17 E) 31

4. What is the value of '4' in the number 2,043,507?

A) Forty

B) Four hundred

C) Four thousand

D) Forty thousand

E) Four hundred thousand

5. Which of the following is true?

A) {0, 2, 6, 9, 12} is a subset of even numbers

B) {-1, 0, 2, 3, 5} is a subset of odd numbers.

C) {-2, -1, 1, 3, 9} is a subset of integers.

D) {2, 3, 5, 7, 27} is a subset of prime numbers

E) {9, 18, 21, 27, 36} is a subset of multiples of 9.

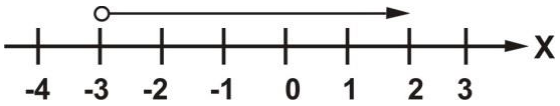
6. If (23 ×82) × 79 = 148,994, find the exact value of (2.3 ×82) × 7.9

A) 1.48994 B) 14.8994 C) 148.994 D) 1489.94 E) 14899.4

7. Convert 206 to a base five numeral.

A) 411_{five} B) 4011_{five} C) 3321_{five} D) 1311_{five} E) 1131_{five}

8. Which of the following inequalities is represented by the number line?



A) $x \leq -3$ B) $x \geq -3$ C) $x < -3$ D) $x > -3$ E) $x = -3$

9. The product of three numbers is 1197. Two of the numbers are 3 and 21. Find the third number

A) 19 B) 57 C) 63 D) 399 E) 1134

10. Simplify: $\begin{pmatrix} -2 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 5 \end{pmatrix}$

A. $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ B. $\begin{pmatrix} -1 \\ 8 \end{pmatrix}$ C. $\begin{pmatrix} -3 \\ 8 \end{pmatrix}$ D. $\begin{pmatrix} -1 \\ -2 \end{pmatrix}$ E. $\begin{pmatrix} -3 \\ -2 \end{pmatrix}$

11. A pencil sells at ¢180.00 and an eraser sells at ¢120.00. how much will you pay if you buy three pencils and four erasers ?

A) ¢1,020.00

B) ¢1,080.00

C) ¢1,200.00

D) ¢1,280.00

E) ¢2,100.00

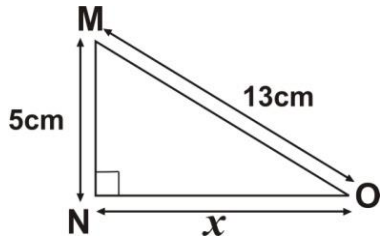
12. Simplify $\frac{0.12 \times 0.08}{2.40}$

A) 0.4 B) 0.04 C) 0.004 D) 0.0004 E) 0.00004

13. How many lines of symmetry has a square ?

A) 2 B) 3 C) 4 D) 6 E) 8

14. In the diagram, MNO is a right-angled triangle. |MO| = 13cm and |MN| is 5cm.



Find the value of x.

A) 3 B) 4 C) 8 D) 12 E) 18

15. Amina spends $^{17}/_{35}$ of her pocket money on transport and food. If she spends $^{2}/_{7}$ on transport only, what fraction does she spend on food ?

A) ¼ B) $^{1}/_{5}$ C) $^{5}/_{7}$ D) $^{15}/_{28}$ E) $^{18}/_{35}$

16. If $x \rightarrow 3x - 4$, what is the image of –2 ?

A) –10 B) –2 C) –1 D) 2 E)10

17. Simplify: $(2ab^2)(3a^2b)$

A) $5a^2b^2$ B) $6a^2b^2$ C) $5a^2b^3$ D) $6a^3b^2$ E) $6a^3b^3$

18. It takes 15 men, 48 days to weed a plot of land. How many men can weed the same plot of land in 16 days, if they work at the same rate?

A) 5 B) 18 C) 32 D) 45 E) 48

19. The ratio 9:x is equivalent to 36:20. What is the value of x ?

A) 4 B) 5 C) 6 D) 8 E) 10

20. In an enlargement, PQ→P′Q′. |PQ| = 3cm and |P′Q′| =15. Calculate the scale factor of the enlargement.

A) $^1/_5$ B) $^2/_3$ C) 5 D) 10 E) 45

21. Simplify: 7a – 3(b – a)

A) 4a–3b B) 6a–3b C) 8a–3b D) 10a–3b E)10a+3b

22. A man deposited an amount of money in his savings account for 5 years. The rate of interest was 14% per annum. If the interest was ¢35,000.00, find the amount deposited.

A) ¢85,000.00

B) ¢50,000.00

C) ¢39,900.00

D) ¢24,500.00

E) ¢15,000.00

23. Arrange the following fractions from the lowest to the highest: ¾ , $^2/_3$ and $^3/_5$.

A) $^3/_5$, $^2/_3$, ¾

B) $^3/_5$, ¾ , $^2/_3$

C) ¾ , $^2/_3$, $^3/_5$

D) ¾ , $^3/_5$, $^2/_3$

E) $^2/_3$, $^3/_5$, ¾

24. The marked price of a radio set is ¢450,000. A discount of 5% of the marked price is allowed. What is the selling price of the radio set?

A) ¢427,000

B) ¢427,500

C) ¢428,571

D) ¢472,500

E) ¢473,684

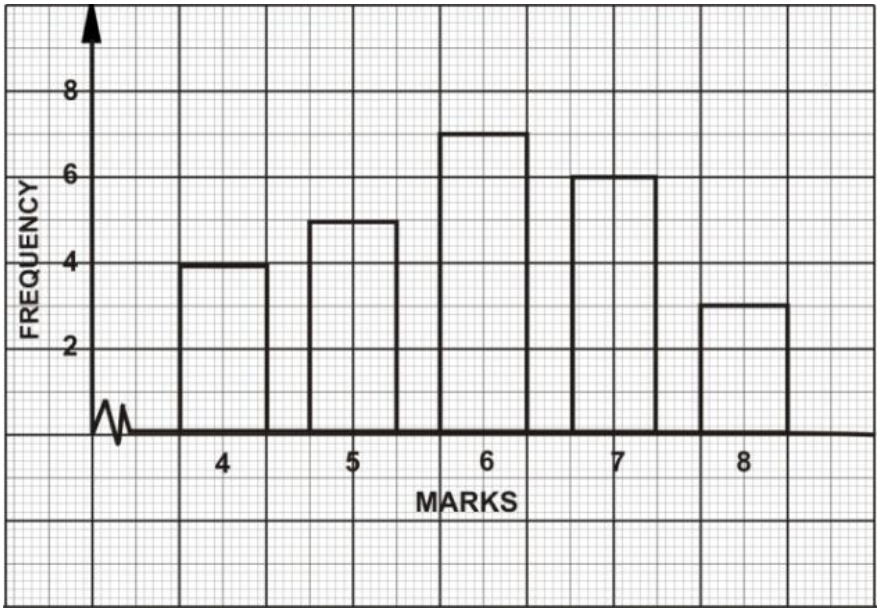
25. If 60% of the pupils in a school is 240, find the total enrolment in the school.

A) 144 B) 160 C) 360 D) 384 E) 400

26. If a * b = 2a – b, evaluate 4 * 3

A) 1 B) 2 C) 3 D) 4 E) 5

The bar chart shows the mark distribution of pupils in a test. Use it to answer questions **27** *and* **28**



27. What is the modal mark?

A) 4 B) 5 C) 6 D) 7 E) 8

28. How many pupils took the test?

A) 5 B) 20 C) 25 D) 30 E) 37

29. A school has a population of 600. Out of this, 120 are girls. What is the probability of meeting a pupil in the school who is a boy?

A) ¼ B) $^1/_5$ C) $^3/_5$ D) $^4/_5$ E) $^{12}/_{25}$

30. If $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$, find k when $k_1 = 1$ and $k_2 = 2$.

A) ½ B) $^2/_3$ C) $^3/_2$ D) 2 E) 3

31. Solve the equation $13x - 2(3x + 4) = 22$.

A) 2 B) 4 C) $^{18}/_{7}$ D) $^{26}/_{7}$ E) $^{30}/_{7}$

32. A rectangular tank is 4m long, 3m wide and 2.5m high. What is the volume of the tank?

A) $24m^3$ B) $30m^3$ C) $36m^3$ D) $48m^3$ E) $60m^3$

33. Write the rule for the mapping:

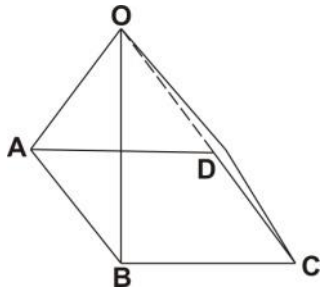
x	1	2	3	4
↓	↓	↓	↓	↓
y	1	3	5	7

A) $x \rightarrow 2x+1$ B) $x \rightarrow 2x-1$ C) $x \rightarrow 2(x+1)$ D) $x \rightarrow 2(x-1)$ E) $x \rightarrow x^2 - 1$

34. If $\mathbf{r} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\mathbf{s} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$, calculate $2\mathbf{r} - 3\mathbf{s}$

A. $\begin{pmatrix} -5 \\ 12 \end{pmatrix}$ B. $\begin{pmatrix} -5 \\ -8 \end{pmatrix}$ C. $\begin{pmatrix} 13 \\ -3 \end{pmatrix}$ D. $\begin{pmatrix} 13 \\ -8 \end{pmatrix}$ E. $\begin{pmatrix} 13 \\ 8 \end{pmatrix}$

35. Name the geometrical figure shown in the diagram.



A) parallelogram

B) triangle

C) cone

D) tetrahedron

E) pyramid

36. A car travels 72 kilometres in an hour. Find its speed in metres per second.

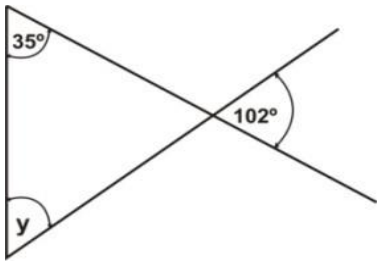
A) $5\ m\ s^{-1}$ B) $20\ m\ s^{-1}$ C) $50\ m\ s^{-1}$ D) $200\ m\ s^{-1}$ E) $1200\ m\ s^{-1}$

37. The circumference of a circular track is 154m. Find the diameter of the track.

(Take $\pi = {}^{22}/_{7}$)

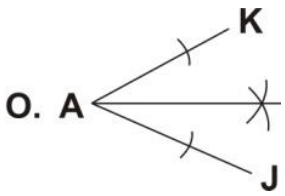
A) 22.0 m B) 24.5 m C) 49.0 m D) 242.0 m E) 484.0 m

38. Find the value of the angle marked y in the diagram.



A) 35° B) 43° C) 67° D) 78° E) 137°

39. Which of the following best describes the construction in the diagram?



A) Constructing a 30° angle
B) Constructing a 60° angle
C) Bisecting a line segment
D) Bisecting a given angle
E) Drawing a perpendicular from a given point.

40. Ama is N years old now. How old will she be in 10 years?

A. (N – 10) years
B. (N + 10) years
C. (10 – N) years
D. 10 N years
E. ${}^{10}/_{N}$ years

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August 2000

MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. E) 10

2. A. {1, 5}

3. D) 17

4. D) Forty thousand

5. C. {-2, -1, 1, 3, 9} is a subset of integers

6. D) 1489.94

7. D) 1311_{five}

8. D) $x > -3$

9. A) 19

10. E. $\begin{pmatrix} -3 \\ -2 \end{pmatrix}$

11. A) ¢1,020.00

12. C) 0.004

13. C) 4

14. D) 12

15. B) $^1/_5$

16. A) –10

17. E) $6a^3b^3$

18. D) 45

19. B) 5

20. C) 5

21. D) 10a–3b

22. B) ¢50,000.00

23. A) $^3/_5, ^2/_3, ¾$

24. B) ¢427,500

25. E) 400

26. E) 5

27. C) 6

28. C) 25

29. D) $^4/_5$

30. B) $^2/_3$

31. E) $^{30}/_7$

32. B) $30m^3$

33. B) $x \rightarrow 2x\text{-}1$

34. D. $\begin{pmatrix} 13 \\ -8 \end{pmatrix}$

35. E) pyramid

36. B) 20 m s^{-1}

37. C) 49.0 m

38. B) 43°

39. D) bisecting a given angle

40. B. (N + 10) years

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MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

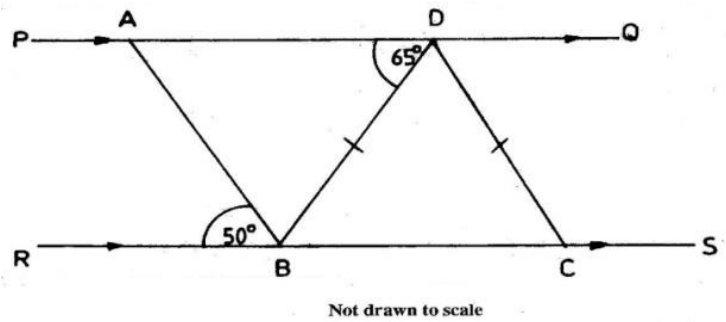
All questions carry equal

1. (a) Simplify: $^2/_3$ of $6\ ¾ \div (2\ ^4/_{15} - 1\ ^2/_3)$.

(b) Solve the equation $\frac{1}{3}(x+3) - 2(x-5) = 4\frac{1}{3}$

(c) If $3y = 2x^2 - 3x + 7$, find y, when x = 5.

2. (a) In the diagram, PADQ and RBCS are parallel lines. $|BD| = |DC|$, angle ADB = 65° and angle ABR = 50°.



Not drawn to scale

(i) Calculate the angle BDC

(ii) Calculate angle ABD

(iii) Find angle BAD

(iv) What type of triangle is triangle ABD ?

(b) Using a ruler and a pair of compasses only, construct triangle XYZ, with |YZ| = 8cm, angle XYZ = 60° and |XY|=9cm.

Measure (i) angle YZX; (ii) |XZ|

3. Ama was granted a loan of ¢800,000.00 by a bank. The rate of interest was 42% per annum.

(a) Calculate

(i) the interest at the end of the year;

(ii) the total amount Ama had to pay at the end of the year.

(b) Ama was able to pay only ¢700,000.00 at the end of the year.

(i) Find how much Ama still owed the bank

(ii) Express the amount Ama owed after paying the ¢700,000.00 to the bank as a percentage of the loan she took from the bank.

4. The following is a record of scores obtained by 30 JSS form 2 pupils in a test marked out of 5.

5, 3, 2, 4, 5, 2,
4, 3, 1, 1, 3, 3,
4, 2, 3, 4, 5, 3,
4, 3, 2, 4, 3, 1,
2, 2, 3, 3, 2, 4.

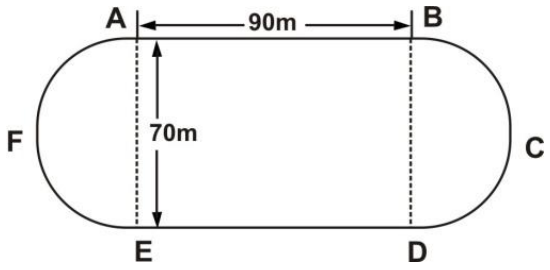
Score (x)	*Tally*	*Frequency (f)*	*fx*
1			
2			
3			
4			
5			
	Total		

a) Copy and complete the table.

b) Find the mean of the distribution.

c) If a pupil is selected at random from the form, what is the probability that he / she scored 4 marks ?

5.



The diagram shows a running track ABCDEFA. AB and ED are the straight sides. The ends AFE and BCD are semi-circular shapes.

|AB| = |ED| = 90m and |AE| = |BD| = 70m.

Find

a) the total length of the two semi-circular ends, AFE and BCD;

b) the perimeter of the running track ABCDEFA;

c) the total area of the running track ABCDEF.
(Take $\pi = {}^{22}/_{7}$)

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August 2000

MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1 (a) $\frac{2}{3}\ of\ 6\frac{3}{4} \div (2\frac{4}{15} - 1\frac{2}{3})$

$\Rightarrow \quad \frac{2}{3}\ of\ \frac{27}{4} \div (\frac{34}{15} - \frac{5}{3})$

$\Rightarrow \quad \frac{2}{3}\ of\ \frac{27}{4} \div (\frac{34\ -\ 25}{15})$

$\Rightarrow \quad \frac{2}{3} \times \frac{27}{4} \div \frac{9}{15}$

$\Rightarrow \quad \frac{2}{3} \times \frac{27}{4} \times \frac{15}{9}$

$\Rightarrow \quad \frac{15}{2}$

$\Rightarrow \quad \underline{\underline{7\frac{1}{2}}}$

1 (b) $\frac{1}{3}(x+3) - 2(x-5) = 4\frac{1}{3}$

$\Rightarrow \quad \frac{1}{3}(x+3) - 2(x-5) = \frac{13}{3}$

$\Rightarrow \quad 3 \times \frac{1}{3}(x+3) - 3 \times 2(x-5) = 3 \times \frac{13}{3}$

$\Rightarrow \quad (x+3) - 6(x-5) = 13$

$\Rightarrow \quad x + 3 - 6x + 30 = 13$

$\Rightarrow \quad x - 6x = 13 - 3 - 30$

$\Rightarrow \quad -5x = -20$

$\Rightarrow \quad \frac{-5x}{-5} = \frac{-20}{-5}$

$\Rightarrow \quad \underline{\underline{x = 4}}$

1 (c) When $x = 5$, we have

$$3y = 2(5)^2 - 3(5) + 7$$
$$\Rightarrow \quad 3y = 2\times25 - 3\times5 + 7$$
$$\Rightarrow \quad 3y = 50 - 15 + 7$$
$$\Rightarrow \quad 3y = 42$$
$$\Rightarrow \quad y = \frac{42}{3}$$
$$\Rightarrow \quad \underline{\underline{y = 14}}$$

2(a) (i) $\angle CBD = \angle ADB = 65°$ (Alternate angles)

Also, $\angle BCD = \angle CBD = 65°$ (base angles of isosceles Δ)

Now taking ΔBCD,

$$\angle BDC + \angle CBD + \angle BCD = 180°$$
$$\Rightarrow \quad \angle BDC + 65° + 65° = 180°$$
$$\Rightarrow \quad \angle BDC + 130° = 180°$$
$$\Rightarrow \quad \angle BDC = 180° - 130°$$
$$\Rightarrow \quad \underline{\underline{\angle BDC = 50°}}$$

(ii) Since $\angle ABR$, $\angle ABD$ and $\angle CBD$ lie on a straight line,

$$\Rightarrow \quad \angle ABR + \angle ABD + \angle CBD = 180°$$
$$\Rightarrow \quad 50° + \angle ABD + 65° = 180°$$
$$\Rightarrow \quad \angle ABD + 115° = 180°$$
$$\Rightarrow \quad \angle ABD = 180° - 115°$$
$$\Rightarrow \quad \underline{\underline{\angle ABD = 65°}}$$

(iii) Taking Δ ABD,

$$\angle BAD + \angle ABD + \angle ADB = 180°$$
$$\Rightarrow \quad \angle BAD + 65° + 65° = 180°$$

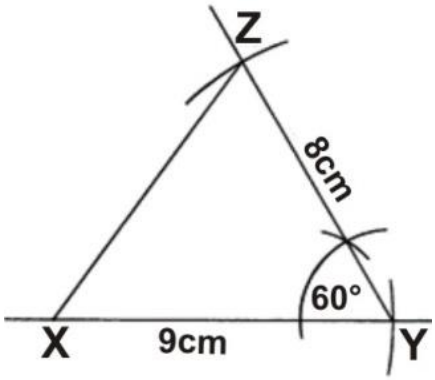
$\Rightarrow \quad \angle BAD + 130° = 180°$

$\Rightarrow \quad \angle BAD = 180° - 130°$

$\Rightarrow \quad \underline{\angle BAD = 50°}$

(iv) Triangle ABD is <u>isosceles</u> (since it has two equal interior angles)

2(b)



(i) angle YZX ≈ <u>67°</u>

(ii) |XZ| ≈ <u>8.5 cm</u>

3(a)(i) Simple interest = Principal × Rate × Time

= 800,000 × 42% × 1

$= 800{,}000 \times \frac{42}{100}$

= 8000 × 42

= 336,000

∴ The simple interest at the end of the year = <u>¢336,000.00</u>

(ii) The amount to be paid

= Principal + S. Interest

= ¢800,000 + ¢336,000

= <u>¢1,136,000</u>

(b) (i) Amount still owed

= Total amount – amount paid

= ¢1,136,000 – ¢700,000

= ¢436,000

(ii) Amount still owed as a percentage of loan taken

$$= \frac{\text{Amount still owed}}{\text{Loan taken (Principal)}} \times 100\%$$

$$= \frac{¢436,000}{¢800,000} \times 100\%$$

$$= \frac{436}{8}\ \%$$

= 54.5% or 54½%

4(a)

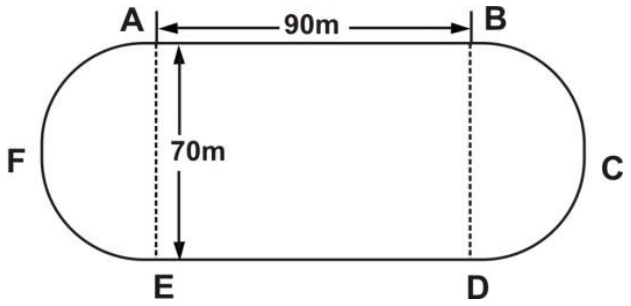
Score (*x*)	*Tally*	*Frequency* (*f*)	*fx*
1	///	3	3
2	~~////~~ //	7	14
3	~~////~~ ~~////~~	10	30
4	~~////~~ //	7	28
5	///	3	15
	Total	30	90

4(b) The mean $= \frac{\Sigma f x}{\Sigma f} = \frac{90}{30} = 3$

4(c) $\text{P(4 marks)} = \frac{\text{Number of pupils that obtained 4 marks}}{\text{Total number of pupils}}$

$$= \frac{7}{30}$$

5(a)



a) Total length of two semi-circular ends AFE and BCD

Approach 1

Total length = 2 semi-circular arcs of diameter 70m

= 2 × length of semi-circular arc

= 2 × ½ (π d)

= π d

= $^{22}/_{7}$ × 70m

= 22 × 10m

= 220 m

5(a)

Approach 2

Total length = AFE length + BCD length

= ½ π d + ½ π d

= (½ × $^{22}/_{7}$ ×70m) + (½ × $^{22}/_{7}$ × 70m)

= 11 × 10m + 11 × 10m

= 110m + 110m

= 220 m

(b) Perimeter of running track ABCDEFA

⇒ Distance around the track

= AB + BCD + ED + AFE

= 90m + 110m + 90m + 110m

= 400 m

(c) Total area of the running track ABCDEF

Approach 1

Total area = Two semi-circles area + rectangle area

= 2 (½ π r^2) + *l w*

$= \pi r^2 + l\ w$

$= {}^{22}/_{7} \times (35)^2 + 90 \times 70$

$= {}^{22}/_{7} \times 35 \times 35 + 90 \times 70$

$= 22 \times 5 \times 35 + 90 \times 70$

$= 3850 + 6300$

$= 10150$

⇒ Total area of the running track ABCDEF = $\underline{10150\ cm^2}$

(c) **Approach 2**

Total area = Area of (AFE + ABDE + BCD)

$= \frac{1}{2}\pi r^2 + l\ w + \frac{1}{2}\pi r^2$

$= [\frac{1}{2} \times {}^{22}/_{7} \times (35)^2] + [90 \times 70] + [\frac{1}{2} \times {}^{22}/_{7} \times (35)^2]$

$= 1925 + 6300 + 1925$

$= 10150$

⇒ Total area of the running track ABCDEF = $\underline{10150\ cm^2}$

August 1999

MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. Set A is called ……of set B, when all the numbers of set A are also members of set B.

A) the universal set

B) the union set

C) the null set

D) a subset

E) an empty set

2. M = {multiples of 3 between 10 and 20}

N = {even numbers between 10 and 20}.

Find M ∩ N.

A. {12, 18}

B. {12, 15, 18}

C. {12, 14, 16, 18}

D. {12, 14, 15, 16, 18}

E. {10, 12, 14, 15, 18, 20}

3. Express 24 as a product of prime factors.

A) 2×3 B) $2^2 \times 3$ C) $2^3 \times 3$ D) $2^2 \times 3^2$ E) $2^2 \times 3^3$

4. Write in standard form 1342.

A) 0.1342×10^{-3}

B) 0.1342×10^{-4}

C) 13.42×10^{2}

D) 1.342×10^{3}

E) 1.342×10^{4}

5. Change 10111_{two} to base ten.

A) 15 B) 17 C) 23 D) 25 E) 29

6. If Y = 595 and Z = 7071, find the sum of Y and Z.

A) 6466 B) 7566 C) 7666 D) 12,021 E) 13,021

7. Simplify: $\frac{1}{2} - \frac{1}{4} + \frac{1}{8}$

A) $\frac{1}{8}$ B) $\frac{1}{6}$ C) $\frac{3}{8}$ D) $\frac{5}{8}$ E) $\frac{7}{8}$

8. A ribbon is 4 m long. How many pieces, each 30 cm long, can be cut from the ribbon?

A) 7 B) 11 C) 12 D) 13 E) 14

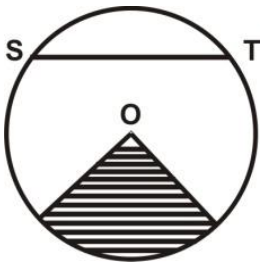
9. Find the integers within the interval $5 < x < 9$

A. {5,6,7} B. {5,6,7,8} C. {5,6,7,8,9} D. {6,7,8} E. {6,7,8,9}

10. Calculate, correct to two decimal places, $0.61 \div 0.8$

A) 0.07 B) 0.08 C) 0.76 D) 0.83 E) 7.62

The diagram below shows a circle with centre O, S and T are points on the circle.
Use it to answer questions **11** *and* **12**



11. What name is given to the shaded region ?

A) sector
B) segment
C) radii
D) arc
E) cone

12. The line ST is called

A) an arc

B) a chord

C) a diameter

D) a radius

E) a segment

13. Simplify: $2 \times 3^2 \times 3^4$

A) 2×3^5 B) 2×3^6 C) 2×3^8 D) 2×9^6 E) 2×9^8

14. Which of the following fractions is the greatest? $^1/_6$, ¼ , ½ , $^1/_3$, $^1/_5$

A) $^1/_6$ B) $^1/_5$ C) ¼ D) $^1/_3$ E) ½

15. The sum of the ages of a man and his wife is 81 years. The ratio of their ages is 5:4. Find the age of the younger person.

A) 30 years B) 36 years C) 45 years D) 51 years E)65 years

16. A car is travelling at 40km per hour. How far does it travel in 2 ½ hours?

A) 16km B) 80km C) 90km D) 100km E) 140km

17. There are 6 girls and 18 boys in a class. What percentage of the class are girls?

A) 14.40% B) 25.00% C) 33.33% D) 66.67% E) 75.00%

18. Find the missing numbers in the sequence 4, 8, 12, _ , _ , _ , 28

A) 14, 16, 22

B) 14, 18, 22

C) 6, 18, 22

D) 16, 20, 24

E) 16, 22, 24

19. A shop increased all its prices by 10%. Calculate the new price for an article which previously sold for ¢7,500.00

A) ¢6,750.00

B) ¢7,575.00

C) ¢7,800.00

D) ¢8,250.00

E) ¢8,350.00

20. Simplify: $7(y + 1) – 2(2y + 3)$

A) $3y – 5$ B) $3y – 2$ C)$3y +1$ D) $3y + 4$ E) $3y + 13$

The table below shows the average monthly rainfall at Nankese from March, 1996 to August, 1996.

Use it to answer question **21** *and* **22**

Month	Mar	Apr	May	Jun	Jul	Aug
Rainfall (mm)	99	145	277	450	267	142

21. Which month recorded the highest amount of rainfall?

A) March B) April C) May D) June E) July

22. What was the total amount of rainfall recorded in April, May and July ?

A) 422mm B) 689mm C) 72mm D)994mm E) 1,139mm

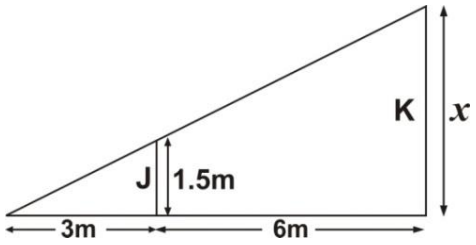
23. Find the median of the following numbers: 1, -1, -5, 3 and -4

A) –1 B) –4 C) –5 D) 1 E) 3

24. There are 15 red and 25 black balls in a bag. Find the probability of selecting a black ball from the bag.

A) $^1/_{25}$ B) $^1/_{15}$ C) $^3/_8$ D) $^3/_5$ E) $^5/_8$

In the diagram below, K is an enlargement of J. Use it answer questions **25** *and* **26**



25. Calculate the scale factor

A) $^1/_3$ B) ½ C) 2 D) 3 E) 9

26. Find the value of x.

A) 3.0m B) 3.5m C) 4.5m D) 6.0m E) 9.0m

27. A trader received a commission of 25% on the sales he made in a month. His commission was ¢180,000.00. Find his total sales for the month.

A) ¢135,000.00
B) ¢150,000.00
C) ¢240,000.00
D) ¢540,000.00
E) ¢720,000.00

28. Kwame travelled from Accra to Kumasi. He travelled $^{3}/_{10}$ of the journey by lorry, $^{2}/_{5}$ of the journey by taxi and the rest by train. What fraction of the journey did he travel by train ?

A) $^{3}/_{10}$ B) $^{3}/_{5}$ C) $^{7}/_{10}$ D) $^{3}/_{25}$ E) $^{22}/_{25}$

29. If 4 (m + 4) = 18, find the value of m.

A) ½ B) $^{3}/_{2}$ C) $^{5}/_{2}$ D) $^{7}/_{2}$ E) $^{17}/_{2}$

Use the mapping below to answer questions **30** *and* **31**

x	1	2	3	4	...
↓	↓	↓	↓	↓	
y	3	5	7	9	...

30. What is the rule for the mapping ?

A) $x \rightarrow 4 - x$ B) $x \rightarrow x-2$ C) $x \rightarrow 2x$ D) $x \rightarrow 2x+1$ E) $x \rightarrow 3x$

31. Find the value of y when x is 7.

A) –3 B) 9 C) 14 D) 15 E) 21

32. A square of side 6cm has the same area as a rectangle of length 9cm. Find the breadth of the rectangle.

A) 3cm B) 4cm C) 6cm D) 24cm E) 36cm

33. Factorize $3r^2s - 9rs^2$

A. rs (3r – s)
B. 3rs(s – 3r)
C. 3rs(r - 3s)
D. r^2s^2 (3r – 9s)
E. $3r^2s^2$(s – 3r)

34. The average mass of 4 boys is 45 kg. When a fifth boy joins them, the average mass changes to 40 kg. What is the mass of the fifth boy?

A) 5 kg B) 10 kg C) 15 kg D) 20 kg E) 25 kg

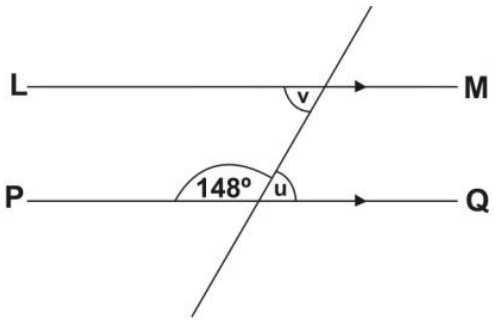
35. E is the point (4, 2) and F the point (2, 1). Calculate the gradient of the straight line EF.

A) – ½ B) –2 C) 2 D) ½ E) 0

36. $\boldsymbol{l} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ and $\boldsymbol{k} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$. Find $\boldsymbol{l} + \boldsymbol{k}$.

A. $\begin{pmatrix} -1 \\ -1 \end{pmatrix}$ B. $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ C. $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ D. $\begin{pmatrix} 7 \\ -1 \end{pmatrix}$ E. $\begin{pmatrix} 7 \\ 1 \end{pmatrix}$

37.



In the diagram, ML and PQ are parallel lines. Find the value of v

A) 32° B) 40° C) 58° D) 140° E) 180°

38. Given the relation $\frac{1}{y} = \frac{1}{c} + \frac{1}{x}$ which of the following expressions is equal to *y*?

A) $c + x$

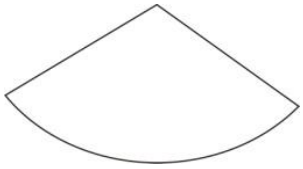
B) $\frac{2}{cx}$

C) $\frac{c + x}{2}$

D) $\frac{cx}{c + x}$

E) $\frac{c+x}{cx}$

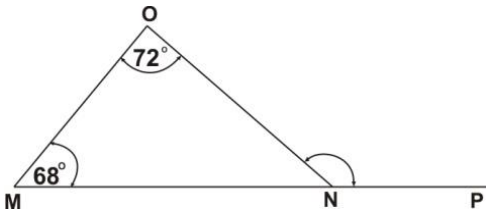
39.



The figure above is the net of

A) a cylinder
B) a triangle
C) a pyramid
D) a cone
E) a sphere

40. In the diagram below, MNO is a triangle. Angle MON = 72° and angle OMN = 68°. Find angle ONP



A) 40° B) 68° C) 72° D) 112° E) 140°

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August 1999

MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. D) a subset

2. A. {12, 18}

3. C) $2^3 \times 3$

4. D) 1.342×10^3

5. C) 23

6. C) 7666

7. C) $^3/_8$

8. D) 13

9. D. {6,7,8}

10. C) 0.76

11. A) sector

12. B) a chord

13. B) 2×3^6

14. E) ½

15. B) 36 years

16. D) 100km

17. B) 25.00%

18. D) 16,20,24

19. D) ¢8,250.00

20. C)3y +1

21. D) June

22. B) 689mm

23. A) –1

24. E) $^{5}/_{8}$

25. D) 3

26. C) 4.5m

27. E) ¢720,000.00

28. A) $^{3}/_{10}$

29. A) ½

30. D) $x \rightarrow 2x+1$

31. D) 15

32. B) 4 cm

33. C. 3rs(r - 3s)

34. D) 20 kg

35. D) ½

36. B. $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

37. A) 32°

38. D) $\frac{cx}{c+x}$

39. D) a cone

40. E) 140°

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MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. (a) Kofi and Kojo were given ¢38,000.00 to share. Kojo had ¢7,500.00 more than Kofi. Find

(i) Kofi's share

(ii) Kojo's share

(b) A trader paid ¢1500.00 for 6 drinking cups. One of the cups got broken. He sold the remaining 5, making a profit of 10%. Calculate

(i) the cost price of each of the six cups;

(ii) the selling price of each of the five cups;

(iii) the profit made on each cup sold.

2. (a) Solve the inequality $\frac{2x-1}{4} - \frac{x-2}{3} > 1$

(b) Find the value of the expression $2x - 3y$, if $x = \frac{1}{3}$ and $y = -\frac{1}{2}$

(c) 25 students in a class took an examination in Mathematics and Science. 17 of them passed in Science and 8 passed in both Mathematics and Science. 3 students did not pass in any of the subjects.

Find

(i) how many passed in Mathematics

(ii) the probability of meeting a student who passed in one subject only.

3. (a) Using a pair of compasses and a ruler only,

(i) construct triangle ABC such that |AB| = 10cm, angle ABC = 30° and |BC| = 8cm. Measure angle ACB.

(ii) Construct a perpendicular from C to meet line AB at D. Measure |CD|.

(b) Calculate the area of triangle ABC.

4. a) (i) Using a scale of 2cm to 2 units on both axes, draw two perpendicular axes OX and OY on a graph sheet.

(ii) On the same graph sheet, mark the x-axis from –10 to 10 and the y-axes from –12 to 12.

(iii) Plot the points A(0, 10), B(-6, -2), C(4, 3) and D(-3,-11).

Use a ruler to join the point A to B and also point C to D.

b) (i) Draw the line x = -2 to meet AB at P and CD at Q.

(ii) Use a protractor to measure angles BPQ and PQC.

(iii) What is the common name given to angles BPQ and PQC ?

(iv) State the relationship between lines AB and CD.

5. The table below shows the marks scored out of 10 by some candidates in a test.

Mark	1	2	3	4	5	6	7	8
Number of candidates	2	3	5	7	8	13	7	5

a) From the above table, find

(i) the modal mark;

(ii) how many candidates took the test;

(iii) the mean mark for the test

b) If 20% of the candidates failed,

(i) how many failed?

(ii) What is the least mark a candidate should score in order to pass?

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MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1 (a) Since Kojo had ¢7,500 more than Kofi,

⇒ (Kojo's) = (¢7,500 + Kofi's) ----------[1]

also Kofi's+ (Kojo's) = ¢38,000 ----------[2]

Then, from equations [1] and [2],

⇒ Kofi's + (¢7,500 + Kofi's) = ¢38,000 ----------[3]

(i) Let f = Kofi's share, then from [3],

⇒ f + (¢7,500 + f) = ¢38,000

⇒ $2f$ + ¢7,500 = ¢38,000

⇒ $2f$ = ¢38,000 – ¢7,500

⇒ $2f$ = ¢ 30,500

⇒ $\frac{2f}{2} = \frac{30{,}500}{2}$

⇒ f = ¢15,250

Therefore Kofi's share is ¢15,250.00

(ii) **Method 1**

Since Kojo had ¢7,500 more than Kofi,

⇒ Kojo's share = ¢7,500 + ¢15,250

= ¢ 22,750.00

(ii) **Method 2**

Kojo's share = Total amt – Kofi's share

= ¢38,000 – ¢15,250

= ¢ 22,750.00

1 (b) (i) If CP of 6 cups = ¢1500.00,

then CP of 1 cup = ¢1,500 ÷ 6

= ¢250

1 (b) (ii) **METHOD 1** (Formula 1)

$$SP = \frac{\text{SP Percentage}}{\text{CP Percentage}} \times CP$$

$$= \frac{110\,\%}{100\,\%} \times ¢\,250$$

$$= 11 \times ¢\,25$$

$$= ¢\,275$$

1 (b) (ii) **METHOD 2** (Formula 2)

SP = CP + Profit

= ¢250 + 10% of ¢250

$$= ¢250 + \frac{10}{100} \times ¢250$$

= ¢ 250 + ¢25

= ¢275

∴ The selling price of each of the five cups = ¢ 275

1 (b) (ii) **METHOD 3** (Equivalent Fractions)

	Percentage	Amount (¢)
CP of 1 cup	**100%**	**250**
SP of 1 cup	100% + 10% = **110%**	**? (*x*)**

From the table $\frac{100}{110} = \frac{250}{x}$

Cross-multiplying, we have $100 \times x = 250 \times 110$

$$\Rightarrow \quad x = \frac{250 \times 110}{100} = 275$$

∴ The selling price of each of the five cups = ¢ 275

1 (b) (ii) **METHOD 4** (If more, less divides …)

From the table above,

If 100% → 250

Then 110% → ? (more)

If more, less (i.e., 100%) divides; hence we have

$\frac{110}{100} \times 250 = 11 \times 25 =$ ¢275

∴ The selling price of each of the five cups = ¢ 275

1 (b) (iii) Profit made on each cup sold

= SP – CP

= ¢275 – ¢250 = ¢25

2 (a) **Approach 1** (Simplifying the LHS first)

$$\frac{2x-1}{4} - \frac{x-2}{3} > 1$$

$$\Rightarrow \quad \frac{3(2x-1) - 4(x-2)}{12} > 1$$

$$\Rightarrow \quad \frac{6x-3-4x+8}{12} > 1$$

$$\Rightarrow \quad \frac{6x-4x-3+8}{12} > 1$$

$$\Rightarrow \quad \frac{2x+5}{12} > 1$$

$$\Rightarrow \quad 2x+5 > 12$$

$$\Rightarrow \quad 2x > 12-5$$

$$\Rightarrow \quad \frac{2x}{2} > \frac{7}{2}$$

$$\Rightarrow \quad \underline{\underline{x > 3\tfrac{1}{2}}} \qquad \text{or} \qquad \underline{\underline{x > 3.5}}$$

2 (a) **Approach 2** (Multiplying by 12 to clear fractions)

$$12\left(\frac{2x-1}{4}\right) - 12\left(\frac{x-2}{3}\right) > 12\times 1$$

$$\Rightarrow \quad 3(2x-1) - 4(x-2) > 12$$

$$\Rightarrow \quad 6x - 3 - 4x + 8 > 12$$

$$\Rightarrow \quad 6x - 4x + 8 - 3 > 12$$

$$\Rightarrow \quad 2x + 5 > 12$$

$$\Rightarrow \quad 2x > 12-5$$

$$\Rightarrow \quad \frac{2x}{2} > \frac{7}{2}$$

$$\Rightarrow \quad \underline{\underline{x > 3\tfrac{1}{2}}} \qquad \text{or} \qquad \underline{\underline{x > 3.5}}$$

2 (b) $2x - 3y$, if $x = \frac{1}{3}$ and $y = -\frac{1}{2}$

$$2(\tfrac{1}{3}) - 3(-\tfrac{1}{2})$$

$$\Rightarrow \quad 2\times\tfrac{1}{3} - 3\times-\tfrac{1}{2}$$

$$\Rightarrow \quad \tfrac{2}{3} - (-\tfrac{3}{2})$$

$$\Rightarrow \quad \tfrac{2}{3} + \tfrac{3}{2}$$

$$\Rightarrow \quad \frac{4 \quad + \quad 9}{6}$$

$$\Rightarrow \quad \frac{13}{6} \quad = \quad \underline{\underline{2\,1/6}}$$

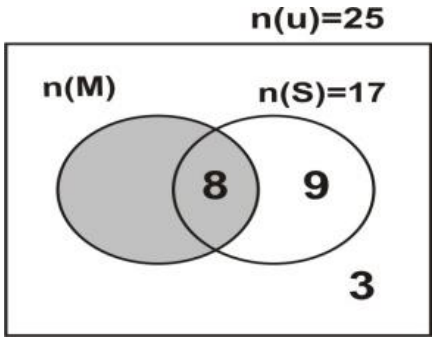
2 (c) (i) Method 1

Let n(U) = No. of students in the class

n(M) = No. of students that passed in Maths

n(S) = No. of students that passed in Science = 17

x = No. of students that passed in Science only (= 17 – 8) = 9



From the diagram above,

$$n(M) + 9 + 3 = 25$$

$$\Rightarrow \quad n(M) + 12 = 25$$

$$\Rightarrow \quad n(M) = 25 - 12$$

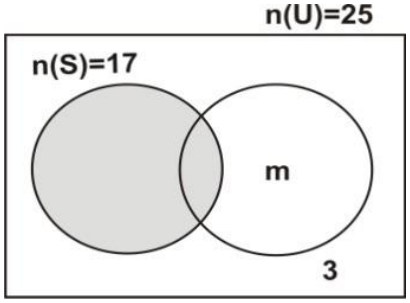
$$\Rightarrow \quad n(M) = \underline{13}$$

2 (c)(i) Method 2

Let n(U) = No. of students in the class

n(S) = No. of students that passed in Science

m = No. of students that passed in Maths only



From the above diagram,

$$n(S) + m + 3 = 25$$
$$\Rightarrow 17 + m + 3 = 25$$
$$\Rightarrow 20 + m = 25$$
$$\Rightarrow m = 25 - 20$$
$$\Rightarrow m = 5$$

But n(Maths)

= n(Maths only) + n(both Maths and Science)

= 5 + 8

= <u>13</u>

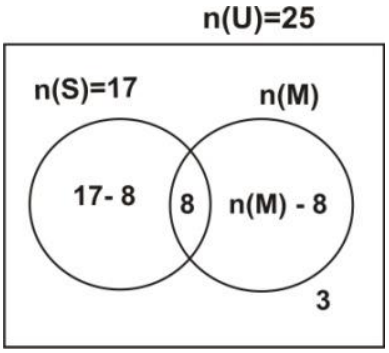
2 (c) (i) Method 3

Let n(U) = No. of students in the class

n(M) = No. of students that passed in Maths

n(S) = No. of students that passed in Science

x = No. of students that passed in Science <u>only</u>



From the diagram above,

$$17 - 8 + 8 + n(M) - 8 + 3 = 25$$
$$\Rightarrow 17 + 0 + n(M) - 8 + 3 = 25$$
$$\Rightarrow 17 - 8 + 3 + n(M) = 25$$
$$\Rightarrow 12 + n(M) = 25$$
$$\Rightarrow n(M) = 25 - 12$$
$$\Rightarrow n(M) = 13$$

$\therefore$ <u>No. of students that passed in Maths is 13</u>

2 (c) (ii) p(one subject only)

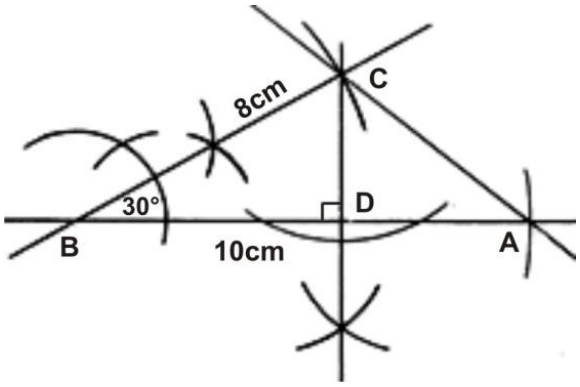
$$= \frac{\text{No. of students who passed one subject only}}{\text{Total No. of students}}$$

$$= \frac{\text{No. of Science only} + \text{No. of Maths only}}{\text{Total No. of students}}$$

$$= \frac{9 + 5}{25}$$

$$= \underline{\underline{14/25}}$$

3 (a)



(i) Angle ACB ≈ $\underline{\underline{98°}}$

(ii) |CD| ≈ $\underline{\underline{3.9 \text{ cm}}}$

(b) Area of triangle = ½ × b × h

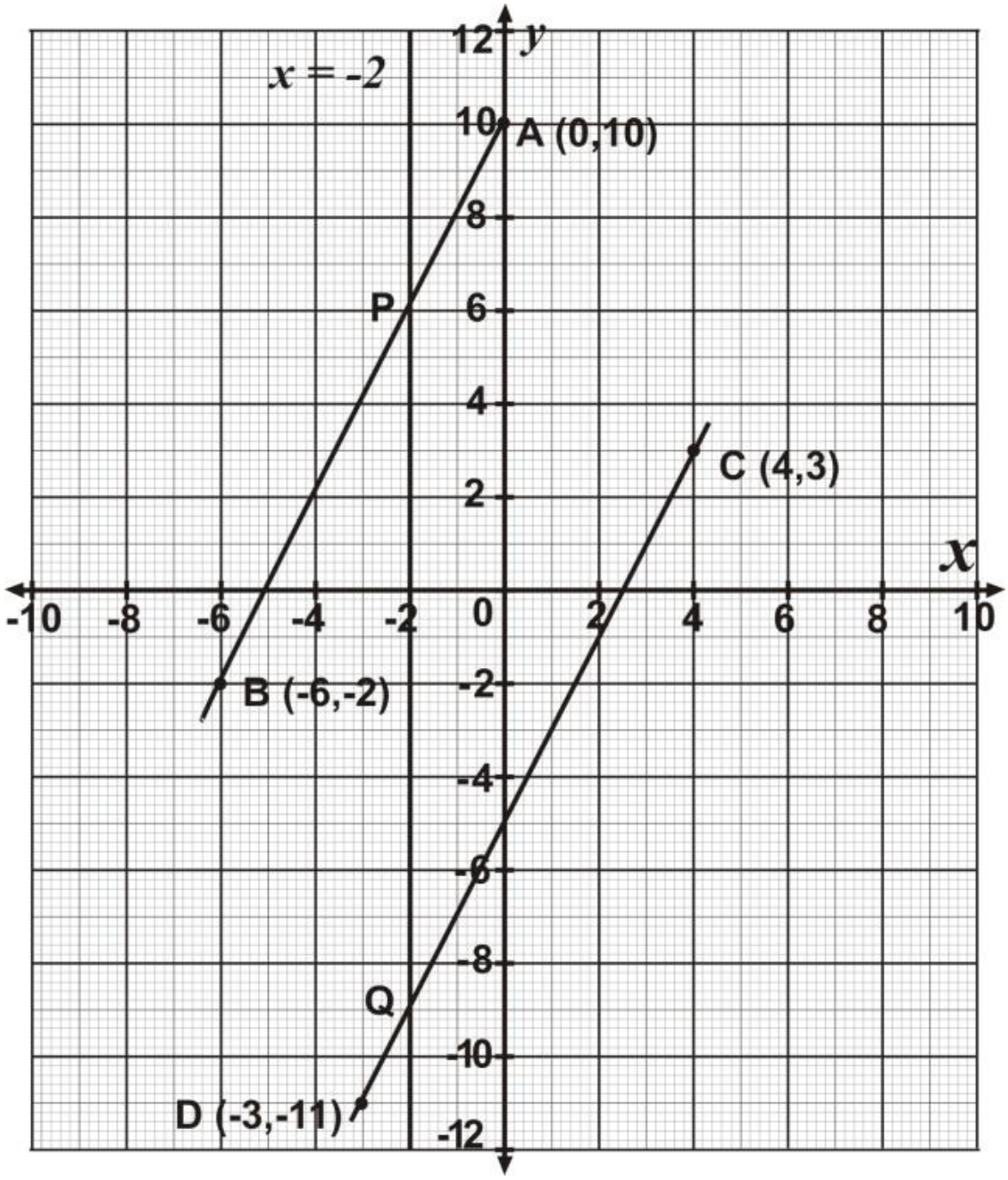
Area of ABC = ½ × |AB| × |CD|

= ½ × 10cm × 3.9cm

= ½ × 39 cm

= $\underline{\underline{19.5 \text{ cm}^2}}$

4 (a), (b)



4(b) (ii) angle BPQ = 25°

angle PQC = 25°

(iii) angles BPQ and PQC are alternate angles

(iv) Lines AB and CD are parallel

5 (a)(i) The mark with the highest frequency is 6

⇒ The modal mark = 6

(ii) 2 +3 + 5 + 7 + 8 + 13 + 7 + 5

= 50 candidates

(iii) **Approach 1**

$$\text{The mean mark} = \frac{\text{The sum of marks}}{\text{Total number of candidates}}$$

$$= \frac{(1\times2)+(2\times3)+(3\times5)+(4\times7)+(5\times8)+(6\times13)+(7\times7)+(8\times5)}{2+3+5+7+8+13+7+5}$$

$$= \frac{2+6+15+28+40+78+49+40}{50}$$

$$= \frac{258}{50} = 5\frac{4}{25} \text{ or } 5.16$$

(iii) **Approach 2**

Marks (x)	**No. of candidates (f)**	**f x**
1	2	2
2	3	6
3	5	15
4	7	28
5	8	40
6	13	78
7	7	49
8	5	40
	$\Sigma f = 50$	**$\Sigma fx = 258$**

$$\text{The mean mark} = \frac{\Sigma f x}{\Sigma f} = \frac{258}{50}$$

$$= 5\frac{4}{25} \text{ or } 5.16$$

5 (b)(i) Number of candidates who failed

Total No. of candidates = 2+3+5+7+8+13+7+5 = 50

Since 20% of the candidates failed

$\Rightarrow$ 20% $\times$ 50 candidates failed

$= \frac{20}{100} \times 50$

= 10 candidates failed

(ii) From the given table, the 10 candidates who failed scored 3 or less, therefore the pass mark is 4

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MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. Two sets which have no common members are known as …

A) Equal sets
B) Equivalent sets
C) Empty sets
D) Disjoint sets
E) Union

2. P = {0, 2, 4, 6} and Q = {1, 2, 4, 5}. Find P ∩ Q.

A. {0, 6}
B. {2, 4}
C. {0, 4}
D. {0, 2, 6}
E. {0, 1, 2, 4, 5}

3. Evaluate ½ [(4 – 1) – (5 – 6)]

A) –4.0 B) 1.0 C) 2.0 D) 3.0 E) 4.0

4. Find the highest(greatest) common factor of 35 and 70.

A) 5 B) 7 C) 10 D) 35 E) 70

5. Write 1101101_{two} in base ten

A) 31 B) 43 C) 108 D) 109 E) 218

6. State the property used in the statement: p (q+r) = pq + pr

A) Associative
B) Commutative
C) Distributive

D) Identity

E) Universal

7. If $n^2 + 1 = 50$, find n

A) 7 B) 24.5 C) 25 D) 49 E) $\sqrt{51}$

8. Evaluate $\dfrac{0.63 \times 0.70}{9.00}$

A) 0.0049 B) 0.049 C) 0.49 D) 4.9 E) 49

9. In the relation $\dfrac{1}{R} = \dfrac{1}{R_1} + \dfrac{1}{R_2}$ if $R_1 = 1$ and $R_2 = 3$, find R

A) ½ B) $^2/_3$ C) ¾ D) $^3/_2$ E) $^4/_3$

10. Arrange the following from the highest to the lowest: $^2/_3$, -9 , $^3/_5$ and 0

A) –9 , $^3/_5$, $^2/_3$, 0

B) –9 , 0 , $^3/_5$, $^2/_3$

C) $^3/_5$, $^2/_3$, 0 , -9

D) $^3/_5$, $^2/_3$, -9 , 0

E) $^2/_3$, $^3/_5$, 0 , -9

11. The total numbers of goals scored each month by a football team are:
3, 4, 8, 2, 4, 6, 4, 8, 7 and 6. What is the mode?

A) 2 B) 4 C) 5 D) 6 E) 8

12. A trader sold 90 oranges at 3 for ¢50.00. How much did she get from selling all the oranges?

A) ¢1,500.00

B) ¢4,500.00

C) ¢6,000.00

D) ¢15,000.00

E) ¢45,000.00

13. A man bought a car for ¢5.5million and sold it for ¢6.5million.
Find the percentage gain.

A) 1.54% B) 1.82% C) 15.38% D) 18.18% E) 84.44%

14. On a map, two towns P and Q are 15.5 cm apart. The scale of the map is 1cm: 4km. Calculate the actual distance between P and Q.

A) 15.5km B) 31km C) 46km D) 60km E) 62km

15. Expand the expression 2(3a + 2b)

A) 6a + 2b B) 5a + 4b C) 6a + 4b D) 10ab E) 12ab

16. Mansah obtained 150 marks out of 240 marks in an English test. What was her percentage score?

A) 33.33% B) 37.5% C) 41.67% D) 62.5% E) 79.1%

17. The marks obtained by six boys in a test are: 14, 20, 25, 15, 28 and 16. Find the mean mark.

A) 19.66 B) 22.00 C) 23.20 D) 26.40 E) 28.00

18. Which of the following is equivalent to $2^2 \times 6^2$

A) 2×3^4 B) $2^2 \times 3^2$ C) $2^2 \times 3^4$ D) $2^4 \times 3$ E) $2^4 \times 3^2$

19. Kwaku had 300 mangoes. He sold 240 of them. What is the percentage of the mangoes left?

A) 8% B) 20% C) 25% D) 26% E) 80%

20. A bag contains 5 black and 6 white balls. What is the probability of picking a white ball?

A) $^1/_{11}$ B) $^1/_6$ C) $^1/_5$ D) $^5/_{11}$ E) $^6/_{11}$

21. Solve for h in the equation $15 - 2h = 6$

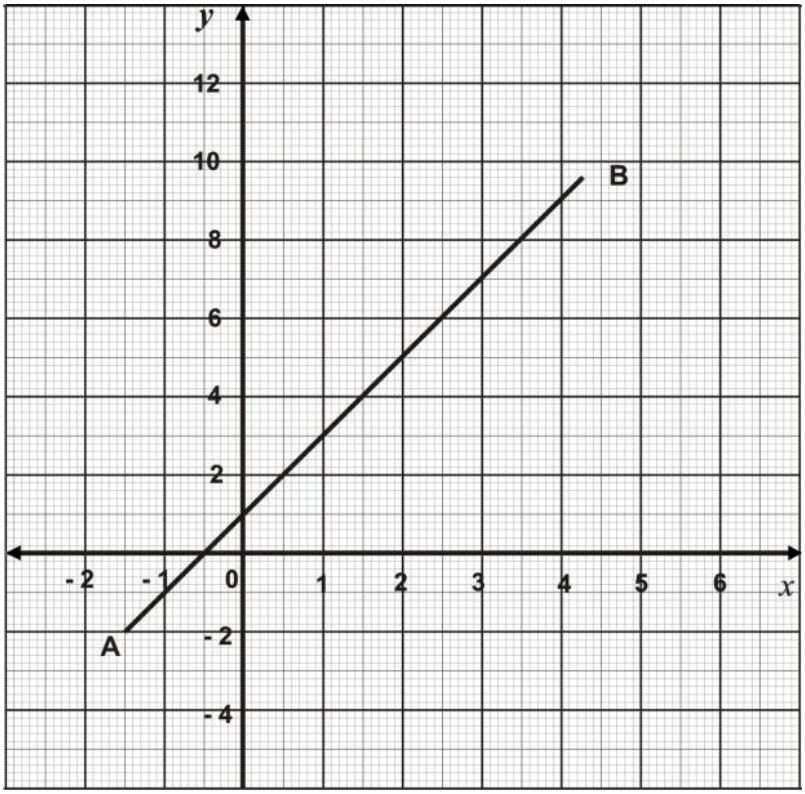
A) –10.5 B) –9.0 C) –4.5 D) 4.5 E) 10.5

22. What is the image of 3 in the mapping $x \rightarrow 3x + 7$?

A) 10 B) 13 C) 16 D) 24 E) 34

23. How many faces has a closed cylindrical object?

A) 1 B) 2 C) 3 D) 4 E) 34



The graph of a straight line AB is shown above.

Use it to answer questions **24** *and* **25**

24. Find the gradient of the line AB

A) 1 B) 2 C) 3 D) 4 E) 5

25. Find the coordinates of the point at which the line cuts the y-axis.

A. (1,0) B. (0,1) C. (-1,0) D. (0,-1) E. (0,0)

26. A man deposited an amount of ¢50,000.00 at a bank for 2 years at a rate of 20%. Find the simple interest.

A) ¢1,000.00
B) ¢2,000.00
C) ¢10,000.00
D) ¢20,000.00
E) ¢200,000.00

27. Kofi received ¢800.00 commission on the sales he made on a particular day. This commission is 20% of sales he made. Calculate his total sales for the day.

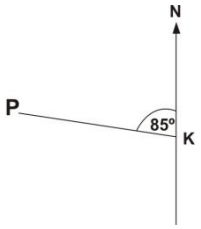
A) ¢1,200.00
B) ¢2,400.00

C) ¢3,200.00

D) ¢4,000.00

E) ¢4,800.00

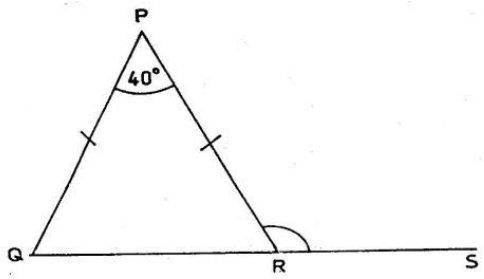
28.



Calculate the bearing of town P from town K in the diagram below.

A) 005° B) 085° C) 095° D) 265° E) 275°

29. In the diagram below, PQR is an isosceles triangle. |PQ| = |PR|, ∠QPR = 40° and QRS is a straight line. Find angle PRS.



A) 40° B) 70° C) 100° D) 110° E) 140°

30. Factorize completely the expression 2xy – 8x + 5y - 20

A. (2x+5)(y–4)

B. (2x–5)(y+4)

C. (2x–5)(y–4)

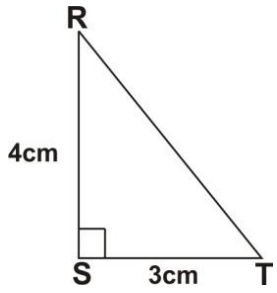
D. (2x+5)(y+4)

E. (x+5)(y–4)

31. Ama is facing east. Through how many degrees should she turn clockwise to face north?

A)90° B)135° C)180° D)225° E)270°

32.



In the triangle RST, < RST is 90°, |RS| = 4cm and |ST| = 3cm. What is the length of RT ?

A) 2cm B) 3cm C) 4cm D) 5cm E) 7cm

33. How many lines of symmetry has a rhombus ?

A) 2 B) 3 C) 4 D) 5 E) 6

34. Given that $\mathbf{a} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $\mathbf{b} - \mathbf{a}$.

A. $\begin{pmatrix} -2 \\ 6 \end{pmatrix}$ B. $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$ C. $\begin{pmatrix} -3 \\ 8 \end{pmatrix}$ D. $\begin{pmatrix} -2 \\ 2 \end{pmatrix}$ E. $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$

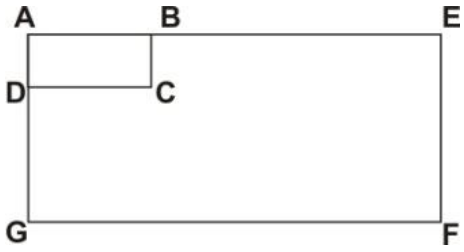
35. If $\mathbf{u} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$, find $2\mathbf{u} + \mathbf{v}$

A. $\begin{pmatrix} 12 \\ 0 \end{pmatrix}$ B. $\begin{pmatrix} 12 \\ -1 \end{pmatrix}$ C. $\begin{pmatrix} 6 \\ 1 \end{pmatrix}$ D. $\begin{pmatrix} 6 \\ 0 \end{pmatrix}$ E. $\begin{pmatrix} 6 \\ -1 \end{pmatrix}$

36. The volume of water in a cuboid is $9m^3$. The length of the cuboid is 3m and the breadth is 2m. Calculate the depth of water in the cuboid.

A) 0.15m B) 0.18m C) 1.5m D) 1.8m E) 6.0m

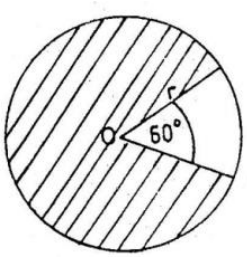
37.



In the diagram above, rectangle AEFG is an enlargement of the rectangle ABCD. If |AB| = 5cm and |AE| = 15cm. What is the scale factor of the enlargement?

A) ⅓ B) ½ C) 2 D) 3 E) 10

38.



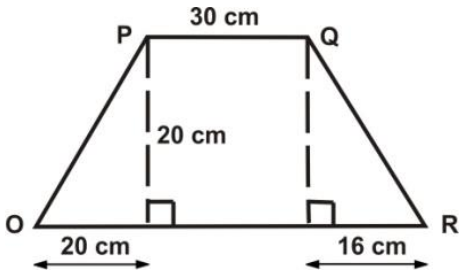
In the diagram, O is the centre of the circle and r is its radius. Calculate the area of the shaded region.

A) $^1/_6\ \pi r^2$ B) $^1/_5\ \pi r^2$ C) $^1/_3\ \pi r^2$ D) $^5/_6\ \pi r^2$ E) $^7/_{12}\ \pi r^2$

39. Which of the following best describes the statement: *"The locus of points 6cm from a fixed point O?"*

A) A parallel line 6cm from O
B) A vertical line 6cm from point O
C) A horizontal line 6 cm from point O
D) An equilateral triangle of side 6cm
E) A circle with centre O and radius 6cm.

40.



OPQR is a trapezium whose height is 20cm. What is the area?

A) 460cm^2 B) 600cm^2 C) 920cm^2 D) 960cm^2 E)1000cm^2

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MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. D) Disjoint sets

2. B. {2, 4}

3. C) 2.0

4. D) 35

5. D) 109

6. C) Distributive

7. A) 7

8. B) 0.049

9. C) ¾

10. E) $^{2}/_{3}$, $^{3}/_{5}$, 0 , -9

11. B) 4

12. A) ¢1,500.00

13. D) 18.18%

14. E) 62km

15. C) 6a + 4b

16. D) 62.5%

17. A) 19.66

18. E) $2^4 \times 3^2$

19. B) 20%

20. E) $^6/_{11}$

21. D) 4.5

22. C) 16

23. C) 3

24. B) 2

25. B. (0,1)

26. D) ¢20,000.00

27. D) ¢4,000.00

28. E) 275°

29. D) 110°

30. A. $(2x+5)(y-4)$

31. E) 270°

32. D) 5cm

33. A) 2

34. E. $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$

35. B. $\begin{pmatrix} 12 \\ -1 \end{pmatrix}$

36. C) 1.5m

37. D) 3

38. D) $^{5}/_{6}\ \pi r^{2}$

39. E) A circle with centre O and radius 6cm

40. D) 960 cm^{2}

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MATHEMATICS

SECTION B
ESSAY
1 hour

1. (a) Simplify the expression $3x^2 + 6xy - 3y^2 + 4x^2 + 8xy + 2y^2$

(b) (i) Solve $3x - 9 \geq 12(x - 3)$

(ii) Illustrate your answer on the number line

(c) If $P = 7$, $a = 16$, $b = 4$ and $c = 3$, evaluate $p^2 - \frac{(a-b)}{C}$

2. a) Using a ruler and a pair of compasses only,

(i) construct a triangle PQR such that |PQ| = 8cm, angle RPQ = 90° and angle PQR = 30°. Measure |RQ|

(ii) construct the perpendicular bisector (mediator) of RQ. Let it meet RQ at O.

b) With O as centre and radius OP, draw a circle. Measure |OP|.

c) What is the special name for the chord RQ?

3. (a) A man travelled from Bakwa to Pabam. The distance between the two towns is 510km. At Pabam he covered additional 40km on official duties. He returned to Bakwa the next day.

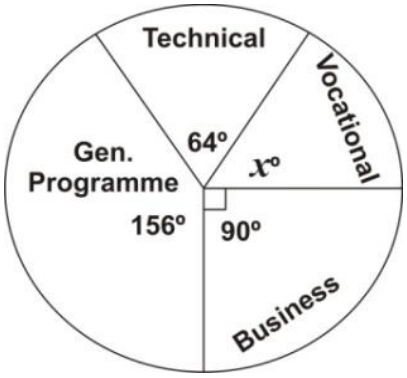
(i) Find the total distance covered by the man

(ii) If the car used one litre of petrol to cover 20km, find the amount of petrol used for the whole journey.

(iii) If a litre of petrol cost ¢522.00, calculate the total cost of petrol used for the whole journey.

(b) A woman sold an article for ¢200,000.00. She made a profit of 25%. Find the cost price of the article

4. (a) The pie chart shows angles representing the number of candidates who applied for admission into four programmes at a senior secondary school. The number of pupils who applied were 1080.



Find:

(i) the angle x^o representing the Vocational programme.

(ii) the number of candidates who applied for Business programme.

(iii) correct to the nearest whole number the percentage of the number of applicants who applied for General Programme.

(b) The data below shows the distribution of the masses of pupils in a school. On a graph paper, draw a bar chart for the distribution.

Masses (in kg)	19	20	21	22	23	24
Frequency	5	9	19	25	18	4

5. (a) Using a scale of 2cm to 2units on both axes, draw two perpendicular axes OX and OY on a graph sheet. On the same graph sheet, mark the x-axis from –8 to 8 and the y-axis from –8 to 8

(i) Plot P(-2, 3) and Q (6, 4). Join PQ

(ii) Draw the image P_1Q_1 of PQ under an anticlockwise rotation through 90° about the origin, where $P \rightarrow P_1$ and $Q \rightarrow Q_1$. Indicate clearly all the co-ordinates.

(iii) Draw the image P_2Q_2 of PQ under a clockwise rotation through 90° about the origin where $P \rightarrow P_2$ and $Q \rightarrow Q_2$. Indicate clearly all the co-ordinates

(b) The base radius of a closed cylinder is 4 m. The height of the cylinder is 7 m. Calculate its total surface area. [Take $\pi = {}^{22}/_{7}$]

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MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1 (a)

$$3x^2 + 6xy - 3y^2 + 4x^2 + 8xy + 2y^2$$

$$= \quad 3x^2 + 4x^2 + 6xy + 8xy - 3y^2 + 2y^2$$

$$= \quad \underline{\underline{7x^2 + 14xy - y^2}}$$

1 (b) (i)

$$3x - 9 \geq 12(x - 3)$$

$$\Rightarrow \quad 3x - 9 \geq 12x - 36$$

Approach 1

$$\Rightarrow \quad 36 - 9 \geq 12x - 3x$$

$$\Rightarrow \quad 27 \geq 9x$$

$$\Rightarrow \quad \frac{27}{9} \geq \frac{9x}{9}$$

$$\Rightarrow \quad 3 \geq x$$

$$\Rightarrow \quad \underline{\underline{x \leq 3}}$$

Turning the '$3 \geq x$' around reverses the $\geq$

1 (b) (i)

Approach 2

$$\Rightarrow \quad 3x - 12x \geq 9 - 36$$

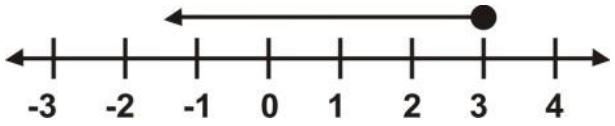
$$\Rightarrow \quad -9x \geq -27$$

$$\Rightarrow \quad \frac{-9x}{-9} \leq \frac{-27}{-9}$$

$$\Rightarrow \quad \underline{\underline{x \leq 3}}$$

[NB: Division by a negative quantity reverses the ≥]

(ii) The number line illustration is shown below:

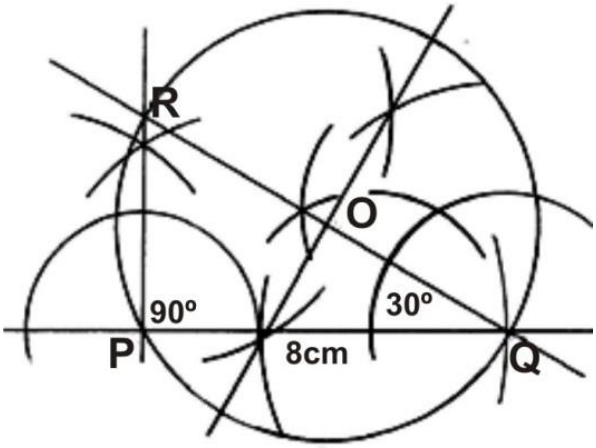


1 (c) $p^2 - \frac{(a-b)}{C}$, if P = 7, a = 16, b = 4 and c = 3

$$\Rightarrow \quad 7^2 - \frac{(16-4)}{3} \qquad \Rightarrow 49 - \tfrac{12}{3}$$

$$\Rightarrow \quad 49 - 4 \qquad = \quad \underline{45}$$

2 (a), (b)



(a) (i) |RQ| ≈ 9.2 cm

(b) |OP| ≈ 4.6 cm

(c) The special name for chord RQ is diameter

3(a) (i)



Total distance covered

= 2 (510km + 40km)

= 2 × 550km

= 1100 km

Alternatively, by expansion,

2 (510km + 40km)

= 1020km + 80km

= 1100 km

(ii) **Method 1** (Simple logic)

1100 ÷ 20 = 55

⇒ The amount of petrol used for the whole journey = 55 litres

(ii) **Method 2** (Proportion)

Distance (km)	Amount of petrol (¢)
20	1
1100	**?**

From the table above, If 20 km → 1 litre

Then 1100 km → ? (more)

If more, less (20km) divides, hence we have

$$\frac{1100km}{20km} \times 1 \text{ litre} = 55 \text{ litres}$$

⇒ The amount of petrol used for the whole journey = 55 litres

(iii) **Method 1** (Simple logic)

522 × 55 = 28,710

⇒ total cost of petrol for the whole journey = ¢28,710.00

(iii) **Method 2** (Proportion)

Amount of petrol (litres)	Cost of petrol (¢)
1	**522**
55	**p**

From the table above,

Approach 1 (Equivalent Fraction)

$$p = \frac{522 \times 55}{1} = 28{,}710$$

$\Rightarrow$ total cost of petrol for the whole journey = ¢28,710.00

(iii) **Approach 2** (if more, less divides ...)

If 1 litre $\rightarrow$ ¢522

Then, 55 litres $\rightarrow$? (more)

If more, less (1 litre) divides, hence we have $\frac{55\text{litres}}{1\text{litre}} \times ¢522$

= ¢28,710.00

3(b) **Method 1** (Formula / Equation)

$$CP = \frac{\text{CP percentage}}{\text{SP percentage}} \times SP$$

$$= \frac{100\,\%}{125\,\%} \times ¢200{,}000$$

$$= \frac{100}{125} \times ¢200{,}000$$

$$= ¢160{,}000.00$$

3(b) **Method 2** (Equivalent fractions)

	Percentage	**Amount (¢)**
SP	100% + 25% **= 125%**	**200,000**
CP	**100%**	**?**

From the table above,

$$\frac{125\%}{100\%} = \frac{¢200{,}000}{x}$$

$$\Rightarrow 125 \times x = ¢200{,}000 \times 100$$

$$\Rightarrow x = \frac{¢200{,}000 \times 100}{125}$$

$$= ¢160{,}000$$

The cost price of the article is ¢ 160,000.00

3(b) **Method 3** (If more, less divides …)

If	125%	→	¢200,000
Then	100%	→	? (less)

If less, more divides, hence we have

$$\frac{100\%}{125\%} \times ¢200{,}000 = ¢160{,}000$$

The cost price of the article is ¢ 160,000.00

4 (a) (i)

$$x° + 90° + 156° + 64° = 360°$$
$$x° + 310° = 360°$$
$$x° = 360° - 310°$$
$$x° = 50°$$

(ii) **Method 1** (Formula)

No. of candidates for business

= fraction of business × Total no. of candidates

$$= \frac{90^o}{360^o} \times 1080$$
$$= 270$$

⇒ 270 candidates applied for the business programme

(ii) **Method 2** (Simple Proportion)

	Angle	Number
Total	360°	1080
Business	90°	x

Approach 1 (Equivalent Fractions)

From the table above,

$$\frac{360}{90} = \frac{1080}{x}$$

Cross multiplying, we have

$$\Rightarrow \quad 360 \times x = 1080 \times 90$$

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$$\Rightarrow \quad x = \frac{1080 \times 90}{360}$$

$$\Rightarrow \quad x = 270$$

270 candidates applied for the business programme

(ii) **Approach 2** (If more, less divides …)

If $360^o \rightarrow$ 1080

Then $90^o \rightarrow$? (less)

If less, more (i.e., 360°) divides; therefore we have

$$\frac{90^o}{360} \times 1080$$

$$= 270$$

270 candidates applied for the business programme

4 (a) (iii) Percentage of applicants who applied for Gen. Prog.

$$= \frac{\text{Angle for Gen. Prog}}{\text{Total angle}} \times 100\%$$

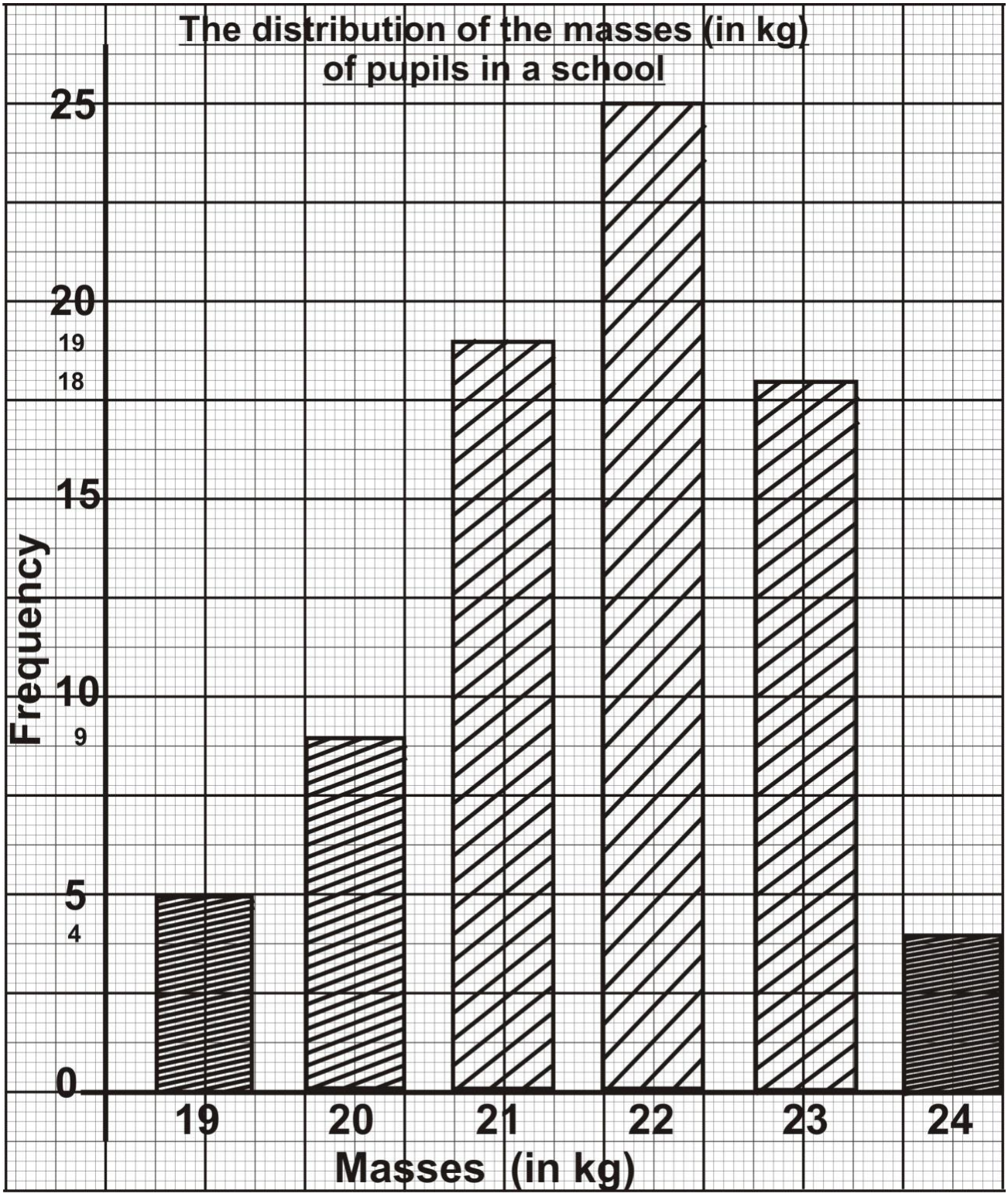
$$= \frac{156}{360} \times 100\%$$

$$= 43\tfrac{1}{3}\ \% \quad \text{or} \quad \approx \quad 43.3\ \%$$

$\Rightarrow$ 43⅓ % of the pupils applied for Gen. programme

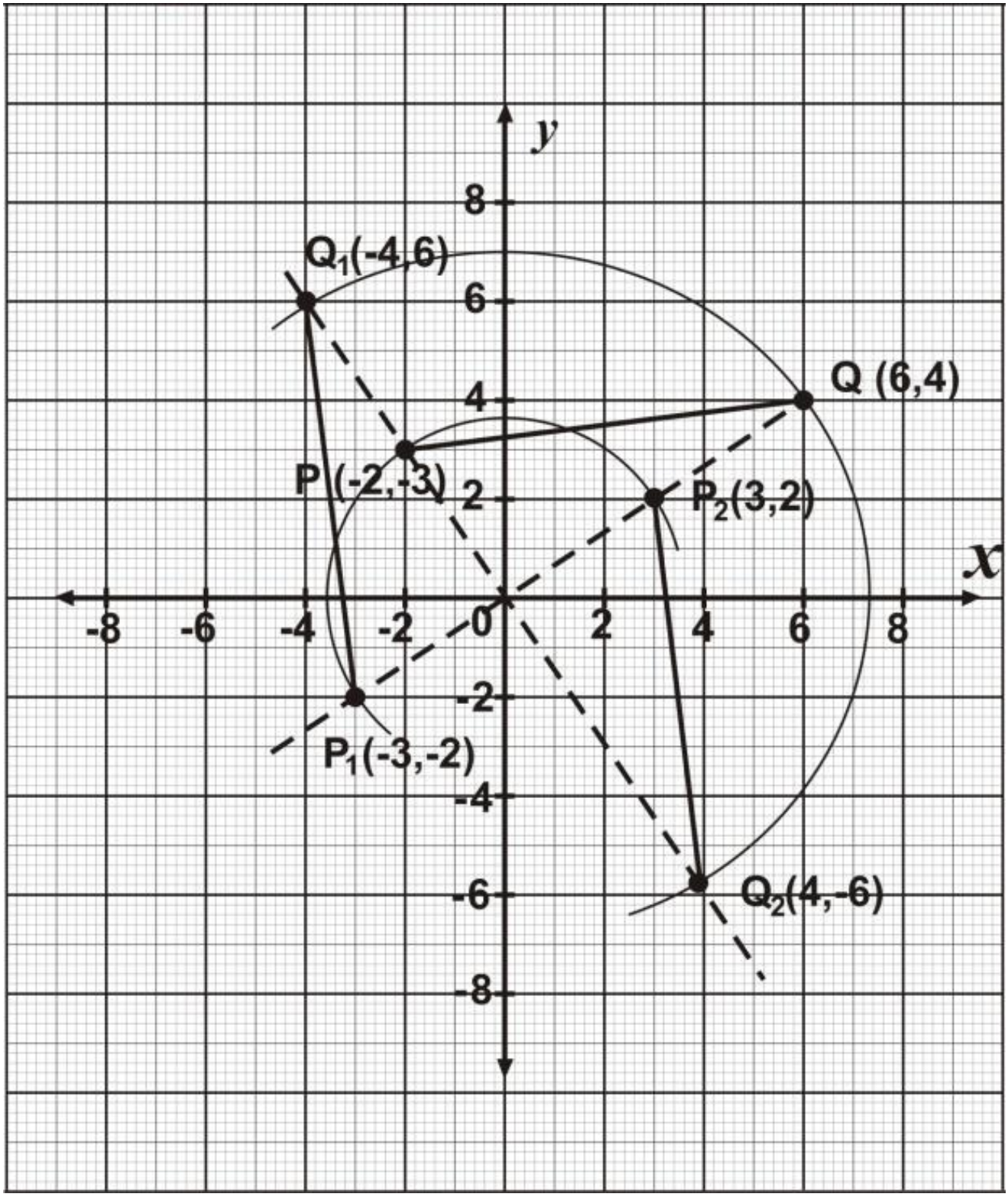
4 (b)

Vertical Axis Scale: 4 cm to 5 pupils


The distribution of the masses (in kg) of pupils in a school
Frequency
25
20
19
18
15
10
9
5
4
0
19
20
21
22
23
24
Masses (in kg)

5 (a)

Approach 1 (By Inspection / Construction)



5 (a)

Approach 2 (The rule / formula)

(ii) Rotating (x, y) about the origin through 90°anticlockwise

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} -y \\ x \end{pmatrix}$$

$$OP\begin{pmatrix} -2 \\ 3 \end{pmatrix} \rightarrow OP_1\begin{pmatrix} -3 \\ -2 \end{pmatrix}, \quad \therefore P_1(-3,-2)$$

$$OQ\begin{pmatrix} 6 \\ 4 \end{pmatrix} \rightarrow OQ_1\begin{pmatrix} -4 \\ 6 \end{pmatrix}, \quad \therefore Q_1(-4,6)$$

∴ Plot and join P_1(-3,-2) and Q_1(-4,6) as the image of line segment PQ under a rotation about the origin through 90° anticlockwise

(iii) Rotating (x, y) about the origin through 90°clockwise

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} y \\ -x \end{pmatrix}$$

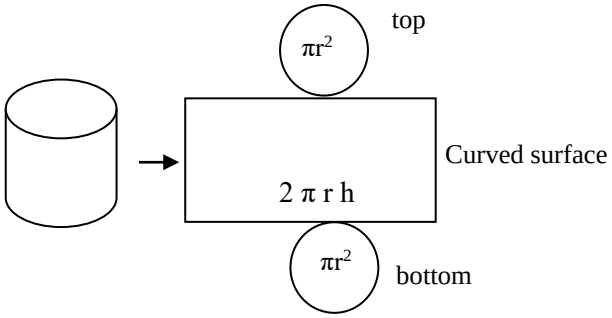
$$OP\begin{pmatrix} -2 \\ 3 \end{pmatrix} \rightarrow OP_2\begin{pmatrix} 3 \\ 2 \end{pmatrix}, \quad \therefore P_2(3, 2)$$

$$OQ\begin{pmatrix} 6 \\ 4 \end{pmatrix} \rightarrow OQ_2\begin{pmatrix} 4 \\ -6 \end{pmatrix}, \quad \therefore Q_2(4,-6)$$

∴ Plot and join P2(3,2) and Q2(4,-6) as the image of line segment PQ under a rotation about the origin through 90° clockwise

(*as shown in the diagram above*)

5 (b)



Total surface area of closed cylinder:

Approach 1

= Area of (Curved surface + 2 circular surfaces)

= $2\pi r h + 2(\pi r^2)$

= $2 \times {}^{22}/_{7} \times 4\text{cm} \times 7\text{cm} + 2 \times {}^{22}/_{7} \times (4\text{cm})^2$

= $2 \times 22 \times 4\text{cm}^2 + 2 \times {}^{22}/_{7} \times 16\text{cm}^2$

= $176\ \text{cm}^2 + \frac{704}{7}\ \text{cm}^2$

= $176\ \text{cm}^2 + 100\frac{4}{7}\ \text{cm}^2$

= $\underline{\underline{276\frac{4}{7}\ cm^2}}$ **or** $\approx$ $\underline{276.57\ \text{cm}^2}$

5 (b) **Approach 2**

= curved surface + top + bottom

= $2\pi r h + \pi r^2 + \pi r^2$

= $[2 \times {}^{22}/_{7} \times 4\text{cm} \times 7\text{cm}] + [{}^{22}/_{7} \times (4\text{cm})^2] + [{}^{22}/_{7} \times (4\text{cm})^2]$

= $176\text{cm}^2 + \frac{352}{7}\ \text{cm}^2 + \frac{352}{7}\ \text{cm}^2$

= $176\ \text{cm}^2 + 50\frac{2}{7}\ \text{cm}^2 + 50\frac{2}{7}\ \text{cm}^2$

= $\underline{\underline{276\frac{4}{7}\ cm^2}}$ **or** $\approx$ $\underline{276.57\ \text{cm}^2}$

5 (b) **Approach 3**

= Areas of (Curved surface + 2 circular surfaces)

= $2\pi r h + 2\pi r^2$

= $2\pi r\ (h + r)$

= $2 \times {}^{22}/_{7} \times 4\text{cm}\ (11\text{cm} + 4\text{cm})$

= $\frac{176}{7}\ \text{cm} \times (11\text{cm})$

= $\frac{176\ cm \times 11\ cm}{7} = \frac{1936\ cm^2}{7}$

= $\underline{\underline{276\frac{4}{7}\ cm^2}}$ **or** $\approx$ $\underline{276.57\ \text{cm}^2}$

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MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

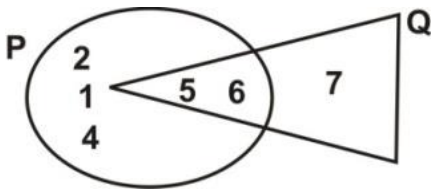
1. Convert 320_{five} to a base ten numeral

A) 25 B) 77 C) 85 D) 86 E) 90

2. P = {2, 4, 6, 8} and Q = {even counting numbers less than 12}.
What is the relationship between P and Q?

A) P = Q B) P ⊂ Q C) P < Q D) Q ⊂ P E) P ∈ Q

3. In the diagram below, P is the set of numbers in the circle and Q is the set of numbers in the triangle.
What is P ∩ Q.?



A. {1, 2, 4}
B. {5, 6}
C. {7}
D. {1, 2, 4, 5, 6, 7}
E. { }

4. The product 6287 x 543 = 3,413,841. What is the value of 628.7 x 5.43 ?

A) 3,413,841
B) 342,384.1
C) 34,138.41

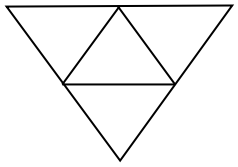
D) 3,413.841

E) 341.3841

5. Express 30cm as a percentage of 2m.

A) 0.5% B) 1.5% C) 6.7% D) 15% E) 66.7%

6.



What solid can be made from this net ?

A) triangle

B) rectangular pyramid

C) triangular prism

D) rectangular prism

E) triangular pyramid

7. Simplify 3(6b – 9a) + 7(6a – 5b)

A) 17b + 6a

B) -17b + 6a

C) 17b+48a

D) 15a – 17b

E) 17b – 15a

8. Arrange the following fractions in descending order $^{9}/_{16}$, $^{5}/_{8}$, 0.62

A) $^{9}/_{16}$, 0.62 , $^{5}/_{8}$

B) 0.62 , $^{5}/_{8}$, $^{9}/_{16}$

C) $^{5}/_{8}$, $^{9}/_{16}$, 0.62

D) $^{5}/_{8}$, 0.62 , $^{9}/_{16}$

E) 0.62 , $^{9}/_{16}$, $^{5}/_{8}$

9. Find the LCM of 18, 42 and 90.

A) 2×3^2

B) $2\times3\times7$

C) $2\times3^2\times5$

D) $2\times3\times5\times7$

E) $2\times3^2\times5\times7$

10. What is the HCF of 18, 42 and 90 ?

A) 21 B) 18 C) 9 D) 6 E) 3

11. Simplify $\begin{pmatrix}4\\5\end{pmatrix} + \begin{pmatrix}-2\\2\end{pmatrix} + \begin{pmatrix}1\\6\end{pmatrix}$

A. $\begin{pmatrix}2\\13\end{pmatrix}$ B. $\begin{pmatrix}3\\13\end{pmatrix}$ C. $\begin{pmatrix}5\\13\end{pmatrix}$ D. $\begin{pmatrix}6\\13\end{pmatrix}$ E. $\begin{pmatrix}7\\13\end{pmatrix}$

12. If $R = \frac{1}{2}(a+b)k$, find k in terms of R, a and b.

A) $\dfrac{a+b}{2R}$

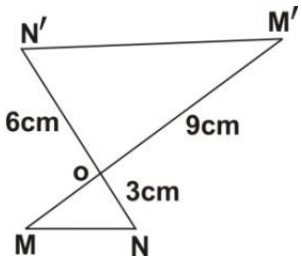
B) $\dfrac{2R}{a+b}$

C) $\dfrac{2R-a}{b}$

D) $\dfrac{2R-b}{a}$

E) $\dfrac{R}{2(a+b)}$

In the figure below, triangle M′ON′ is an enlargement of triangle MON, with centre O.



13. Find the scale factor of the enlargement

A) 9 B) 3 C) 2 D) –2 E) –3

14. Find |OM|

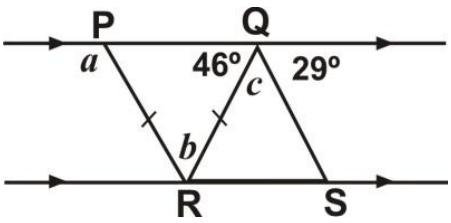
A) 9 B) 6 C) 4.5 D) 3 E) 1.5

15. Find the image of the point (6, 3) when translated by the vector $\begin{pmatrix} -4 \\ -1 \end{pmatrix}$

A. (-2,-2) B. (2,-2) C. (-2,2) D. (2,2) E. (10,4)

16. A student spends $^{17}/_{35}$ of his pocket money on transport and fruits. He spends $^{5}/_{6}$ of the remainder on sweets. What fraction of his pocket money does he spend on sweets?

A) $^{4}/_{7}$ B) $^{3}/_{7}$ C) $^{17}/_{35}$ D) $^{18}/_{35}$ E) $^{17}/_{42}$



In the diagram above, PQ is parallel to RS and |PR| = |QR|.

Use the diagram to answer questions **17** *to* **19**

17. What is ***a*** ?

A) 29 B) 46 C) 75 D) 88 E) 134

18. Find **b.**

A) 29 B) 46 C) 88 D) 92 E) 105

19. Find the value of **c**.

A) 29 B) 46 C) 75 D) 88 E) 105

20. If $p = 7$, $q = 5$ and $r = 3$, find the value of $p^2 + q - r^3$

A) 10 B) 27 C) 51 D) 57 E) 81

21. Simplify $35x^5y^3 \div 7xy^2$

A) $5x^6y^5$ B) $5x^4y$ C) $5x^6y$ D) $5x^4y^5$ E) $5x^4y^2$

22. The sum of two numbers is 240. If one of them is y, find the other number.

A) 120y B) 240 + y C) 240 – y D) 240y E) 240/y

23. Make n the subject of the relation $2n + 5 = 7a$

A) $n = \frac{1}{2}(7a+5)$

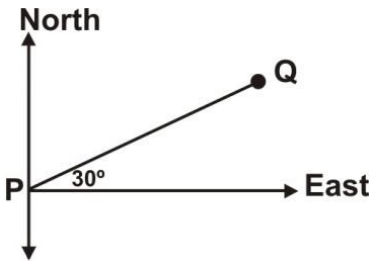
B) $n = \frac{1}{2}(7a-5)$

C) $n = 2(7a+5)$

D) $n = 2(7a-5)$

E) $n = 7a-5$

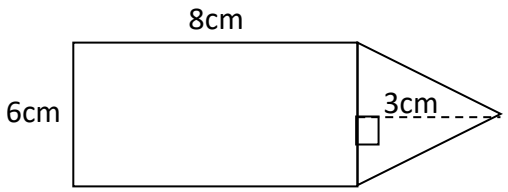
24.



Find the bearing of Q from P in the diagram above.

A) 030° B) 060° C) 070° D) 090° E) 150°

25.



The figure above is made up of a rectangle and a triangle. The dimensions of the rectangle are 8cm and 6cm. The triangle has 6cm as its base and 3cm as its height.

Find the area of the figure.

A) $33cm^2$ B) $48cm^2$ C) $51cm^2$ D) $57cm^2$ E) $66cm^2$

26. The marks obtained by ten boys in a test are: 0, 1, 3, 3, 5, 7, 8, 9, 9 and 9.

Find the median score.

A) 3 B) 5 C) 6 D) 7 E) 8

27. The marks obtained by ten boys in a test are: 0, 1, 3, 3, 5, 7, 8, 9, 9 and 9.

Calculate the mean score.

A) 4.4 B) 5.4 C) 6.0 D) 6.4 E) 9.0

28. The marks obtained by ten boys in a test are: 0, 1, 3, 3, 5, 7, 8, 9, 9 and 9.

What is the probability that a boy chosen at random scored 3?

A) $^1/_{27}$ B) $^1/_{18}$ C) $^1/_{10}$ D) $^1/_5$ E) $^1/_3$

29. Express 57_{ten} as a base two (binary) numeral.

A) 101011_{two}
B) 100111_{two}
C) 11010_{two}
D) 110111_{two}
E) 111001_{two}

30. The scale of a map is 1: 100,000. What is the distance (in km) between two towns 4cm apart on the map?

A) 0.04 B) 0.4 C) 4.0 D) 40 E) 400

31. If 22% of a rope is 55m long, find the full length of the rope.

A) 12.1m B) 25m C) 121m D) 250m E) 2500m

32. Adwoa and Ama share an amount of ¢6,000.00 in the ratio 3:2. Find Adwoa's share.

A) ¢2,000.00
B) ¢2,400.00
C) ¢3,000.00
D) ¢3,600.00
E) ¢4,000.00

33. Find the truth set of $5x - 8 \leq 2x + 4$

A. $\{x \geq 4\}$ B. $\{x \geq -4\}$ C. $\{x \leq 4\}$ D. $\{x \leq -4\}$ E. $\{x = 4\}$

34. The diameter of a circular tray is 28cm. Find the area of the tray.

(Take $\pi = {}^{22}/_{7}$)

A) 44 cm^2 B) 88 cm^2 C) 154 cm^2 D) 616 cm^2 E) 2464 cm^2

35. If $\mathbf{a} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ -5 \end{pmatrix}$, find $\mathbf{a} + \mathbf{b}$

A. $\begin{pmatrix} 2 \\ 8 \end{pmatrix}$ B. $\begin{pmatrix} 8 \\ 8 \end{pmatrix}$ C. $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$ D. $\begin{pmatrix} 8 \\ -8 \end{pmatrix}$ E. $\begin{pmatrix} 8 \\ -2 \end{pmatrix}$

36.

x	0	1	2	3	4
y	0	1	4	9	16

What is the rule for the mapping above ?

A) $x \rightarrow x+3$ B) $x \rightarrow x+1$ C) $x \rightarrow x-1$ D) $x \rightarrow x+2$ E) $x \rightarrow x^2$

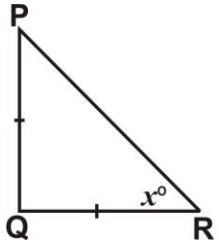
37. If 50 oranges cost ¢2,500.00 how many oranges can be bought for ¢15,000.00?

A) 3 B) 30 C) 60 D) 300 E) 360

38. A farmer has 1853 pineapple suckers. He plants 17 pineapples in a row. How many rows can he plant?

A) 17 B) 19 C) 91 D) 109 E) 190

39.



In triangle PQR, |PQ| = |QR|, and angle PQR = 90°. Find x

A) 30° B) 45° C) 60° D) 90° E) 180°

40. If a × b × c = 1197 and a = 21, b = 3, find c.

A) 19 B) 49.9 C) 57 D) 63 E) 399

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MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. C) 85

2. B) $P \subset Q$

3. B. {5, 6}

4. D) 3,413.841

5. D) 15%

6. E) triangular pyramid

7. D) 15a – 17b

8. D) $^5/_8$, 0.62 , $^9/_{16}$

9. E) $2\times3^2\times5\times7$

10. D) 6

11. B. $\begin{pmatrix} 3 \\ 13 \end{pmatrix}$

12. B) $\frac{2R}{a+b}$

13. D) –2

14. C) 4.5

15. D. (2, 2)

16. B) $^{3}/_{7}$

17. E) 134

18. C) 88

19. E) 105

20. B) 27

21. B) $5x^4y$

22. C) 240 – y

23. B) n = ½ (7a-5)

24. B) 060°

25. D) $57cm^2$

26. C) 6

27. B) 5.4

28. D) $^{1}/_{5}$

29. E) 111001_{two}

30. C) 4.0

31. D) 250m

32. D) ¢3,600.00

33. C. $\{x \leq 4\}$

34. D) 616 cm^2

35. E. $\begin{pmatrix} 8 \\ -2 \end{pmatrix}$

36. E) $x \rightarrow x^2$

37. D) 300

38. D) 109

39. B) 45°

40. A) 19

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MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. (a) If $P = 4$, $a = 16$, $b = -5$ and $c = 3$, evaluate $p^2 - \frac{(a-b)}{C}$

(b) Solve the inequality $5x - 3(x - 1) \geq 39$. Illustrate your answer on the number line.

(c) If $x = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$ and $y = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$

Find (i) $x + 2y$ (ii) $3x - y$

2. Using a ruler and a pair of compasses only,

a) (i) construct a triangle ABC such that |AB| = 8cm, angle ABC = 60° and |BC| = 8cm.

(ii) What type of triangle is triangle ABC?

b) construct the bisector of angle BAC to meet |BC| at D. Measure |AD|.

c) construct the perpendicular bisector of |BA| to meet |AD| at O.

d) Using O as centre and radius OD, draw a circle to touch the three sides of the triangle.

3. (a) If $2y - 5x + 10 = 0$, find:

(i) y, when x = 2; (ii) x, when y = 5

(b) (i) Using a scale of 2cm to 1 unit on both axes, draw two perpendicular lines OX and OY on a graph sheet.

(ii) On the same graph sheet, mark the *x*-axis from –5 to 5 and the *y*-axis from –6 to 6.

(iii) Plot on the same graph sheet, the points A (0,-5) and B(4, 5). Join AB using a ruler.

(iv) Find the gradient of the line AB

(v) Measure the acute angle that the line AB makes with the *x*-axis, using a protractor.

4. (a) The table below shows the distribution of the ages (in years) of children who were treated in a clinic in a day.

Age (in years)	1	2	3	4	5
Frequency	6	4	2	3	5

Find

(i) the mean age

(ii) the modal age

(b) Draw a bar chart for the distribution.

5. (a) The volume of a cylinder is $220cm^3$. The radius of the cross-section is 2.5cm. Find the height of the cylinder. (Take $\pi = {}^{22}/_{7}$)

(b) Each of the interior angles of a regular polygon is 140°. How many sides does it have?

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MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1 (a)

$p^2 - \frac{(a-b)}{C}$, if P = 4, a = 16, b = -5 and c = 3

$\Rightarrow \quad 4^2 - \frac{[16-(-5)]}{3} \qquad \Rightarrow \quad 4^2 - \frac{(16+5)}{3}$

$\Rightarrow \quad 16 - \frac{21}{3} \qquad \Rightarrow \quad 16 - 7$

$= \quad \underline{\underline{9}}$

(b)

$5x - 3(x-1) \geq 39$

$\Rightarrow \quad 5x - 3x + 3 \geq 39$

$\Rightarrow \quad 2x + 3 \geq 39$

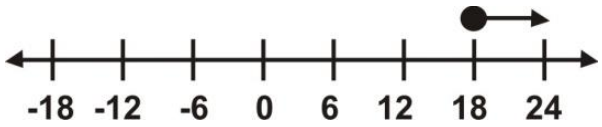
$\Rightarrow \quad 2x \geq 39 - 3$

$\Rightarrow \quad 2x \geq 36$

$\Rightarrow \quad \frac{2x}{2} \geq \frac{36}{2}$

$\Rightarrow \quad \underline{\underline{x \geq 18}}$

The number line illustration is shown below:



(c) (i)

$$x + 2y$$

$$= \begin{pmatrix} -3 \\ 2 \end{pmatrix} + 2\begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} -3 \\ 2 \end{pmatrix} + \begin{pmatrix} 2\times 4 \\ 2\times -1 \end{pmatrix}$$

$$= \begin{pmatrix} -3 \\ 2 \end{pmatrix} + \begin{pmatrix} 8 \\ -2 \end{pmatrix}$$

$$= \begin{pmatrix} -3+8 \\ 2+(-2) \end{pmatrix}$$

$$= \begin{pmatrix} -3+8 \\ 2-2 \end{pmatrix} = \underline{\underline{\begin{pmatrix} 5 \\ 0 \end{pmatrix}}}$$

(ii)

$$3x - y$$

$$= 3\begin{pmatrix} -3 \\ 2 \end{pmatrix} - \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

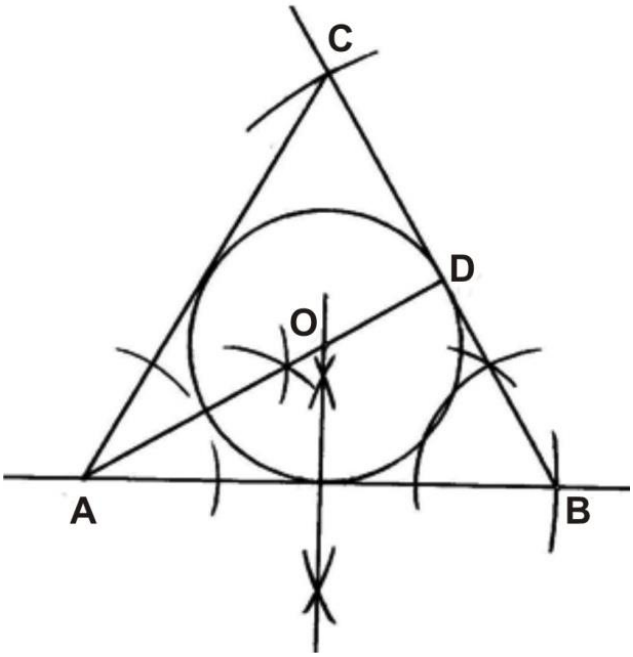
$$= \begin{pmatrix} 3\times -3 \\ 3\times 2 \end{pmatrix} - \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} -9 \\ 6 \end{pmatrix} - \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} -9-4 \\ 6-(-1) \end{pmatrix}$$

$$= \begin{pmatrix} -9-4 \\ 6+1 \end{pmatrix} = \underline{\underline{\begin{pmatrix} -13 \\ 7 \end{pmatrix}}}$$

2.



a) (ii) ABC is an <u>equilateral triangle</u>

b) |AD| ≈ <u>6.9 cm</u>

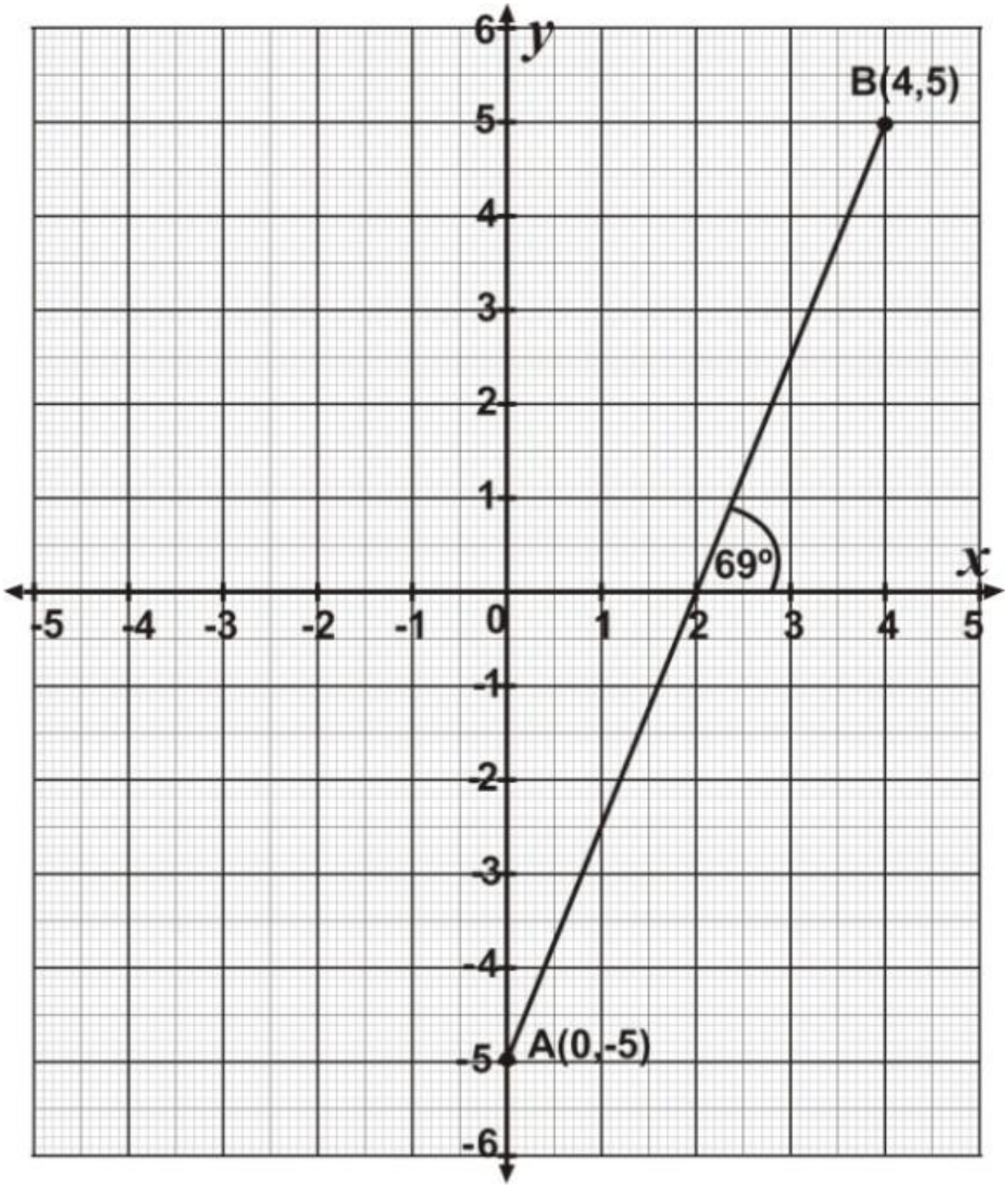
3 (a) (i) When $x = 2$, we have

$$2y - 5(2) + 10 = 0$$
$$\Rightarrow \quad 2y - 10 + 10 = 0$$
$$\Rightarrow \quad 2y + 0 = 0$$
$$\Rightarrow \quad 2y = 0$$
$$\Rightarrow \quad y = {}^{0}/_{2}$$
$$\Rightarrow \quad \underline{y = 0}$$

(ii) When $y = 5$, we have

$$2(5) - 5x + 10 = 0$$
$$\Rightarrow \quad 10 - 5x + 10 = 0$$
$$\Rightarrow \quad 20 - 5x = 0$$
$$\Rightarrow \quad 20 = 5x$$
$$\Rightarrow \quad \frac{20}{5} = x$$
$$\Rightarrow \quad 4 = x$$
$$\Rightarrow \quad \underline{x = 4}$$

3 (b)



(iv) The gradient of line AB

$$= \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - (-5)}{4 - 0} = \frac{10}{4} = \underline{2.5}$$

or

$$= \frac{y_1 - y_2}{x_1 - x_2} = \frac{-5 - 5)}{0 - 4} = \frac{-10}{-4} = \underline{2.5}$$

(v) The acute angle that the line AB makes with the x-axis $\approx$ 69°

4 (a) (i) **Approach 1**

$$\text{The mean age} = \frac{\text{The sum of ages}}{\text{The total number of children}}$$

$$= \frac{(1\times6)+(2\times4)+(3\times2)+(4\times3)+(5\times5)}{6+4+2+3+5}$$

$$= \frac{6+8+6+12+25}{6+4+2+3+5}$$

$$= \frac{57}{20} = 2\tfrac{17}{20}\ years \text{ or } 2.85 \text{ years}$$

4 (a) (i) **Approach 2**

Age (years) (x)	**Frequency (f)**	**f x**
1	6	6
2	4	8
3	2	6
4	3	12
5	5	25
	$\Sigma f = 20$	**$\Sigma fx = 57$**

$$\text{The mean age} = \frac{\Sigma f x}{\Sigma f} = \frac{57}{20}$$

$$= 2\tfrac{17}{20}\ years \text{ or } 2.85 \text{ years}$$

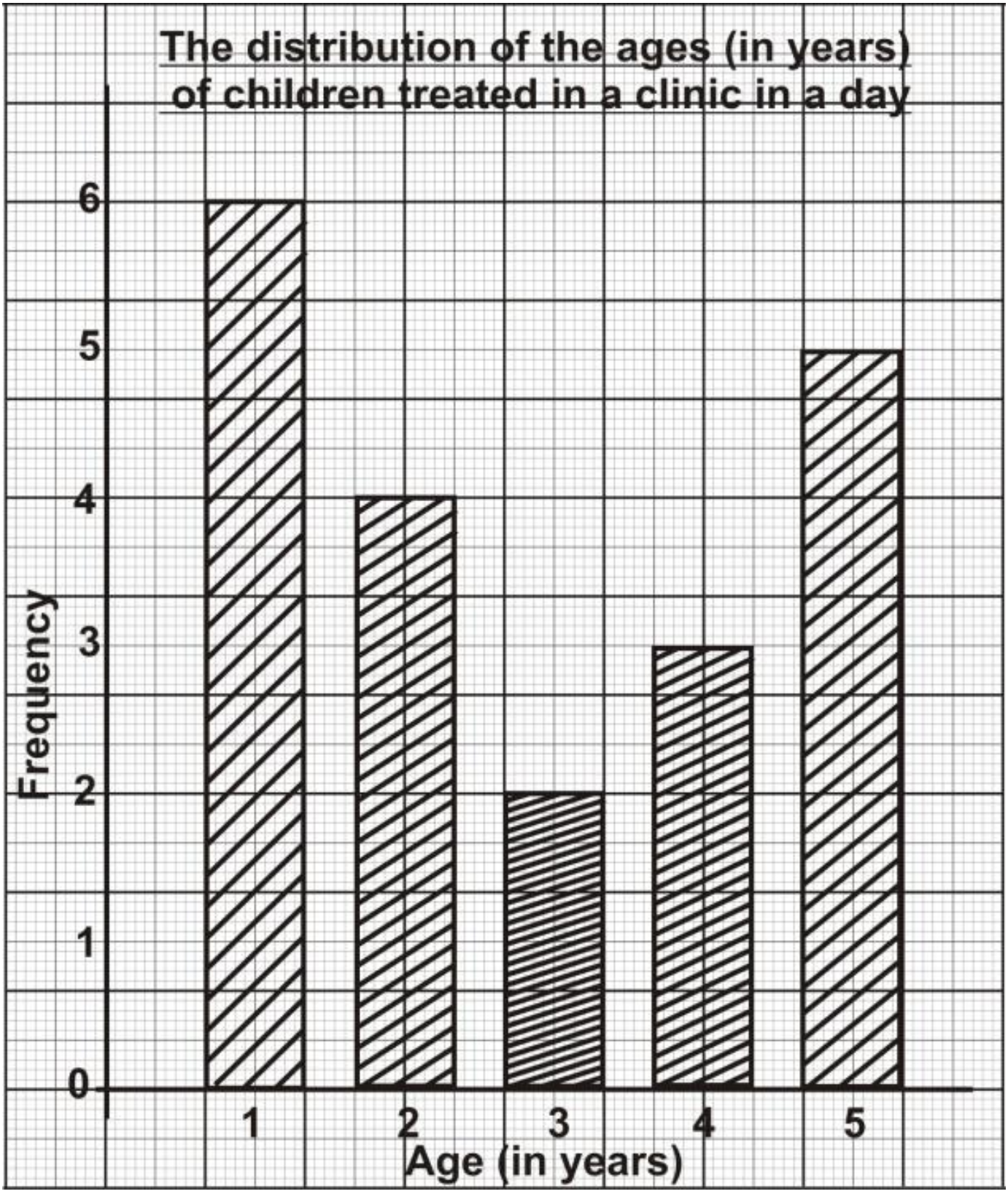
(ii) The modal age = the most-occurring age

= 1 year

4 (b)

Vertical AxisScale: 3 cm to 1 child

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5 (a) Volume of cylinder = Area of base × height

Approach 1 (Substitution first)

$$V = \pi r^2 \times h$$

$$\Rightarrow \quad 220\text{cm}^3 = \tfrac{22}{7} \times (2.5cm)^2 \times h$$

$$\Rightarrow \quad 220\text{cm}^3 = \frac{22}{7} \times 2.5cm \times 2.5cm \times h$$

$$\Rightarrow \quad \frac{220cm^3 \times 7}{22 \times 2.5cm \times 2.5cm} = h$$

$$\Rightarrow \quad \frac{10cm^3 \times 7}{2.5cm \times 2.5cm} = h$$

$$\Rightarrow \quad \frac{70\ cm^3}{6.25\ cm^2} = h$$

$$\Rightarrow \quad \frac{70\ cm^3}{6\frac{1}{4}\ cm^2} = h$$

$$\Rightarrow \quad \frac{70\ cm^3}{25/4\ cm^2} = h$$

$$\Rightarrow \quad \frac{4 \times 70\ cm^3}{25\ cm^2} = h$$

$$\Rightarrow \quad \frac{4 \times 14\ cm^3}{5\ cm^2} = h$$

$$\Rightarrow \quad \frac{56\ cm^3}{5\ cm^2} = h$$

$$\Rightarrow \quad 11\tfrac{1}{5}\ cm = 11.2\text{ cm} = h$$

$\Rightarrow$ The height of the cylinder = $11^1/_5$ cm or 11.2cm

Alternatively, from the 5th step

$$\frac{70\ cm^3}{6.25\ cm^2} = h$$

$$\Rightarrow \quad \frac{7000\ cm^3}{625\ cm^2} = h$$

$$\Rightarrow \quad \frac{56\ cm^3}{5\ cm^2} = h$$

$$\Rightarrow \quad 11\tfrac{1}{5}\ cm = 11.2\text{ cm} = h$$

⇒ The height of the cylinder = $11\frac{1}{5}$ cm or 11.2cm

5 (a) **Approach 2** (making *h* the subject first)

$V = \pi r^2 \times h$

$\Rightarrow \quad \frac{V}{\pi r^2} = h$

$\Rightarrow \quad \frac{220\,cm^3}{\frac{22}{7} \times (2.5\,cm)^2} = h$

$\Rightarrow \quad \frac{7 \times 220\ cm^3}{22 \times 2.5cm \times 2.5\,cm} = h$

$\Rightarrow \quad \frac{7 \times 10\ cm^3}{2.5\,cm \times 2.5\,cm} = h$

$\Rightarrow \quad \frac{70\ cm^3}{6.25\ cm^2} = h$

$\Rightarrow \quad \frac{7000\ cm^3}{625\ cm^2} = h$

$\Rightarrow \quad \frac{56\ cm^3}{5\ cm^2} = h$

$\Rightarrow \quad 11\frac{1}{5}\ cm = h$ or

$\Rightarrow \quad 11.2\ \text{cm} = h$

⇒The height of the cylinder = $11\frac{1}{5}$ cm or 11.2cm

5 (b) **METHOD 1** (Using relation between exterior angle and no. of sides)

Since interior angle = 140°,

⇒ Exterior angle = 180° - 140° = 40°

Therefore Number of sides (n) = $\frac{360^o}{40^o}$

= 9 sides.

5 (b) **METHOD 2** (Using relation between interior angle and no. of sides)

For a regular polygon, each interior angle $= \frac{180^0(n-2)}{n}$, where n = no. of sides

$\Rightarrow$ $140° = \frac{180^0(n-2)}{n}$

$\Rightarrow$ $140° \text{ n} = 180° (\text{n} - 2)$

$\Rightarrow$ $140° \text{ n} = 180°\text{n} - 360°$

$\Rightarrow$ $360° = 180° \text{ n} - 140°\text{n}$

$\Rightarrow$ $360° = 40° \text{ n}$

$\Rightarrow$ $\frac{360°}{40^0} = \text{n}$

$\Rightarrow$ $9 = \text{n}$

$\therefore$ The polygon has 9 sides

August 1996

MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. U = {0, 1}. How many subsets have U?

A) 1 B) 2 C) 3 D) 4 E) 5

2. The following addition is done in base ten. What number represents ABC?

$$\begin{array}{r} 225 \\ 343 \\ ABC \\ \hline 1000 \end{array}$$

A) 324 B) 242 C) 423 D) 432 E) 234

*Use the mapping below to answer questions **3** and **4***

$2^3 \longrightarrow 8$

$2^2 \longrightarrow 4$

$2^1 \longrightarrow 2$

$2^0 \longrightarrow \mathbf{a}$

$2^{-1} \longrightarrow \mathbf{b}$

3. The value of **a** is

A) 0 B) ½ C) 1 D) 16 E) 32

4. What is the value of **b** ?

A) –2 B) ¼ C) ½ D) 1 E) 2

5. Write two hundred thousand and fifty seven in figures

A) 20,057

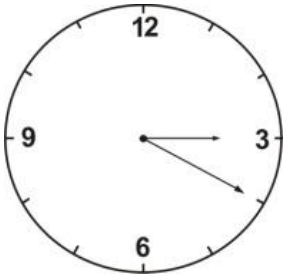
B) 200,057

C) 2,000,057

D) 20,000,057

E) 200,570

6.



What time is the clock above showing?

A) 4:30 B) 4:15 C) 3:40 D) 3:20 E) 3:04

7. Kojo, Ebo and Ama share ¢14,000.00 among themselves. Kojo had twice as much as Ebo and Ebo also had twice as much as Ama. How much did Ebo get?

A) ¢8,000.00

B) ¢6,000.00

C) ¢4,000.00

D) ¢3,000.00

E) ¢2,000.00

8. At what rate of simple interest will ¢5,000.00 amount to ¢ 7,500 if saved for 5 years ?

A) 5% B) 6 ⅔ % C) 7 ½ % D) 10% E) 12 ½ %

9. Simplify $^{1}/_{3}$ - ½ + $^{2}/_{5}$

A) $^{17}/_{30}$ B) $^{13}/_{30}$ C) $^{7}/_{30}$ D) $^{-7}/_{30}$ E) $^{-17}/_{30}$

10. If $2^{x} = 8$, what is the value of x ?

A) 0 B) 1 C) 2 D) 3 E) 4

The relation between the Celsius (C) and Fahrenheit (F) scale of temperature is given by:

$C = \frac{5}{9}(F - 32)$

11. If C is 40, F will be

A) 104.0 B) 78.4 C) 72.0 D) 65.6 E) 40.0

12. If C is –40, find the value of F

A) –104.0 B) –78.4 C) –72.0 D) –65.6 E) –40.0

13. The ages in years of 9 children at a birthday party are:2, 3, 3, 3, 4, 5, 5, 5 and 6.
What is the mean age?

A) 3.0 B) 3.5 C) 4.0 D) 4.5 E) 5.0

14. The ages in years of 9 children at a birthday party are:
2, 3, 3, 3, 4, 5, 5, 5 and 6. What is the median age?

A) 2.0 B) 3.0 C) 3.5 D) 4.0 E) 5.0

15. The ages in years of 9 children at a birthday party are: 2, 3, 3, 3, 4, 5, 5, 5 and 6.
If a child is picked at random, what is the probability that he is 5 years old?

A) $^2/_9$ B) $^1/_3$ C) $^4/_9$ D) $^5/_9$ E) $^2/_3$

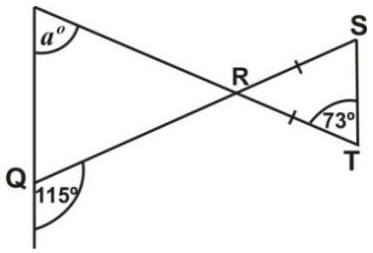
16. P (2, 5) and Q (-2, 3) are points in the Cartesian plane, find the vector PQ

A. $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ B. $\begin{pmatrix} -4 \\ -2 \end{pmatrix}$ C. $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$ D. $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$ E. $\begin{pmatrix} 0 \\ -2 \end{pmatrix}$

17. Find K in the vector equation $\begin{pmatrix} 3 \\ 4 \end{pmatrix} + K\begin{pmatrix} 3 \\ 4 \end{pmatrix} = -\begin{pmatrix} 3 \\ 4 \end{pmatrix}$

A) -3 B) -2 C) -1 D) – ¾ E) 2

18.



In the above diagram, |RS| = |RT|. Find a

A) 146° B) 81° C) 73° D) 65° E) 34°

19. If $a^2 - b^2 = (a + b)(a - b)$, evaluate $9.32^2 - 0.68^2$

A) 87.32 B) 86.4 C) 74.65 D) 10.0 E) 8.64

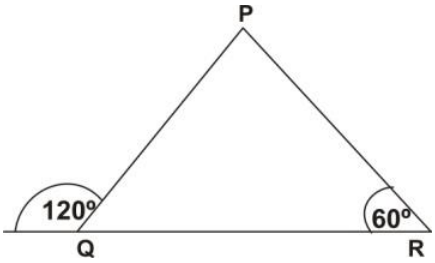
20. Simplify $\frac{6^2}{2^2 \times 3}$

A) 1 B) 2 C) 3 D) 6 E) 9

21. The sum of the ages of Kwaku and Kojo is 30 years. Kwaku is 4 years older than Kojo. How old is Kojo?

A) 19 B) 17 C) 13 D) 11 E) 9

22. What type of triangle is PQR?



A) Equilateral
B) Isosceles
C) scalene
D) right-angled
E) obtuse-angled

Use the equation y = (x + 2) (x – 2) to answer questions **23** *and* **24**.

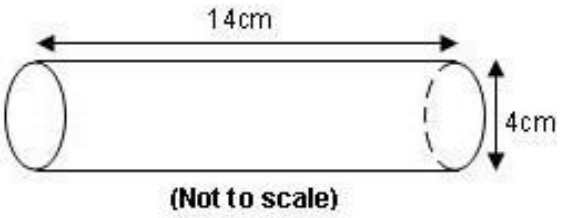
23. If x = -1, find y.

A) –4 B) –3 C) 3 D) 4 E) 9

24. Find the set of values of x for which y is zero.

A. {0} B. {0,2} C. {0,-2} D. {-2,2} E. {2}

Use the information below to answer questions **25** *to* **27**



In the diagram above, the cylinder has diameter 4cm and length 14cm.

(Take $\pi = {}^{22}/_{7}$)

25. Find the circumference of the base.

A) ${}^{44}/_{7}$ cm B) ${}^{88}/_{7}$ cm C) ${}^{176}/_{7}$ cm D) 44 cm E) 176 cm

26. The area of the base is

A) $176cm^2$ B) $44cm^2$ C) ${}^{176}/_{7}$ cm^2 D) ${}^{88}/_{7}$ cm^2 E) ${}^{44}/_{7}cm^2$

27. What is the volume of the cylinder?

A) $176cm^3$ B) $44cm^3$ C) ${}^{176}/_{7}$ cm^3 D) ${}^{88}/_{7}$ cm^3 E) ${}^{44}/_{7}cm^3$

28. A woman bought 210 oranges for ¢650.00. She sold all of them at 3 for ¢20.00. How much profit did she make?

A) ¢350.00

B) ¢450.00

C) ¢550.00

D) ¢650.00

E) ¢750.00

29. The ratio of boys to girls in a school is 9:11. If there are 400 pupils in the school, how many boys are there?

A) 80 B) 120 C) 180 D) 220 E) 280

30. Express the product of 162.5 x 0.5 in standard form.

A) 81.25×10^{-1}

B) 81.25×10

C) 8.125×10^{-1}

D) 8.125×10

E) 0.8125×10^{-2}

31. What is 16% of ¢500,000.00?

A) ¢80 B) ¢8,000 C) ¢80,000 D) ¢420,000 E) ¢492,000

32. Simplify $\frac{0.24 \times 14.3}{5.2}$

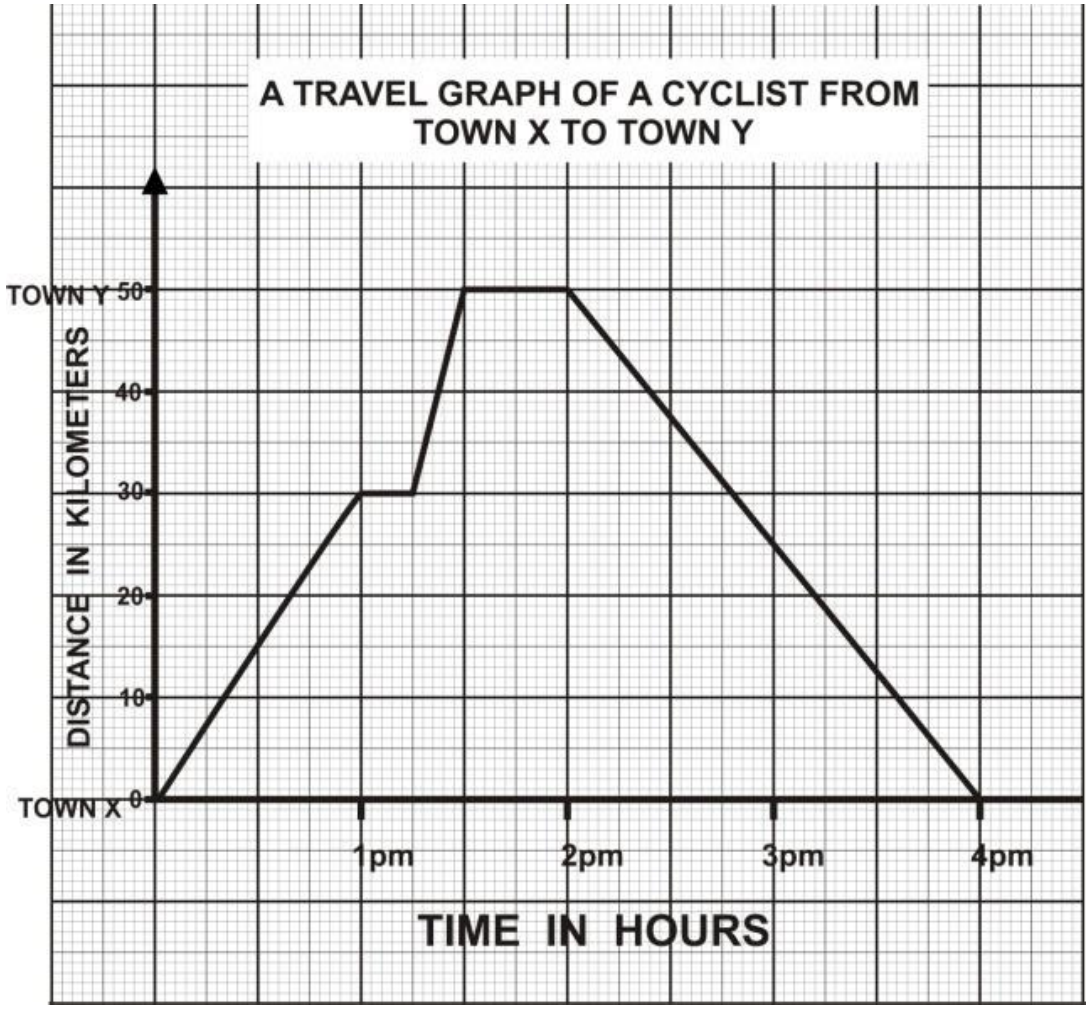
A) 66 B) 6.60 C) 0.70 D) 0.66 E) 0.60

33. The product of 2x and 3 is 138. Find x

A) 23 B) 33 C) 68 D) 69 E) 138

Use the graph below to answer questions **34** *to* **36**

The travel graph describes the journey of a cyclist from town **X** to town **Y**



34. What was the average speed for the return journey from town Y to town X

A) 100 km h^{-1}

B) 50 km h^{-1}

C) 33.33 kmh^{-1}

D) 25 km h^{-1}

E) 20 km h^{-1}

35. State the period within which he travelled to town Y after his first rest?

A) 1:00 – 2:00 pm

B) 1:00 – 4:00 pm

C) 1:15 – 1:30 pm

D) 1:30 – 2:00 pm

E) 2:00 – 4:00 pm

36. How many minutes did the cyclist spend at town Y?

A) 15 min B) 20 min C) 30 min D) 45 min E) 60 min

37. 200 bottles of equal capacity hold 350 litres of water. How much water does each bottle hold ?

A) 1750 litres

B) 175 litres

C) 17.5 litres

D) 1.75 litres

E) 0.17 litres

38. Find the solution set of $2x - 4 > -6$

A. $\{x = -5\}$ B. $\{x < 5\}$ C. $\{x > 5\}$ D. $\{x < -5\}$ E. $\{x > -5\}$

39. A rectangular field 50 meters wide and x metres long requires 260 meters of fencing. Which of the following statements is true?

A) $x + 100 = 260$

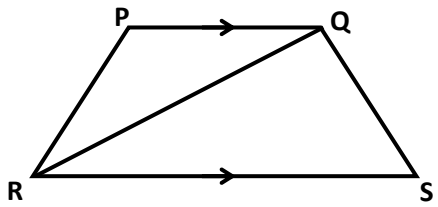
B) $2x + 50 = 260$

C) $4x + 200 = 260$

D) $2x + 100 = 260$

E) $4x + 100 = 260$

40.



In the diagram above, PQSR is a trapezium. PQ is parallel to RS. $\angle PQR = \angle QRS$. What type of triangle is triangle RQS?

A) Isosceles

B) scalene

C) equilateral

D) Right–angled

E) obtuse-angled

August 1996

MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. D) 4

2. D) 432

3. C) 1

4. C) ½

5. B) 200,057

6. D) 3:20

7. C) ¢4,000.00

8. D) 10%

9. C) $^{7}/_{30}$

10. D) 3

11. A) 104.0

12. E) –40.0

13. C) 4.0

14. D) 4.0

15. B) $^{1}/_{3}$

16. B. $\begin{pmatrix} -4 \\ -2 \end{pmatrix}$

17. B) -2

18. B) 81°

19. B)86.4

20. C) 3

21. C) 13

22. A) Equilateral

23. B) –3

24. D. {-2,2}

25. B) $^{88}/_{7}$ cm

26. D) $^{88}/_{7}$ cm^2

27. A) 176 cm^3

28. E) ¢750.00

29. C) 180

30. D) 8.125 x 10

31. C) ¢80,000

32. D) 0.66

33. A) 23

34. D) 25 km h^{-1}

35. C) 1:15 – 1:30 pm

36. C) 30 min

37. D) 1.75 litres

38. E. $\{x > -5\}$

39. D) $2x + 100 = 260$

40. B) scalene

August 1996

MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. (a) Kofi is n years old now

(i) How old was he 5 years ago?

(ii) How old will he be 10 years from now?

(ii) If his age in 10 years time will be four times his age 5 years ago, how old is he now?

(b) Convert 2342_{five} to a base ten numeral

(c) Given that $f = \frac{vu}{v+u}$, find v, if f = 20 and u = 5

2. (a) A man deposited ¢350,000.00 in his account in a bank. A simple interest of 4% per annum was paid on his deposit. Calculate the total amount at the end of 4 years.

(b) The cost of sending a telegram is ¢500 for the first 12 words and ¢25.00 for every extra word.

Find the cost of sending a telegram containing 20 words.

3. The table below shows the distribution of pupils in a JSS form one (1) class who speak some of the Ghanaian languages.

Ghanaian Language	**No. of students who speak the language**
Nzema	5
Ga	20
Twi	30
Ewe	25
Fante	10

(a) (i) Draw a pie chart for the distribution.

(ii) What is the modal Ghanaian language?

(b) If a pupil is selected at random from the form, what is the probability that he speaks Ga?

4. Using a ruler and a pair of compasses only,

a) Construct triangle ABC such that |AB| = 6cm, |AC| = 10cm and |BC| = 8cm.
Measure angle ABC.

b) Construct the perpendicular bisectors (mediators) of |AB| and |BC|. Let the bisectors meet at O.

c) Construct a circle with centre O and radius OA. Measure the radius of the circle.

5. a) Using a scale of 2cm to 2 units on both axes draw two perpendicular axis, OX and OY on a graph sheet.

b) On this graph sheet, mark the x-axis from –4 to 10 and the y-axis from –6 to 12

c) Plot on the same graph sheet the points A (2, 1), B(3, 4) and C(4, 2). Join the points to form a triangle ABC.

d) Draw an enlargement $A_1B_1C_1$ of triangle ABC with scale factor 2 from the origin O(0, 0), such that $A \rightarrow A_1$, $B \rightarrow B_1$ and $C \rightarrow C_1$. Indicate the co-ordinates of triangle $A_1B_1C_1$. Show all lines of transformation.

e) Using the x-axis as mirror line, draw the image $A_2B_2C_2$ of triangle ABC, where $A \rightarrow A_2$, $B \rightarrow B_2$ and $C \rightarrow C_2$. Indicate the co-ordinates of triangle $A_2B_2C_2$.

August 1996

MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1 (a)

(i) Kofi's age 5 years ago = $n - 5$

(ii) Kofi's age in 10 years' time = $n + 10$

Kofi's age in 10 years' time = 4 × (Kofi's age 5 years ago)

$\Rightarrow n + 10 = 4 \times (n - 5)$

$\Rightarrow n + 10 = 4n - 20$

$\Rightarrow 20 + 10 = 4n - n$

$\Rightarrow 30 = 3n$

$\Rightarrow 30 \div 3 = n$

$\Rightarrow 10 = n$

Therefore Kofi is 10 years old now

(b) Converting 2342_{five} to a base ten numeral

2	3	4	2
5^3	5^2	5^1	5^0

$\Rightarrow (2 \times 5^3) + (3 \times 5^2) + (4 \times 5^1) + (2 \times 5^0)$

$= (2 \times 125) + (3 \times 25) + (4 \times 5) + (2 \times 1)$

$= 250 + 75 + 20 + 2$

$= 347$

(c) If f = 20 and u = 5, we have

$$20 = \frac{v(5)}{v+5}$$

$\Rightarrow$ 20 (v + 5) = 5v

$\Rightarrow$ 20v + 100 = 5v

$\Rightarrow$ 20v – 5v = - 100

$\Rightarrow$ 15v = - 100

$\Rightarrow$ $\frac{15v}{15} = \frac{-100}{15}$

$\Rightarrow$ v = $\frac{-20}{3}$

$\Rightarrow$ v = $\underline{\underline{-6\frac{2}{3}}}$

2 (a) Total amount = Principal + S. Interest

= P + P × R × T

= 350,000 + (350,000 × 4% × 4)

= 350,000 + $(350{,}000 \times \frac{4}{100} \times 4)$

= 350,000 + (3,500 × 4 × 4)

= 350,000 + 56,000

= $\underline{\underline{¢\ 406{,}000}}$

(b) Cost of the first 12 words = ¢500

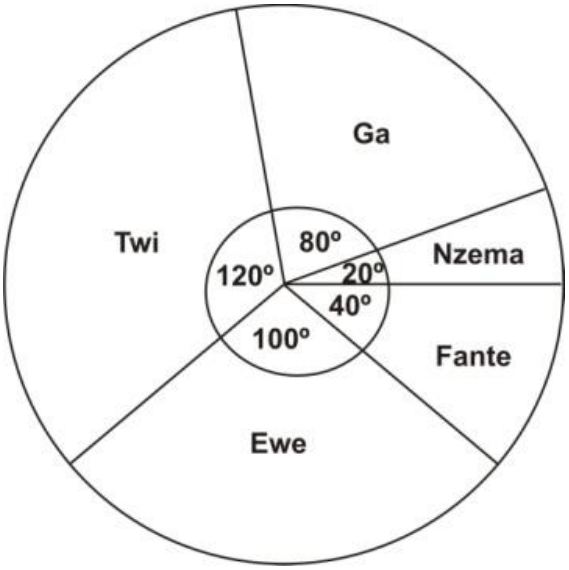
Remaining 8 words = 8 × ¢25 = ¢200

Cost of sending 20 words = ¢500 + ¢200 = $\underline{\underline{¢700}}$

3(a) (i)

Ghanaian Language	No. of students	Angle of sector
Nzema	5	$\frac{5}{90} \times 360^{o} = 20^{o}$
Ga	20	$\frac{20}{90} \times 360^{o} = 80^{o}$
Twi	30	$\frac{30}{90} \times 360^{o} = 120^{o}$
Ewe	25	$\frac{25}{90} \times 360^{o} = 100^{o}$
Fante	10	$\frac{10}{90} \times 360^{o} = 40^{o}$
TOTAL	**90**	**360°**

Pie chart showing the distribution of pupils in JSS form 1 who speak some of the Ghanaian Languages



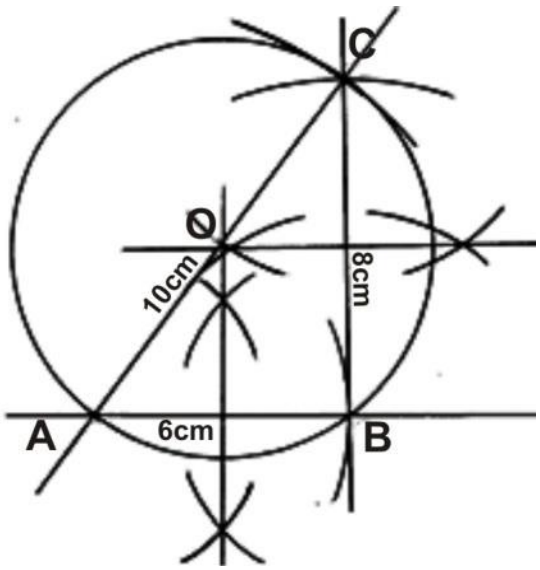
(a) (ii) The modal Ghanaian Language is **Twi**

(b) P (student speaks Ga) = $\frac{\text{Number of students who speak Ga}}{\text{Total number of students}}$

$= \frac{20}{5+20+30+25+10}$

$= \frac{20}{90} = \underline{\underline{\frac{2}{9}}}$

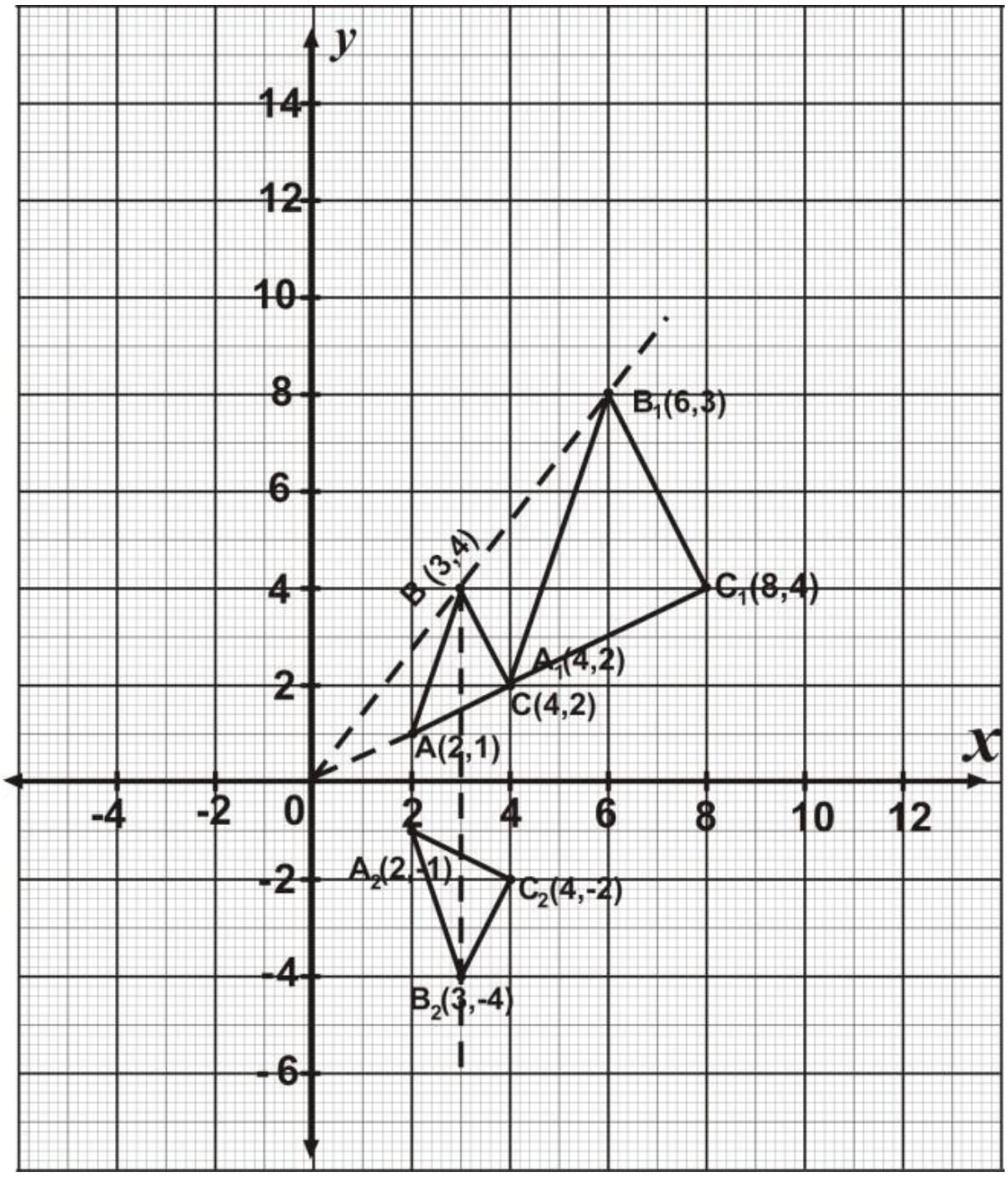
4.



(a) Angle ABC ≈ 90°

(c) Radius of circle ≈ 5 cm

5. **Approach 1** (By Inspection / Construction)



Approach 2 (The rule / formula)

(d) Enlarging (x, y) from (0,0) by scale factor k

$$\begin{pmatrix} x \\ y \end{pmatrix} \to k\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} kx \\ ky \end{pmatrix}$$

$$OA\begin{pmatrix} 2 \\ 1 \end{pmatrix} \to OA_1\begin{pmatrix} 2\times 2 \\ 2\times 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \quad \therefore A_1(4,2)$$

$$OB\begin{pmatrix}3\\4\end{pmatrix} \rightarrow OB_1\begin{pmatrix}2\times 3\\2\times 4\end{pmatrix} = \begin{pmatrix}6\\8\end{pmatrix}, \quad \therefore\ B_1(6,8)$$

$$OC\begin{pmatrix}4\\2\end{pmatrix} \rightarrow OC_1\begin{pmatrix}2\times 4\\2\times 2\end{pmatrix} = \begin{pmatrix}8\\4\end{pmatrix}, \quad \therefore\ C_1(8,4)$$

∴ Plot and join $A_1(4,2)$, $B_1(6,8)$ and $C_1(8,4)$ as the image of triangle ABC under an enlargement by scale factor 2 from the origin (0,0)

(e) Reflecting (x, y) in the x-axis

$$\begin{pmatrix}x\\y\end{pmatrix} \rightarrow \begin{pmatrix}x\\-y\end{pmatrix}$$

$$OA\begin{pmatrix}2\\1\end{pmatrix} \rightarrow OA_2\begin{pmatrix}2\\-1\end{pmatrix}, \quad \therefore\ A_2(2,-1)$$

$$OB\begin{pmatrix}3\\4\end{pmatrix} \rightarrow OB_2\begin{pmatrix}3\\-4\end{pmatrix}, \quad \therefore\ B_2(3,-4)$$

$$OC\begin{pmatrix}4\\2\end{pmatrix} \rightarrow OC_2\begin{pmatrix}4\\-2\end{pmatrix}, \quad \therefore C_2(4,-2)$$

∴ Plot and join $A_2(2,-1)$, $B_2(3,-4)$ and $C_2(4,-2)$ as the image of triangle ABC under a reflection in the x- axis

August 1995

MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. P = {prime numbers less than 20} and Q = {odd numbers less than 10}.

Find $P \cap Q$

A. {2, 3}

B. {1, 3, 5, 7, 11)

C. {3, 5, 7, 9}

D. {3, 5, 7}

E. {3, 5, 7, 11}

2. Convert 104_{ten} to a binary numeral

A) 1101000

B) 1010100

C) 1101100

D) 1011010

E) 1110100

3. What is the HCF of 48, 30 and 18 ?

A) 2 B) 3 C) 5 D) 6 E) 9

4. Express 34m 5cm 6mm in millimetres

A) 340506 mm

B) 342506 mm

C) 34056 mm

D) 30456 mm

E) 34565 mm

5. Write 356.07 in standard form

A) 35.607 x 10

B) 35.607 x 10^2

C) 3.5607 x 10^2

D) 3.5607 x 10^{-2}

E) 0.35607 x 10^3

6. Divide (1 ½ + ¼) by (1 ½ - ¼)

A) $^5/_7$ B) 1 C) 1 $^2/_5$ D) 1 ¾ E) 3

7. Which property is illustrated by the statement a × (b+c) = a×b + a×c

A) Inverse

B) Identity

C) Commutative

D) Distributive

E) Associative

8. How many edges has a cube?

A) 4 B) 6 C) 8 D) 12 E) 18

9. What is the mode of the following numbers:

4, 5, 3, 3, 4, 2, 7, 6, 5, 4, 4 and 1 ?

A) 3 B) 4 C) 5 D) 6 E) 7

10. What is the rule for the mapping below ?

x	1	2	3	4	5
↓	↓	↓	↓	↓	↓
y	2	4	8	16	32

A) $y = 2x+2$ B) $y = 2^x$ C) $y = x+2$ D) $y = x+1$ E) $y = 2x$

11. Solve $4x - 6 < -2$

A) $x < 1$ B) $x > 1$ C) $x < -1$ D) $x > -1$ E) $x < 4$

12. In a school, 80 pupils wrote an examination and 64 of them passed. What is the percentage of pupils who passed?

A) 8% B) 16% C) 20% D) 64% E) 80%

13. The interior angle of a regular polygon is 120°. How many sides has the polygon?

A) 3 B) 4 C) 5 D) 6 E) 8

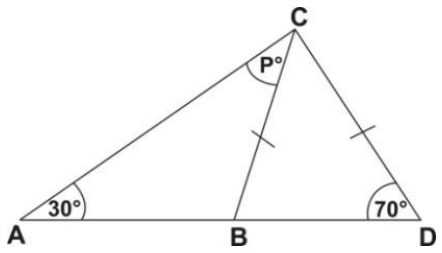
14. Kofi throws a die. What is the probability that he throws the number 2?

A) 1 B) $^2/_3$ C) ½ D) $^1/_3$ E) $^1/_6$

15. What is the image of - 4 under the mapping $x \to \frac{1}{2}x - 2$?

A) 4 B) 2 C) 0 D) –2 E) –4

16.



Find the value of P in the diagram above

A) 110 B) 90 C) 70 D) 50 E) 40

17. Make *a* subject of the relation $P = 2(a + b)$

A) $a = \dfrac{P - 2b}{2}$

B) $a = \dfrac{P + 2b}{2}$

C) $a = \dfrac{2b - P}{2}$

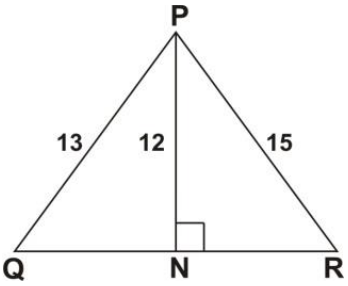
D) $a = \dfrac{P - b}{2}$

E) $a = \dfrac{P - 2}{b}$

18. What is the value of x in the relation $5^x = 125$?

A) 2 B) 3 C) 4 D) 5 E) 6

19. Calculate the length of QR in triangle PQR.



A) 11 B) 14 C) 16 D) 17 E) 25

20. Five cards are numbered 1 to 5. A card is picked at random. What is the probability that it has an even number?

A) 1 B) $^4/_5$ C) $^3/_5$ D) $^2/_5$ E) $^1/_5$

21. If $\mathbf{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$, find $3\mathbf{u} + 2\mathbf{v}$

A. $\begin{pmatrix} 8 \\ 9 \end{pmatrix}$ B. $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$ C. $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ D. $\begin{pmatrix} 4 \\ 9 \end{pmatrix}$ E. $\begin{pmatrix} -4 \\ -9 \end{pmatrix}$

22. Arrange the following fractions in descending order ¾ , $^5/_8$, $^4/_5$, $^{13}/_{20}$

A) $^4/_5$, ¾ , $^{13}/_{20}$, $^5/_8$

B) $^4/_5$, $^5/_8$, ¾ , $^{13}/_{20}$

C) $^5/_8$, $^4/_5$, ¾ , $^{13}/_{20}$

D) $^{4}/_{5}, {}^{5}/_{8}, {}^{13}/_{20}, ¾$

E) $^{13}/_{20}, {}^{5}/_{8}, {}^{4}/_{5}, ¾$

23. How many lines of symmetry has a square?

A) 0 B) 1 C) 2 D) 3 E) 4

24. If $(x, y) \rightarrow (x, 2y)$, find the image of (2 ½, - ¼) under the same mapping.

A. (2 ½, -2)

B. (2 ½, - ½)

C. (2, 2)

D. (2, - ¼)

E. (2, 2 ½)

25. The ages (in years) of eight boys are:

14, 14.5, 15, 12, 11.5, 13, 10.5 and 13.5.

What is their average age?

A) 14 B) 13 C) 12 D) 11 E) 10

26. Convert 133_{five} to a base ten numeral.

A) 23 B) 25 C) 31 D) 40 E) 43

27. Find the simple interest on ¢15,000.00 at rate of 20% per annum for 5 years.

A) ¢10,000.00

B) ¢15,000.00

C) ¢30,000.00

D) ¢50,000.00

E) ¢90,000.00

28. List all members of the set {x: 2<x<8, x is an integer}

A. {3,4,5}

B. {2,3,4,5,6,7,8}

C. {2, 2 ½,3,4,5,6,7,8}

D. {3, 3 ½,3,4,5,6,7,8}

E. {3,4,5,6,7}

29. Simplify 348.94 - 188.34

A) 60.60 B) 60.68 C) 160.60 D) 200.60 E) 206.60

30. Find the difference between the values of $(2d)^2$ and $2d^2$ when d=3

A) 16 B) 18 C) 24 D) 28 E) 54

31. The heights of two boys are in the ratio 4:5. The shorter boy is 80cm. What is the height of the taller boy?

A) 100cm B) 150cm C) 164cm D) 180cm E) 200cm

32. Factorize completely 5xy + 10ny

A. 5y(x + n)

B. 5y(x + 2n)

C. 5xy(1 + 2n)

D. 5(xy + 2ny)

E. x (5x + 10n)

33. If 13x – 12 = 5x + 60, find x

A) –9 B) –6 C) 4 D) 6 E) 9

34. The length of a spring when a mass of n kg is hung on it is L = (74+15n) mm. What is the length of the spring when a mass of 1.20kg is hanged on it?

A) 75.20 mm

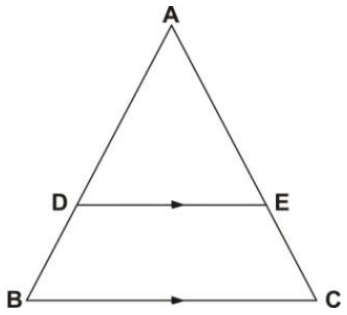
B) 89.00 mm

C) 92.00mm

D) 97.00 mm

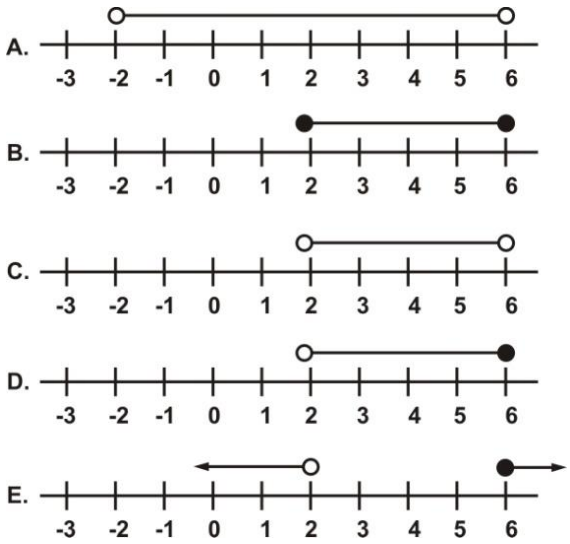
E) 104.00mm

35. In the figure below, triangle ABC is an enlargement of triangle ADE. If |AE| = 20cm and |EC| = 10cm. What is the scale factor of the enlargement?



A) ½ B) $^{3}/_{2}$ C) 2 D) $^{5}/_{2}$ E) 3

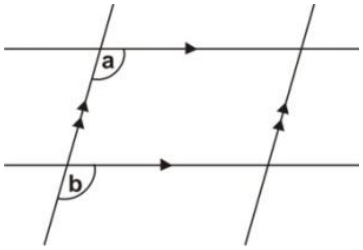
36. Which of the number lines below represents the inequality $2 < x \leq 6$?



37. A cylinder has a radius 6cm and height 7cm. Find its volume. (Take $\pi = {}^{22}/_{7}$)

A) $132cm^3$ B) $264cm^3$ C) $294cm^3$ D) $792cm^3$ E) $924cm^3$

38.



The angles a and b in the figure above are

A) interior opposite

B) corresponding

C) vertically opposite

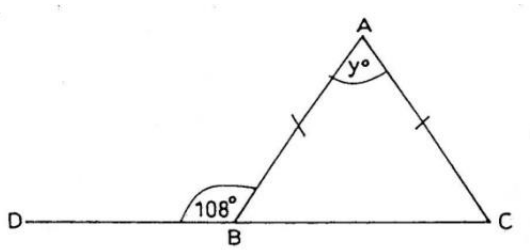
D) acute

E) alternate

39. A man travelled a distance of 1.2km in 30 minutes. What distance can he cover in 55 minutes travelling at the same speed?

A) 1.375km B) 2.02km C) 2.2km D) 12.02km E) 13.25km

40.



In the diagram above, ΔABC is an isosceles triangle. ∠ABD is 108°. Find the value of y.

A) 72 B) 60 C) 48 D) 36 E) 24

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MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. D. {3, 5, 7}

2. A) 1101000

3. D) 6

4. C) 34056 mm

5. C)3.5607×10^2

6. C) $1\ ^2/_5$

7. D) Distributive

8. D) 12

9. B) 4

10. B) $y = 2^x$

11. A) $x < 1$

12. E) 80%

13. D) 6

14. E) $^1/_6$

15. E) –4

16. E) 40

17. A) $a = \frac{P - 2b}{2}$

18. B) 3

19. B) 14

20. D) $^{2}/_{5}$

21. D. $\binom{4}{9}$

22. A) $^{4}/_{5}$, ¾ , $^{13}/_{20}$, $^{5}/_{8}$

23. E) 4

24. B. (2 ½, - ½)

25. B) 13

26. E) 43

27. B) ¢15,000.00

28. E. {3,4,5,6,7}

29. C) 160.60

30. B)18

31. A) 100cm

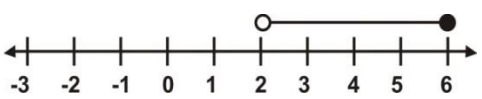
32. B. $5y(x + 2n)$

33. E) 9

34. C) 92.00 mm

35. B) $^{3}/_{2}$

36. D.



37. D) 792 cm^3

38. B) corresponding

39. C) 2.2km

40. D) 36

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MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. (a) There are 20 students in a hostel. 16 of them are fluent in French and 10 of them are fluent in English. Each student is fluent in at least one of the two languages

(i) Illustrate this information on a Venn diagram

(ii) How many students are fluent in both English and French?

(b) The sum of the ages of two brothers: Kofi and Kweku is 35. Kofi's age is two-thirds of Kwaku's age. Find their ages.

2. (a) Using a ruler and a pair of compasses only

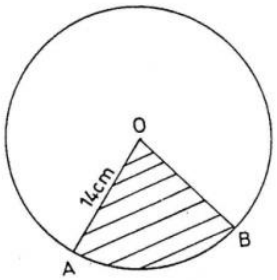
(i) construct triangle ABC such that |AB| = 9cm, angle BAC = 60° and angle ABC = 45°.

(ii) construct a line from the point C perpendicular to line AB and let it meet AB at P. Measure |CP| and |AP|

(b) What is the value of angle ACP?

3. (a) Mansah earns a salary of ¢10,000.00 per month as a sales girl. In addition to the salary, she is given a commission of 1.5% of whatever sales she makes in a month.
In January this year, she made sales of ¢7,500,000.00. What was the total amount Mansah earned at the end of January?

(b) The diagram below shows a circle with centre O and radius 14cm. The shaded region AOB is a sector with angle AOB = 72°.



Find:

(i) The length of the minor arc AB

(ii) The area of the shaded sector AOB [Take $\pi = {}^{22}/_{7}$]

4. (a) Using a scale of 2cm to 1 unit on both axes, draw two perpendicular axes, OX and OY on a graph sheet.

(b) On the graph sheet, mark the x-axis from –5 to 5 and the y-axis from –6 to 6

(c) Plot on the same graph sheet the points A(1, 1 ½), B(4, 1 ½) and C (1, 4). Join the points to form a triangle. What type of triangle have you drawn ?

(d) Draw the image triangle $A_1B_1C_1$ of ABC under a reflection in the y-axis, where $A \rightarrow A_1$, $B \rightarrow B_1$ and $C \rightarrow C_1$. Label the vertices and the co-ordinates clearly.

(e) Draw the image triangle $A_2B_2C_2$ of triangle ABC under an enlargement with scale factor –1 with the centre of enlargement as the origin (0,0), where $A \rightarrow A_2$, $B \rightarrow B_2$ and $C \rightarrow C_2$. Show all lines of enlargement. Label the vertices and co-ordinates clearly

(f) What single transformation maps $A_1B_1C_1$ onto $A_2B_2C_2$ where $A_1 \rightarrow A_2$, $B_1 \rightarrow B_2$ and $C_1 \rightarrow C_2$?

5. (a) The data below shows the distribution of the ages of workers in a factory.

Age (in years)	**No. of workers**
19	3
24	7
29	8
34	4
39	5
44	3

(i) How many workers are there in the factory?

(ii) What is the modal age of the distribution?

(iii) Calculate the mean age of the workers, correct to one decimal place.

(b) (i) Make T the subject of the relation $I = \frac{P \times T \times R}{100}$

(ii) If I = ¢40,000.00, P = ¢64,000.00 and R = 25%, find the value of T in years

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MATHEMATICS

SECTION B
ESSAY

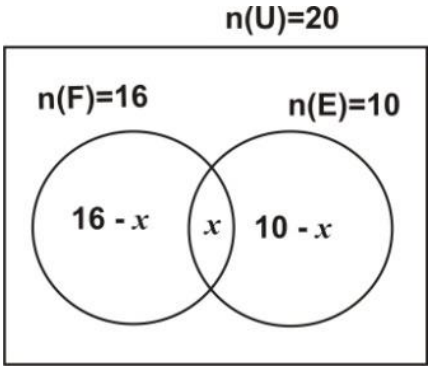
SOLUTIONS

1(a) (i) Let n(U) = No. of students in the hostel

n(F) = No. of students who are fluent in French

n(E) = No. of students who are fluent in English

x = No. of students who are fluent in both French and English



(ii) From the Venn diagram in (i) above,

$16 - x + x + 10 - x + 0 = 20$

$\Rightarrow 16 + 0 + 10 - x + 0 = 20$

$\Rightarrow 26 - x = 20$

$\Rightarrow 26 - 20 = x$

$\Rightarrow x = 26 - 20$

$\Rightarrow \underline{x = 6}$

$\therefore$ 6 students are fluent in both English and French

1(b)

Kofi's + Kweku's = 35 ----------**[1]**

And Kofi's = $\frac{2}{3}$ × Kweku's ----------**[2]**

⇒ Putting [2] in [1], we have

⇒ $(\frac{2}{3} \times \text{Kweku's}) + \text{Kweku's} = 35$ ----------**[3]**

Letting Kweku's age = w, **[3]** becomes

$$\frac{2}{3} \times w + w = 35$$

Approach 1 (Multiplying through by 3 to clear fractions first)

⇒ $3 \times (\frac{2}{3} w) + 3(w) = 3(35)$

⇒ $2w + 3w = 105$

⇒ $5w = 105$

⇒ $w = 105 \div 5$

⇒ $w = 21$

Therefore Kweku is 21 years old

and Kofi's age = two-thirds of 21 years

$= \frac{2}{3} \times 21\text{years}$

$= 2 \times 7$ years

= 14 years

1(b)

Approach 2 (Simplifying the LHS first)

$$\frac{2}{3} \times w + w = 35$$

⇒ $\frac{2w}{3} + \frac{w}{1} = 35$

⇒ $\frac{2w + 3w}{3} = 35$

⇒ $\frac{5w}{3} = 35$

⇒ $5w = 35 \times 3$

⇒ $5w = 105$

⇒ $\frac{5w}{5} = \frac{105}{5}$

⇒ $w = 21$

Therefore Kweku is 21 years old

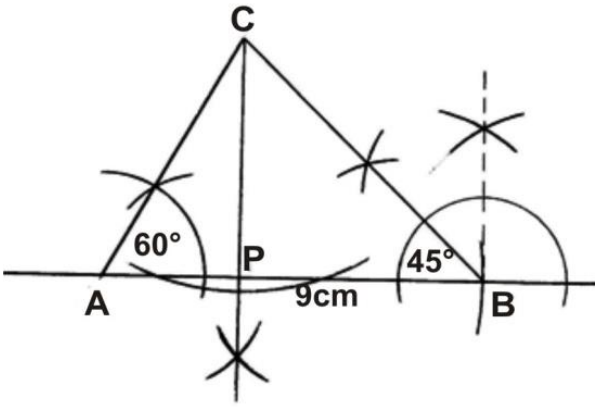
and Kofi's age = two-thirds of 21 years

$= \frac{2}{3} \times 21 \text{years}$

$= 2 \times 7 \text{ years}$

$= \underline{14 \text{ years}}$

2 (a)



(a) **(ii)** |CP| ≈ 5.7 cm

|AP| ≈ 3.3 cm

(b) angle ACP ≈ 30°

3 (a) Total amount earned = Salary + Commission

= ¢10,000 + 1.5% of Sales

$= ¢10{,}000 + (\frac{1.5}{100} \times ¢7{,}500{,}000)$

= ¢10,000 + ¢112,500

= ¢ 122,500

3 (b) (i) Length of minor arc = fraction of circumference

= Minor arc AB fraction × Circumference.

$= \frac{\theta}{360^\circ} \times 2\pi r$

$= \frac{72^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 14cm$

$= \frac{1}{5} \times 2 \times 22 \times 2cm$

$= \frac{88}{5}$ cm

$= \underline{\underline{17\frac{3}{5}}}$ cm or $\underline{\underline{17.6 \text{ cm}}}$

(ii) Area of the shaded sector AOB = fraction of area

= Minor area AOB fraction × area

$= \frac{\theta}{360^\circ} \times \pi r^2$

$= \frac{72^\circ}{360^\circ} \times \frac{22}{7} \times (14cm)^2$

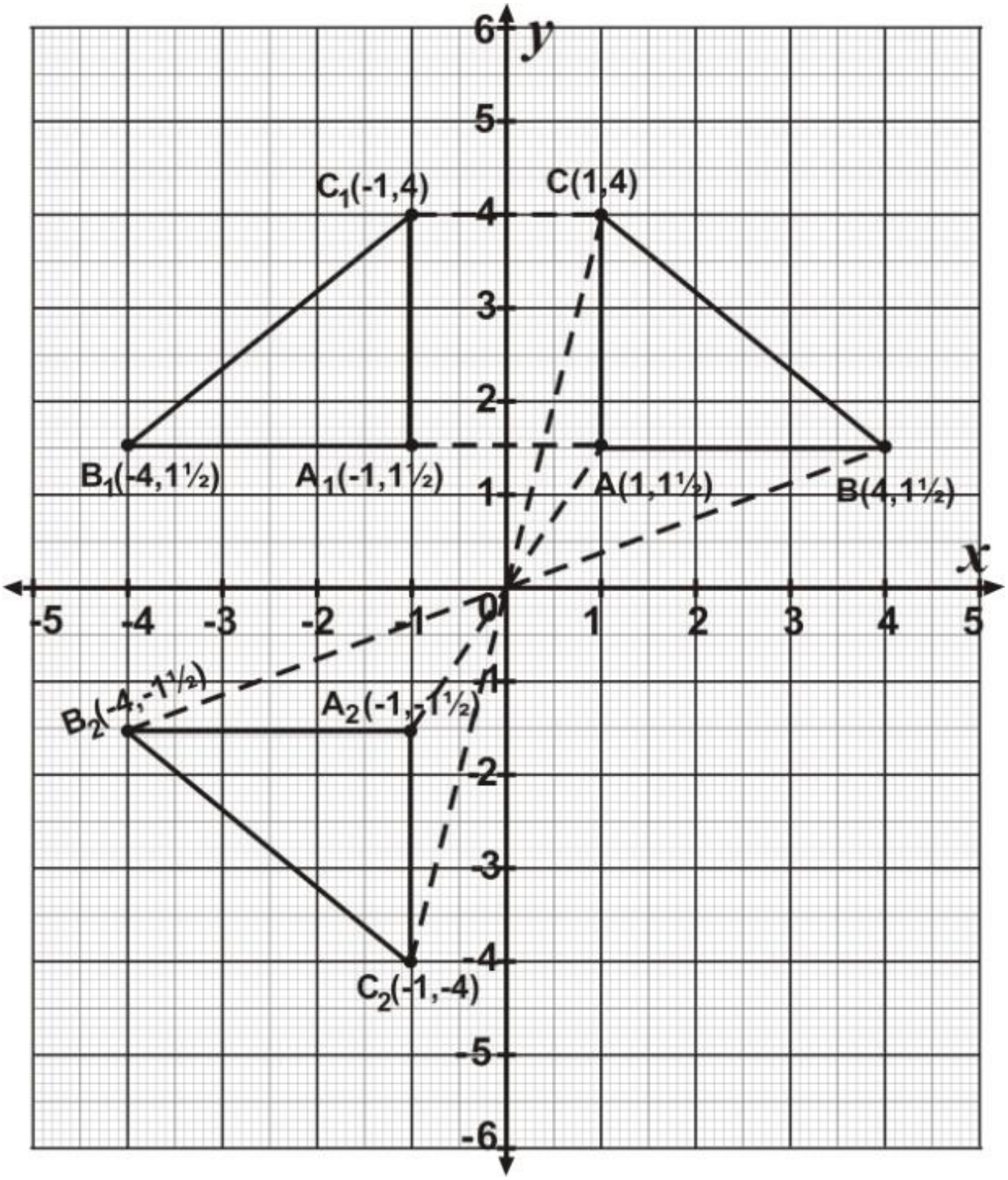
$= \frac{72^\circ}{360^\circ} \times \frac{22}{7} \times 14cm \times 14cm$

$= \frac{1}{5} \times 22 \times 2cm \times 14cm$

$= \frac{616cm^2}{5}$

$= \underline{\underline{123\frac{1}{5} cm^2}}$ or $\underline{\underline{123.2 \ cm^2}}$

4 (a) **Approach 1** (By Inspection / Construction)



Approach 2 (The rule / formula)

4 (d) Reflecting (x, y) in the y-axis

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} -x \\ y \end{pmatrix}$$

$$OA\begin{pmatrix} 1 \\ 1\frac{1}{2} \end{pmatrix} \rightarrow OA_1\begin{pmatrix} -1 \\ 1\frac{1}{2} \end{pmatrix}, \qquad \therefore A_1(-1,\ 1\tfrac{1}{2})$$

$$OB\begin{pmatrix} 4 \\ 1\frac{1}{2} \end{pmatrix} \rightarrow OB_1\begin{pmatrix} -4 \\ 1\frac{1}{2} \end{pmatrix}, \qquad \therefore B_1(-4, 1\tfrac{1}{2})$$

$$OC\begin{pmatrix}1\\4\end{pmatrix} \rightarrow OC_1\begin{pmatrix}-1\\4\end{pmatrix}, \qquad \therefore C_1(-1,4)$$

∴ Plot and join A_1(-1,1½), B_1(-4,1½) and C_1(-1,4) as the image of triangle ABC under a reflection in the y axis.

4 (e) Enlarging $(x, y)$from (0,0) by scale factor k

$$\begin{pmatrix}x\\y\end{pmatrix} \rightarrow k\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}kx\\ky\end{pmatrix}$$

$$OA\begin{pmatrix}1\\1\frac{1}{2}\end{pmatrix} \rightarrow OA_2\begin{pmatrix}-1\times1\\-1\times1\frac{1}{2}\end{pmatrix} = \begin{pmatrix}-1\\-1\frac{1}{2}\end{pmatrix}, \quad \therefore A_2(-1,-1\tfrac{1}{2})$$

$$OB\begin{pmatrix}1\\1\frac{1}{2}\end{pmatrix} \rightarrow OB_2\begin{pmatrix}-1\times4\\-1\times1\frac{1}{2}\end{pmatrix} = \begin{pmatrix}-4\\-1\frac{1}{2}\end{pmatrix}, \quad \therefore B_2(-4,-1\tfrac{1}{2})$$

$$OC\begin{pmatrix}1\\4\end{pmatrix} \rightarrow OC_2\begin{pmatrix}-1\times1\\-1\times4\end{pmatrix} = \begin{pmatrix}-1\\-4\end{pmatrix}, \quad \therefore C_2(-1,-4)$$

∴ Plot and join A_2(-1,-1½), B_2(-4,-1½) and C_2(-1,-4) as the image of triangle ABC under an enlargement by scale factor -1 from the origin (0,0)

4(f) The single transformation that maps $A_1B_1C_1$ onto $A_2B_2C_2$ is reflection in the *x*-axis

5 (a) (i) Total no. of workers = 3 + 7 + 8 + 4 + 5 + 3

= 30

(ii) Modal age = the most occurring age

= 29 years

(iii) Finding Mean age using the table,

Age in years (*x*)	**No. of workers (f)**	**f *x***
19	3	57
24	7	168
29	8	232
34	4	136
39	5	195
44	3	132
	Σf = 30	**Σf*x*= 920**

$$\text{The mean age} = \frac{\Sigma f x}{\Sigma f} = \frac{920}{30}$$

$$= \underline{\underline{30\tfrac{2}{3}\ years}} \approx \underline{\underline{30.67 \text{ years}}}$$

5 (b) (i)

$$I = \frac{P \times T \times R}{100}, \qquad \text{making } T \text{ the subject}$$

$$\Rightarrow \quad 100\ I = P\ T\ R \qquad \text{Cross-multiplying}$$

$$\Rightarrow \quad \frac{100\ I}{P\ R} = \frac{P\ T\ R}{P\ R}$$

$$\Rightarrow \quad \frac{100\ I}{P\ R} = T$$

$$\Rightarrow \quad \underline{\underline{T = \frac{100\ I}{P\ R}}}$$

(ii) $\mathbf{T = \frac{100\ I}{P\ R}}$

$$= \frac{100 \times 40{,}000}{64{,}000 \times 25}$$

$$= \frac{5}{2}$$

$$= \underline{\underline{\textbf{2 ½ years}}}$$

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August 1994

MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. Which of the following is the set of factors of 12?

A. {12, 6, 4, 3, 2, 1}

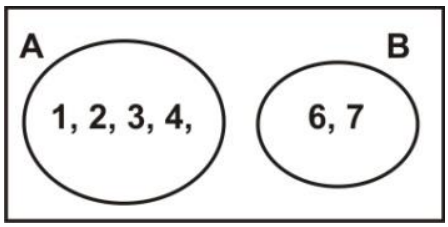
B. {12, 6, 4, 3, 2}

C. {12, 6, 4, 2}

D. {6, 4, 2, 1}

E. {6, 4, 3, 2}

2. Which of the following describes the relationship between the sets A and B in the Venn diagram below?



A) $A \subset B$

B) $A \cap B = 5$

C) $A \cap B = \phi$

D) $A \cup B = \{1,2,3,4,5,6,7\}$

E) $B \subset A$

3. Mark is 30 years old. Yaw is half as old as Mark. Paul is 10 years older than Yaw. How old is Paul?

A) 30 years

B) 25 years

C) 20 years

D) 15 years

E) 10 years

4. How many lines of symmetry does a rectangle have ?

A) 1 B) 2 C) 3 D) 9 E) 11

5. Which of the following is not a prime number ?

A) 3 B) 5 C) 7 D) 9 E) 11

6. If x = {1, 2, 3, 4, 5}, find the truth set of $2x + 1 < 7$

A. {1, 2} B. {2, 3} C. {1, 2, 3} D. {3} E. {2}

7. In how many years will ¢5,000.00 yield a simple interest of ¢1,000.00 at a rate of 5% per annum.

A) 4 years
B) 5 years
C) 10 years
D) 25 years
E) 30 years

8. Make *m* the subject of the relation $f = \frac{t}{s} - m$

A) $m = \frac{t}{s} + f$

B) $m = \frac{t}{s} - f$

C) $m = \frac{fs - t}{s}$

D) $m = f - \frac{t}{s}$

E) $m = -f - \frac{t}{s}$

9. Which property of arithmetic operation is illustrated by the statement:

$a \times (b+c) = ab + ac$?

A) Addition
B) Association
C) Commutative
D) Multiplication
E) Distributive

10. Simplify $15^9 \div 15^7$

A) 30 B) 63 C) 225 D) 240 E) 625

11. If $^1/_x = 1\frac{1}{2}$, find x.

A) $3\frac{1}{2}$ B) $^3/_2$ C) $^4/_3$ D) $^2/_3$ E) $^1/_3$

12. Calculate the size of an exterior angle of a regular pentagon.

A) 72° B) 90° C) 108° D) 360° E) 540°

13. Factorize $x^2 - 5x + 6$

A. $(x + 3)(x - 2)$
B. $(x - 2)(x - 3)$
C. $(x + 1)(x - 6)$
D. $(x + 2)(x + 3)$
E. $(x + 6)(x - 1)$

14. Kwame, Atsu and Kojo shared a profit of ¢500,000.00 in the ratio 1:4:3 respectively. How much did Atsu get?

A) ¢62,500.00
B) ¢125,000.00
C) ¢187,500.00

D) ¢250,000.00

E) ¢312,500.00

15. Use the identity $a^2 – b^2 = (a + b)(a – b)$ to evaluate $83^2 - 17^2$

A) 660 B) 6,600 C) 7,178 D) 7,600 E) 8,317

16. What is the probability of obtaining a prime number when a fair die is thrown once?

A) ⅔ B) ½ C) ⅓ D) $^1/_6$ E) $^5/_6$

17. The following are the scores obtained by girls in a beauty contest:

12, 16, 19, 14, 17, 8, 11, 19.

What is the probability of obtaining a score of 19?

A) $^1/_9$ B) $^1/_8$ C) ¼ D) $^{19}/_{53}$ E) $^{19}/_{108}$

18. Express 25 as a percentage of 75.

A) 300% B) 100% C) 50% D) 33.3% E) 25%

19. Find the value of *m* given that $m = (3^6 \div 3^4) \times 27$

A) 3^1 B) 3^2 C) 3^5 D) 3^6 E) 3^{13}

20. Express 0.0043216 in standard form

A) 4.3216×10^{-4}

B) 4.3216×10^{-3}

C) 4.3216×10

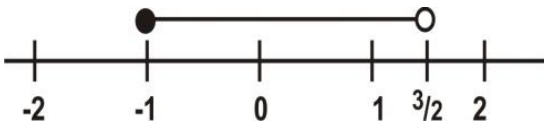
D) 4.3216×10^3

E) 4.3216×10^4

21. Given that a = 2 and b = 3, evaluate (2a + b) (a – 2b)

A) 28 B) 8 C) –7 D) –8 E) –28

22.



Which of the following is illustrated on the number line above?

A) $-1 < x < {}^{3}/_{2}$

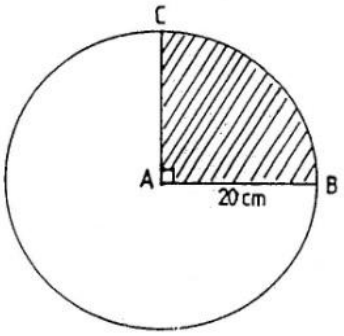
B) $-1 \leq x < {}^{3}/_{2}$

C) $-1 \leq x \leq {}^{3}/_{2}$

D) $-1 < x \leq {}^{3}/_{2}$

E) ${}^{3}/_{2} \leq x \leq -1$

23.



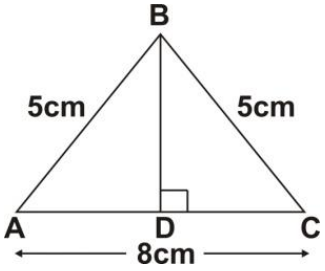
In the diagram above, A is the centre of the circle with radius 20cm. If angle BAC is 90°, find the perimeter of the shaded sector.

[Take $\pi = 3.14$]

A) 31.4cm B) 31.8cm C) 40.0cm D) 51.4cm E) 71.4

24. In triangle ABC, |AB| = |BC| = 5cm, and |AC| = 8cm.

Find |BD|.



A) 3cm B) 4cm C) 9cm D) 33cm E) 41cm

25. Write ¢35,632.00 correct to the nearest thousand cedis.

A. ¢40,000.00
B. ¢36,000.00
C. ¢35,600.00
D. ¢35,000.00
E. ¢30,000.00

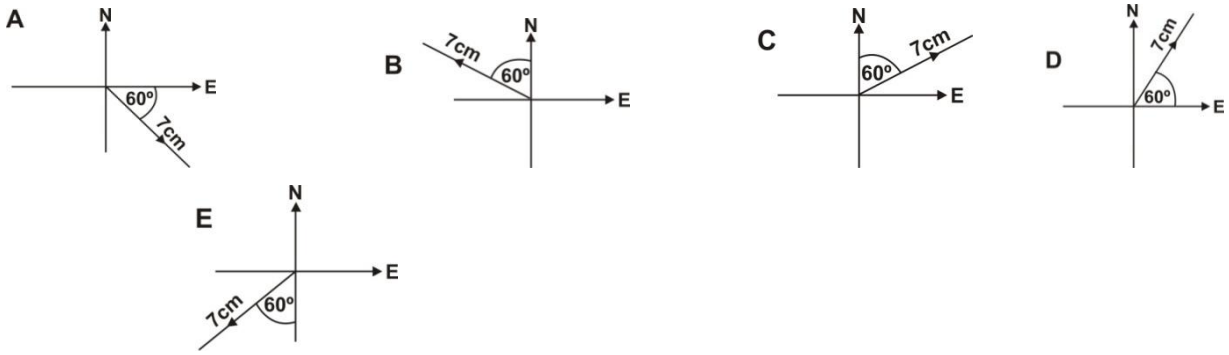
26. Solve for x if $2x + 5 = 3x - 7$

A) 12 B) 2 C) $^2/_3$ D) –2 E) –12

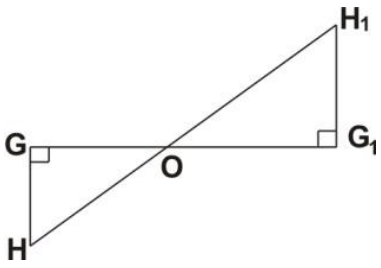
27. Evaluate $(\,^2/_3 - ¼) \div \,^5/_6$

A) ½ B) $^1/_5$ C) $^2/_3$ D) –2 E) -12

28. A boy walked 7km on a bearing 060°. Which of the following diagrams shows his direction.



29.



In the diagram, G′OH′ is an enlargement of triangle GOH with scale factor k. If |GG′| = 5cm, |GO| = 2cm and |G′H′| = 1cm. What is the value of k ?

A) $-\frac{5}{2}$ B) $-\frac{3}{2}$ C) $-\frac{2}{5}$ D) $\frac{2}{3}$ E) $\frac{3}{2}$

30. If $14_x = 9_{ten}$, find x.

A) Nine B) Eight C) Seven D) Six E)Five

31. How many faces has a cuboid ?

A) 6 B) 8 C) 12 D) 16 E) 32

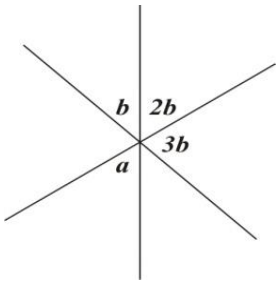
32. Which of the following is a factor of the expression ac – 2bc + ad – 2bd?

A) c – d B) a – 2b C) a + b D) a + 2b E) c + 2d

33. Three boys weeded a piece of land in 4 hours. How long would it take 18 boys to weed the same piece of land weeding at the same rate?

A) ⅔ h B) 3 ⅓ h C) 4 ⅔ h D) 22 h E) 24 h

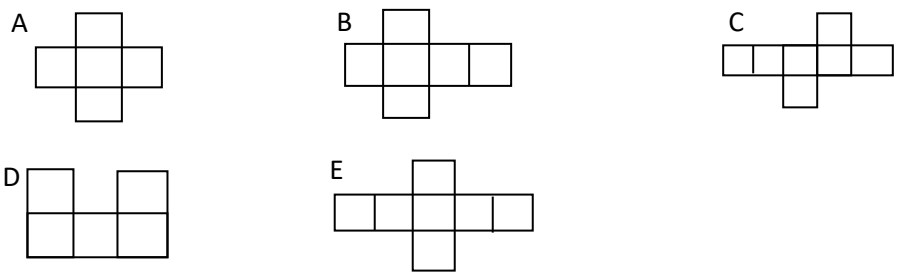
34.



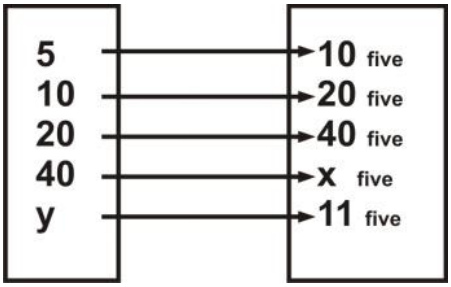
Find the size of the angle marked a in the diagram above.

A) 30° B) 60° C) 90° D) 120° E) 180°

35. Which of the following is the net of a cube?



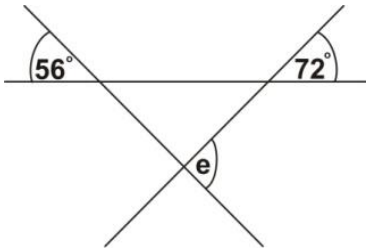
36. Use the mapping below to answer question 36



Find the value of x

A) 30_{five} B) 100_{five} C) 110_{five} D) 120_{five} E)130_{five}

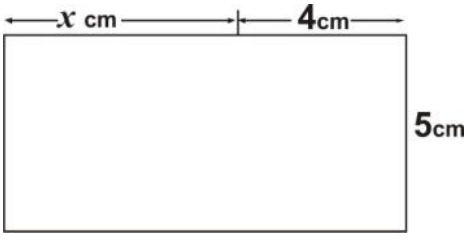
37.



In the diagram above, find the size of the angle marked e^o.

A) 52° B) 56° C) 72° D) 108° E) 128°

38.



If the area of the figure below is $60cm^2$, find *x*.

A) 8cm B) 9cm C)12cm D)16cm E) 20cm

39. The mean of the numbers 4, 3, 3, x is 5, find x

A) 20 B) 10 C) 5 D) 4 E) 3

40. If a number is selected at random from the table below, what is the probability that the number is 5?

Number	1	3	5	7	9
Frequency	25	15	8	10	2

A) $^{8}/_{25}$ B) $^{4}/_{25}$ C) $^{1}/_{12}$ D) $^{2}/_{15}$ E) $^{5}/_{8}$

August 1994

MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1.	A.	{12, 6, 4, 3, 2, 1}
2.	C)	A∩B = ∅
3.	B)	25 years
4.	B)	2
5.	D)	9
6.	A.	{1, 2}
7.	A)	4 years
8.	B)	$m = \frac{t}{s} - f$
9.	E)	Distributive
10.	C)	225
11.	D)	$^2/_3$
12.	A)	72°
13.	B.	$(x - 2)(x - 3)$
14.	D)	¢250,000.00
15.	B)	6,600

16. B) ½

17. C) ¼

18. D) 33.3%

19. B) 3^2

20. B) 4.3216×10^{-3}

21. E) –28

22. B) $-1 \leq x < 3/2$

23. E) 71.4 cm

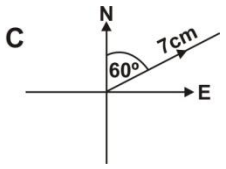
24. A) 3cm

25. B. ¢36,000.00

26. A) 12

27. A) ½

28.



29. B) $-3/2$

30. E) Five

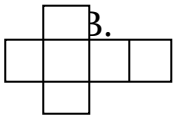
31. A) 6

32. B) a – 2b

33. A) ⅔ hour

34. B) 60°

35. B.



36. E) 130_{five}

37. E) 128°

38. A) 8cm

39. B) 10

40. D) $^{2}/_{15}$

August 1994

MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt **four** *questions* **only** *from this section*

All working must be clearly shown.

The use of calculators is not allowed

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. (a) Multiply $(a - b)$ by $(2b - a)$

(b) Find the truth set of $2x - 6 \leq 5(3 - x)$.

Illustrate your answer on a number line.

(c) Given that $u = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and $v = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$, find $\frac{1}{3}(u + \frac{1}{2}v)$

2. (a) A ladder leans against a wall. The end of the ladder touches the wall 12m from the ground. The foot of the ladder is 9m away from the foot of the wall.

(i) What is the length of the ladder?

(ii) Calculate the angle that the ladder makes with the wall.

(b) Given that $\pi = 3.14$ and $g = 20$. Find the value of F in the relation $F = \frac{3}{4}\pi g^2$

3. Using a ruler and a pair of compasses only,

a) construct triangle PQR in which |PQ| = 8cm, angle QPR = 45° and angle PQR = 90°. Measure |QR|

b) construct the mediator of line PQ to meet line PR at the point S.

c) With S as the centre and radius 3cm, construct a circle.

4. Using a scale of 2cm to 1unit on both axis, draw two perpendicular lines OX and OY on a graph sheet. Mark the x-axis from –5 to 5 and the y-axis from –6 to 6. Mark the origin O. Draw on the same graph sheet, indicating in each case, the co-ordinates of all the vertices,

(i) The square ABCD where A(1,2), B(4, 2), C(4,5) and D(1,5)

(ii) Using the y-axis as a mirror line draw the image $A_1B_1C_1D_1$ of square ABCD where $A \rightarrow A_1$, $B \rightarrow B_1$, $C \rightarrow C_1$ and $D \rightarrow D_1$

(iii) Draw an enlargement $A_2B_2C_2D_2$ of the square ABCD with scale factor –1 from O, such that $A \rightarrow A_2$, $B \rightarrow B_2$, $C \rightarrow C_2$ and $D \rightarrow D_2$

(iv) What single transformation maps $A_2B_2C_2D_2$ onto the square $A_1B_1C_1D_1$?

5. The following table gives the distribution of sales of soft drinks sold by the Akrokyere JSS canteen in one week.

Soft Drink	**Number of bottles sold**
Fanta	13
Pepsi	21
Mirinda	14
Coca-cola	17
Sprite	7

Draw a pie chart to illustrate the sales.

August 1994

MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1 (a) $(a - b)(2b - a)$

$2ab - a^2 - 2b^2 + ab$

$\Rightarrow \quad 2ab + ab - a^2 - 2b^2$

$\Rightarrow \quad \underline{3ab - a^2 - 2b^2}$

(b) $2x - 6 \leq 5(3 - x)$

$\Rightarrow \quad 2x - 6 \leq 15 - 5x$

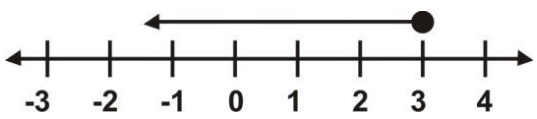
$\Rightarrow \quad 2x + 5x \leq 15 + 6$

$\Rightarrow \quad 7x \leq 21$

$\Rightarrow \quad \frac{7x}{7} \leq \frac{21}{7}$

$\Rightarrow \quad \underline{\underline{x \leq 3}}$

The number line illustration is shown below:



(c) $\frac{1}{3}(\mathbf{u} + \frac{1}{2}\mathbf{v})$

$= \quad \frac{1}{3}\left[\begin{pmatrix} -2 \\ 3 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 2 \\ 6 \end{pmatrix}\right]$

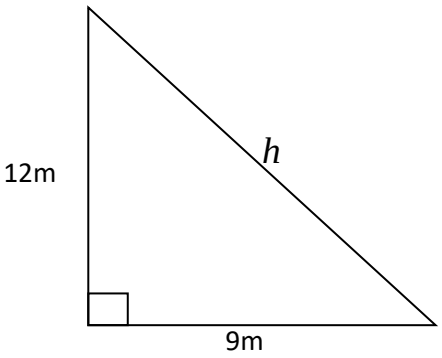
$$= \frac{1}{3}\left[\begin{pmatrix}-2\\3\end{pmatrix}+\begin{pmatrix}\frac{1}{2}\times 2\\ \frac{1}{2}\times 6\end{pmatrix}\right]$$

$$= \frac{1}{3}\left[\begin{pmatrix}-2\\3\end{pmatrix}+\begin{pmatrix}1\\3\end{pmatrix}\right]$$

$$= \frac{1}{3}\begin{pmatrix}-2+1\\3+3\end{pmatrix} \quad = \quad \frac{1}{3}\begin{pmatrix}-1\\6\end{pmatrix}$$

$$= \begin{pmatrix}\frac{1}{3}\times -1\\ \frac{1}{3}\times 6\end{pmatrix} \quad = \quad \underline{\underline{\begin{pmatrix}-\frac{1}{3}\\2\end{pmatrix}}}$$

2 (a) (i) From the Pythagorean Theorem, if h is the hypotenuse of a right-angled triangle and a and b are the other two sides, then $h^2 = a^2 + b^2$



From the diagram above, if h = length of ladder, we have

$h^2 = (12\text{m})^2 + (9\text{m})^2$

$h^2 = 144\text{m}^2 + 81\text{m}^2$

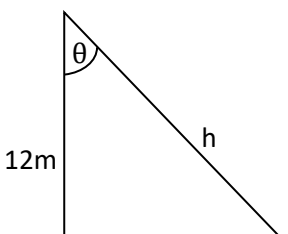
$h^2 = 225\text{m}^2$

$h = \sqrt{225m^2}$

$h = \underline{15\text{ m}}$

The length of the ladder is 15m

(a) (ii)



$$\tan\theta = \frac{\text{Opposite side}}{\text{adjacent side}}$$

From the diagram above,

$\Rightarrow \quad \tan\theta = \frac{9\,\text{m}}{12\,\text{m}}$

$\Rightarrow \quad \tan\theta = \frac{3}{4}$

$\Rightarrow \quad \theta = \tan^{-1}\left(\frac{3}{4}\right)$

$\Rightarrow \quad \underline{\underline{\theta \approx 36.87^{o}}}$ (*Using tables or a scientific calculator*)

(b) $F = \frac{3}{4}\pi g^{2}$,

given that $\pi = 3.14$ and $g = 20$

$= \quad \frac{3}{4} \times 3.14 \times 20^{2}$

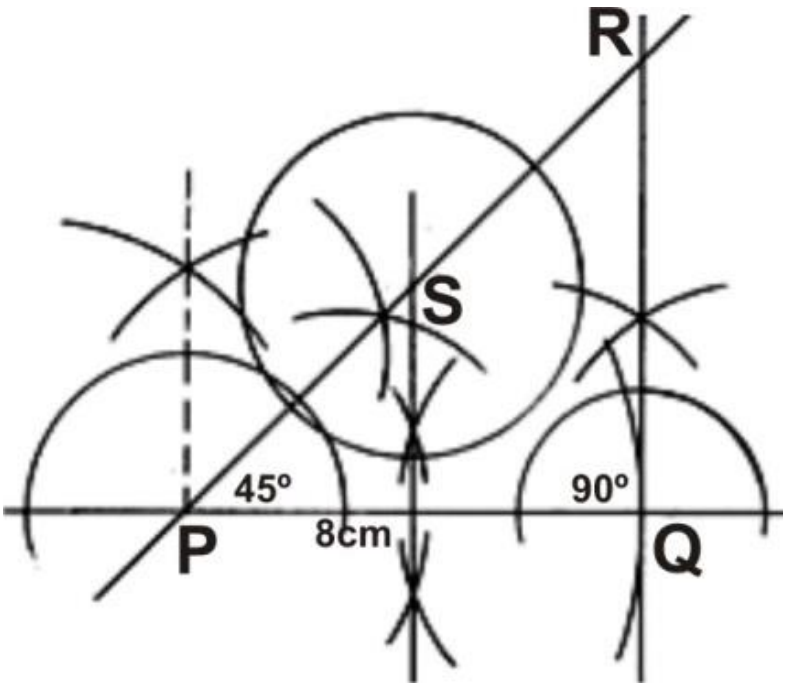
$= \quad \frac{3}{4} \times 3.14 \times 400$

$= \quad 3 \times 3.14 \times 100$

$= \quad 3 \times 314$

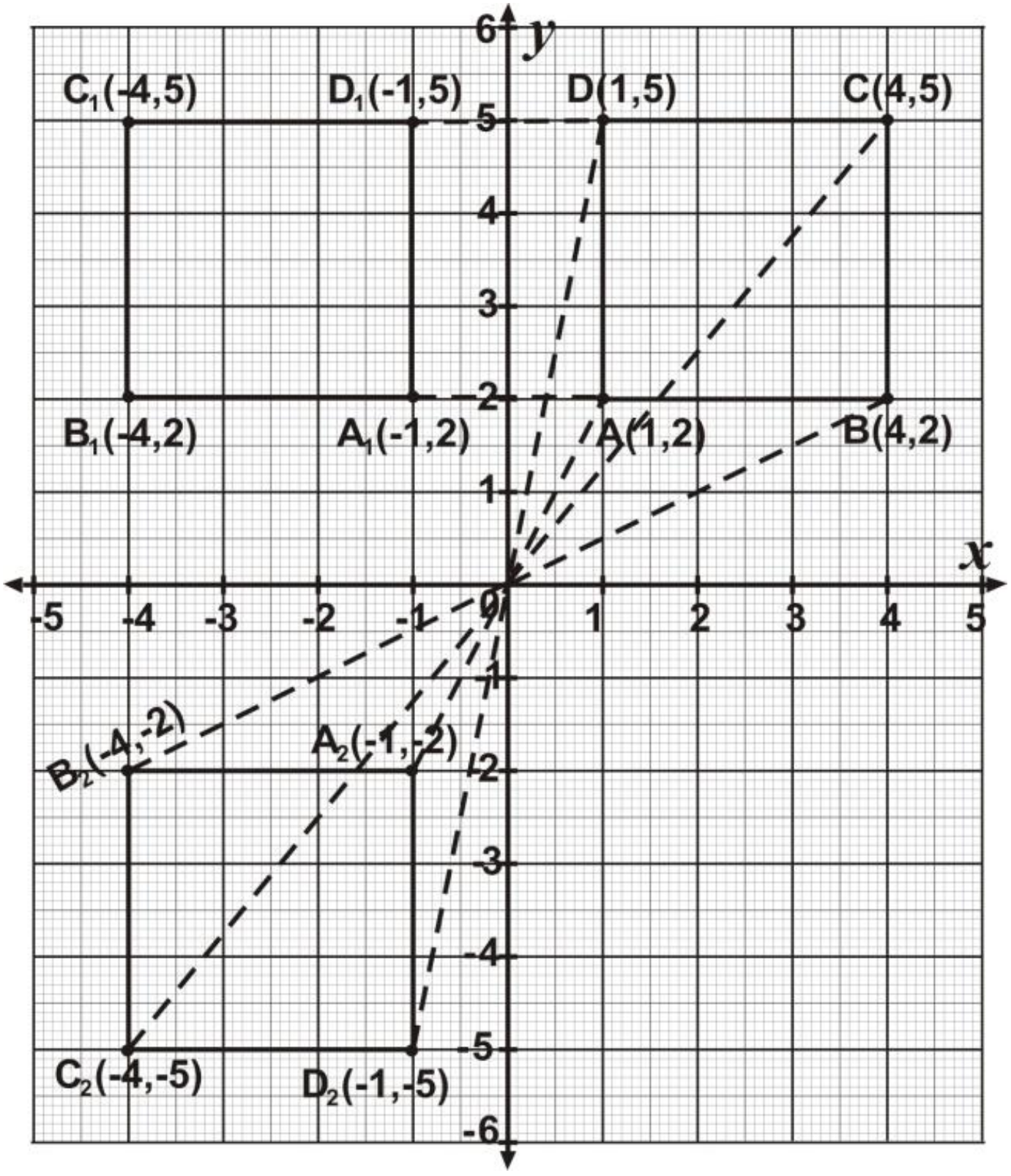
$= \quad \underline{\underline{942}}$

3.



(a) |QR| = 8 cm

4. **Approach 1** (By Inspection / Construction)



4. (ii) **Approach 2** (The rule / formula)

Reflecting (x, y) in the y-axis

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} -x \\ y \end{pmatrix}$$

$$OA\begin{pmatrix} 1 \\ 2 \end{pmatrix} \rightarrow OA_1\begin{pmatrix} -1 \\ 2 \end{pmatrix}, \quad \therefore \ A_1(-1,2)$$

$$OB\begin{pmatrix}4\\2\end{pmatrix} \rightarrow OB_1\begin{pmatrix}-4\\2\end{pmatrix}, \quad \therefore B_1(-4, 2)$$

$$OC\begin{pmatrix}4\\5\end{pmatrix} \rightarrow OC_1\begin{pmatrix}-4\\5\end{pmatrix}, \quad \therefore C_1(-4, 5)$$

$$OD\begin{pmatrix}1\\5\end{pmatrix} \rightarrow OD_1\begin{pmatrix}-1\\5\end{pmatrix}, \quad \therefore D_1(-1,5)$$

∴ Plot and join A_1(-1,2), B_1(-4,2), C_1(-4,5) and D(-1,5) as the image of ABCD under a reflection in the y axis.

(iii) **<u>Enlargement from (0,0) by scale factor k</u>**

$$\begin{pmatrix}x\\y\end{pmatrix} \rightarrow k\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}kx\\ky\end{pmatrix}$$

$$OA\begin{pmatrix}1\\2\end{pmatrix} \rightarrow OA_2\begin{pmatrix}-1\times1\\-1\times2\end{pmatrix} = \begin{pmatrix}-1\\-2\end{pmatrix}, \quad \therefore A_2(-1,-2)$$

$$OB\begin{pmatrix}4\\2\end{pmatrix} \rightarrow OB_2\begin{pmatrix}-1\times4\\-1\times2\end{pmatrix} = \begin{pmatrix}-4\\-2\end{pmatrix}, \quad \therefore B_2(-4,-2)$$

$$OC\begin{pmatrix}4\\5\end{pmatrix} \rightarrow OC_2\begin{pmatrix}-1\times4\\-1\times5\end{pmatrix} = \begin{pmatrix}-4\\-5\end{pmatrix}, \quad \therefore C_2(-4,-5)$$

$$OD\begin{pmatrix}1\\5\end{pmatrix} \rightarrow OD_2\begin{pmatrix}-1\times1\\-1\times5\end{pmatrix} = \begin{pmatrix}-1\\-5\end{pmatrix}, \quad \therefore D_2(-1,-5)$$

<u>∴ Plot and join A_2(-1,-2), B_2(-4,-2), C_2(-4,-5) and D_2 (-1,-5)</u> as the image of triangle ABCD under an enlargement by scale factor -1 from the origin (as shown above)

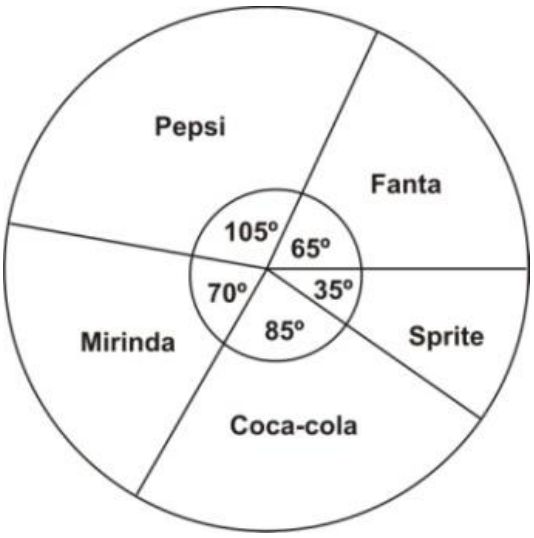
(iv) The single transformation that maps $A_2B_2C_2$ D_2 onto $A_1B_1C_1D_1$ is <u>reflection in the x-axis</u>

5.

Soft Drink	**Number of bottles sold**	**Angle of sector**
Fanta	13	$\frac{13}{72} \times 360^o = 65^o$
Pepsi	21	$\frac{21}{72} \times 360^o = 105^o$

Mirinda	14	$\frac{14}{72} \times 360^{o} = 70^{o}$
Coca-cola	17	$\frac{17}{72} \times 360^{o} = 85^{o}$
Sprite	7	$\frac{7}{72} \times 360^{o} = 35^{o}$
TOTAL	**72**	**360°**

Pie chart showing the distribution of sales of soft drinks sold by the Akrokyere JSS canteen in one week



August 1993

MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. Expand (2a + b) (a + 2b)

A) $2a^2 + 2b^2$
B) $2a^2 + b^2$
C) $5a^2 + 2a^2$
D) $2a^2 + 2a + 4ab^2$
E) $2a^2 + 5ab + 2b^2$

2. Find the missing number in the following binary operation:

$$\begin{array}{r} 1100110 \\ - \ \underline{xxxxxxx} \\ \underline{111011} \end{array}$$

A) 111011 B) 101001 C) 100011 D) 101110 E)101011

3. If x = {1, 3, 5, 7, 9, 11, 13, 15}, find the truth set of $x - 3 \geq 10$.

A. {15}
B. {13,15}
C. {11,13,15}
D. {9,11,13,15}
E. {7,9,11,13,15}

4. Which of these has the least number of lines of symmetry ?

A) an equilateral triangle
B) a rectangle
C) a square

D) a circle

E) an isosceles triangle

5. Find 2% of ¢2,000.00

A) ¢40.00 B) ¢50.00 C) ¢100.00 D) ¢800.00 E) ¢5,000.00

6. Find the highest common factor of 18, 36 and 120.

A) $2^2\times3^3\times5$ B) $2\times3\times5$ C) 2×3 D) $2^3\times2^2$ E) $3^2\times2^2$

7. Arrange the fractions ¾, $^2/_3$, $^3/_5$ in ascending order.

A) $^2/_3$, ¾ , $^3/_5$

B) ¾ , $^2/_3$, $^3/_5$

C) $^3/_5$, $^2/_3$, ¾

D) $^3/_5$, ¾ , $^2/_3$

E) ¾ , $^3/_5$, $^2/_3$

8. Make *b* the subject of the relation $\frac{1}{a}=\frac{1}{b}+c$

A) $b=\frac{ac-1}{c}$

B) $b=\frac{a}{a+ac}$

C) $b=\frac{c}{1-ac}$

D) $b=\frac{a}{1-ac}$

E) $b=\frac{1-ac}{a}$

9. A man has three children whose ages are 9 years, 12 years and 24 years. Find the ratio of their ages.

A) 1:2:3 B) 1:2:4 C) 2:3:6 D) 3:4:8 E)3:6:8

10. Ten students in Kwamekrom JSS took 9 days to weed the school compound. How long would 15 students take to weed the compound if they worked at the same rate?

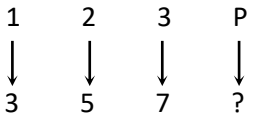
A) 5 days B) 6 days C) 13 ½ days D) 14 days E) 16 days

11. Which of the following inequalities is / are true?

I: $\frac{5}{6} > \frac{3}{4}$ II: $\frac{3}{4} > \frac{3}{5}$ III: $\frac{3}{8} > \frac{1}{2}$

A. I only
B. II only
C. III only
D. I and II only
E. I and III only

12. Find the image of P in the mapping below



A) P^2+2 B) P^2+1 C) $2P+1$ D) $P+2$ E) p^2-2

13. A car travelled a distance of 50km in an hour. What distance did it travel in 30 minutes at the same speed?

A) 1,500km B) 100km C) 80km D) 25km E) 20km

14. In an enlargement, the area of the object was multiplied by 144 to get the area of the image. Find the scale factor of the enlargement.

A) 12 B) 36 C) 48 D) 72 E) 144

15. Mr. Yevu saved ¢2,500.00 at a simple interest rate of 25% per annum for 4 years. Calculate the interest he earned on his savings.

A) ¢625.00
B) ¢2,500.00
C) ¢3,125.00

D) ¢5,000.00

E) ¢10,000.00

The table below gives the number of goals scored by a football team in a league season.

No. of goals scored in a match	0	1	2	3	4	5
Frequency	1	7	6	4	1	1

16. Find the number of goals scored by the team.

A) 41 B) 40 C) 20 D) 19 E) 15

17. What is the mean number of goals scored by the team?

A) 7 B) 6 C) 4 D) 2 E) 1

18. What is the total number of matches played in the league season?

A) 6 B) 7 C) 19 D) 20 E) 40

19. The probability of obtaining a head when a coin is tossed is $^1/_3$. What is the probability of obtaining a tail?

A) 1 B) ⅔ C) ½ D) ⅓ E) 0

20. Given that **a** = $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and **b** = $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$, find **a** + 2**b**

A. $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ B. $\begin{pmatrix} 2 \\ 13 \end{pmatrix}$ C. $\begin{pmatrix} 0 \\ -2 \end{pmatrix}$ D. $\begin{pmatrix} 6 \\ 13 \end{pmatrix}$ E. $\begin{pmatrix} 2 \\ -7 \end{pmatrix}$

21. Find in base ten the value of 4 in 143_{five}.

A) 48 B) 40 C) 25 D) 20 E) 9

22. If the interior angle of a regular polygon is 120°, how many sides does it have?

A) 5 B) 6 C) 7 D) 8 E) 9

23. Simplify $(3\frac{1}{2} + 7) \div (4\frac{1}{3} - 3)$

A) $6\frac{7}{8}$ B) 7 C) $7\frac{7}{8}$ D) $10\frac{1}{2}$ E) 14

24. Find the solution set of $n - \frac{2}{3} > \frac{1}{3} - n$

A. $\{n: n > -\frac{1}{6}\}$

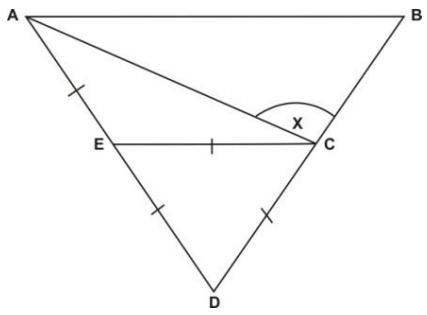
B. $\{n: n = 0\}$

C. $\{n: n > \frac{1}{3}\}$

D. $\{n: n > \frac{1}{2}\}$

E. $\{n: n > 1\}$

25. In the diagram below, |AE| = |ED| = |DC| = |CE|. Calculate the size of the angle marked x.



A) 150° B) 120° C) 90° D) 60° E) 30°

26. Five times a number is four more than the number. Find the number.

A) $\frac{3}{2}$ B) 1 C) $\frac{2}{3}$ D) $\frac{1}{2}$ E) –1

27. A basket contains 450 oranges, if each orange costs ¢15.00, find the total cost of the oranges.

A) ¢30.00 B) ¢465.00 C) ¢435.00 D) ¢675.00 E) ¢6,750.00

28. A bottle of soft drink costs ¢200.00. The commission paid on one bottle is 2% of the cost price. Find the commission paid on 24 bottles of the soft drink.

A) ¢96.00 B) ¢296.00 C) ¢400.00 D) ¢4,704.00 E) ¢4,800.00

29. If $\frac{9}{21}$ is equivalent to $\frac{1+x}{7}$, find the value of *x*.

A) 1 B) 2 C) 4 D) 5 E) 6

30. Write 39.975km correct to three significant figures

A) 39km B) 39.975km C) 49km D) 40.0km E) 40.9km

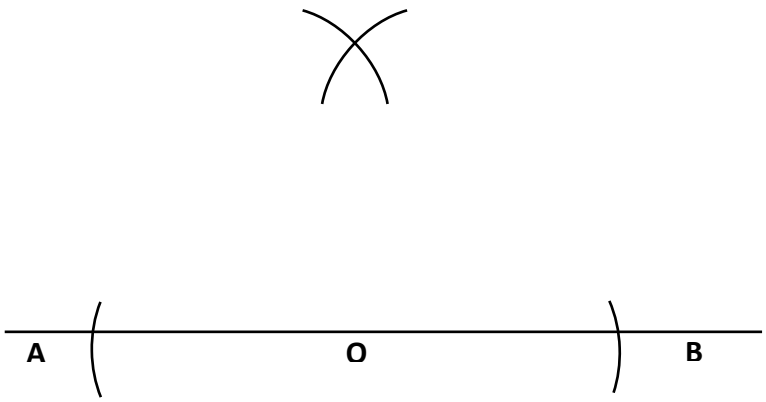
31. Find x if $^1/_x + {}^1/_3 = 1$

A) $^{-3}/_2$ B) $^{-2}/_3$ C) $^2/_3$ D) $^3/_4$ E) $^3/_2$

32. The bearing of Aboku from Bebeka is 055°. What is the bearing of Bebeka from Aboku?

A) 035° B) 055° C) 125° D) 235° E) 305°

33.



The diagram above shows the construction of :

A) the perpendicular bisector of the line

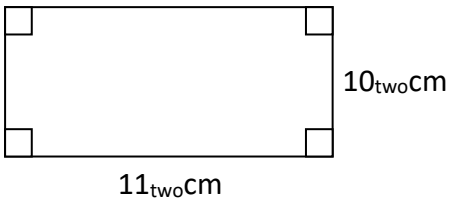
B) an angle of 45° at the point A

C) an angle of 45° at the point B

D) an angle of 90° at the point O

E) an angle of 90° at the point B

34. The dimensions of the rectangle are given in base two. Find its perimeter.



A) 100_{two} cm

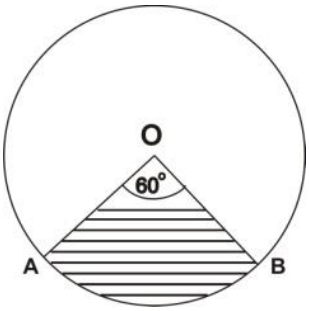
B) 101_{two} cm

C) 110_{two} cm

D) 1001_{two} cm

E) 1010_{two} cm

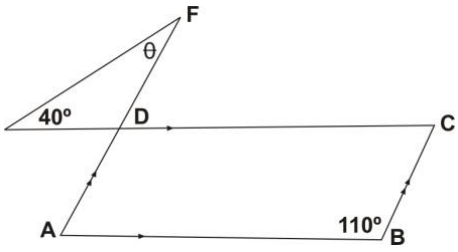
35.



The area of circle, centre O, is $120cm^2$. Angle AOB is 60°. Find the area of sector AOB.

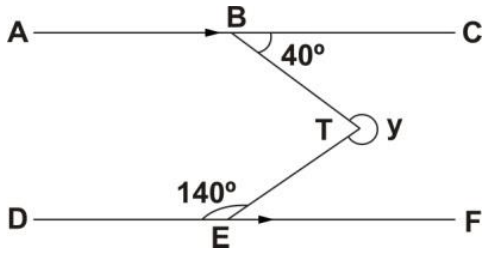
A) $2cm^2$ B) $3cm^2$ C) $6cm^2$ D) $20cm^2$ E) $60cm^2$

36. In the diagram, ABCD is a parallelogram; EC and AF are straight lines. Angle ABC = 110° and angle DEF = 40°. Find the angle marked Θ



A) 30° B) 40° C) 50° D) 70° E) 110°

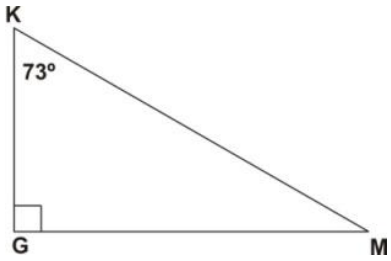
37. In the diagram below AC // DF. Angle CBT is 40° and angle DET is 140°



Find the value of the reflex angle marked y.

A) 320° B) 280° C) 220° D) 100° E) 80°

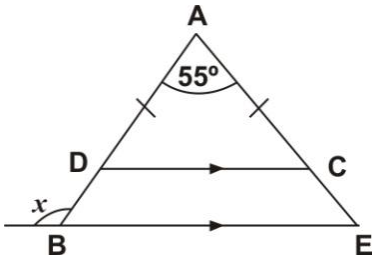
38.



In the diagram above, the angle of elevation of K from M is

A) 17° B) 73° C) 90° D) 107° E) 163°

39. In the diagram below, ACD is an isosceles triangle in which |AD| = |AC| and DC is parallel to BE, find the value of the angle marked x.



A) 55° B) 62.5° C) 110° D) 117.5° E) 125°

40. Three girls Ama, Adjoa and Abena measured the length of the sides of 3 right-angled triangles as follows.

Ama's measurements were 80mm, 40mm, 50mm. Adjoa's measurements were 50mm, 120mm, 130mm. Abena's measurements were 20mm, 30mm and 40mm.

Whose measurement(s) was / were correct ?

A) Ama’s only
B) Abena’s and Ama’s
C) Adjoa’s and Abena’s
D) Ama’s and Adjoa’s only
E) Adjoa’s only

August 1993

MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. E) $2a^2 + 5ab + 2b^2$

2. E) 101011

3. B. {13,15}

4. E) An isosceles triangle

5. A) ¢40.00

6. C) 2×3

7. C) $^3/_5, ^2/_3, ¾$

8. C) $b = \frac{c}{1-ac}$

9. D) 3:4:8

10. B) 6 days

11. D. I and II only

12. C) 2P+1

13. D) 25km

14. A) 12

15. B) ¢2,500.00

16. B) 40

17. D) 2

18. D) 20

19. B) ⅔

20. B. $\begin{pmatrix} 2 \\ 13 \end{pmatrix}$

21. A) 48

22. B) 6

23. C) $7\ ^{7}/_{8}$

24. D. {n: n > ½}

25. C) 90°

26. B) 1

27. E) ¢6,750.00

28. A) ¢96.00

29. B) 2

30. D) 40.0km

31. E) $^{3}/_{2}$

32. D) 235°

33. D) an angle of 90° at the point O

34. E) 1010_{two} cm

35. D) 20 cm^2

36. A) 30°

37. B) 280°

38. A) 17°

39. D) 117.5°

40. E) Adjoa’s only

August 1993

MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt four questions only from this section

All working must be clearly shown.

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. (a) Simplify $\frac{2a+4b}{3} - \frac{3(a-b)}{2}$

(b) Solve $5(a-5) - \frac{1}{2}(2a+6) = 4$

(c) If $r = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $q = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$, calculate $6(r+2q)$

2. Using a ruler and a pair of compasses only,

a) construct a triangle ABC such that |BA| = 10cm, angle ABC = 90° and angle BAC = 30°. Measure the length BC.

b) (i) Bisect the angle ACB to meet BA at D.

(ii) What type of triangle is CDA?

c) Calculate the area of triangle ABC

3. Olu bought a radio for ¢65,000.00. After one year the radio was valued at 75% of the cost price.

(a) What was the price of the radio after one year?

(b) If he sold the radio for ¢55,700.00, calculate his profit or loss over the cost price.

4. (a) Using a scale of 2cm to 1unit on both axes, draw two perpendicular lines OX and OY on a graph sheet.

(b) On this graph sheet, mark the x-axis from –5 to 5 and the y-axis from –6 to 6.

(c) Plot on the same graph sheet the points A(-2, 4) and B(4, -5). Join the points A and B with the help of a ruler.

(d) Using the graph , find

(i) the gradient (slope) of the line AB;

(ii) the value of x, when $y = 0$;

(iii) the value of y when $x = 2$

(e) Plot on the same graph sheet the points C(-3,-1) and D(3,3). Join the points C and D. with the help of a protractor, measure the angle between the lines AB and CD. What is the gradient of the line CD?

5. The ages of 20 school children were recorded as follows:

13, 9, 15, 17, 13, 9, 11, 9, 11, 15,

17, 15, 11, 9, 9, 11, 15, 11, 11, 11.

(a) Make a frequency table for the data using the ages of 9, 11, 13, …

(b) Use your table to calculate the mean age

(correct to the nearest whole number)

August 1993

MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1 (a) $\dfrac{2a+4b}{3} - \dfrac{3(a-b)}{2}$

$\Rightarrow \dfrac{2a+4b}{3} - \dfrac{3a-3b}{2}$

$\Rightarrow \dfrac{2(2a+4b) - 3(3a-3b)}{6}$

$\Rightarrow \dfrac{4a + 8b - 9a + 9b}{6}$

$\Rightarrow \dfrac{4a - 9a + 8b + 9b}{6}$

$\Rightarrow \dfrac{-5a + 17b}{6}$

$= \dfrac{17b - 5a}{6}$ or

Or $\frac{1}{6}(17b - 5a)$

1 (b) $5(a - 5) - \frac{1}{2}(2a + 6) = 4$

Approach 1 (Expanding first)

$$5a - 25 - a - 3 = 4$$

$$\Rightarrow \quad 5a - a = 4 + 25 + 3$$

$$\Rightarrow \quad 4a = 32$$

$$\Rightarrow \quad \frac{4a}{4} = \frac{32}{4}$$

$$\Rightarrow \quad a = \underline{\underline{8}}$$

1 (b) **Approach 2** (Multiplying by 2 first to clear fractions)

$$2 \times 5(a - 5) - 2 \times \tfrac{1}{2}(2a + 6) = 2(4)$$

$$\Rightarrow \quad 10(a - 5) - (2a + 6) = 2(4)$$

$$\Rightarrow \quad 10a - 50 - 2a - 6 = 8$$

$$\Rightarrow \quad 10a - 2a = 8 + 50 + 6$$

$$\Rightarrow \quad 8a = 64$$

$$\Rightarrow \quad a = 64 \div 8$$

$$\Rightarrow \quad a = \underline{\underline{8}}$$

1 (c)

$$6\ (\mathbf{r} + 2\mathbf{q})$$

$$= 6\left[\begin{pmatrix}3\\1\end{pmatrix} + 2\begin{pmatrix}-2\\1\end{pmatrix}\right]$$

$$= 6\left[\begin{pmatrix}3\\1\end{pmatrix} + \begin{pmatrix}2\times -2\\2\times 1\end{pmatrix}\right]$$

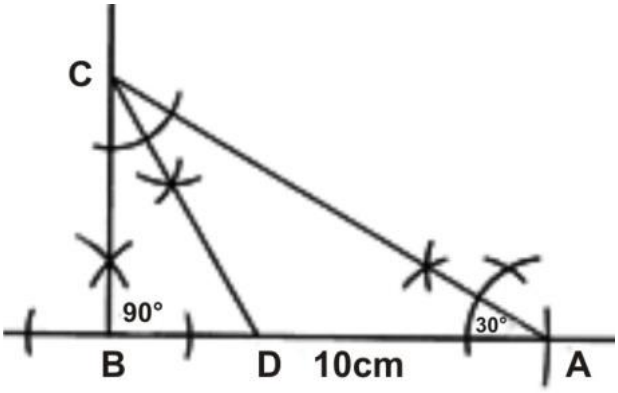
$$= 6\left[\begin{pmatrix}3\\1\end{pmatrix} + \begin{pmatrix}-4\\2\end{pmatrix}\right]$$

$$= 6\begin{pmatrix}3+(-4)\\1+2\end{pmatrix}$$

$$= 6\begin{pmatrix}3-4\\1+2\end{pmatrix} = 6\begin{pmatrix}-1\\3\end{pmatrix}$$

$$= \begin{pmatrix}6\times -1\\6\times 3\end{pmatrix} = \underline{\underline{\begin{pmatrix}-6\\18\end{pmatrix}}}$$

2 (a)



(a) |BC| ≈ <u>5.7 cm</u>

(b) (ii) CDA is an isosceles triangle

(c) Area of triangle = ½ × base × height

$= \frac{1}{2} \times 10cm \times 5.7cm$

$= \frac{1}{2} \times 57\ cm^2$

$= {}^{57}/_{2}\ cm^2$

= <u>28 ½ cm²</u> or <u>28.5 cm²</u>

3 (a) Price after one year = 75% of the cost price

$= \frac{75}{100} \times ¢65{,}000$

= 75 × ¢650

= <u>¢48,750</u>

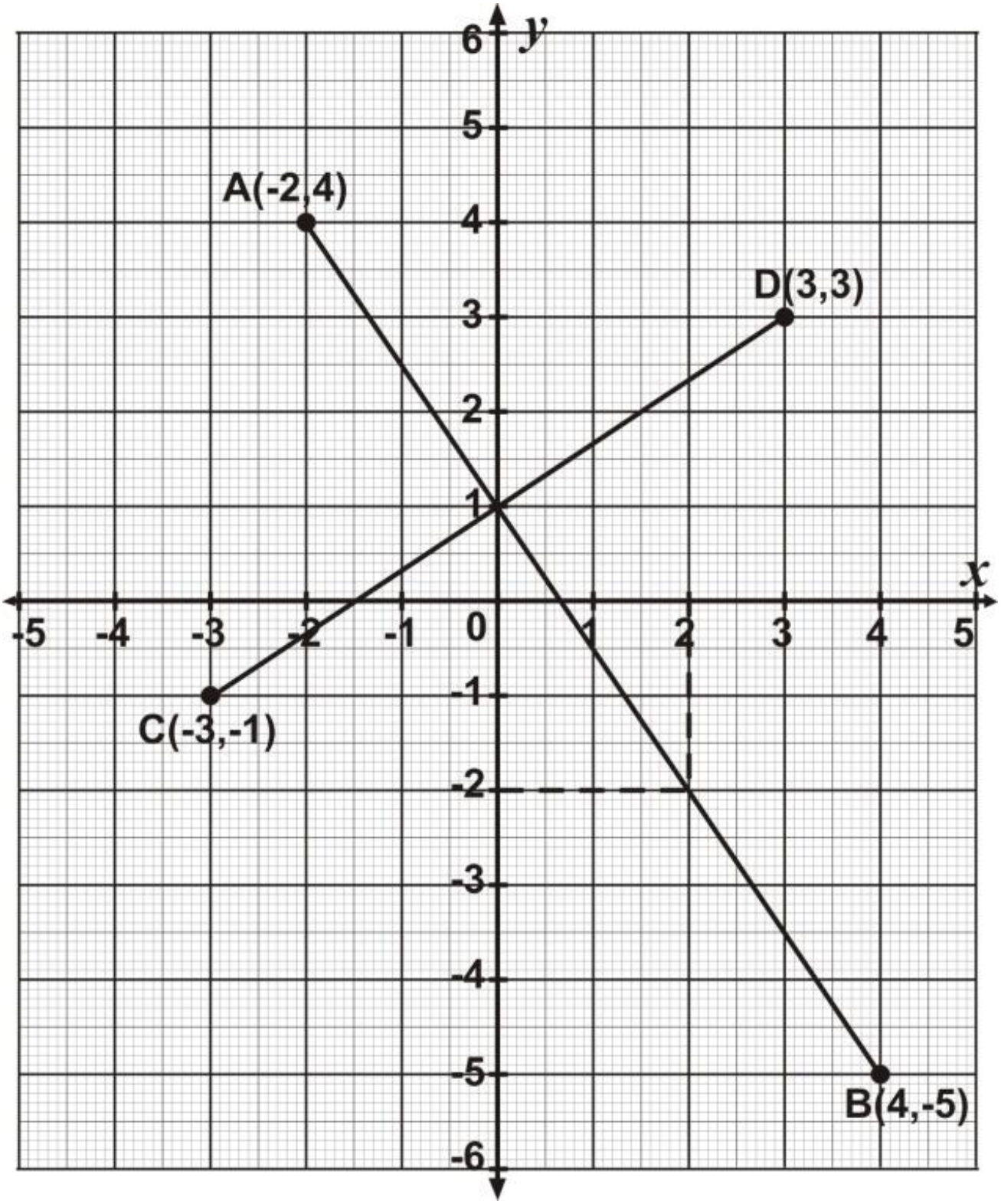
(b) Since price (value) after one year = CP = ¢48,750

Profit = SP – CP

= ¢55,700 – ¢48,750 = ¢6,950

⇒ He made a profit of <u>¢6,950</u>

4.



4 (d) From the graph, using the points A (-2, 4) and B (4, -5),

(d) (i) The gradient of line AB

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{-5 - 4}{4 - (-2)} = \frac{-9}{6} = \frac{-3}{2}$$

$$= \underline{\underline{-1\tfrac{1}{2}}} \quad \text{or } \underline{-1.5}$$

(d) (i) **Approach 2**

$$\frac{y_1 - y_2}{x_1 - x_2} = \frac{4 - (-5)}{-2 - 4} = \frac{9}{-6} = \frac{-3}{2}$$

$$= \underline{\underline{-1\frac{1}{2}}} \quad \text{or} \quad \underline{\underline{-1.5}}$$

(d) (ii) When $y = 0$, $\underline{\underline{x \approx 0.7}}$

(d) (iii) When $x = 2$, $\underline{\underline{y = -2}}$

(e) From the graph, the gradient of line CD

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - (-1)}{3 - (-3)} = \frac{4}{6} = \underline{\underline{\frac{2}{3}}}$$

(e) **Or:**

$$\frac{y_1 - y_2}{x_1 - x_2} = \frac{-1 - 3}{-3 - 3} = \frac{-4}{-6} = \underline{\underline{\frac{2}{3}}}$$

5 (a) Frequency Table

Age	Tally	Frequency
9	𝍸	5
11	𝍸 //	7
13	//	2
15	////	4
17	//	2

5 (b)

Age (x)	Tally	Frequency (f)	fx

9	~~////~~	5	45
11	~~////~~ //	7	77
13	//	2	26
15	////	4	60
17	//	2	34
		Σf= 20	**Σfx = 242**

From the frequency table,

The mean age $= \frac{\Sigma f x}{\Sigma f} = \frac{242}{20}$

$= \underline{\underline{12\frac{1}{10}\,years}}$ or <u>12.1 years</u>

August 1992

MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

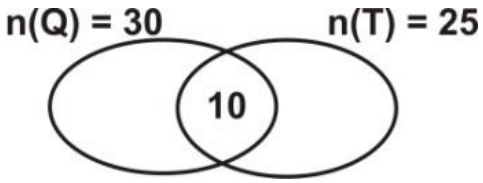
1. Find the set of prime factors of 12

A. {3} B. {2, 3} C. {3, 4} D. {2, 6} E. {2,3,4,6,12}

2. If P = {7, 11, 13} and Q = {9, 11, 13}. Find P U Q.

A. {7}
B. {9}
C. {7, 9}
D. {7, 9, 11}
E. {7, 9, 11, 13}

3. In the diagram, set Q has 30 members and set T has 25 members. Q ∩ T has 10 members. Find the number of members of Q U T.



A) 35 B) 45 C) 55 D) 65 E) 75

4. Find the value of $\sqrt{6\frac{1}{4}}$

A) 5.0 B) 4.9 C) 2.5 D) 2.4 E) 1.2

5. Convert 39_{ten} to a base five numeral.

A) 100111 B) 1110 C) 234 D) 124 E) 103

6. Find the least whole number which must be added to 207 to make it divisible by 17

A) 0 B) 3 C) 13 D) 14 E) 15

7. Simplify $2 \times (3\frac{1}{3} + 1\frac{1}{6})$

A) $2\frac{1}{3}$ B) $4\frac{1}{4}$ C) $6\frac{2}{3}$ D) 8 E) 9

8. If $8.51 \div 2.3 = 3.7$, find the value of $85.1 \div 2.3$

A) 0.037 B) 0.37 C) 3.7 D) 37 E) 370

9. Express 0.625 as a fraction in its lowest term.

A) $\frac{7}{8}$ B) $\frac{3}{4}$ C) $\frac{5}{8}$ D) $\frac{1}{2}$ E) $\frac{1}{3}$

10. An amount of money is shared between Kofi and Ama in the ratio 3:5. If Ama received ¢4,650.00, what is Kofi's share?

A) ¢930.00 B) ¢1,550.00 C) ¢1,743.75 D) ¢2,790.00 E) ¢2,906.25

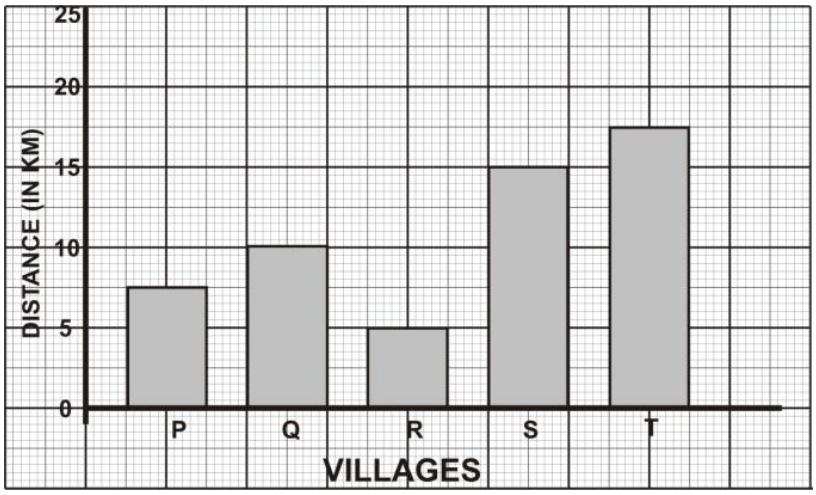
11. If \$1.00 = ¢340.00, what was the cedi value of an article which cost \$6.50?

A) ¢6,630.00
B) ¢2,380.00
C) ¢2,210.00
D) ¢346.50
E) ¢333.50

12. Find the simple interest on ¢28,000.00 at 3 ½ % per annum for 6 months.

A) ¢490.00
B) ¢560.00
C) ¢980.00
D) ¢4,000.00
E) ¢5,880.00

The bar chart shows the distances of 5 villages, P, Q, R, S and T from a market town. Use it to answer questions 13 and 14



13. Which village is farthest from the market town?

A) P B) Q C) R D) S E) T

14. How much farther is village Q than village R from the market town?

A) 2 km B) 3 km C) 4km D) 5km E) 6km

15. There are 20 beads in a box. Some are red and some green. The chance that one bead taken at random from the box is red is ¼. Find the number of red beads in the box.

A) 16 B) 15 C) 10 D) 5 E) 4

16. A bag contains 12 mangoes of which 4 are not ripe. What is the chance of picking at random a ripe mango from the bag?

A) ⅛ B) ¼ C) ⅓ D) ½ E) ⅔

17. Find the value of $p^2 - 6p + 9$ when $p = -2$

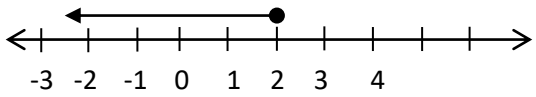
A) –7 B) 1 C) 12 D) 13 E) 25

18. If $x = 2$, find the value of q in the equation $3x - 4 = x + q$

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A) 8 B) 1 C) 0 D) –1 E) –8

19.



Which of the following inequalities is shown on the number line above, where P is a real number.?

A) $p \geq 2$ B) $p > 2$ C) $p = 2$ D) $p \leq 2$ E) $p < 2$

*Use the mapping below to answer questions **20** to **21***

-2	-1	0	2	3	4	...	*x*
↓	↓	↓	↓	↓	↓		
y	-1	1	5	7	9	...	21

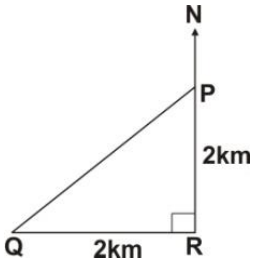
20. Find the value of *x*

A) –7 B) 5 C) 6 D) 7 E) 10

21. Find the value of *y*

A) 3 B) 1 C) –1 D) –3 E) -5

22. In the diagram, find the bearing of P from Q.



A) 045° B) 090° C) 135° D) 180° E) 270°

23. A polygon has 10 sides. Which of the following gives the sum of its interior angles?

A) 2 x 180°

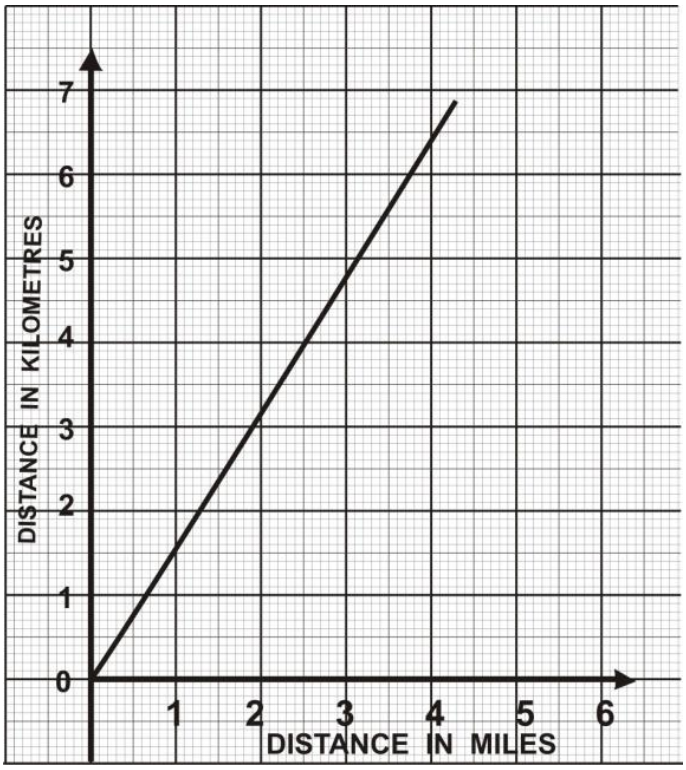
B) 4 x 180°

C) 6 x 180°

D) 8 x 180°

E) 10 x 180°

The diagram shows the conversion graph for miles and kilometres.



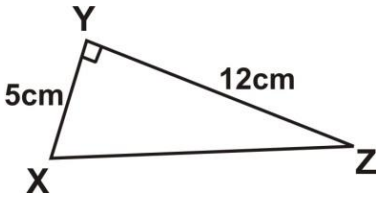
24. Find in kilometres, the equivalent of 4 miles

A) 2.5 B) 3 C) 3.5 D) 6 E) 6.4

25. Express 4 kilometres in miles

A) 6.4 B) 6 C) 3.5 D) 3 E) 2.5

The diagram below is a right-angled triangle.Use it to answer questions **26** *and* **27**



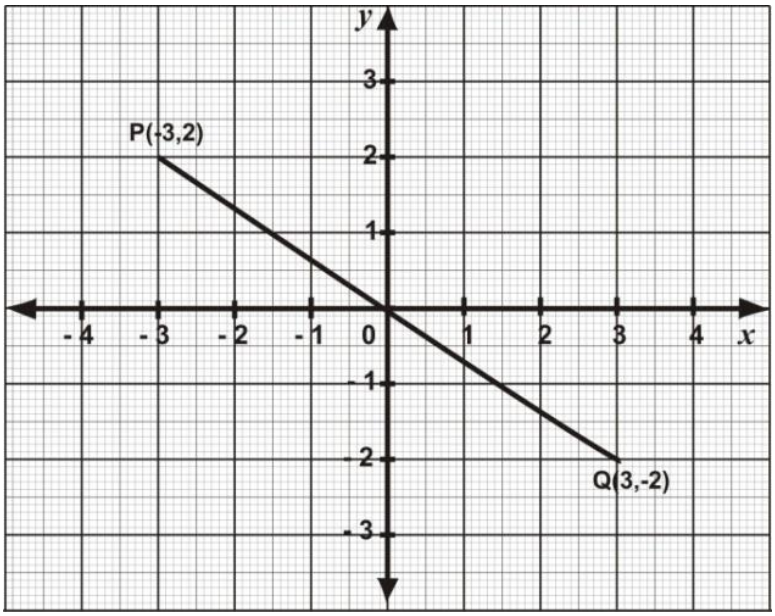
26. Find |XZ|

A) 2.4cm B) 7cm C) 13cm D) 17cm E) 60cm

27. Find tan ∠YXZ

A) $^{5}/_{13}$ B) $^{5}/_{12}$ C) $^{12}/_{13}$ D) $^{12}/_{5}$ E) $^{13}/_{5}$

28. The diagram below shows two points P and Q in the number plane. Find the vector PQ.

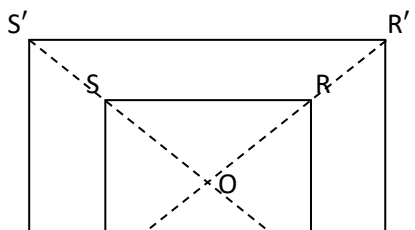


A. $\begin{pmatrix} 6 \\ 4 \end{pmatrix}$ B. $\begin{pmatrix} 6 \\ -4 \end{pmatrix}$ C. $\begin{pmatrix} -6 \\ 4 \end{pmatrix}$ D. $\begin{pmatrix} 4 \\ -6 \end{pmatrix}$ E. $\begin{pmatrix} -4 \\ -6 \end{pmatrix}$

29. Find the length of the vector P = $\begin{pmatrix} -5 \\ 12 \end{pmatrix}$

A) 7 B) 10 C) 13 D) 17 E) 25

30.



In the diagram, square P′Q′R′S′ is an enlargement of square PQRS from centre O. The area of PQRS is 4cm^2 and the area of P′Q′R′S′ is 9cm^2. Find the scale factor of the enlargement.

A) $^2/_3$ B) $^4/_9$ C) 1½ D) $^{-2}/_3$ E) $^{-9}/_4$

31. The volume of a cube is 27cm^3. Find the area of one of its faces.

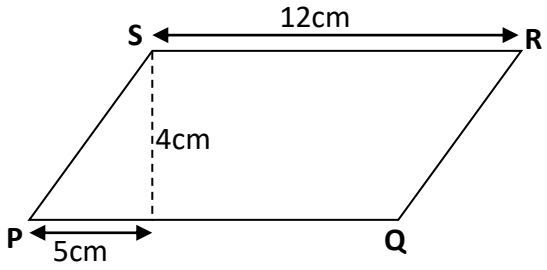
A) 3cm^2 B) 6cm^2 C) 9cm^2 D) 18cm^2 E) 54cm^2

32. A cylinder is of height 3 cm and radius 2 cm. Find its curved surface area.

[Take $\pi = 3.142$]

A) 7 cm^2 B) 12 cm^2 C) 18 cm^2 D) 38 cm^2 E) 54 cm^2

33.



Find the area of the parallelogram PQRS

A) 20cm^2 B) 21cm^2 C) 48cm^2 D) 60cm^2 E) 240cm^2

34. If one-third of a number is added to one-fifth of the same number, the result is 8. Find the number.

A) 3 B) 5 C) 15 D) 40 E) 45

35. If 1: x is equivalent to 6 ¼ :25, find x.

A) 4 B) 5 C) 6.25 D) 24 E) 100

36. Express $^{3}/_{8}$ as a percentage.

A) 0.373% B) 12% C) 25% D) 37 ½% E) 40%

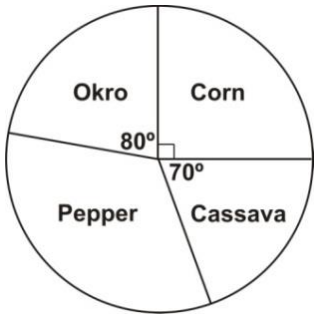
37. Akosua buys 480 pineapples for ¢24,000.00. She sells all the pineapples for ¢28,000.00. Find her profit percent.

A) 13.9% B) 16.7% C) 20% D) 40% E) 83.3%

38. How many edges has a cuboid ?

A) 4 B) 6 C) 8 D) 10 E) 12

The pie chart shows the distribution of crops on a farm of area 250 hectares.



39. Find the area of the plot with corn.

A) 48.6ha B) 55.3ha C) 62.5ha D) 83.3ha E) 125.0ha

40. What fraction of the farm is planted with pepper?

A) ⅓ B) ¼ C) $^{1}/_{5}$ D) $^{2}/_{9}$ E) $^{7}/_{32}$

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MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. B. {2, 3}

2. E. {7, 9, 11, 13}

3. B) 45

4. C) 2.5

5. D) 124

6. D) 14

7. E) 9

8. D) 37

9. C) $^{5}/_{8}$

10. D) ¢2,790.00

11. C) ¢2,210.00

12. A) ¢490.00

13. E) T

14. D) 5km

15. D) 5

16. E) ⅔

17. E) 25

18. C) 0

19. D) $p \leq 2$

20. E) 10

21. D) –3

22. A) 045°

23. D) $8 \times 180°$

24. E) 6.4

25. E) 2.5

26. C) 13cm

27. D) $^{12}/_{5}$

28. B. $\begin{pmatrix} 6 \\ -4 \end{pmatrix}$

29. C) 13

30. C) 1½

31. C) $9cm^2$

32. D) $38cm^2$

33. C) 48 cm^2

34. C) 15

35. A) 4

36. D) 37 ½%

37. B) 16.7%

38. E) 12

39. C) 62.5ha

40. A) ⅓

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MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

Attempt four questions only from this section

All working must be clearly shown.

Marks will not be awarded for correct answers without corresponding working.

All questions carry equal

1. (a) Solve $5 - 2x > x + 2$, where x is a real number.
Illustrate your result on the number line.

(b) Find the truth set of the equation: $\frac{2}{3}(3y - 1) - (y + 2) = \frac{1}{3}$

(c) Factorize completely: $mp + np - mt - nt$

(d) Make *t* the subject of the relation $v = u + at$

2. A landlady rented out her house for ¢240,000.00 for one year. During the year she paid 15% of the rent as income tax. She also paid 25% of the rent as property tax and spent ¢10,000.00 on repairs.

Calculate:

a) The landlady's total expenses.

b) The remainder of the rent after the landlady's expenses.

c) The percentage of the rent she spent on repairs.

3. (a) Using a scale of 2cm to 1 unit on both axes, draw perpendicular lines OX and OY on a graph sheet.

(b) On this graph sheet, mark the x-axis from –5 to 5 and the y-axis from –6 to 6

(c) Plot on the same graph sheet the points A(1,1), B(4,3) and C(2,5). Join the points A,B and C to form a triangle.

(d) Using the y-axis as the mirror line, draw the image $A_1B_1C_1$ of the triangle ABC, such that $A \rightarrow A_1$, $B \rightarrow B_1$ and $C \rightarrow C_1$. Write down the co-ordinates of A_1, B_1 and C_1.

(e) Using the x-axis as the mirror line, draw the image $A_2B_2C_2$ of triangle ABC where $A \rightarrow A_2$, $B \rightarrow B_2$ and $C \rightarrow C_2$.

4. The table below gives the frequency distribution of the marks obtained in a class test by a group of 64 pupils.

Marks (out of ten)	Frequency
2	9
3	14
4	13
5	10
6	5
7	8
8	2
9	3

(a) Draw a bar chart for the distribution

(b) A pupil is chosen at random from the class. What is the probability that the pupil obtained 7 marks?

5. Using a ruler and a pair of compasses only,

(a) draw |PQ| = 9 cm

(b) construct a perpendicular to PQ at Q

(c) construct angle QPS = 60° at the point P on PQ such that |PS| = 6.5cm

(d) construct a line parallel to PQ through S. let the perpendicular through Q and the parallel through S, meet at R. Measure |PR|.

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MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1 (a)

$$5 - 2x > x + 2$$

$$\Rightarrow \quad 5 - 2 > x + 2x$$

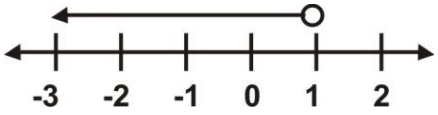
$$\Rightarrow \quad 3 > 3x$$

$$\Rightarrow \quad 1 > x$$

$$\Rightarrow \quad \underline{\underline{x < 1}}$$

Turning the '$1 > x$' around reverses the $>$

The number line illustration is shown below:



1 (b)

$$\frac{2}{3}(3y - 1) - (y + 2) = \frac{1}{3}$$

Approach 1 (Expanding and grouping first)

$$\Rightarrow \quad ⅔ \times 3y - ⅔ \times 1 - y - 2 = ⅓$$

$$\Rightarrow \quad 2y - ⅔ - y - 2 = ⅓$$

$$\Rightarrow \quad 2y - y = ⅓ + ⅔ + 2$$

$$\Rightarrow \quad y = 1 + 2$$

$$\Rightarrow \quad \underline{\underline{y = 3}}$$

1 (b) **Approach 2** (Multiplying by 3 first – to clear fractions)

$3 \times \frac{2}{3}(3y-1) - 3 \times (y+2) = 3 \times \frac{1}{3}$

$\Rightarrow 2(3y-1) - 3(y+2) = 1$

$\Rightarrow$ 6y – 2 – 3y – 6 = 1

$\Rightarrow$ 6y – 3y = 1 + 2 + 6

$\Rightarrow$ 3y = 9

$\Rightarrow$ y = 9 ÷ 3

$\Rightarrow$ y = 3

1 (c) mp + np – mt – nt

= p (m + n) – t(m + n)

= (m + n) (p – t)

1 (d) $v = u + at$, making *t* the subject

Placing terms containing *t* on one side:

$v - u = at$

$\Rightarrow \frac{v-u}{a} = \frac{at}{a}$

$\Rightarrow \frac{v-u}{a} = t$

$\Rightarrow t = \frac{v-u}{a}$

2 (a) Total expenses = Income tax + property tax + repairs

= (15% of ¢240,000) + (25% of ¢240,000) + ¢10,000

$= \left(\frac{15}{100} \times ¢240{,}000\right) + \left(\frac{25}{100} \times ¢240{,}000\right) + ¢10{,}000$

= ¢36,000 + ¢60,000 + ¢10,000

= ¢106,000

⇒ The landlady's total expenses = ¢106,000.00

2 (b) The remainder = Total Amount – Expenses

= ¢240,000.00 – ¢106,000.00

= ¢134,000.00

2 (c) Percentage of the rent spent on repairs

$$= \frac{\text{Amount on repairs}}{\text{Total income from rent}} \times 100\%$$

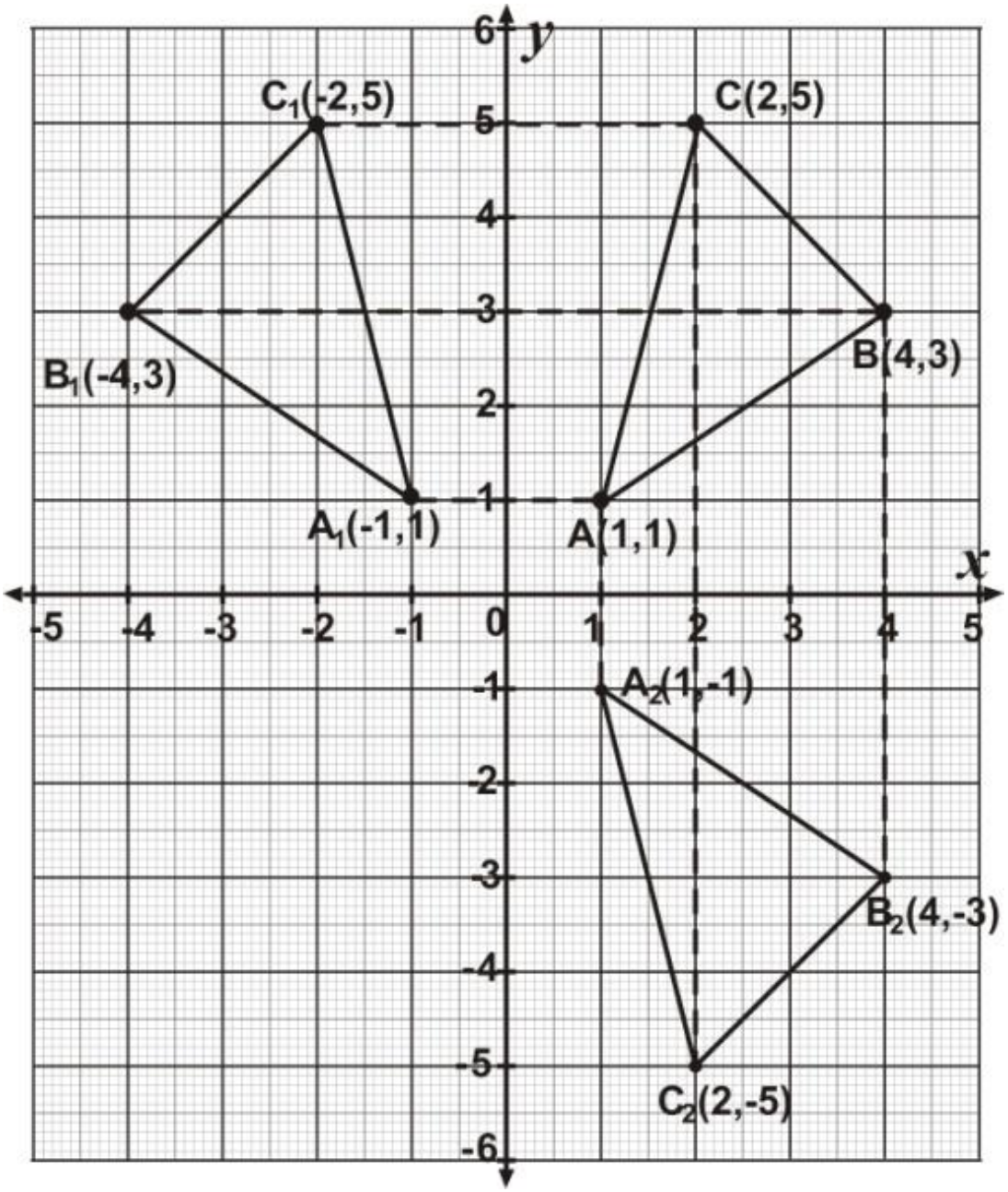
$$= \frac{10{,}000}{240{,}000} \times 100\%$$

$$= \frac{25}{6}\%$$

$$= \underline{4\tfrac{1}{6}\%} \quad \text{or} \quad \approx \quad \underline{4.167\%}$$

3.

Approach 1 (By Inspection / Construction)



3 (d) **Approach 2** (The rule / formula)

Reflecting (x, y) in the y-axis

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} -x \\ y \end{pmatrix}$$

$$OA\begin{pmatrix} 1 \\ 1 \end{pmatrix} \rightarrow OA_1\begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad \therefore A_1(-1, 1)$$

$$OB\begin{pmatrix} 4 \\ 3 \end{pmatrix} \rightarrow OB_1\begin{pmatrix} -4 \\ 3 \end{pmatrix}, \quad \therefore B_1(-4, 3)$$

$$OC\begin{pmatrix} 2 \\ 5 \end{pmatrix} \rightarrow OC_1\begin{pmatrix} -2 \\ 5 \end{pmatrix}, \quad \therefore C_1(-2, 5)$$

∴ Plot and join A_1(-1,1), B_1(-4,3) and C_1(-2,5) as the image of triangle ABC under a reflection in the y axis.

3 (e) Reflection in the x-axis

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} x \\ -y \end{pmatrix}$$

$$OA\begin{pmatrix} 1 \\ 1 \end{pmatrix} \rightarrow OA_2\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \therefore A_2(1, -1)$$

$$OB\begin{pmatrix} 4 \\ 3 \end{pmatrix} \rightarrow OB_2\begin{pmatrix} 4 \\ -3 \end{pmatrix}, \quad \therefore B_2(4, -3)$$

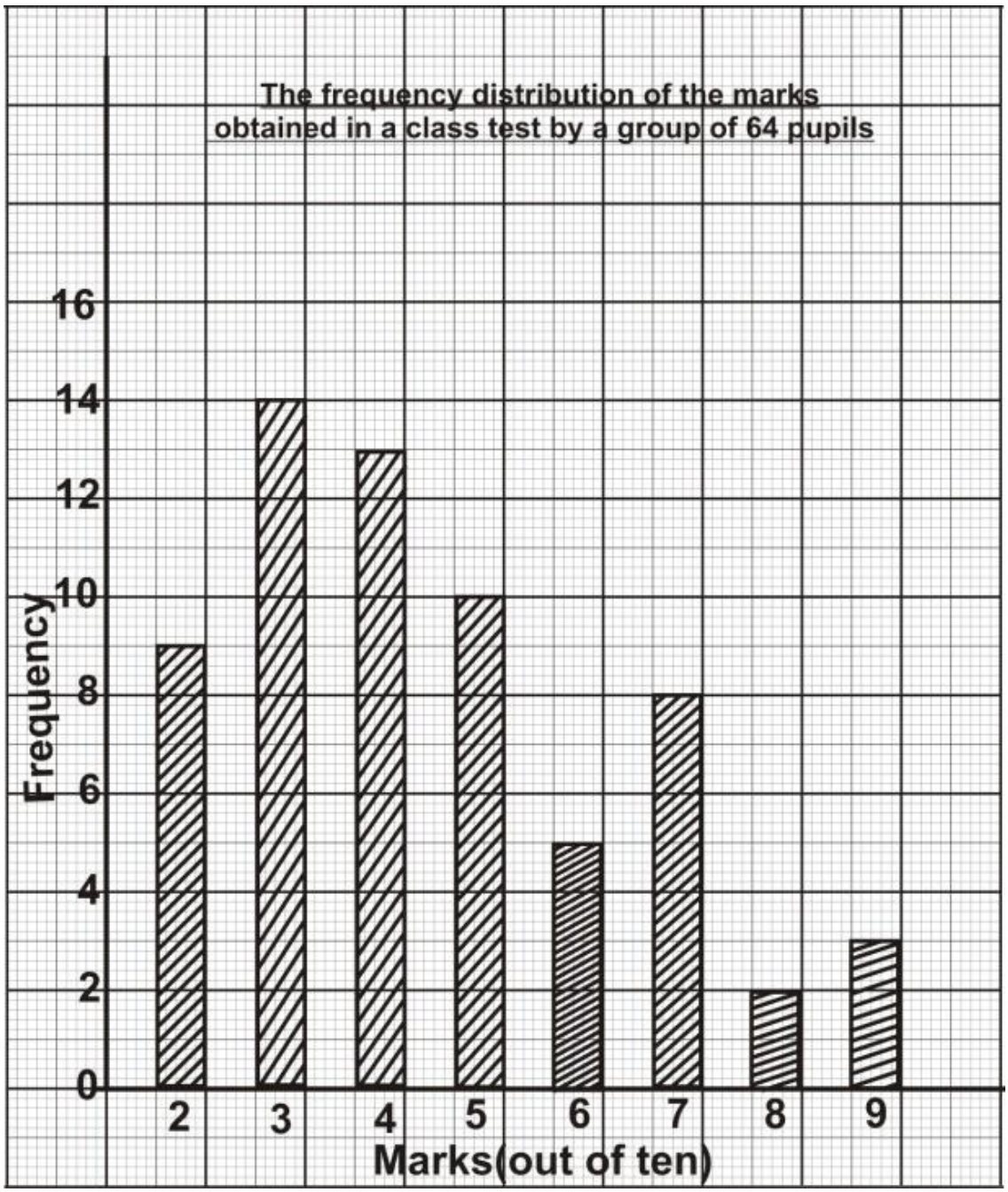
$$OC\begin{pmatrix} 2 \\ 5 \end{pmatrix} \rightarrow OC_2\begin{pmatrix} 2 \\ -5 \end{pmatrix}, \quad \therefore C_2(2, -5)$$

∴ Plot and join A_2(1,-1), B_2(4,-3) and C_2(2,-5) as the image of triangle ABC under a reflection in the x- axis

(*as shown in the diagram above*)

4 (a) Bar chart

Vertical Axis Scale: **2 cm to 2 pupils (1cm to 1 pupil)**

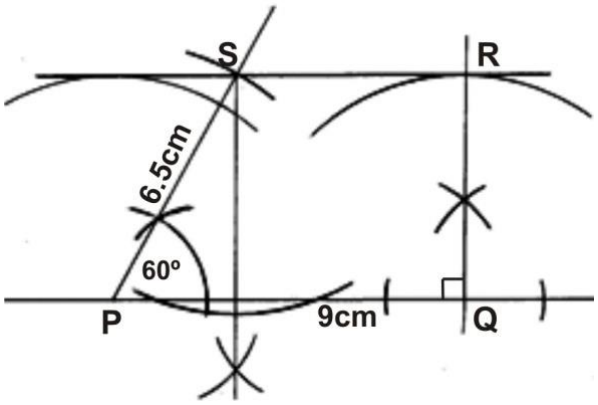


4 (b) P (7 marks) $= \dfrac{\text{Number of pupils that obtained 7 marks}}{\text{Total number of pupils}}$

$= \dfrac{8}{9+14+13+10+5+8+2+3}$

$= \dfrac{8}{64} = \underline{\underline{\dfrac{1}{8}}}$

5.



5 (d) |PR| = 10.6 c m

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MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. P = {1, 2, 3, 8, 10}, Q = {8, 1, x, 3, 2}. If P = Q, what is the value of x?

A) 1 B) 2 C) 3 D) 8 E) 10

2. If Y = {house, tree} and V = {cat, house, tree} which of the following is true of Y and V?

A) Y = V B) Y ⊂ V C) V ⊂ Y D) V ∈ Y E) Y∈V

3. The following addition is in base ten. Find the missing addend.

$$\begin{array}{r} 2345 \\ 1045 \\ **** \\ \hline 5110 \end{array}$$

A) 1300 B) 1720 C) 2765 D) 4065 E) 9500

4. Given that $2^2 \times 3^2 \times 7 = 252$, find the least number that should be multiplied by 252 to make the product a perfect square.

A) 2 B) 3 C) 6 D) 7 E) 252

5. Write 4687.02 in standard form

A) 46.8702×10^3

B) 46.8702×10^4

C) 4.68702×10^5

D) 4.68702×10^3

E) 0.468702×10^4

6. Convert 11001_{two} to a decimal numeral

A) 6 B) 7 C) 14 D) 25 E) 50

7. A boy spent $^{2}/_{7}$ of his pocket money on transport and $^{1}/_{5}$ on sweets. What fraction of his pocket money does he spend on transport and sweets?

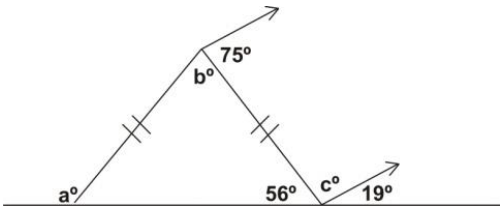
A) $^{2}/_{35}$ B) $^{3}/_{35}$ C) $^{1}/_{7}$ D) $^{17}/_{35}$ E) $^{18}/_{35}$

8. The product of three numbers is 1197. Two of the numbers are 3 and 19. Find the third number.

A) 21 B) 54 C) 210 D) 544 E) 1175

9. Find the GCF (HCF) of $2^3 \times 3^2$ and $2^3 \times 3^4$.

A) 8 B) 9 C) 72 D) 81 E) 648



10. Find the value of a°

A) 68° B) 75° C) 105° D) 112° E) 124°

11. What is the value of b° ?

A) 68° B) 75° C) 105° D) 112° E) 124°

12. What is the value of c° ?

A) 68° B) 75° C) 105° D) 112° E) 124°

13. Simplify $2ab^2 \times 3a^2b$

A) $5a^3b^3$ B) $5a^2b^2$ C) $6a^3b^3$ D) $5a^2b^2$ E) $36ab$

14. In the relation $v = u + at$, find v when $u = 6$, $a = 10$ and $t = 2$

A) 18 B) 26 C) 32 D) 48 E) 120

15. The marks obtained by 10 children in a mental drill are:
0, 1, 3, 3, 5, 7, 8, 9, 9, 9. What is the modal mark?

A) 3 B) 5 C) 7 D) 8 E) 9

16. The marks obtained by 10 children in a mental drill are:
0, 1, 3, 3, 5, 7, 8, 9, 9, 9. Find the median mark.

A) 3 B) 5 C) 6 D) 7 E) 8

17. The marks obtained by 10 children in a mental drill are:
0, 1, 3, 3, 5, 7, 8, 9, 9, 9. Calculate the mean mark.

A) –54 B) 5.4 C) 10 D) 54 E) 540

18. The marks obtained by 10 children in a mental drill are: 0, 1, 3, 3, 5, 7, 8, 9, 9, 9.
What is the probability that a child chosen at random scored 3 marks?

A) $^{2}/_{54}$ B) $^{3}/_{54}$ C) $^{2}/_{10}$ D) $^{3}/_{10}$ E) $^{1}/_{3}$

19. A trader received a commission of 12 ½ % on sales made in a month. His commission was ¢35,000.00. Find his total sales for the month.

A) ¢36,250.00
B) ¢59,750.00
C) ¢245,000.00
D) ¢280,000.00
E) ¢315,000.00

20. A map of a large town is drawn to the scale of 1:100,000. What is the distance in kilometres (km) represented by a line segment 4cm along the map?

A) 0.04km B) 0.4km C) 4km D) 40km E) 400km

21. Adjoa and Ama share ¢600.00 between them in the ratio 3:2. Find Adjoa's share.

A) ¢200.00 B) ¢240.00 C) ¢300.00 D) ¢360.00 E)¢400.00

22. Simplify: $\frac{2a}{3} - \frac{a-b}{2}$

A) $a - 3b$ B) $\frac{a-3b}{6}$ C) $\frac{a+3b}{6}$ D) $a + 3b$ E) $a - b$

23. If $a^6 \div a^4 = 64$. Find a

A) 8 B) 10 C) 16 D) 20 E) 24

24. Find one-hundredth of 1.0756

A) 107.56 B) 10.756 C) 0.01756 D) 0.010756 E) 0.001756

25. What is the rule for this mapping

x	1	2	3	4	5
↓	↓	↓	↓	↓	↓
y	1	3	5	7	9

A) $x \rightarrow 2x-1$ B) $x \rightarrow 2(x-1)$ C) $x \rightarrow 2x+1$ D) $x \rightarrow 2(x+1)$ E) $x \rightarrow 2^x-1$

26. The circumference of a circular track is 440m. Find the diameter of this track.

[Take $\pi = {}^{22}/_{7}$]

A) 70m B) 140m C) 280m D) 690m E) 1400m

27. If $\mathbf{r} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{s} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$, calculate $6(\mathbf{r} + 2\mathbf{s})$

A. $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$ B. $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ C. $\begin{pmatrix} 7 \\ 3 \end{pmatrix}$ D. $\begin{pmatrix} -6 \\ 18 \end{pmatrix}$ E. $\begin{pmatrix} 6 \\ 18 \end{pmatrix}$

28. Which of the following would you use to measure an angle?

A) Ruler

B) A pair of compasses

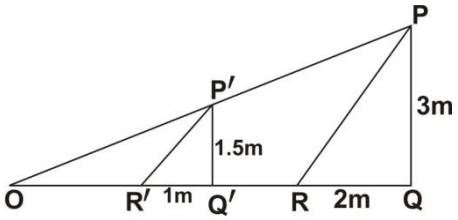
C) A set square

D) A protractor

29. Express the ratio of 64cm to 48cm in its simplest form.

A) 3:4 B) 4:3 C) 16:12 D) 12:16 E) 64:48

30.



In the diagram, P→P′, Q→Q′, R→R′, where P′Q′R′ is an enlargement.

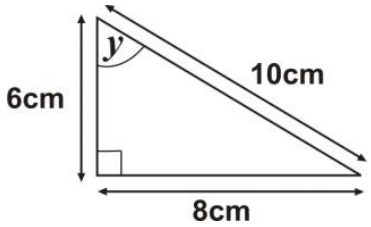
What is the scale factor of this enlargement?

A) –2 B) ${}^{1}/_{3}$ C) ½ D) 2 E) 3

31. If |PR| = 3.6m, what is |P′R′| ?

A) –7.2m B) –1.8m C) 1.2m D) 1.8m E) 7.2m

32. Find the tangent of the angle marked Y in the diagram



A) $^{3}/_{5}$ B) ¾ C) $^{4}/_{5}$ D) $^{4}/_{3}$ E) $^{5}/_{3}$

33. Kojo paid ¢270,000.00 for a TV set after he had been given a discount of 10%. Find the marked price.

A) ¢300,000.00
B) ¢297,000.00
C) ¢280,000.00
D) ¢260,000.00
E) ¢243,000.00

In a secondary school class, 23 pupils study Economics, 6 pupils study both Government and Economics. 48 pupils study either Government or Economics or both.

34. What is the total number of pupils who study Government?

A) 17 B) 23 C) 24 D) 25 E) 31

35. How many pupils study only Government?

A) 17 B) 23 C) 24 D) 25 E) 31

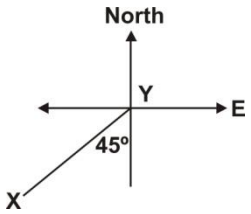
36. How many pupils study only Economics?

A) 17 B) 23 C) 24 D) 25 E) 31

37. When a certain number is subtracted from 10 and the result is multiplied by 2, the final result is 4. Find the number.

A) 8 B) 12 C) 16 D) 24 E) 32

38. From the diagram below, calculate the bearing of point X from Y

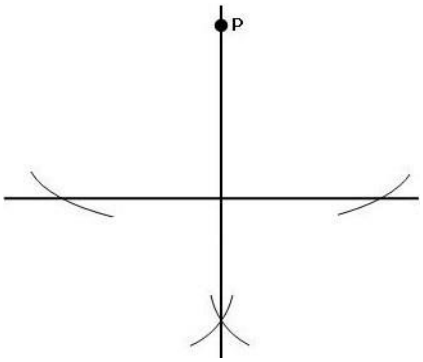


A) 035° B) 045° C) 135° D) 145° E) 225°

39. If 22% of the length of a rope is 55cm, find the full length of the rope.

A) 12.1cm B) 25cm C) 121cm D) 250cm E)2500cm

40. Which of these best describes the given construction?



A) Bisecting a line

B) Constructing the bisector of a line segment.

C) Constructing the perpendicular to a line

D) Constructing a perpendicular to a given line from a point outside the line.

E) Constructing a perpendicular to a given line through a point on the line.

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August 1991

MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. E) 10

2. B) $Y \subset V$

3. B) 1720

4. D) 7

5. D) 4.68702×10^3

6. D) 25

7. D) $^{17}/_{35}$

8. A) 21

9. C) 72

10. E) 124°

11. A) 68°

12. C) 105°

13. C) $6a^3b^3$

14. B) 26

15. E) 9

16. C) 6

17. B) 5.4

18. C) $^{2}/_{10}$

19. D) ¢280,000.00

20. C) 4km

21. D) ¢360.00

22. C) $\frac{a+3b}{6}$

23. A) 8

24. D) 0.010756

25. A) $x \rightarrow 2x\text{-}1$

26. B) 140 m

27. D. $\begin{pmatrix} -6 \\ 18 \end{pmatrix}$

28. D) A protractor

29. B) 4:3

30. D) 2

31. D) 1.8m

32. D) $^{4}/_{3}$

33. A) ¢300,000.00

34. E) 31

35. D) 25

36. A) 17

37. A) 8

38. E) 225°

39. D) 250cm

40. D) Constructing a perpendicular to a given line from a point outside the line.

August 1991

MATHEMATICS

SECTION B

ESSAY

1 hour

[60 marks]

1. (a) If X = {Prime numbers less than 13} and Y = {odd numbers less than 13}

(i) List the members of X and Y

(ii) List the members of {X ∩ Y} and {X U Y}

(b) Three school children share some oranges as follows: Akwasi gets 1/3 of the total, and the remainder is shared between Abena and Jantuah in the ratio 3:2. If Jantuah gets 24 oranges, how many does Akwasi get?

2. Using a ruler and a pair of compasses only,

a) construct the triangle XYZ, in which |YZ| = 6cm, angle XYZ = 60° and |XZ| = 9cm. Measure |XY|

b) (i) construct the mediator of YZ.

(ii) draw a circle, centre X and radius 5cm. Measure |YA|, where A is the point of intersection of the mediator and the circle in the triangular region XYZ

3. (a) Solve the equation: $\frac{2x-1}{3} - \frac{x-2}{4} = 1$

(b) Given that $m = -2$ and $n = ¾$, find the value of

(i) $m^2 (n - 1)$ (ii) $n^2 - \frac{3}{m}$

(c) Factorize completely 2ap + aq + bq – 2bp

4. (a) The following table shows the distribution of votes in an election for class prefect.

Name	**Number of votes**
Acquaye	6
Borquaye	12
Commey	18

(i) Draw a pie chart to illustrate the distribution.

(ii) What fraction of the votes was cast for Borquaye?

(b) The heights in cm of 10 school children are as follows:

165, 165, 155, 159, 174,

154, 169, 155, 155, 150

Make a frequency table for this data

Use your table to find the mode and median of the distribution.

August 1991

MATHEMATICS

SECTION B
ESSAY
ESSAY

SOLUTIONS

1 (a)

(i) X = {2, 3, 5, 7, 11}

Y = {1, 3, 5, 7, 9, 11}

(ii) $X \cap Y$ = {3, 5, 7, 11}

$X \cup Y$ = {1, 2, 3, 5, 7, 9, 11}

1 (b) Since Akwasi gets ⅓ of the total, remaining fraction = 1 – ⅓ = ⅔

⇒ the remainder = ⅔ of total

⇒ Abena's + Jantuah's = ⅔ of total

	Abena	**Jantuah**
Ratio	3	**2**
No. of oranges	***a***	**24**

From the table above, $\frac{3}{a} = \frac{2}{24}$

⇒ $3 \times 24 = 2 \times a$ ⇒ $2 \times a = 24 \times 3$,

⇒ $a = \frac{24 \times 3}{2} = 36$

⇒ Abena gets 36 oranges

Hence, ⅔ of the total → (36 + 24) oranges = 60 oranges

	Fraction	No. of oranges
Abena + Jantuah	⅔	36 + 24 **= 60**
Akwasi	⅓	**?**

Approach 1

Now, if ⅔ → 60 oranges

Then ⅓ → ? (less)

If less, more (i.e., ⅔) divides, hence we have

$$\frac{1/3}{2/3} \times 60\,\text{oranges} = \frac{1}{3} \div \frac{2}{3} \times 60\,\text{oranges}$$

$$= \frac{1}{3} \times \frac{3}{2} \times 60\,\text{oranges}$$

$$= \underline{30\ \text{oranges}}$$

$$\Rightarrow \text{Akwasi gets } \underline{30\ \text{oranges}}$$

Approach 2

$$\frac{2/3}{1/3} = \frac{60\,\text{oranges}}{x}$$

Cross-multiplying, we have $\frac{2}{3}x = \frac{1}{3} \times 60\,\text{oranges}$

Multiplying both sides by $^3/_2$ (to remove $^2/_3$) :

$$\frac{3}{2} \times \left(\frac{2}{3}x\right) = \frac{3}{2} \times \left(\frac{1}{3} \times 60\,\text{oranges}\right)$$

$$x = \frac{3}{2} \times \frac{1}{3} \times 60\,\text{oranges}$$

$$x = \underline{30\ \text{oranges}}$$

$$\Rightarrow \text{Akwasi gets } \underline{30\ \text{oranges}}$$

Approach 3

Let the total no.= T, then ⅔ of T = 60 oranges

$$\Rightarrow \frac{2}{3} \times T = 60$$

Solving for T by multiplying through by $\frac{3}{2}$, we have

$$\tfrac{3}{2} \times \tfrac{2}{3} T = \tfrac{3}{2} \times 60 \qquad \Rightarrow \quad T = \tfrac{3}{2} \times 60 = 90$$

⇒ The total no. of oranges = 90 oranges

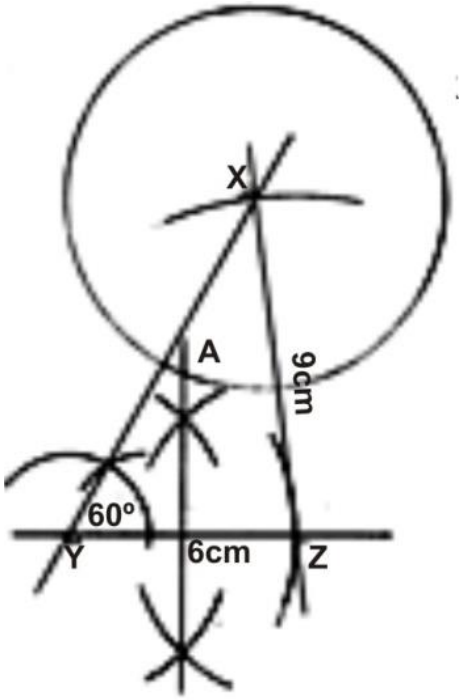
⇒ Akwasi's share = ⅓ of total

= ⅓ × 90 oranges

= <u>30 oranges</u>

⇒ Akwasi gets <u>30 oranges</u>

2)



(a) |XY| ≈ <u>10.3 cm</u>

(b) |YA| ≈ <u>5.7 cm</u>

3 (a)

$$\frac{2x-1}{3} - \frac{x-2}{4} = 1$$

Approach 1 (Simplifying the LHS first)

$$\frac{4(2x-1)-3(x-2)}{12} = 1$$

$$\Rightarrow \quad \frac{8x-4-3x+6}{12} = 1$$

$$\Rightarrow \quad \frac{5x+2}{12} = 1$$

$$\Rightarrow \quad 5x + 2 = 1 \times 12$$

$$\Rightarrow \quad 5x = 12 - 2$$

$$\Rightarrow \quad 5x = 10$$

$$\Rightarrow \quad \frac{5x}{5} = \frac{10}{5}$$

$$\Rightarrow \quad \underline{\underline{x = 2}}$$

3 (a)

Approach 2 (Multiplying by 12 – to clear fractions)

$$12 \times \left(\frac{2x-1}{3}\right) - 12 \times \left(\frac{x-2}{4}\right) = 12 \times 1$$

$$\Rightarrow \quad 4(2x-1) - 3(x-2) = 12$$

$$\Rightarrow \quad 8x - 4 - 3x + 6 = 12$$

$$\Rightarrow \quad 8x - 3x = 12 + 4 - 6$$

$$\Rightarrow \quad 5x = 10$$

$$\Rightarrow \quad x = 10 \div 5$$

$$\Rightarrow \quad \underline{\underline{x = 2}}$$

3 (b) Given that $m = -2$ and $n = \frac{3}{4}$

(i)

$$m^2 (n-1)$$

$$\Rightarrow \quad (-2)^2 \left(\tfrac{3}{4} - 1\right)$$

$$= \quad 4 \left(-\tfrac{1}{4}\right)$$

$$= \quad \underline{\underline{-1}}$$

(ii)

$$n^2 - \frac{3}{m}$$

$$\Rightarrow \quad \left(\frac{3}{4}\right)^2 - \frac{3}{-2}$$

$$= \quad \frac{9}{16} + \frac{3}{2}$$

$$= \quad \frac{9 \ + \ 24}{16}$$

$$= \quad \frac{23}{16} \quad = \quad \underline{\underline{1\,^{7}\!/_{16}}}$$

3 (c)

$$2ap + aq - bq - 2bp$$

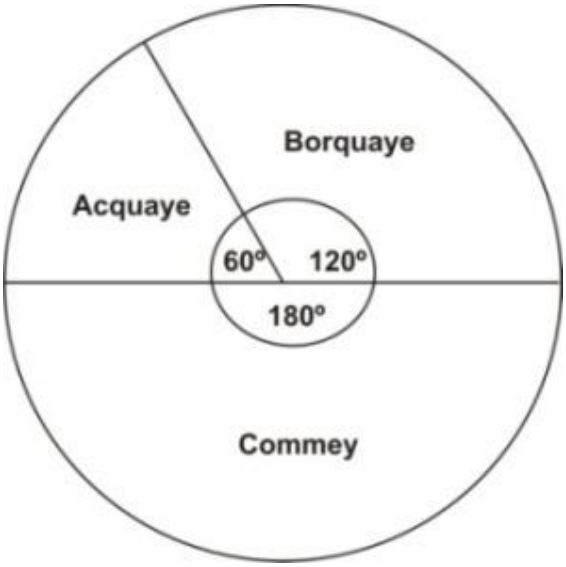
$$= \quad a\,(2p + q) - b\,(q + 2p)$$

$$= \quad \underline{(2p + q)\,(a - b)}$$

4 (a) (i)

Name	**Number of votes**	**Angle of sector**
Acquaye	**6**	$\frac{6}{36} \times 360^{o} = 60^{o}$
Borquaye	**12**	$\frac{12}{36} \times 360^{o} = 120^{o}$
Commey	**18**	$\frac{18}{36} \times 360^{o} = 180^{o}$
TOTAL	**36**	**360°**

Pie chart showing the distribution of votes in an election for class prefect



4 (a) (ii) Borquaye's fraction

Approach 1

$$\frac{\text{Borquaye's number of votes}}{\text{Total number of votes}} = \frac{12}{36} = \frac{1}{3}$$

4 (a) (ii) **Approach 2**

$$\frac{\text{Borquaye's angle}}{\text{Total angle}} = \frac{120}{360} = \frac{1}{3}$$

4 (b) (i)

Frequency table

Height (in cm)	Tally	Frequency
150	/	1
154	/	1
155	///	3
159	/	1
165	//	2
169	/	1
174	/	1

(ii) (α) From the table, the mode = 155 cm

(ii) (β) **Finding the median**

Approach 1

Listing all the given heights, we have

150, 154, 155, 155, 155, 159, 165, 165, 169, 174

The middle heights (in cm) are 155 and 159

$\therefore$ The median $= \dfrac{155\,\text{cm} + 159\,\text{cm}}{2}$

$= \dfrac{314}{2}\,\text{cm} = 157\ \text{cm}$

(ii) (β) **Approach 2**

The median

= the height (in cm) at the ½(Σf+1)th position

$\Rightarrow$ the ½ (10+1)th position

= ½ (11) = 5½th position

The height at the 5½th position

= the average of the heights at the 5th and 6th positions

$= \dfrac{155\,cm + 159\,cm}{2} = 157\,\text{cm}$

August 1990

MATHEMATICS

SECTION A
OBJECTIVE TEST
1 hour

1. If P = {7, 9, 13}, Q = {1, 7, 13}. Find P ∩ Q

A. {1, 7, 13}

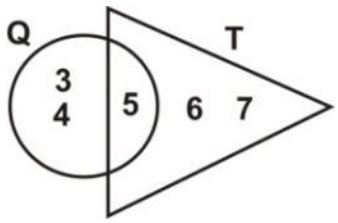
B. {1, 9, 13}

C. {7, 13}

D. {7, 9, 13}

E. {13}

2. In the diagram, Q is the set of numbers inside the circle and T is the set of numbers inside the triangle. Find Q U T.



A. {5}

B. {6, 7}

C. {3, 4, 5}

D. {5, 6, 7}

E. {3, 4, 5, 6, 7}

3. Given that (23 × 82) × 79 = 148,994, find the exact value of (2.3 × 82) × 7.9

A) 14.8994 B) 148.994 C) 1489.94 D) 14899.4 E) 148994.0

4. Convert 25_{ten} to a base two numeral

A) 1000 B) 100111 C) 10101 D) 11001 E) 11100

5. If $(3.14 \times 18) \times 17.5 = 3.14 \times (3a \times 17.5)$. Find the value of a.

A) 3.0 B) 5.8 C) 6.0 D) 9.0 E) 18.0

6. If 26039 oranges are shared equally among 13 women, how many oranges does each woman receive?

A) 23 B) 1203 C) 230 D) 2003 E) 2300

7. Mr. Mensah withdrew some money from the bank. He gave ½ of it to his sons and ⅓ to his daughter. If he had ¢500.00 left, how much did he take from the bank?

A) ¢600.00 B) ¢750.00 C) ¢1500.00 D) ¢2000.00 E) ¢3000.00

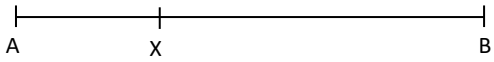
8. Simplify ½ (1 ½ + ¾ ÷ ¼)

A) 1 ½ B) 2 ¼ C) 2 ¾ D) 4 ½ E) 6

9. If 21:2x = 7:10, find x.

A) 3 B) 2 ¼ C) 15 D) 35 E) 50

10.



X is a point on the line segment AB such that |AB| = 10cm and |AX| = 4cm. Find the ratio |AX| : |XB|

A) 5:3 B) 3:2 C) 5:4 D) 1:1 E) 2:3

11. In an examination, 60% of the candidates passed. The number that passed was 240. How many candidates failed?

A) 140 B) 160 C) 360 D) 400 E) 600

12. A table which cost ¢2,400.00 to manufacture was sold for ¢3,000.00. Find the profit percent.

A) 80% B) 25% C) 24% D) 20% E) 11.1%

13. If $a = 2^2 \times 2^3 \div 2^4$, find the value of a

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A) 2^9 B) 2^5 C) 2^2 D) 2 E) 2^0

14. The distance between two towns is 12875km.
Express this distance in standard form.

A) 1.2875×10^3 km

B) 1.2875×10^4 km

C) 12.875×10^3 km

D) 128.75×10^2 km

E) 12.875×10^4 km

The pie chart shows the monthly expenditure of Mr. A

whose monthly income is ¢18,000.00.



15. What fraction of Mr. A's income is spent on food?

A) $^1/_6$ B) ¼ C) $^1/_3$ D) $^2/_5$ E) ½

16. How much does Mr. A spend on rent?

A) ¢90.00

B) ¢450.00

C) ¢4,500.00

D) ¢9,000.00

E) ¢16,200.00

17. What is the size of the angle that represents savings?

A) 40° B) 60° C) 130° D) 230° E) 320°

18. Find the missing addend

20.45
****
19.18
44.30

A) 83.39 B) 19.69 C) 25.12 D) 23.85 E) 4.67

19. Remove the brackets: a – 2(b – 3c)

A) a–2b–3c

B) a–2b - 6c

C) a–2b+6c

D) a+2b+6c

E) a–2b+3c

20. Simplify $\frac{2a}{3} - \frac{a-b}{2}$

A) a – 3b B) $\frac{a-3b}{6}$ C) $\frac{a+3b}{6}$ D) $a + 3b$ E) $a - b$

21. If q = ut + ½ ft, find q when u = 20, t = 10 and f = 15

A) 350 B) 275 C) 237.5 D) 55 E) 42.5

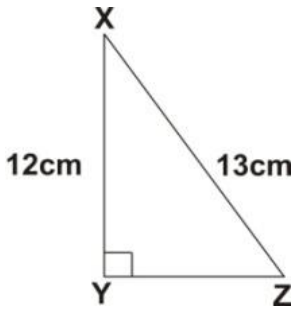
22. If $\frac{3}{15}$ is equivalent to $\frac{45}{a}$, find a

A) 225 B) 150 C) 135 D) 30 E) 9

23. Find the least common multiple of 7, 14 and 18

A) 71418 B) 1764 C) 252 D) 126 E) 98

24.



In triangle XYZ, |XZ| = 13cm, |XY| = 12cm and angle XYZ = 90°. Find |YZ| if the area of the triangle is $30cm^2$.

A) 25 cm B) 14 cm C) 12.5 cm D) 5 cm E) 1 cm

25. Write 1204_{five} as a number in base ten.

A) 9996 B) 179 C) 39 D) 35 E) 19

26. Multiply (2x + y) by (2x – y)

A) $4x^2–4xy–y^2$

B) $4x^2+xy–y^2$

C) $4x^2–xy–y^2$

D) $4x^2+y^2$

E) $4x^2–y^2$

27. A watchman was paid a basic wage of ¢250.00 a day. If he worked every day in the month, calculate his basic wage for February 1988.

A) ¢6250.00

B) ¢7200.00

C) ¢7250.00

D) ¢7750.00

E) ¢8750.00

28. A tank contains 250 litres of water. If 96 litres are used, what percentage of the original quantity is left?

A) 61.6% B) 60.5% C) 59.0% D) 54.2% E) 38.4%

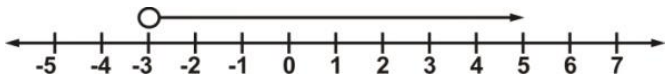
29. Evaluate $10 \div \left(3\frac{1}{2} + 1\frac{1}{5}\right)$

A) $2\frac{6}{47}$ B) $2\frac{1}{6}$ C) 4 D) $4\frac{2}{35}$ E) $4\frac{7}{10}$

30. A bag contains 24 marbles, 10 of which are blue and the rest green. A boy picks a marble at random from the bag. What is the probability that he picks a green marble?

A) $^1/_{14}$ B) $^7/_{17}$ C) $^5/_{12}$ D) $^7/_{12}$ E) $^7/_{10}$

31.



Which of the following inequalities is represented on the number line, where n is a real number?

A) $n \geq -3$ B) $n < -3$ C) $n \leq -3$ D) $n > -3$ E) $n = 3$

32. What is the name of the line segment drawn to join any two points on the circumference of a circle?

A) arc B) chord C) radius D) sector E) segment

Use the mapping to answer questions **33** and **34**

½	→	1	→	3.14
1	→	2	→	6.28
2	→	4	→	12.56
3	→	6	→	***x***

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33. Find the value of ***x***

A) 9.42 B) 12 C) 18 D) 18.84 E) 25.12

34. Find the value of ***y***

A) 2 B) 5 C) 7 D) 9 E) 10

35. The area of a square is $49cm^2$. Find the perimeter of the square.

A) 7cm B) 14cm C) 28cm D) 49cm E) 196cm

36. The least number in a set of real numbers is 24 and the greatest is 30. Which of the following is the correct interpretation of the statement?

A) $24 \leq x \leq 30$

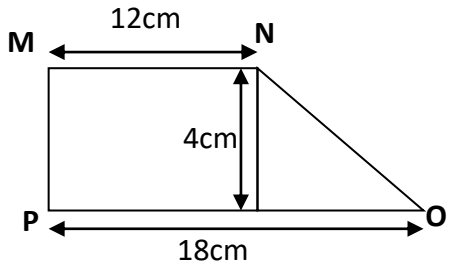
B) $24 < x < 29$

C) $23 < x < 29$

D) $24 < x < 30$

E) $23 \leq x \leq 29$

37.



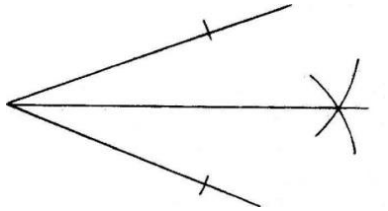
Find the area of the trapezium MNOP.

A) $120cm^2$ B) $72cm^2$ C) $60cm^2$ D) $48cm^2$ E) $38cm^2$

38. The length of a field, 1.2 km long is represented on a map by a line 40 mm long. What is the scale of the map?

A) 1:100 B) 1:300 C) 1:1000 D) 1:3000 E) 1:30000

39.



The diagram shows the construction for:

A) Copying a given angle

B) Bisecting a line segment

C) Drawing a perpendicular to a given line

D) Bisecting a given angle

E) Drawing an angle of 30°

40. Simplify $\begin{pmatrix} 2 \\ -3 \end{pmatrix} - \begin{pmatrix} -1 \\ 5 \end{pmatrix}$

A. $\begin{pmatrix} -3 \\ 8 \end{pmatrix}$ B. $\begin{pmatrix} 3 \\ -8 \end{pmatrix}$ C. $\begin{pmatrix} -3 \\ -8 \end{pmatrix}$ D. $\begin{pmatrix} 3 \\ 8 \end{pmatrix}$ E. $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$

August 1990

MATHEMATICS

SECTION A
OBJECTIVE TEST

SOLUTIONS

1. C. {7, 13}

2. E. {3, 4, 5, 6, 7}

3. C) 1489.94

4. D) 11001

5. C) 6.0

6. D) 2003

7. E) ¢3000.00

8. B) 2 ¼

9. C) 15

10. E) 2:3

11. B) 160

12. B) 25%

13. D) 2

14. B) 1.2875×10^4

15. C) $^1/_3$

16. C) ¢4,500.00

17. A) 40°

18. E) 4.67

19. C) a–2b+6c

20. C) $\frac{a+3b}{6}$

21. B) 275

22. A) 225

23. D) 126

24. D) 5cm

25. B) 179

26. E) $4x^2–y^2$

27. C) ¢7250.00

28. A) 61.6%

29. A) $2\frac{6}{47}$

30. D) $^{7}/_{12}$

31. D) n > -3

32. B) chord

33. 33. D) 18.84

34. 34. B) 5

35. C) 28cm

36. A) $24 \leq x \leq 30$

37. C) 60 cm^2

38. E) 1:30000

39. D) bisecting a given angle

40. B. $\begin{pmatrix} 3 \\ -8 \end{pmatrix}$

August 1990

MATHEMATICS

SECTION B
ESSAY
1 hour

[60 marks]

1. List the members of each of the sets

B = {Whole numbers from 20 to 30}. D = {factors of 63}

(a) List the members of (i) $B \cap D$

(ii) $B \cup D$

(b) In a class of 60 students, 46 passed Mathematics and 42 passed English language. Everybody passed at least one of the two subjects.

(i) Illustrate this information on a Venn diagram

(ii) How many students passed in both subjects?

2. (a) Factorize completely $3a^2 + 2ab - 12ac - 8bc$

(b) Solve $\frac{x}{4} + \frac{3}{5} = \frac{3x}{5} - 2$

(c) Find the solution set of $x + 3 > 19 - 3x$, where x is a real number.

Illustrate your answer on the number line.

3. (a) Using a ruler and a pair of compasses only

(i) Construct triangle PQR such that |PQ| = 6cm, |QR| = 4cm and angle PQR = 90°

(ii) Construct the perpendicular bisectors of PQ and QR. Name the intersection O.

(iii) Draw a circle O as centre and OQ as radius

(b) Measure (i) |PR| (ii) angle QPR

4. The following is the result of a survey conducted in a class of a junior secondary school to find the favourite soft drink of each pupil in the class.

Soft Drink	**Number of pupils**
Coca-cola	6
Pepsi-cola	5
Pee-cola	8
Fanta	3
Muscatella	5
Mirinda	4
Club cola	6
Sprite	3

(a) Draw a bar chart showing this information, using a scale of 2cm to 1unit on the vertical axis.

(b) How many pupils are in the class?

(c) What is the percentage of pupils who prefer club-cola?

August 1990

MATHEMATICS

SECTION B
ESSAY

SOLUTIONS

1 (a)

B = {20, 21, 22, 23, …, 30}

D = {1, 3, 7, 9, 21, 63}

(i) $B \cap D$ = {21}

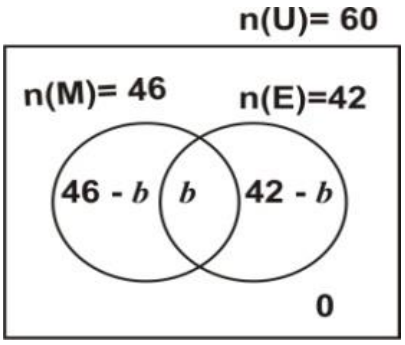
(ii) $B \cup D$ = {1, 3, 7, 9, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 63}

1 (b) (i) Let n(U) = No. of students in the class

n(M) = No. of students that passed Maths

n(E) = No. of students that passed English

b = No. of students that passed both Maths and English



(b) (ii) From the diagram above,

46 – b + b + 42 – b + 0 = 60

$\Rightarrow \quad 46 + 0 + 42 - b + 0 = 60$

$\Rightarrow \quad 46 + 42 - b = 60$

$\Rightarrow \quad 88 - b = 60$

$\Rightarrow \quad 88 - 60 = b$

$\Rightarrow \quad \underline{b = 28}$

$\therefore$ 28 students passed in both subjects

2 (a)

$3a^2 + 2ab - 12ac - 8bc$

$= a(3a + 2b) - 4c(3a + 2b)$

$= \underline{(3a + 2b)(a - 4c)}$

2 (b)

$$\frac{x}{4} + \frac{3}{5} = \frac{3x}{5} - 2$$

Approach 1 (Grouping and simplifying first)

$$\Rightarrow \quad 2 + \frac{3}{5} = \frac{3x}{5} - \frac{x}{4}$$

$$\Rightarrow \quad \frac{10 + 3}{5} = \frac{12x - 5x}{20}$$

$$\Rightarrow \quad \frac{13}{5} = \frac{7x}{20}$$

$$\Rightarrow \quad 13 \times 20 = 7x \times 5$$

$$\Rightarrow \quad \frac{13 \times 20}{7 \times 5} = \frac{7x \times 5}{7 \times 5}$$

$$\Rightarrow \quad \frac{52}{7} = x$$

$$\Rightarrow \quad \underline{\underline{x = 7\frac{3}{7}}} \quad \text{or}$$

$$\Rightarrow \quad \underline{\underline{x \approx 7.429}}$$

2 (b) **Approach 2** (Multiplying by 20 first – to clear fractions)

$$\frac{x}{4}+\frac{3}{5}=\frac{3x}{5}-2$$

$$20\times\left(\frac{x}{4}+\frac{3}{5}\right)=20\times\left(\frac{3x}{5}-2\right)$$

$$\Rightarrow \quad 20\times\frac{x}{4}+20\times\frac{3}{5}=20\times\frac{3x}{5}-20\times 2$$

$$\Rightarrow \quad 5\times x+4\times 3=4\times 3x-20\times 2$$

$$\Rightarrow \quad 5x+12=12x-40$$

$$\Rightarrow \quad 40+12=12x-5x$$

$$\Rightarrow \quad 52=7x$$

$$\Rightarrow \quad 7x=52$$

$$\Rightarrow \quad x=52\div 7$$

$$\Rightarrow \quad \underline{\underline{x=7\tfrac{3}{7}}} \text{ or}$$

$$\Rightarrow \quad \underline{\underline{x\approx 7.429}}$$

2 (c)

$$x+3>19-3x$$

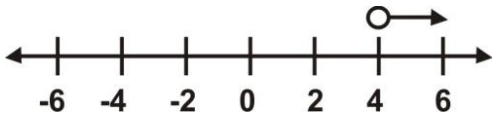
$$\Rightarrow \quad x+3x>19-3$$

$$\Rightarrow \quad 4x>16$$

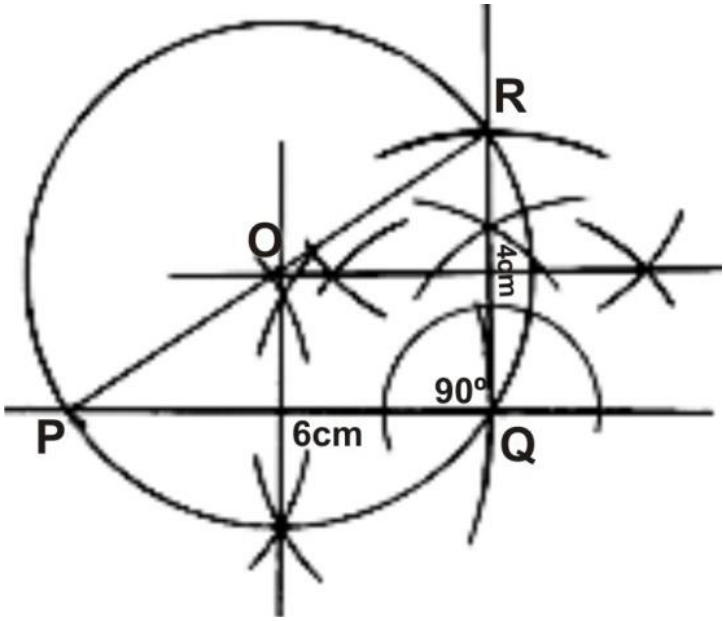
$$\Rightarrow \quad x>16\div 4$$

$$\Rightarrow \quad \underline{\underline{x>4}}$$

The number line illustration is shown below:



3 (a)

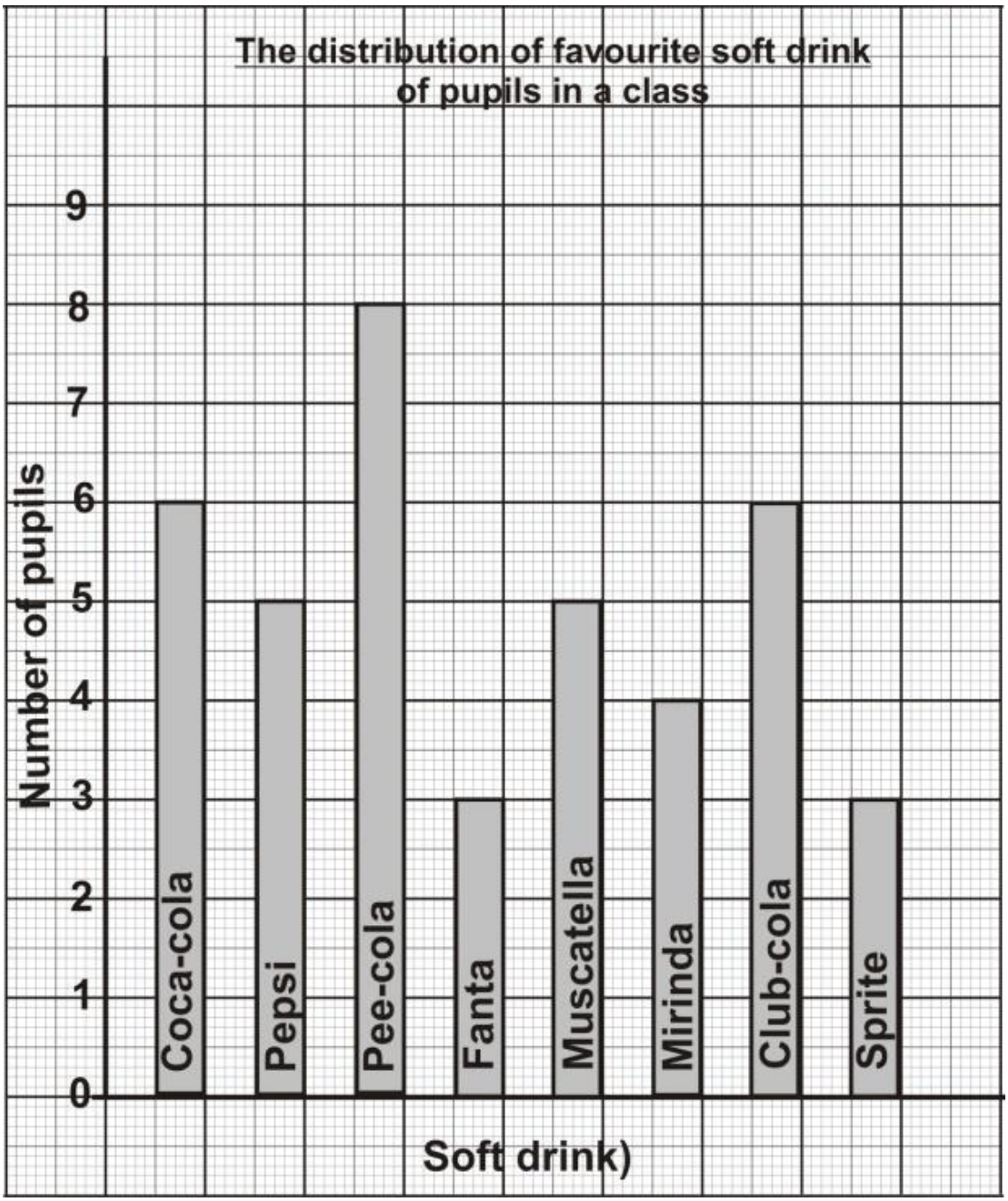


(b) **(i)** |PR| ≈ 7.2 cm

(ii) angle QPR ≈ 34°

4 (a)

Vertical AxisScale: 2 cm to 1 pupil



4 (b) Total number of pupils

$= 6 + 5 + 8 + 3 + 5 + 4 + 6 + 3$

$= \underline{40}$

(c) Percentage of pupils who prefer Club-cola

$$= \frac{\text{No. of pupils who prefer club-cola}}{\text{Total no. of pupils}} \times 100\%$$

$$= \frac{6}{40} \times 100\% \qquad = \underline{15\ \%}$$

⇒15% of the pupils prefer Club-cola

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